

ACTION RIGIDITY: COMPANION DERIVATIONS

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ABSTRACT. This companion preserves the shared-apparatus, polynomial, rational, meromorphic, geometric, and acquisition regimes accompanying *Action recovery from projective matrix curves*. It includes all mathematical inputs of the preceding expanded edition, with their own hypotheses and proofs. Additional arguments give a finite-level continuity lemma, a quantitative four-anchor calibration certificate, explicit determinant perturbation bounds, and precise period and branched-fibre conventions. The principal article gives the arbitrary-rank singular spectrum theorem; the present volume provides the other observation regimes and their derivations without conflating their nuisance assumptions.

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1. INTRODUCTION

An action enters a clocked observation through a factor $T - x$. Unknown selection changes its amplitude, normalization removes total mass, and unknown readout channels conceal the identity of the component. Recovering x requires separating these three losses of information. A fourth distinction is equally important: a detector that can be rechosen independently at every site is not the same nuisance as one shared over an observation domain. This paper studies the resulting observation quotients, including their singular statistical boundary.

The first main result concerns spatial sharing. Suppose the latent odds at two clocks are $r_j(z) = k_j(T_j - x(z))$, where the positive constants k_j are unknown and common to the sites, while the two full-rank readout channels may vary with z . If the continuous action is not constant on any open set, the observed matrix fields determine it and both constants. Theorem 2.2 proves this without supplying labels or excluding isolated profile coincidences. The proof identifies the exact obstruction to spatial calibration: a false choice of clock constants allows directly matched action values at at most one level and interchanged values at at most two levels. Four nondegenerate calibration sites therefore determine the clock constants by a finite calculation. On a simple surface, this gives a sharp two-clock geometric threshold for a shared detector and

constant unknown reference. The one-clock lower bound uses positive full-rank channels and distinct metrics, while keeping that reference and detector fixed.

The second main result describes the price of near rank deficiency. At one site with binary channels, a constant relative detector and three clocks, write $\rho = |\det U \det V|$. Theorem 3.2 proves, uniformly against all candidates in the closed forward model, the action bound

$$|x - x'| \leq C \min\{1, \|P - P'\|_*/\rho\}.$$

The key step is to take the determinant after clearing the common pole of the matrix curve. Its coefficient error has no extra inverse power of the visible contrast. This yields honest confidence intervals with locally adaptive expected length of order $\min\{1, (\sqrt{N}\rho)^{-1}\}$ and a minimum-residual estimator with squared risk of order $\min\{1, (N\rho^2)^{-1}\}$. Theorems 3.3 and 3.5 give matching uniform risk bounds and matching honest-length lower bounds at explicit least-favourable parameters. Along the lower-bound family both observed marginals remain exactly fixed; only a joint interaction of size ρ changes. Thus two equally weak channels of contrast ϵ have transition $N\epsilon^4 \asymp 1$, rather than an unspecified loss of conditioning. At zero contrast the family is exactly indistinguishable, and honest intervals need not shrink. The estimator does not receive a lower bound on ρ .

The third result is an intrinsic detector classification. On a compact real Riemann surface, replace the logarithmic action curve by primitives of normalized third-kind differentials. A finite-dimensional detector space consists of primitives of meromorphic differentials with a fixed pole divisor. Theorem 4.2 gives necessary and sufficient tests involving integral residue charges and normalized second-kind directions. The tests use periods as well as principal parts. Theorem 4.3 supplies the uniform clock budget $\deg D + 2g + 4$. On the sphere this reduces exactly to the rational polar budget; on positive-genus surfaces it gives genuinely nonrational detector examples. Theorem 4.4 further retains the original factor $T - x$ under a finite meromorphic clock map, allowing, for example, elliptic-integral detector primitives. The budget is sufficient, not asserted minimal for every meromorphic subspace. The sharper polynomial count uses the additional structure of oriented intervals.

1.1. The role of the observation quotient. There are three different quotients in these conclusions. At a single site the logarithmic action curve is taken modulo a detector function space; its secants and tangents give exact fibres and local inversion. For shared apparatus the nuisance consists of functions of the clock index alone on the product of clock and spatial domains; spatial contrasts and the residual permutation ambiguity determine a different threshold. At finite sample size the relevant object is the modulus of the full matrix observation map, after allowing its unknown channels. The product theorem computes that modulus rather than inserting an unknown condition number into a statistical bound.

These distinctions also delimit the geometric conclusions. Once an experiment determines boundary distances, a separate geometric theorem is needed to determine a metric. The surface rigidity input is the Pestov–Uhlmann theorem [33]. The finite noisy metric statements later in the paper use the specified local stability class and yield confidence and stability certificates modulo boundary-fixing diffeomorphisms. They do not claim an efficient inversion algorithm for an arbitrary infinite-dimensional metric class, a new global boundary-rigidity theorem, or a new partial-data geometric theorem. The new sharp threshold concerns which selective observations determine the geometric theorem’s input.

1.2. Relation to earlier work and organization. Observable matrix pencils belong to the established identification theory of latent models [21, 23]. We use that factorization and resolve its remaining permutations by spatial sharing; the factorization itself is not a novelty claim. Modulus-based recovery and honest adaptation have classical precedents in [41, 42, 43]. Our

statistical contribution is the explicit uniform determinant-product modulus and its attaining unknown-channel family, not a general two-point lower-bound method. Normalized differentials and divisor degrees are classical tools [44, 45]; our additional statement classifies which unknown meromorphic detector subspaces preserve an action quotient. Closest-predecessor comparisons are given again at the corresponding theorems.

Sections 2–4 prove these three results. The subsequent part retains the pointwise polynomial and nonpolynomial inverses, rational examples, complete fibres, finite-record covering reconstruction, geometric certificates, raw acquisition bounds and exact lift constructions. Their observation hypotheses are not changed by the shared-apparatus result. The expanded edition additionally preserves the companion observation regimes and historical derivations, with their own hypotheses. No conclusion from one regime is silently used as an observation premise in another.

2. SPATIAL SELF-CALIBRATION WITH TWO CLOCKS

A clock count has meaning only relative to the indices over which the nuisance may vary. We first treat a shared apparatus. The readout channels remain unknown and may vary with the observation site.

Let Q be a connected smooth domain without isolated points, and fix $T_1 < T_2$. At $z \in Q$ let $x(z) < T_1$ be continuous. We assume that x is not constant on any nonempty open subset of Q . Let $U(z)$ and $V(z)$ be continuous probability-column matrices, with two columns and full column rank. For two unknown constants $k_1, k_2 > 0$ put

$$r_j(z) = k_j(T_j - x(z)), \quad p_j(z) = \frac{r_j(z)}{1 + r_j(z)}, \quad P_j(z) = U(z) \operatorname{diag}(p_j(z), 1 - p_j(z)) V(z)^\top. \quad (2.1)$$

The matrices P_j are the normalized projective observations. In the two-arm apparatus this model follows from a constant unknown reference τ and a shared relative detector law c :

$$k_j = \frac{\exp c(T_j)}{T_j - \tau}, \quad \sup_Q x < \tau < T_1.$$

Only its values at the two clocks enter; no degree bound on c is needed. Sharing includes relative throughput, the constant term of c . An independently varying throughput is a different nuisance class.

Lemma 2.1 (Two mixtures determine their latent factors). *At a site where $p_1 \neq p_2$, the pair (P_1, P_2) determines the two channel columns and the two probability vectors, up to one simultaneous interchange of the components. Hence it determines (r_1, r_2) up to simultaneous inversion.*

Proof. The column and row spaces of either rank-two matrix are those of U and V . Choose orthonormal bases A, B of these spaces and set $S_j = A^\top P_j B$. With $H = A^\top U$ and $G = B^\top V$, both invertible, we have

$$S_2 S_1^{-1} = H \operatorname{diag} \left(\frac{p_2}{p_1}, \frac{1 - p_2}{1 - p_1} \right) H^{-1}.$$

The eigenvalues are distinct. Its eigenlines recover the lines of the columns of U ; requiring column sum one recovers the actual columns. Then $U^\top P_j = \operatorname{diag}(p_j, 1 - p_j) V^\top$. The row sums give the weights and normalization gives V . The only choice is the common permutation of the two eigenlines. $\square$

This is the ordinary observable-pencil argument, not a new latent factorization theorem; compare [21, 23]. The issue below is whether its unobserved permutations can conceal another action field.

Theorem 2.2 (Two-clock spatial rigidity). *Among the models (2.1), the two observed matrix fields determine x and k_1, k_2 uniquely. No channel labelling is supplied. Points where the two mixture profiles coincide are allowed. The conclusion also holds if a competing action is merely continuous; it need not satisfy the nowhere-local-constancy condition separately.*

Proof. The equality $r_1 = r_2$ is the linear equation $k_1(T_1 - x) = k_2(T_2 - x)$. It has at most one solution in x , since $T_1 \neq T_2$. Its preimage has empty interior. Thus the set Q_* on which Lemma 2.1 applies is open and dense. Write a competing model with primes. At every point of Q_* one of two alternatives holds. If the components agree in order, then

$$x' = a_j x + T_j(1 - a_j), \quad a_j = k_j/k'_j, \quad j = 1, 2. \quad (2.2)$$

Unless $(k'_1, k'_2) = (k_1, k_2)$, consistency of these two expressions is a nonzero linear equation in x , and therefore holds at at most one action value. Indeed an identically zero equation would first give $a_1 = a_2 = a$, then $(T_1 - T_2)(1 - a) = 0$, hence $a = 1$. If the components are interchanged, then

$$x' = T_j - \frac{b_j}{T_j - x}, \quad b_j = (k_j k'_j)^{-1} > 0. \quad (2.3)$$

Consistency requires

$$(T_1 - T_2)(T_1 - x)(T_2 - x) - b_1(T_2 - x) + b_2(T_1 - x) = 0. \quad (2.4)$$

This polynomial is quadratic with nonzero leading coefficient, so it has at most two roots. If the clock constants differed, $x(Q_*)$ would consequently be contained in a set of at most three points. Continuity and density would give the same conclusion for $x(Q)$, contrary to the hypothesis. Thus $k'_j = k_j$.

With these constants fixed, the direct alternative gives $x' = x$. The swapped alternative remains possible only at the at most two roots of (2.4). The complement of their preimages and of the single profile-collision preimage is dense: each preimage is closed with empty interior, and there are finitely many. On this dense set $x' = x$. Continuity proves equality on Q . No continuation of eigenvectors through a multiple eigenvalue has been assumed. $\square$

Proposition 2.3 (Finite calibration and local stability). *Four sites with distinct true action values and noncoincident mixture profiles determine k_1, k_2 from their two-clock matrices. A finite procedure consists of at most 2^4 labelling choices and linear systems of size 4×2 . On a fixed correctly labelled regular neighbourhood, two sites with distinct actions already give a smooth local inverse for their actions and the clock constants. The local inverse is Lipschitz on compact subsets with positive channel, eigenvalue, action-separation and clock-constant margins.*

Proof. For each simultaneous-inversion choice at the four sites, denote the chosen odds by s_{ij} and solve

$$s_{i1}h_1 - s_{i2}h_2 = T_1 - T_2, \quad i = 1, \dots, 4. \quad (2.5)$$

Keep only solutions $h_j > 0$ whose resulting actions $T_j - s_{ij}h_j$ agree and lie in the declared action range. The true choice is retained. If another retained solution had different clock constants, the proof of Theorem 2.2 would allow its four distinct action values to fall in a set of at most three values. This is impossible. For the true choice, any two rows with different actions are independent: the determinant of their un-signed odds matrix is $k_1 k_2 (T_1 - T_2)(x_i - x_l)$. Thus the retained clock constants are unique, even when more than one labelling choice gives them.

For two locally labelled sites put $a = (r_{11} - r_{21})/(r_{12} - r_{22})$. Direct substitution gives

$$k_1 = \frac{r_{11} - ar_{12}}{T_1 - T_2}, \quad k_2 = k_1/a, \quad x_i = T_1 - r_{i1}/k_1. \quad (2.6)$$

All denominators are nonzero. The spectral normalization in the lemma is smooth near a fixed simple spectrum, and these formulas are smooth. Bounds on their derivatives give the stated compact-class estimate. $\square$

The four-site assertion calibrates the two clock constants, not an arbitrary finite set of action values at the exceptional swapped roots. Continuity supplies those action values in Theorem 2.2. The two-site assertion is local and does not supply global labels.

Lemma 2.4 (Boundary distances have no open plateau). *For a smooth simple metric on a compact manifold of dimension at least two, the boundary distance is not constant on any nonempty open subset of $(\partial M \times \partial M) \setminus \text{diag}$.*

Proof. An open plateau would contain a product $A \times B$ of disjoint open boundary sets. Fix $y \in A$. First variation with respect to the first endpoint says that the initial velocity of the geodesic from y to every $z \in B$ is orthogonal to $T_y \partial M$. It must be the unique inward unit normal. Uniqueness of the geodesic equation and simplicity then give one exit endpoint, whereas B contains more than one point. This contradiction proves the assertion. $\square$

Corollary 2.5 (Sharp shared-apparatus geometric clock threshold). *Let M be a compact surface with a smooth simple metric g . Observe (2.1) at every distinct boundary pair with $x = d_g$, a constant unknown reference delay larger than the diameter, and a shared relative detector law. Two clocks determine g up to a boundary-fixing diffeomorphism. One clock does not suffice, even with a fixed constant reference, a fixed constant relative detector law, and strictly positive full-rank channels. The indistinguishable one-clock metrics can be chosen in a small scaling family near any fixed simple metric with strict apparatus margins.*

Proof. The lemma and Theorem 2.2 determine the boundary distance. The boundary-distance rigidity theorem of Pestov–Uhlmann [33] gives the claimed isometry. This is the only geometric rigidity input.

For necessity fix one clock T , a constant reference τ , and a constant relative throughput, all with strict margins. Put $g_t = (1 + t)^2 g_0$. Then $x_t = (1 + t) d_{g_0}$ changes, while the reference and detector remain fixed. Let $w_t = (p_t, 1 - p_t)^\top$ be its two mixture weights. Start with fixed strictly positive full-rank probability matrices U, V . Set

$$S_t = I + (w_0 - w_t) \mathbf{1}^\top, \quad U_t = U S_t, \quad V_t = V \text{diag}(w_0) S_t^{-\top} \text{diag}(w_t)^{-1}. \quad (2.7)$$

Since $\mathbf{1}^\top(w_0 - w_t) = 0$, S_t is invertible with $S_t^{-1} = I - (w_0 - w_t) \mathbf{1}^\top$, $\mathbf{1}^\top S_t = \mathbf{1}^\top$ and $S_t w_t = w_0$. Consequently U_t, V_t have column sums one, and

$$U_t \text{diag}(w_t) V_t^\top = U \text{diag}(w_0) V^\top.$$

Uniformly small $|t|$ preserves strict positivity and full rank, because the weights have a uniform positive lower bound. The matrices S_t themselves need not be stochastic; the actual channel matrices U_t, V_t are positive stochastic matrices. All fields are smooth off the boundary diagonal. Different t give different boundary distances, so the metrics are not boundary-fixing isometric. $\square$

2.1. What is shared, and what remains a gauge. The pointwise polynomial threshold and the shared-apparatus threshold concern different experiments. In the former the detector can be rechosen at every boundary pair. In the latter,

$$\log r_j(z) = b_j + \log(T_j - x(z)), \quad b_j = \log k_j,$$

is taken modulo functions of the clock index alone. Spatial contrasts remove this nuisance, but leave an unobserved inversion at every site. The linear and quadratic obstruction calculation removes that last ambiguity. It is not enough to impose a constant reference while allowing the relative throughput to vary independently with z .

For an unrestricted shared detector, the two clocks identify k_j , not its unsampled values or the reference separately: any admissible τ' can be accompanied by $c'(T_j) = \log k_j + \log(T_j - \tau')$. For a constant relative detector, $k_j = a/(T_j - \tau)$, the recovered ratio $R = k_1/k_2 > 1$ instead gives $\tau = (RT_1 - T_2)/(R - 1)$ and $a = k_1(T_1 - \tau)$. The assertion about geometry does not require recovering this otherwise unobservable apparatus decomposition.

The closest predecessors are the latent-pencil method and blind calibration, not a new boundary-distance theorem. The additional statement is the sharp change of the geometric clock threshold under shared nuisance, with unknown channels, unresolved labels and actual one-clock metric alternatives. Classical boundary rigidity is invoked only after the selective experiment has determined its input.

3. THE RANK-DEGENERATE STATISTICAL TRANSITION

We now keep the reference and detector unknown at a single site and allow the channels to approach rank deficiency. Three clocks suffice for a constant relative detector, but the statistical price of that identification diverges at a computable rate. The divergence is a product of the two channel contrasts, not a fixed inverse constant.

Fix $T_1 < T_2 < T_3$ and bounds

$$0 < D_0 < \tau_- < \tau_+ < T_1, \quad 0 < a_- < 1 < a_+ < \infty. \quad (3.1)$$

Let $\mathcal{K}$ consist of all $\omega = (U, V, x, \tau, a)$ with U, V two-by-two probability-column matrices (including their rank-deficient boundary), $x \in [0, D_0]$, $\tau \in [\tau_-, \tau_+]$ and $a \in [a_-, a_+]$. The observation is

$$p_j = \frac{a(T_j - x)}{a(T_j - x) + (T_j - \tau)}, \quad P_j = U \operatorname{diag}(p_j, 1 - p_j) V^\top. \quad (3.2)$$

Write $\|P - P'\|_* = \max_{1 \leq j \leq 3} \|P_j - P'_j\|_F$ and

$$\rho(\omega) = |\det U \det V|. \quad (3.3)$$

All constants in this section depend only on the fixed bounds and clocks, except when an additional dependence is stated.

Lemma 3.1 (Observable rank and latent rank). *For any two-column probability channels, probability weights $\pi_1, \pi_2 \geq p_0$ with $\pi_1 + \pi_2 = 1$, and $P = U \operatorname{diag}(\pi) V^\top$,*

$$\sigma_2(P) \geq p_0 \sigma_2(U) \sigma_2(V), \quad \sigma_2(P) \leq \sqrt{2} \min\{\sigma_2(U), \sigma_2(V)\}.$$

Here a zero second singular value is allowed. In the binary model (3.2), if $s(P) = \max_j |\det P_j|$, then

$$c\rho \leq s(P) \leq \rho/4. \quad (3.4)$$

For any two binary probability matrices, $|\det P - \det Q| \leq \|P - Q\|_F$.

Proof. The lower product inequality follows by restricting the three factors to their rank-two column and row spaces. For the upper inequality, replace U by its best rank-one approximation and use $\|V\|_{\text{op}} \leq \sqrt{2}$ and $\|\operatorname{diag}(\pi)\|_{\text{op}} \leq 1$; the analogous argument applies to V . In the binary case $\det P_j = p_j(1 - p_j) \det U \det V$. Compactness of the scalar apparatus bounds gives $p_j, 1 - p_j \geq p_0 > 0$, proving the comparison. Finally the Frobenius norm of the gradient of the determinant of a two-by-two matrix equals the Frobenius norm of that matrix. Every matrix on the segment joining two probability matrices has norm at most one. Integration along the segment proves the last assertion. $\square$

Theorem 3.2 (A uniform one-sided inverse modulus). *For every $\omega \in \mathcal{K}$ with $\rho(\omega) > 0$ and every $\omega' \in \mathcal{K}$, with no rank condition on the candidate,*

$$|x - x'| + |\tau - \tau'| + |\log a - \log a'| \leq C \min \left\{ 1, \frac{\|P - P'\|_*}{\rho(\omega)} \right\}. \quad (3.5)$$

In particular its constant is uniform as either true channel loses rank; all deterioration is displayed by ρ^{-1} .

Proof. Put $C_b = U_{:,b} V_{:,b}^\top$, $r = a/(1+a)$, $h = rx + (1-r)\tau$ and $\gamma = \|C_1 - C_2\|_F$. The exact matrix curve has the representation

$$\begin{aligned} P(T) &= A + B/(T - h), & A &= rC_1 + (1-r)C_2, \\ B &= r(1-r)(\tau - x)(C_1 - C_2). \end{aligned} \quad (3.6)$$

Both r and $1-r$ are bounded below, $\tau - x$ is bounded below, and h stays in a fixed compact interval to the left of T_1 . Taking marginals of $C_1 - C_2$ gives the differences of the channel columns. For binary probability columns their determinant is the difference of their first entries. Consequently $\rho \leq C\gamma$. Moreover

$$R := \frac{\langle P_1 - P_2, P_2 - P_3 \rangle_F}{\|P_2 - P_3\|_F^2} = \frac{T_2 - T_1}{T_3 - T_2} \frac{T_3 - h}{T_1 - h}, \quad c\gamma \leq \|P_2 - P_3\|_F \leq C\gamma. \quad (3.7)$$

Set $\delta = \|P - P'\|_*$. It suffices to prove the assertion for $\delta \leq c_0\rho$, with a sufficiently small fixed $c_0 > 0$; for the remaining pairs boundedness gives the result after enlarging C . In this small-error case the candidate difference $P'_2 - P'_3$ has norm comparable to γ . The ratio in (3.7) changes by at most $C\delta/\gamma$. Its right-hand side is a smooth strictly increasing function of h whose derivative is bounded below on the compact pole interval. Applying the same identity to the candidate gives

$$|h - h'| \leq C\delta/\gamma. \quad (3.8)$$

This first step uses the visible matrix contrast, not its determinant.

Consider the affine matrix polynomial

$$K(T) = (T - h)P(T) = r(T - x)C_1 + (1-r)(T - \tau)C_2.$$

Its determinant is

$$d(T) = \det K(T) = r(1-r) \det U \det V (T - x)(T - \tau). \quad (3.9)$$

We claim that all three coefficients of $d - d'$ are bounded by $C\delta$, uniformly in the channel ranks. A naive estimate of the pole error before taking the determinant would not prove this claim. Let $Q(T)$ be the affine matrix polynomial interpolating the true P_1, P_2 . Interpolating $(T_i - h')P_i$ at $i = 1, 2$ yields $\widetilde{K} = K - (h' - h)Q$. Both K and Q have the form $s(T)C_1 + t(T)C_2$ with uniformly bounded affine scalar coefficients. The identity

$$\det(sC_1 + tC_2) = st \det U \det V$$

therefore bounds the coefficient error from $\det \widetilde{K} - d$ by $C\rho(|h' - h| + |h' - h|^2) \leq C\delta$, using (3.8), the bounded pole interval and $\rho \leq C\gamma$. On the other hand $K' - \widetilde{K}$ is the affine interpolant of $(T_i - h')(P'_i - P_i)$, so its coefficients have norm at most $C\delta$. The determinant is Lipschitz on the bounded set of such affine matrix coefficients. This proves the claim.

The leading coefficient of d has absolute value at least $c\rho$. Taking c_0 smaller ensures that the leading coefficient of d' has absolute value at least $c\rho/2$. Since $d(x) = 0$, the claim gives $|d'(x)| \leq C\delta$. But the root ordering and apparatus gap imply

$$|d'(x)| = |r'(1-r') \det U' \det V'| |x - x'| |x - \tau'| \geq c\rho |x - x'|.$$

Evaluating instead at τ gives $|\tau - \tau'| \leq C\delta/\rho$. Finally $r = (\tau - h)/(\tau - x)$, so (3.8) gives $|r - r'| \leq C\delta/\rho$. The log-odds map on the compact interval of r proves the estimate for $\log a$. $\square$

The cancellation in the affine determinant polynomial is what retains one power of the rank product. Replacing it by a generic spectral condition number would lose the sharp degeneration order.

3.1. Honest intervals and locally adaptive diameter. At each clock take n independent readouts from its four-entry categorical law and denote the empirical matrix by $\hat{P}_j$. For coverage, independence is needed within each clock, not between clocks. For $0 < \zeta < 1$ put

$$\eta = \left(\frac{2}{n} \log \frac{24}{\zeta} \right)^{1/2}, \quad \mathcal{C} = \{\omega \in \mathcal{K} : \max_j \|P_j(\omega) - \hat{P}_j\|_F \leq \eta\}. \quad (3.10)$$

If $\mathcal{C} \neq \emptyset$, report the interval I from its smallest to its largest x coordinate. Otherwise report model inconsistency and the empty set. Write $w = \text{diam}(I)$, with $\text{diam}(\emptyset) = 0$ only for the loss calculations below. No action estimate is assigned to an empty set.

Theorem 3.3 (Rank-adaptive confidence and risk). *Uniformly over the closed model $\mathcal{K}$, including singular channels, $\Pr_\omega\{x \in I\} \geq 1 - \zeta$. For each true $\rho > 0$, on the same coverage event,*

$$w \leq C \min\{1, \eta/\rho\}, \quad w \leq \min \left\{ D_0, \frac{C\eta}{(\hat{s} - \eta)_+} \right\}, \quad \hat{s} = \max_j |\det \hat{P}_j|. \quad (3.11)$$

A zero denominator in the second expression means the bound D_0 . For fixed ζ , with $N = 3n$,

$$\mathbb{E}_\omega w \leq C_\zeta \min\{1, (\sqrt{N}\rho)^{-1}\}. \quad (3.12)$$

A minimum-residual estimator $\hat{x}$ over the full compact model, using no rank threshold, satisfies

$$\mathbb{E}_\omega |\hat{x} - x|^2 \leq C \min\{1, (N\rho^2)^{-1}\}. \quad (3.13)$$

At $\rho = 0$ the guarantees are coverage and the bounded parameter-range bounds, not a shrinking diameter.

Proof. Hoeffding's inequality applied to the twelve empirical entries and a union bound give $e := \max_j \|\hat{P}_j - P_j\|_F \leq \eta$ with probability at least $1 - \zeta$. The truth then belongs to $\mathcal{C}$. Every feasible candidate is at distance at most 2η from the true matrix tuple. Theorem 3.2, applied separately to two extreme candidates, gives the first diameter bound. Lemma 3.1 gives $s(P) \geq \hat{s} - \eta$ and $s(P) \leq \rho/4$, which gives the observable version after a change of constant.

For the expectation bound do not discard the exceptional event. Every feasible candidate, on every sample outcome, has observation distance at most $\eta + e$ from the truth. Thus the same one-sided inverse bound gives $w \leq C \min\{1, (\eta + e)/\rho\}$ whenever the set is nonempty; the inequality also holds with the declared empty-set convention. For each clock, $\mathbb{E}\|\hat{P}_j - P_j\|_F^2 \leq 1/n$, whence $\mathbb{E}e^2 \leq 3/n$. It follows that $\mathbb{E}w \leq C \min\{1, (\eta + \sqrt{3/n})/\rho\}$, proving (3.12). A residual minimizer exists by compactness and differs from the truth in observation norm by at most $2e$. The same argument, squared and integrated, proves (3.13). Fixed choices among minimizers are measurable, since the sample space is finite. $\square$

Rational grids of the simplex, action and throughput factors give a finite exhaustive implementation. Padding the residual tolerance by a forward Lipschitz constant times the grid radius and the action interval by the grid radius preserves coverage. The same proof replaces η by $\eta + O(\delta_{\text{grid}})$ in the diameter bounds. No inverse constant or unknown rank floor is needed by this procedure.

3.2. A least-favourable family with exactly preserved marginals. The product in the upper bound cannot be replaced by either individual channel contrast. This can be proved inside strictly positive channels, without invoking a generic singular model.

Choose interior x_0, τ_0 and let $h = (x_0 + \tau_0)/2$, $d = h - x_0 > 0$. For $0 \leq t \leq t_0$, with a fixed sufficiently small $t_0 > 0$, put

$$\begin{aligned} x_t &= x_0 + t, & \tau_t &= \tau_0 - t, & a_t &= 1, & \lambda_t &= 1 - t/d, \\ U_{t,:1} &= m + \epsilon_u e_0 / (2\lambda_t), & U_{t,:2} &= m - \epsilon_u e_0 / (2\lambda_t), \\ V_{t,:1} &= m + \epsilon_v e_0 / (2\lambda_t), & V_{t,:2} &= m - \epsilon_v e_0 / (2\lambda_t), & m &= \frac{1}{2}(1, 1)^\top, & e_0 &= (1, -1)^\top. \end{aligned} \quad (3.14)$$

Take $0 \leq \epsilon_u, \epsilon_v \leq \epsilon_*$, where $\epsilon_* > 0$ is fixed and small. All channel entries and all joint cell probabilities have a uniform positive lower bound. Positive contrasts give full rank, and $\rho(\omega_t) = \epsilon_u \epsilon_v / \lambda_t^2$.

Lemma 3.4 (Exact cancellation of the marginal information). *At every clock, the two observed marginals in (3.14) are independent of t , and*

$$P_j(\omega_t) - P_j(\omega_0) = \frac{\epsilon_u \epsilon_v}{4} (\lambda_t^{-2} - 1) e_0 e_0^\top. \quad (3.15)$$

In particular the difference is the same at all clocks. Its Frobenius norm is comparable to $\epsilon_u \epsilon_v t$ on $[0, t_0]$. If either contrast is zero the entire family is observationally identical, not merely tangent-indistinguishable.

Proof. The mixing weights satisfy $p_j(t) = \frac{1}{2} + \lambda_t(p_j(0) - \frac{1}{2})$. The channel mean is therefore $m + \epsilon_u(p_j(0) - \frac{1}{2})e_0$, independent of t , and similarly for V . For two conditionally independent readings, subtracting the outer product of these marginals from the joint law gives $p_j(t)(1 - p_j(t))\epsilon_u \epsilon_v e_0 e_0^\top / \lambda_t^2$. Using $p_j(t) = 1/2 + \lambda_t v_j$ cancels the v_j^2 term and proves (3.15). Finally $\|e_0 e_0^\top\|_F = 2$ and $\lambda_t^{-2} - 1$ is comparable to t on the fixed small interval. $\square$

Theorem 3.5 (Sharp singular transition and honest-diameter lower bound). *Assume the $N = 3n$ records are independent across clocks as well. There is $\rho_* > 0$ such that for every $0 \leq \rho_0 \leq \rho_*$, with $\mathcal{K}(\rho_0) = \{\omega \in \mathcal{K} : \rho(\omega) \geq \rho_0\}$,*

$$c \min\{1, (N\rho_0^2)^{-1}\} \leq \inf_{\tilde{x}} \sup_{\omega \in \mathcal{K}(\rho_0)} \mathbb{E}_\omega |\tilde{x} - x|^2 \leq C \min\{1, (N\rho_0^2)^{-1}\}. \quad (3.16)$$

At $\rho_0 = 0$ the minimum on the right is interpreted as one. For any random interval honest at level $1 - \zeta$ over $\mathcal{K}$, where $0 < \zeta < 1/4$, and any positive $\epsilon_u, \epsilon_v \leq \epsilon_$, its expected length at the base point of (3.14) is at least*

$$c_\zeta \min\{1, (\sqrt{N}\epsilon_u \epsilon_v)^{-1}\}. \quad (3.17)$$

The same nonvanishing lower bound holds when one contrast is zero. Thus the uniformly informative boundary is $\sqrt{N}\rho \asymp 1$; for equally weak contrasts ϵ it is $N\epsilon^4 \asymp 1$.

Proof. The upper bound is Theorem 3.3. For the lower bound write $\rho_b = \epsilon_u \epsilon_v$. The positive cell-probability lower bound and Lemma 3.4 give

$$\text{KL}(\mathbb{P}_t^{(N)}, \mathbb{P}_0^{(N)}) \leq CN\rho_b^2 t^2,$$

using $\text{KL}(P, Q) \leq \sum (P - Q)^2 / Q$ for each categorical law and addition under independence. Choose $t = c_1 \min\{t_0, (\sqrt{N}\rho_b)^{-1}\}$ with a sufficiently small fixed $c_1 > 0$. The total variation distance is at most $1/2$. The test selecting the nearer of x_0, x_t turns any estimator into a two-point test; its larger squared risk is at least $t^2/16$. For a specified $0 < \rho_0 \leq \rho_*$ choose $\epsilon_u = \epsilon_v = \sqrt{\rho_0}$. Both alternatives have rank product at least ρ_0 , since $\lambda_t \leq 1$, and are in the fixed compact model

when ρ_* is small enough. This proves the uniform lower bound. If $\rho_0 = 0$, choose a zero contrast and two fixed distinct t ; their laws agree exactly.

For an honest interval J , coverage at both alternatives implies

$$\Pr_0\{x_0, x_t \in J\} \geq 1 - 2\zeta - \|\mathbb{P}_t^{(N)} - \mathbb{P}_0^{(N)}\|_{\text{TV}}.$$

Decrease c_1 , now depending on ζ , so that this is bounded below by a positive constant. On this event the interval has length at least t . This proves (3.17). At zero contrast the variation distance is zero and t can be fixed. $\square$

3.3. The observation modulus and its predecessors. For a true parameter ω define the one-sided action modulus

$$\Omega_\omega(\delta) = \sup\{|x' - x| : \omega' \in \mathcal{K}, \|P(\omega') - P(\omega)\|_* \leq \delta\}.$$

The determinant argument computes a uniform upper bound $C \min\{1, \delta/\rho\}$; the family (3.14) attains its order, including its nonzero value at zero noise when a contrast vanishes. A residual confidence set has diameter bounded by twice this modulus at its observation radius. Thus exact fibres, weak identification and confidence diameter are three scales of the same observation quotient in this experiment. This assertion does not identify the moduli of all detector spaces or replace the geometric stability theorem used later.

The use of moduli to compare deterministic recovery and statistical risk is classical; see Donoho [41]. Cai–Low [42] develop honest adaptation through between-class moduli for linear functionals. Dufour [43] identifies obstructions to uniformly informative inference near nonidentification. Those general principles are not claimed as new here. The additional calculation is the uniform product modulus for this nonlinear unknown-channel experiment, the exact marginal-preserving family, and the matching transition in both squared risk and honest expected length. The compact action range gives nonshrinking bounded intervals at the singularity, not the unbounded intervals of an unbounded parameter experiment. The result is for the constant-relative-detector, binary, three-clock model; higher-degree and higher-rank singular strata are not silently assigned the same exponent.

4. AN INTRINSIC DETECTOR THEOREM ON A COMPACT DOUBLE

The polar tests have a formulation which does not require a rational coordinate. The ambient object is a compact Riemann surface with a real structure. Periods enter the obstruction test, and the canonical divisor enters the clock budget. Thus principal parts alone no longer give the answer.

Let X be a connected compact Riemann surface of genus g , equipped with an antiholomorphic involution σ . Choose disjoint open arcs I, J in its real locus and a real point $b \notin I \cup J$. A real analytic parameter on I is fixed for differentiating action locations. The arc J is the clock domain; choose a base clock $T_* \in J$. A differential ω is called real if $\sigma^*\bar{\omega} = \omega$. An imaginary-period normalization means that the real part of every period on the punctured surface is zero. The terminology specifies the normalization, not the location of its residues: the residues below are real.

Lemma 4.1 (Normalized action differentials). *For distinct real points x, b , there is a unique meromorphic differential $\omega_{x,b}$ with simple poles of residues $+1, -1$ at x, b , no other poles, and imaginary periods. It is real and depends smoothly on x away from b . The differential*

$$B_x = \partial_x \omega_{x,b}$$

has a nonzero double principal part at x , zero residue, no other poles, and imaginary periods. It is independent of the choice of b . A change of local action parameter rescales B_x by a nonzero real factor.

Proof. The prescribed-principal-part theorem for differentials supplies a meromorphic differential with these two residues: their sum is zero. This is the usual Riemann–Roch consequence, recalled in [44]. Choose a homology basis of $2g$ cycles avoiding the poles. The real-linear map from holomorphic differentials to the real parts of their periods on these cycles is injective. Indeed, if these real parts vanish, the real part of a local primitive is a globally defined harmonic function on X , hence constant. Its differential is zero, so the holomorphic differential is zero. Both real dimensions are $2g$, and the map is therefore an isomorphism. Subtract the unique holomorphic differential cancelling the real parts of the prescribed periods. Small loops around the poles already have imaginary periods, because their integrals are $2\pi i$ times real residues. This proves existence. The same harmonic argument proves uniqueness.

Conjugation by σ preserves the specified poles, residues and normalization, and hence preserves the differential by uniqueness. The local principal-part construction and the fixed finite-dimensional period correction depend smoothly on x in a coordinate neighbourhood; uniqueness joins these constructions. In a local coordinate z the moving simple principal part is $dz/(z-x)$. Its derivative has principal part $dz/(z-x)^2$. The principal part at b is fixed, so differentiation removes it. All periods remain imaginary. The difference obtained from two choices of b is a holomorphic differential with imaginary periods and must vanish. The chain rule proves the final assertion. $\square$

The imaginary-period convention is the real-normalized convention applied after multiplication by i . Normalized differentials on compact and degenerating curves are studied, for example, in [45]; only the fixed-surface existence and uniqueness just proved are used here.

Define the action curve on the clock arc by

$$L_x(T) = \int_{T_*}^T \omega_{x,b}, \quad T \in J. \quad (4.1)$$

The integral is along J and is real. Let D be an effective divisor whose support avoids J . Let E be a finite-dimensional real subspace of the real meromorphic differentials with poles bounded by D , and put

$$\mathcal{F}_E = \left\{ c + \int_{T_*}^T e : c \in \mathbb{R}, e \in E \right\}. \quad (4.2)$$

Constants are included, and the primitive is only required on J . The observed scalar law is $f = e + L_x - L_y$, with $e \in \mathcal{F}_E$ and ordered distinct $x, y \in I$. We call this detector space regularly separating if this representation is unique and its parameter map into functions on J has injective differential at every such pair. In particular this definition includes the nuisance coefficients.

Let $S = \text{supp } D \cap I$ and define the finite charge set

$$\mathcal{Z}_2(S) = \{c \in \mathbb{Z}^S \setminus \{0\} : \sum_s c_s = 0, \sum_s \max(c_s, 0) \leq 2\}.$$

For $c \in \mathcal{Z}_2(S)$ let ω_c be the unique imaginary-period differential with simple principal parts of residues c_s at $s \in S$, and no other poles. Equivalently, $\omega_c = \sum_s c_s \omega_{s,b}$; the pole at b cancels.

Theorem 4.2 (Divisor–period classification). *The detector space $\mathcal{F}_E$ is regularly separating if and only if*

$$\omega_c \notin E \quad (c \in \mathcal{Z}_2(S)), \quad (4.3)$$

$$E \cap \text{span}_{\mathbb{R}}\{B_s : s \in A\} = \{0\} \quad (A \subset S, 1 \leq |A| \leq 2). \quad (4.4)$$

These are finitely many membership and intersection tests in a finite-dimensional space of meromorphic differentials. They use the full normalized differentials, including their holomorphic period corrections, rather than just their principal parts.

Proof. A scalar collision, differentiated on the clock arc, gives a member of E equal, up to sign, to

$$\omega_{x,b} - \omega_{y,b} - \omega_{x',b} + \omega_{y',b}.$$

By analytic continuation the equality is one of meromorphic differentials on X . Its residue divisor is $\delta_x - \delta_y - \delta_{x'} + \delta_{y'}$. Every noncancelled pole must lie in S . If this divisor is nonzero, its coefficients belong to $\mathcal{Z}_2(S)$, and normalization identifies the differential with ω_c . If the divisor is zero, equality of the signed measures $\delta_x - \delta_y$ and $\delta_{x'} - \delta_{y'}$, with both pairs off the diagonal, gives $x = x'$ and $y = y'$. The remaining nuisance difference is constant, and the original collision makes that constant zero.

Conversely every $c \in \mathcal{Z}_2(S)$ is such a secant divisor. For positive mass two, list its positive atoms with multiplicity as p_1, p_2 and its negative atoms as n_1, n_2 ; use the ordered pairs (p_1, n_1) and (n_2, p_2) . For positive mass one, write $c = \delta_p - \delta_n$ and use $(p, z), (n, z)$, where $z \in I \setminus \{p, n\}$. Both pairs are nondegenerate. If $\omega_c \in E$, integration and the included constants give a scalar collision. This proves the exact-identification criterion.

A parameter tangent, differentiated in the clock variable, is a member of E plus $uB_x - vB_y$. Nonzero double principal parts at distinct points cannot cancel. Every action with nonzero tangent coefficient must therefore belong to S . The tangent obstruction is precisely (4.4); a one-point obstruction is realized by choosing any different second action with zero tangent coefficient. Once again constants remove the integration ambiguity. If no obstruction occurs, both action coefficients vanish, and the original zero tangent then forces every nuisance coefficient to vanish. $\square$

The theorem applies on positive-genus surfaces even though no global rational coordinate exists. For example, take E to be any real subspace of holomorphic differentials on X and $D = 0$. Both tests are empty, so all these detector spaces are regularly separating. More generally, differentials with poles only outside the action arc are admissible. Coupled subspaces with poles on I require the full tests: replacing a normalized differential by another differential with the same principal parts can change its membership in E .

Theorem 4.3 (A divisor clock budget). *If the tests of Theorem 4.2 hold, any*

$$N = \deg D + 2g + 4 \tag{4.5}$$

distinct clocks in J identify the nuisance and ordered unequal actions, and the finite evaluation map is an immersion. Its inverse is locally smooth. On every compact set of coefficient and action parameters away from the diagonal and the ends of the coordinate arcs, it has a uniform Lipschitz constant.

Proof. Let H be a difference of two scalar laws, or a scalar parameter tangent. If it vanishes at N ordered clocks, Rolle's theorem on the real arc gives at least $N - 1$ distinct zeros of its differential. This differential extends meromorphically to X . Its poles are bounded by D together with at most four additional pole orders: four simple moving poles for a secant, or two double moving poles for a tangent. The auxiliary pole at b cancels. Its pole degree is at most $\deg D + 4$.

A nonzero meromorphic differential on a compact surface of genus g has zero degree minus pole degree equal to $2g - 2$. Thus the number of its distinct zeros is at most $\deg D + 2g + 2$. Since $N - 1 = \deg D + 2g + 3$, the differential must vanish identically. A sampled zero then gives $H = 0$. The two tests in Theorem 4.2 give injectivity and full differential rank respectively. A nonsingular coordinate minor gives a smooth local inverse. For completeness, on a compact parameter set the

local inverse estimates have a finite subcover. Pairs not covered by a common sufficiently small local neighbourhood have a positive minimum observed separation by compactness and injectivity. Combining the two bounds gives the asserted global Lipschitz estimate on that set. $\square$

This is a sufficient budget, not a universal minimal-clock theorem. If $d = \dim \mathcal{F}_E$, immersion necessarily requires $N \geq d + 2$. There can be a gap between this dimensional lower bound and (4.5). A smaller optimal count requires extra structure; the polynomial oriented-interval theorem supplies exactly such structure. The two-clock shared-nuisance theorem in Section 2 instead changes the observation domain and cannot be obtained by optimizing a pointwise zero count.

4.1. The rational theorem as the genus-zero case. On the sphere with its real structure, $\omega_{x,b} = dT/(T-x) - dT/(T-b)$, and hence $L_x - L_y$ is $\log(T-x) - \log(T-y)$ up to a constant on a clock interval lying to the right of the actions. Also $B_s = dT/(T-s)^2$. Theorem 4.2 therefore gives the charge and double-pole classification for rational derivatives.

More precisely, let D_f be the degree of the finite common denominator of a rational derivative space, and let ℓ be its largest order at infinity, with $\ell = -\infty$ for the zero space. Passing from rational functions to differentials adds the possible pole at infinity of order $\max\{\ell + 2, 0\}$. The minimal ambient pole divisor has degree

$$\deg D = D_f + \max\{\ell + 2, 0\}.$$

Thus (4.5) becomes $D_f + 6 + \max\{\ell, -2\}$, exactly the rational pole budget. The additional $2g$ in the intrinsic theorem is the canonical-divisor contribution, while the normalization in the obstruction test records the period geometry. Neither contribution is visible in a list of rational partial fractions.

4.2. Predecessors and the additional statement. Riemann–Roch, the residue constraint, period normalization and the degree of the canonical divisor are classical; see [44, 45]. The theorem does not propose a new existence theorem for normalized differentials. Its additional statement is the necessary-and-sufficient finite obstruction classification of unknown detector subspaces for this action quotient, together with a finite clock immersion theorem expressed in intrinsic divisor data. This extends the rational criterion to nonrational compact doubles and shows precisely where principal-part tests need period information.

4.3. Nonrational detectors with the original affine action factor. The intrinsic curve need not be used to change the physical action law. A finite meromorphic clock map gives a second version that retains the original factor $T - x$ while allowing nonrational detector primitives. Let $\pi : X \rightarrow \mathbb{P}^1$ be a nonconstant meromorphic map of degree m , commuting with the real structures. Let a real arc J_X map diffeomorphically onto a real clock interval J_0 , and let $I_0 \subset \mathbb{R}$ be an open interval strictly to its left. Use real meromorphic detector differentials E with poles bounded by an effective divisor D avoiding J_X . Their primitives on J_X , expressed in the clock coordinate $T = \pi(z)$, need not have rational derivatives in T . The scalar law is now exactly

$$f(T) = e(T) + \log(T-x) - \log(T-y), \quad x \neq y \in I_0.$$

Put $S_\pi = \pi(\text{supp } D) \cap I_0$ and define

$$R_c^\pi = \sum_{s \in S_\pi} c_s \frac{d\pi}{\pi - s}, \quad B_s^\pi = \frac{d\pi}{(\pi - s)^2}.$$

Here c runs through the same finite charge set $\mathcal{Z}_2(S_\pi)$. The notation $d\pi$ denotes the meromorphic differential of the clock map, not an integration variable.

Theorem 4.4 (Branched-clock detector classification). *For the original logarithmic action law in this clock chart, regular separation holds if and only if*

$$R_c^\pi \notin E \quad (c \in \mathcal{Z}_2(S_\pi)), \quad E \cap \text{span}\{B_s^\pi : s \in A\} = \{0\} \quad (A \subset S_\pi, 1 \leq |A| \leq 2).$$

Under these tests, any $\deg D + 4m + 2g$ distinct clocks in J_0 give an injective immersion, with a smooth local inverse and compact-class Lipschitz bounds as above. Branch values in the action interval are allowed. The count is sufficient.

Proof. Differentiate a scalar collision on the clock arc and pull it back to X . The action contribution is R_c^π with signed charges at the four action values, after cancellations. At a point of ramification index e_p over an action s , the differential $d\pi/(\pi - s)$ has residue e_p . Distinct action fibres are disjoint. A nonzero charge therefore produces a pole at every point of its fibre; membership in E requires, in particular, that its action value belong to S_π . Zero charges give identical ordered pairs. The realization of every nonzero charge by two ordered unequal pairs is the same as in Theorem 4.2. This proves the secant criterion in both directions. For tangents, B_s^π has pole order $e_p + 1$ at each point over s , and no poles elsewhere. Poles in distinct fibres cannot cancel, which proves the tangent criterion, including one-fibre obstructions. Constants again remove the primitive ambiguity.

For the clock budget a secant contribution has pole degree at most $4m$: each noncancelled action fibre contributes at most m . There is no pole over infinity, since the total charge is zero. For a tangent, each of the two possible fibres contributes $\sum_{p \rightarrow s} (e_p + 1) = m + \#\pi^{-1}(s) \leq 2m$. These formulas also prove the assertions at ramified action values. Adding D , the pole degree of the differentiated collision or tangent is at most $\deg D + 4m$. Its zero degree, if the differential is nonzero, is at most $\deg D + 4m + 2g - 2$. Rolle's theorem for the stated number of clocks forces it to vanish. The classification and the compact inverse argument finish the proof. $\square$

For a concrete positive-genus example take a nonsingular real elliptic curve $Y^2 = (T - a)(T - b)(T - c)$ with $a < b < c$ and clock interval strictly to the right of c , on the positive real sheet. The clock map $\pi = T$ has degree two. The real holomorphic differential dT/Y spans a detector derivative space with $D = 0$. Its primitive is an elliptic integral and has a nonrational derivative in the clock coordinate. The two obstruction families are absent, so constants and this elliptic primitive form a regularly separating detector space for the unchanged logarithmic action law. Ten clocks are a sufficient uniform design. This example and the theorem concern genuinely nonrational detectors, not only a replacement of the action curve by an Abelian integral.

5. INTRODUCTION AND MAIN RESULTS

An action occurs in a clocked observation as the zero of an affine factor $T - W$. A selective detector multiplies that factor by an unknown function of the deadline. We ask which detector spaces permit recovery of the zero after both total mass and component identity have been lost, and how much geometry can be recovered from finitely many such observations. The logarithmic action curve modulo the detector space connects the two questions: its secants encode exact action information, its tangents control inversion, and its nontrivial fibres describe which geometric changes can be hidden by the apparatus.

There are two main conclusions. First, for finite-dimensional detector spaces whose derivatives are rational, regular identification is classified by a finite list of residue and double-pole obstructions (Theorem 8.1). This is a coefficient-space test, not a search over unknown actions. Under principal-part closure it gives all maximal admissible enlargements in a fixed ambient divisor class (Corollary 8.2). The same pole data, including order at infinity, give one uniform clock budget (Theorem 8.3). Positivity of an annihilator is neither assumed nor necessary.

Second, finite selective readout records yield geometric confidence sets. In a two-arm travel-time experiment, the propagation action is a boundary distance and the longer reference action is unknown. The reference may change with the endpoint pair; both readout channels and all relative detector coefficients may change as well. A compact forward model gives data-derived action intervals without supplying a channel signal floor to the estimator (Theorem 12.1). On a boundary mesh of diameter h , let w_h be the largest reported interval width. On the precise local boundary-rigidity stability class described in Section 13, every feasible metric satisfies

$$\|\hat{g} - \psi^* g\|_{C^2} \leq C(w_h + 2Lh)^\mu, \quad 0 < \mu < 1,$$

for a boundary-fixing diffeomorphism ψ . Theorem 13.1 proves this implication directly from finitely many categorical records, with no interpolation of the unknown nuisance fields. Its geometric input is the stability theorem of Stefanov–Uhlmann [39], with its hypotheses retained explicitly.

On regular channel classes the certificate has width of order $\sqrt{\log(m_h/\zeta)/n}$ for n readouts in each cell. A balanced surface design with total readout budget N gives the sufficient C^2 pullback rate $(\log(N/\zeta)/N)^{\mu/4}$. For a finite-dimensional geometric prior, Theorem 14.2 instead proves matching N^{-1} bounds for mean squared parameter loss on a finitely selected boundary design. The alternatives used in its lower bound are actual metrics. Theorem 14.3 also realizes the missing-clock fibre by non-boundary-isometric simple metrics: $q + 3$ clocks are worst-case sharp for the endpoint-dependent reference class. Finally Theorem 15.1 charges all raw attempts, including failures. Its certificate uses observed acceptance counts, rather than treating retained readouts as free preparations.

The underlying exact action theory remains useful beyond the two-arm application. Theorem 5.1 gives the sharp polynomial inverse with unknown visible count and a possibly nontrivial component covering. Theorems 7.2 and 7.4 give finite and generic analytic clock designs; prescribed singularities, a positive-kernel-free example, and action divisors are treated in Section 7. The complete polynomial fibres appear in Theorem 11.1. The finite-record covering theorem recovers monodromy, while the geometric constructions retain their exact observation domains and full primitives. The boundary-metric experiment needs only the pointwise action inverse, not a supplied covering.

5.1. Observation model. Let $T_1 < \dots < T_J$ be real deadlines and let Q be a compact connected smooth domain, possibly with boundary or corners, with a fixed finite coordinate atlas. Fix an integer $q \geq 0$. At $z \in Q$ the observations are positive projective classes of $u \times v$ matrices

$$M_j = U \operatorname{diag} \left\{ a_b \exp \left(\sum_{\nu=1}^q \theta_{b,\nu} T_j^\nu \right) (T_j - W_b) \right\}_{b=1}^B V^\top, \quad a_b > 0, \quad W_b < T_1. \quad (5.1)$$

The columns of U, V are probability vectors and have full column rank. The channels may depend on z but not on j . The two readings are independent conditional on the component and endpoint. The visible count $B \leq \min(u, v)$ is not supplied. The normalized matrices are

$$P_j = \frac{M_j}{\mathbf{1}^\top M_j \mathbf{1}} = U \operatorname{diag}(\pi_{j,b}) V^\top. \quad (5.2)$$

The joint matrix, not each row or column, is normalized. All coefficients are locally C^m , $m \geq 0$; local enumerations may differ by permutations. The tuple of a component is $(W_b, U_{:,b}, V_{:,b}, \log a_b, \theta_{b,1}, \dots, \theta_{b,q})$. It defines a finite covering of Q , not necessarily a connected one. No enumeration or trivialization is observed. Put

$$\alpha_b = \log a_b - B^{-1} \sum_c \log a_c, \quad \vartheta_{b,\nu} = \theta_{b,\nu} - B^{-1} \sum_c \theta_{c,\nu}, \quad \Xi = (W, U, V, \alpha, \vartheta). \quad (5.3)$$

These are global C^m fields on the covering: the fibrewise sums are permutation invariant and the local centred fields have the same transition permutations as the original tuples.

Theorem 5.1 (Sharp selective-detector rigidity). *Suppose (5.1) holds. At every endpoint assume the tuples $(W_b, \theta_{b,1}, \dots, \theta_{b,q})$ are pairwise distinct and the actions are not all equal. Any $q + 3$ distinct deadlines determine B , the component covering and Ξ , up to a covering isomorphism over Q . The complete continuous ambiguity is*

$$a_b \mapsto h(z)a_b, \quad \theta_{b,\nu} \mapsto \theta_{b,\nu} + \rho_\nu(z), \quad (5.4)$$

where $h > 0$ and h, ρ_ν are C^m functions on Q , common to all components of a fibre. No action changes. On simply connected Q the covering is trivial and two global enumerations differ by one constant permutation.

On each reference neighbourhood specified in Theorem 10.3, the inverse has a locally Lipschitz C^m extension to real matrix fields, in the ambient centred affine parameter space modulo the same covering isomorphisms. There is no loss of spatial derivatives. The rank and covering stratum of this extension are fixed; Theorem 11.3 separately treats their recovery from finite records under quantitative signal assumptions.

The count is optimal over this observation class. With $q + 2$ deadlines, strictly positive two-component full-rank models with unequal actions have an action-changing continuous ambiguity. In particular, four clocks are necessary and sufficient in the worst log-affine case.

The optimality is a worst-case statement, not a requirement for every larger model. Invisible duplicates and arbitrarily weak components cannot be counted by a fixed signal threshold. Equal actions among some components are allowed when other coefficients separate them and one unequal-action pair remains.

5.2. Relation to earlier methods. The latent spectral step belongs to the established theory of multiway identification and observable operators [21, 22, 23]. Here the deadline is a controlled index, the two channels are unknown and may vary with the endpoint, and the weights follow a restricted action–detector law. We use exact full-rank and joint-separation hypotheses rather than a generic identifiability assertion. The matrix factorization itself is not claimed as a new tensor theorem.

After that step, the scalar data have the form

$$f(T) = P(T) + \log(T - x) - \log(T - y), \quad P \in \mathcal{P}_q, \quad x, y < T_1. \quad (5.5)$$

We write $L_x(T) = \log(T - x)$ throughout. Divided differences and Chebyshev systems are classical [24, 26]. They give the two-moment proof below, but not the positive-kernel-free example of Section 7. The general secant criterion is not a characterization of arbitrary moment measures. Its finite-design argument belongs beside the analytic experiment theory of Sontag [38] and the finite-measurement inverse theory of Alberti–Santacesaria [35]. The finite residue and double-pole tests of Theorem 8.1 give an additional structural classification within the rational-derivative category. Finite-rate-of-innovation and Prony methods also recover finite singular supports [28, 29]; their linear moment or convolution samples are not our projective matrices or sampled logarithms with an additive unknown detector space.

The covering construction uses local spectral perturbation and covering space theory [30, 31]. Its additional task is to match actual channel columns across independently chosen nonorthogonal minor coordinates. The finite-record theorem supplies edge permutations and monodromy from observations; it does not assume a reference covering. The noisy pairwise-permutation experiment of [37] is different from this vertex-readout experiment. Blind bilinear calibration in [36] likewise provides an important comparison, but not an inverse for the present nonlinear action and

two-channel nuisance law. These distinctions are detailed at the relevant theorems rather than used to claim novelty for spectral decomposition itself.

Generating functions and exact-lift composition are standard [27, 25]. Section 18 replaces the common-lift premise by literal identities of reconstructed lifts on an observed invariant domain; its elementary group criterion does not certify an unobserved invariant domain or an exact root from a noisy compatibility residual. For the exact surface application, Pestov–Uhlmann’s boundary-distance theorem supplies the metric step [33]. For finite noisy observations we use the stated local stability estimate of [39] and prove the complete acquisition, confidence and geometric error chain. The partial-data metric results in [34] solve different geometric inverse problems and are not inferred from the finite-dimensional clock theorem.

6. ORIENTED INTERVALS AND DETECTOR QUOTIENTS

Lemma 6.1 (Two moments of an interval). *Let I be an open interval, $w : I \rightarrow (0, \infty)$ continuous, and $r : I \rightarrow \mathbb{R}$ continuous. On $x \neq y$ the map*

$$\mathcal{M}(x, y) = \left(\int_y^x w(s) ds, \int_y^x w(s)r(s) ds \right)$$

is injective on all oriented intervals if and only if r is strictly monotone. If w, r are smooth and r is strictly monotone, it is a local smooth diffeomorphism and

$$\det D\mathcal{M}(x, y) = w(x)w(y)(r(x) - r(y)). \quad (6.1)$$

Proof. Choose $F' = w$, put $u = F(y)$, $d = F(x) - F(y)$, and $R = r \circ F^{-1}$ on $I' = F(I)$. The first moment is d . For $d > 0$ the second is $H_d(u) = \int_u^{u+d} R(t) dt$. If R is strictly increasing, $H'_d(u) = R(u+d) - R(u) > 0$; strict decrease reverses the inequality. The two moments determine d, u , hence x, y . The sign of d separates orientations, and reversing endpoints handles $d < 0$.

Conversely, injectivity makes each H_d injective on the interval $\{u : u, u+d \in I'\}$, hence strictly monotone. A constant interval of R would give a constant interval of H_d for small d , and is excluded. Choose $u_0 < u_1$ with $R(u_0) \neq R(u_1)$. Exhaust I' by compact intervals whose interiors contain these two points. On any such compact interval, $H_d/d \rightarrow R$ uniformly as $d \downarrow 0$. For all sufficiently small d the sign of $H_d(u_1) - H_d(u_0)$ is thus fixed by $R(u_1) - R(u_0)$, and fixes the monotonicity orientation of H_d on its entire interval of definition. For arbitrary $v_0 < v_1$ in I' , choose an exhaustion interval containing all four points in its interior. Passing to the limit in the inequality for $H_d(v_0), H_d(v_1)$ shows that R has the same nonstrict monotonicity on all of I' . Equality at two distinct points would make it constant between them, already excluded. Thus it is strictly monotone. Differentiation gives the determinant. $\square$

This is a theorem about moving endpoints and a known density, not about recovery of arbitrary measures from two moments. Its global conclusion does not follow merely from its nonsingular Jacobian.

6.1. A sampled positive-kernel criterion. Let $J = d + 2$, let $\mathcal{D} \subset \mathbb{R}^J$ have dimension d and contain $\mathbf{1}$, and choose a basis c^0, c^1 of $\mathcal{D}^\perp$. Define

$$k_a(s) = \sum_{j=1}^J \frac{c_j^a}{T_j - s}, \quad a = 0, 1. \quad (6.2)$$

Assume the kernel space has a member positive on $I \subset (-\infty, T_1)$, and choose $k_0 > 0$.

Theorem 6.2 (Positive-kernel detector classification). *For $\ell_j = e_j + \log(T_j - x) - \log(T_j - y)$, $e \in \mathcal{D}$ and $x \neq y$ in I , the ordered actions and sampled nuisance vector are globally identifiable if and only if k_1/k_0 is strictly monotone. Equivalently,*

$$k_0(s)k_1(t) - k_1(s)k_0(t) \quad (6.3)$$

has one strict sign for $s < t$ in I . The condition is independent of the annihilator basis. Its failure gives distinct nondegenerate action intervals and nuisance vectors with exactly the same scalar data.

Proof. Since $\log(T - x) - \log(T - y) = -\int_y^x (T - s)^{-1} ds$, the contrasts $(-c^0 \cdot \ell, -c^1 \cdot \ell)$ are the two interval moments. Equal contrasts are sufficient as well as necessary for the logarithmic action vectors to differ by an element of $\mathcal{D}$. Apply Lemma 6.1, then subtract to recover e . A basis change multiplies (6.3) by its nonzero determinant. Between two bases with positive first kernel, the new ratio is a real fractional-linear transform of the old one with a nowhere-zero denominator; its derivative with respect to that ratio has constant nonzero sign. This also proves ratio-level invariance. $\square$

The hypothesis on a positive kernel is not universal. General function spaces and higher-dimensional annihilators are treated below. This scalar theorem alone supplies neither spectral separation of a latent mixture nor recovery of function coefficients from a rank-deficient evaluation map.

6.2. Sharp polynomial clock rigidity. For distinct nodes use

$$[t_0, \dots, t_n]f = \sum_i \frac{f(t_i)}{\prod_{h \neq i} (t_i - t_h)}.$$

It annihilates polynomials of degree less than n , and induction from the first difference gives

$$[t_0, \dots, t_n] \frac{1}{T - s} = \frac{(-1)^n}{\prod_i (t_i - s)}. \quad (6.4)$$

Theorem 6.3 (The sharp scalar inverse). *For $q \geq 0$, $q + 3$ distinct values of (5.5) determine $x \neq y < T_1$ and P globally, with a locally smooth inverse. With only $q + 2$ values, action-changing one-parameter families give the same data. Their entire fibres are parametrized in Theorem 11.1.*

Proof. Take the consecutive order- $q + 1$ differences

$$\begin{aligned} d_0 &= (-1)^{q+2} [T_1, \dots, T_{q+2}]f, & w_0(s) &= \prod_{i=1}^{q+2} (T_i - s)^{-1}, \\ d_1 &= (-1)^{q+2} [T_2, \dots, T_{q+3}]f, & w_1(s) &= \prod_{i=2}^{q+3} (T_i - s)^{-1}. \end{aligned} \quad (6.5)$$

They equal $\int_y^x w_0, \int_y^x w_1$. These two annihilators are independent and span the polynomial evaluation annihilator. Moreover

$$\frac{w_1(s)}{w_0(s)} = \frac{T_1 - s}{T_{q+3} - s}, \quad \left(\frac{w_1}{w_0} \right)'(s) = \frac{T_1 - T_{q+3}}{(T_{q+3} - s)^2} < 0. \quad (6.6)$$

The interval lemma recovers x, y ; subtraction and interpolation recover P . It also proves smoothness. Directly, the scalar Jacobian has columns $1, T, \dots, T^q, -(T - x)^{-1}, (T - y)^{-1}$. Multiplying a row relation by $(T - x)(T - y)$ gives a polynomial of degree at most $q + 2$ vanishing at $q + 3$ nodes. Evaluation at x, y forces the two pole coefficients to vanish, and then the polynomial coefficients vanish. At $q + 2$ nodes the columns $1, T, \dots, T^q, -(T - x)^{-1}$ alone are independent by the same argument with denominator $T - x$. The implicit function theorem

solves for these parameters while a nearby y varies. Strict probability and action margins persist on a small segment. $\square$

For $d_0 > 0$, choose $F' = w_0$ and solve the strictly decreasing equation

$$\int_u^{u+d_0} \frac{w_1(F^{-1}(t))}{w_0(F^{-1}(t))} dt = d_1. \quad (6.7)$$

Then $y = F^{-1}(u)$ and $x = F^{-1}(u + d_0)$. The condition $u, u + d_0 \in F((-\infty, T_1))$ gives the bracketing interval. Reverse orientation for negative d_0 . No action sorting is used. Once y is known, the single contrast $\int_y^x w_0$ is strictly increasing in x , including $x = y$, so $q + 2$ nodes identify every remaining component. The first contrast vanishes exactly at equal actions.

6.3. A nonpolynomial positive-kernel family.

Proposition 6.4 (Exponential log-detectors). *Fix a known $\lambda > 0$ and clocks $T_j = j$, $1 \leq j \leq 4$. Replace the polynomial in (5.1) by $\beta_b e^{\lambda T_j}$, where β_b is unknown. If the (W_b, β_b) are distinct, $W_b < 1$, and the actions are not all equal, four projective matrices determine the actions, channels, relative weights and relative β_b . Four are necessary in the worst case. The local enumerations give the intrinsic component covering. This does not assert recovery of an unknown shape parameter λ .*

Proof. Put $a = e^\lambda > 1$ and $B_0 = 3 + 2/(a - 1) > 3$. The annihilators of $\text{span}\{\mathbf{1}, (a, a^2, a^3, a^4)\}$

$$c^0 = (a, -a - 1, 1, 0), \quad c^1 = (0, a, -a - 1, 1)$$

have kernels

$$k_0(s) = \frac{(a-1)(B_0-s)}{(1-s)(2-s)(3-s)} > 0, \quad \frac{k_1}{k_0}(s) = \frac{(1-s)(B_0+1-s)}{(4-s)(B_0-s)}.$$

For $s < 1$, the logarithmic derivative of the latter ratio is

$$-\frac{3}{(1-s)(4-s)} + \frac{1}{(B_0-s)(B_0+1-s)} < 0.$$

The second term is smaller than $1/((3-s)(4-s))$, which is smaller than the magnitude of the first. The positive-kernel theorem identifies an unequal pair and its nuisance coefficients. With one action known, the first positive contrast identifies every other action, even an equal one.

Spectral separation must also be checked. For $f(T) = \alpha + \beta e^{\lambda T} + L_x - L_y$, with $x \neq y$,

$$e^{-\lambda T} f'(T) = \lambda \beta + (x - y) \frac{e^{-\lambda T}}{(T - x)(T - y)}.$$

The nonconstant term is strictly monotone on $T > \max(x, y)$; hence f' has at most one zero. Three equal values of f are impossible. For $x = y$, distinct β gives the same separation conclusion. The two reduced operators at clocks two and three therefore have distinct joint vectors. The matrix calculation in Lemma 9.1, scalar inversion and centring recover the full model and its covering.

For three clocks, the columns $1, e^{\lambda T}, -(T - x)^{-1}$ are independent. The derivative of a putative linear combination, multiplied by $e^{-\lambda T}$, is a constant plus a multiple of $e^{-\lambda T}(T - x)^{-2}$, a strictly decreasing positive function. Three zeros would give two derivative zeros and force all coefficients to vanish. The implicit function theorem solves for (α, β, x) as a nearby y varies. Fixed positive full-rank two-component channels turn this scalar level set into identical projective matrices, proving the lower bound. $\square$

Known positive affine changes of the clock coordinate give the same example on four equally spaced clocks. A small positive baseline keeps these detector functions valid retention probabilities on the finite observation interval.

7. DETECTOR FUNCTION SPACES AND STABLE FINITE DESIGNS

A criterion on one sampled subspace does not by itself decide whether a function space admits a useful design. We separate these questions. Let $K_T = [\tau_0, \tau_1]$, with $\tau_0 < \tau_1$, be a clock interval, and let $I \subset (-\infty, \tau_0)$ be an open action interval. Let $\mathcal{F} \subset C(K_T)$ be a finite-dimensional real space containing constants, with any fixed coefficient norm. Set

$$L_x(T) = \log(T - x), \quad \mathcal{A} = \{(x, y) \in I^2 : x \neq y\}, \quad \Gamma(x, y) = L_x - L_y.$$

The exact functional observation is $e + \Gamma(x, y)$, $e \in \mathcal{F}$. Its image in $C(K_T)/\mathcal{F}$ is a secant of the logarithmic action curve.

Definition 7.1. The detector space is *regularly separating on I* if

$$\Gamma(a) - \Gamma(a') \in \mathcal{F} \implies a = a' \quad (a, a' \in \mathcal{A}), \quad (7.1)$$

$$-\frac{u}{T-x} + \frac{v}{T-y} \in \mathcal{F} \implies u = v = 0 \quad ((x, y) \in \mathcal{A}). \quad (7.2)$$

The second condition is a regularity condition, not part of set-theoretic identification. It excludes a vanishing first-order signal in the quotient.

Theorem 7.2 (Function-space criterion and finite deadline design). *If $\mathcal{F}$ is regularly separating, then for every compact $K \subset \mathcal{A}$ there are finitely many distinct deadlines in (τ_0, τ_1) and a constant $c_K > 0$ such that*

$$\max_j |(e - e')(T_j) + \Gamma(a)(T_j) - \Gamma(a')(T_j)| \geq c_K(\|e - e'\|_{\mathcal{F}} + |a - a'|) \quad (7.3)$$

for all $a, a' \in K$ and all $e, e' \in \mathcal{F}$. Evaluation is injective on $\mathcal{F}$. The finite observation map is an immersion near each point of K and has a locally smooth left inverse, including a smooth local extension from ambient sampled data.

Conversely, functional identification implies (7.1). A differentiable local left inverse at every admissible pair, from any collection of point evaluations, implies (7.2). Thus these conditions characterize regular identification and the existence of stable finite designs on compact classes, rather than merely restating a criterion for one chosen design.

Proof. Consider every nonzero difference $e + \Gamma(a) - \Gamma(a')$, normalized by $t = \|e\|_{\mathcal{F}} + |a - a'|$. We show that the closure of this family in $C(K_T)$ is compact and does not contain zero. Given a sequence, pass to subsequences so that a, a' converge in K and the normalized coefficient vector e/t converges. If $t \rightarrow \infty$, the logarithmic part divided by t converges to zero, whereas $\|e/t\|_{\mathcal{F}} \rightarrow 1$. If $t \rightarrow t_0 \in (0, \infty)$, the limit is a normalized secant difference plus a member of $\mathcal{F}$. Its vanishing would force $a = a'$ by (7.1), then $e = 0$, contradicting $t_0 > 0$. If $t \rightarrow 0$, the actions have a common limit $a = (x, y)$ and $(a - a')/t$ has a convergent subsequence with limit (u, v) . Taylor's formula, uniformly on K_T , gives the limit

$$e_0 - \frac{u}{T-x} + \frac{v}{T-y}, \quad \|e_0\|_{\mathcal{F}} + |(u, v)| = 1.$$

It cannot vanish by (7.2) and the independence of a basis of $\mathcal{F}$. These alternatives also prove sequential compactness; the logarithm has uniformly bounded parameter derivatives on the compact action and clock sets. In the case of a bounded sequence of t with no limit, take a further convergent subsequence first.

Let $\kappa > 0$ be the minimum supremum norm on this compact closure. For each member choose an interior deadline where its absolute value exceeds $\kappa/2$. Continuity in the function and compactness give finitely many such evaluations detecting every member by at least $\kappa/4$. Repeated deadlines may be removed. This proves (7.3). To secure the immersion simultaneously, include the compact normalized tangent family

$$\left\{ e - \frac{u}{T-x} + \frac{v}{T-y} : a \in K, \|e\|_{\mathcal{F}} + |(u,v)| = 1 \right\}$$

in the same finite-subcover construction. Condition (7.2) excludes zero from it. The sampled derivative then has full column rank. A nonsingular coordinate minor and the inverse function theorem give the stated smooth local left inverse. Taking $a = a'$ in (7.3) proves injective evaluation on $\mathcal{F}$.

If (7.1) fails, the corresponding nuisance difference makes even the full functions identical. If (7.2) fails, a nonzero tangent vector, supplemented by its nuisance coefficients, lies in the kernel of every sampled derivative. The chain rule contradicts a differentiable left inverse. $\square$

The proof distinguishes two kinds of separation. A nonsingular tangent map need not separate distant secants; exact injectivity alone need not provide a differentiable inverse. The clock set and c_K depend on the function space and the compact action class, not just on its dimension. The theorem does not assert that an arbitrary prescribed set of that many clocks works. It allows annihilators of any dimension and contains no assumption on the sign of their Cauchy kernels.

For a proposed enlargement $\mathcal{F} \subset \tilde{\mathcal{F}}$, the exact test is particularly transparent: no nontrivial secant difference in (7.1), and no nonzero tangent combination in (7.2), may enter $\tilde{\mathcal{F}}$. This gives the boundary of *regular* admissible enlargements. It is not a claim that there is one largest admissible detector space.

7.1. A uniform generic clock bound for analytic spaces. The preceding compactness construction does not bound the number of clocks by the dimension. Analyticity supplies such a bound, and permits one clock set to work on the entire admissible action interval.

Lemma 7.3 (Finite-order zeros and generic evaluations). *Let $\mathcal{P}$ be an open subset of $\mathbb{R}^p$, J_0 an open interval, and $H : \mathcal{P} \times J_0 \rightarrow \mathbb{R}$ smooth. Suppose every function $H(\theta, \cdot)$ is nonzero and real analytic. If $N > p$, then for almost every $(T_1, \dots, T_N) \in J_0^N$ there is no θ for which $H(\theta, T_j) = 0$ for all j . The same assertion holds on countably many parameter charts of dimension at most p .*

Proof. At an incidence point where all the sampled values vanish, let r_j be the finite order of the zero at T_j . For the subset of incidence points with this same order vector, all equations $\partial_T^{r_j-1} H(\theta, T_j) = 0$ hold. Their derivative matrix with respect to the N clock variables is diagonal with nonzero entries $\partial_T^{r_j} H(\theta, T_j)$. The implicit function theorem therefore places this subset locally in a graph $(T_1, \dots, T_N) = \varphi(\theta)$. There are countably many order vectors, and second countability gives countably many such local graphs covering each subset. A smooth image of a p -dimensional coordinate domain in $\mathbb{R}^N$, $p < N$, has N -dimensional Lebesgue measure zero: restrict to countably many compact coordinate boxes where the map is Lipschitz and cover each box by small cubes. The countable union of these images is null. Analyticity is used to exclude an infinite-order zero of a nonzero function, not to assume that all zeros are simple. Countably many parameter charts do not change the conclusion. $\square$

Theorem 7.4 (Generic analytic deadline design). *Suppose $\dim \mathcal{F} = d$, every member of $\mathcal{F}$ is real analytic on an open connected clock interval above I , and (7.1)–(7.2) hold there. Then almost every ordered set of $d+5$ distinct clocks in any nonempty open subinterval identifies every $e \in \mathcal{F}$ and every $(x, y) \in \mathcal{A}$ globally, with a locally smooth inverse. For each such clock set, (7.3) holds on every compact $K \subset \mathcal{A}$. Thus the exceptional clock set is independent of the unknown action*

pair and nuisance coefficients. The count is a uniform sufficient bound, not a sharp count for each analytic space. Smooth inversion here is local in parameter space; the uniform estimates are on compact action classes. No properness at the ends of I is asserted.

Proof. Choose a basis of $\mathcal{F}$ and consider

$$H(e, a, a'; T) = e(T) + \Gamma(a)(T) - \Gamma(a')(T).$$

Outside the trivial set $e = 0, a = a'$, the functional secant condition makes this a nonzero analytic function of T . Its parameter space has dimension $d + 4$ and is covered by countably many open coordinate charts. Lemma 7.3 with $N = d + 5$ shows that almost every clock tuple excludes every nontrivial scalar collision. This includes nuisance-only differences and hence full-rank evaluation on $\mathcal{F}$.

For the derivative consider $H(e, x, y, u, v; T) = e(T) - u/(T - x) + v/(T - y)$, with nonzero joint coefficient vector (e, u, v) . Normalize one nonzero coefficient to one; the finitely many such charts have dimension $d + 3$, including x, y . The tangent condition makes these functions nonzero. The lemma again excludes their simultaneous sampled zeros for almost every clock tuple (already $d + 4$ clocks suffice for this derivative step). Removing the null sets from both arguments gives injectivity and full derivative rank at every admissible point. Distinctness and ordering only discard diagonals and restrict to an open chamber. The local inverse theorem and the normalized-secant compactness argument prove the final statements. No parameter-dependent exceptional set is exchanged with an uncountable union: the incidence argument has already included all parameters before projecting to clock space. $\square$

The clock hierarchy is now explicit: $d + 5$ generic clocks for regularly separating analytic spaces, $d + 4$ arbitrary clocks for the protected-pole class below, and the sharp $d + 2 = q + 3$ for polynomial log-detectors. These are different design conclusions, not interchangeable dimension counts. The analytic theorem is a scalar design theorem; latent spectral separation must still be verified, as it is in the applications.

7.2. Logarithmic detectors with protected singularities. Let $a_1, \dots, a_k < \tau_0$ be distinct known numbers outside I . For $q \geq 0$ consider the function space

$$\mathcal{F}_{q,A} = \mathcal{P}_q + \text{span}\{\log(T - a_1), \dots, \log(T - a_k)\}. \quad (7.4)$$

The locations a_i are fixed features of the detector family, not unknown actions. Polynomial differentiation and residues show that this sum has dimension $q + k + 1$.

Theorem 7.5 (Protected-pole rigidity). *For (7.4), any $q + k + 5$ distinct deadlines in K_T determine $x \neq y$ in I and $e \in \mathcal{F}_{q,A}$ from $e(T_j) + \log(T_j - x) - \log(T_j - y)$. The inverse is locally smooth and is uniformly Lipschitz on every compact separated action class, with unbounded nuisance differences allowed as in (7.3).*

Replace the polynomial log-detectors in (5.1) by members of this space. If each pair of components differs in its action or nonconstant detector coefficients, and the actions are not all equal, the same $q + k + 5$ clocks determine the visible count, both channels, absolute actions and detector coefficients modulo a common member of $\mathcal{F}_{q,A}$. The component covering is intrinsic as before. The bound is sufficient; no optimality for this enlarged family is claimed.

Proof. The difference of two competing scalar observations is

$$H(T) = P(T) + \sum_i \beta_i \log(T - a_i) + L_x - L_y - L_{x'} + L_{y'}.$$

Its derivative is rational plus P' . Multiply by the product of its distinct pole factors, of degree at most $k + 4$. The resulting polynomial has degree at most $q + k + 3$ (also when $q = 0$). If H

vanishes at $q + k + 5$ clocks, Rolle's theorem gives at least $q + k + 4$ zeros of H' above every pole. Thus that numerator is identically zero. At each protected pole the residue is β_i , so all β_i vanish. At the moving poles the remaining identity is

$$\delta_x - \delta_y = \delta_{x'} - \delta_{y'}.$$

Since both intervals are nondegenerate, equality of these signed point masses gives $x = x'$ and $y = y'$. Finally $P' = 0$ and the sampled zeros give $P = 0$.

For the derivative of the scalar parameter map, a putative dependence has the form

$$H(T) = P(T) + \sum_i \beta_i \log(T - a_i) - u/(T - x) + v/(T - y).$$

After differentiation, the two moving poles are at most double; the common denominator still has degree at most $k + 4$. The same degree bound and Rolle argument apply. The double-pole coefficients give $u = v = 0$, then the simple protected residues and the constant give $\beta_i = 0$ and $P = 0$. The sampled Jacobian has full column rank. The normalized-secant compactness argument in the preceding proof, now for this fixed sampled map, gives the claimed compact-class Lipschitz bound. The same residue arguments establish (7.1)–(7.2) at the function level.

For latent separation, a difference of two log-weight trajectories minus a constant has only two moving poles. Its differentiated numerator has degree at most $q + k + 1$. Already $q + k + 3$ equal sampled values force it to be constant identically. Protected and moving residues then force equal actions and equal nonconstant coefficients, contrary to the hypothesis. Thus the joint spectral trajectories are distinct at our clocks, and Lemma 9.1 applies with all the available reduced operators. A pair with nonzero projection modulo the sampled detector space has unequal actions; a zero projection for unequal actions is excluded by the same two-pole argument. Recover one unequal pair by the scalar inverse. With one action fixed, the corresponding one-action map is injective and has a nonvanishing derivative modulo the detector space, again by its protected poles, even when the new action equals the anchor. Recover all remaining actions, interpolate the nuisance coefficients, and centre them fibrewise. The spectral columns provide the same covering and matching construction as before. $\square$

Proposition 7.6 (Identification with no positive annihilator). *Let $a < b$, $I = (a, b)$, and*

$$\mathcal{F} = \text{span}\{1, h\}, \quad h(T) = \log \frac{T - a}{T - b}, \quad T > b.$$

Any six distinct clocks above b identify ordered unequal actions in I and both nuisance coefficients, with a locally smooth inverse. The associated latent model is also identifiable under the distinct (W_b, β_b) and unequal-action hypotheses. For every finite set of clocks, every nonzero sampled annihilator kernel changes sign on (a, b) . Thus no two-dimensional subspace of that annihilator satisfies the positive-kernel hypothesis.

Proof. For a scalar collision, $H' = \beta h' + (L_x - L_y - L_{x'} + L_{y'})'$ has at most six simple poles. Its total residue is zero, so it is $O(T^{-2})$ at infinity. Its numerator after multiplication by the six-factor denominator has degree at most four. Six sampled zeros of H force five zeros of H' , hence $H' = 0$. The residues at a, b give $\beta = 0$, and the moving residues determine the ordered pair. The constant is fixed by one sample. For the parameter derivative, $H' = \beta h' + u/(T - x)^2 - v/(T - y)^2$ has a denominator of degree six and is again $O(T^{-2})$; the same argument proves full derivative rank. For spectral separation, the corresponding two-moving-pole derivative has numerator degree at most two, so four equal clock values already force identical action-coefficient pairs. The remaining latent reconstruction follows the preceding theorem.

If c annihilates 1 and the sampled h , then its Cauchy kernel satisfies

$$\int_a^b \sum_j \frac{c_j}{T_j - s} ds = \sum_j c_j \log \frac{T_j - a}{T_j - b} = 0. \quad (7.5)$$

A nonzero c has a nonzero rational kernel, by its distinct pole residues. This kernel is continuous on $[a, b]$, so zero integral forces both signs. In particular it cannot be positive throughout (a, b) . $\square$

This example belongs to a fixed two-dimensional function space, not to a detector space chosen after observing the data. Its larger annihilator does useful work that no positive-kernel reduction can do.

7.3. Several action thresholds and the common-divisor gauge. The same invariant has a higher-dimensional form. Replace $T - W_b$ in the model by a monic polynomial

$$D_b(T) = \prod_{\ell=1}^n (T - W_{b,\ell}), \quad W_{b,\ell} \in I,$$

with known degree $n \geq 1$, and take detector logarithms in $\mathcal{F}_{q,A}$. Repeated roots are allowed for the exact statement.

Theorem 7.7 (Action divisors). *Suppose the pairs consisting of D_b and the nonconstant detector coefficients are distinct. With $q + k + 4n + 1$ distinct clocks, the projective matrices determine the channels, the relative detector coefficients and the reduced polynomial family*

$$E_b = D_b/G, \quad G = \gcd(D_1, \dots, D_B),$$

where all polynomials are monic. They also determine $\deg G$. The complete polynomial ambiguity is to replace G by any monic polynomial $\tilde{G}$ of the same degree with roots in I . Consequently all action multisets are absolutely identifiable whenever $G = 1$. In addition there is the common additive detector gauge.

Proof. For two proposed log-ratios the moving logarithmic derivatives have at most $4n$ distinct poles, with integer residues. Together with the k protected poles, the differentiated numerator has degree at most $q + k + 4n - 1$. The stated samples force it to vanish identically. Its protected residues recover the relative detector coefficients, and its moving residues recover the reduced rational function D_b/D_c . Monicity fixes the rational multiplicative constant. The same argument with at most $2n$ moving poles proves joint spectral separation; equal spectral trajectories would force equal polynomial and nonconstant detector data. The component matrices are therefore recovered first.

Fix component 1 and write the recovered ratios in reduced form. Let H be the monic least common multiple of their denominators. At any root w , writing m_b for its multiplicity in D_b , the multiplicity in H is $\max_b(m_1 - m_b)_+ = m_1 - \min_b m_b$. Hence $H = D_1/G$ and $(D_b/D_1)H = D_b/G = E_b$. Since n is known, $\deg G = n - \deg E_1$. Every alternative family with the same ratios has these same reduced polynomials, and so is $\tilde{G}E_b$ with a common monic factor of that degree. Conversely this replacement multiplies every branch weight at each clock by $\tilde{G}(T)/G(T) > 0$ and leaves the projective matrices unchanged. This proves completeness of the gauge. $\square$

The theorem identifies unordered roots through their divisor, not a smooth enumeration through a multiple-root collision. Smooth rootwise stability requires the corresponding root-separation and fixed multiplicity assumptions. For $n = 1$ the condition $G = 1$ is exactly that not all actions coincide. The sharp $q + 3$ polynomial clock count uses the additional interval structure and improves the general bound.

8. A FINITE POLAR CLASSIFICATION OF DETECTOR SPACES

The functional criterion admits a finite algebraic classification in a natural category larger than polynomial and logarithmic detectors. Let J_0 be an open clock interval, let I be an open interval strictly to its left, and let $\mathcal{F} \subset C^1(J_0)$ be finite dimensional and contain constants. In this section every derivative of an element of $\mathcal{F}$ is a rational function without poles on J_0 . Put

$$E = \{e' : e \in \mathcal{F}\} \subset \mathbb{R}(T).$$

Let S be the finite set of poles occurring in E and $S_I = S \cap I$. Nonreal poles are understood in conjugate pairs. Partial fractions identify E with a subspace of a finite real coefficient space, including its polynomial part. Thus membership and intersection tests in E are finite linear algebra on a specified basis.

For $c \in \mathbb{Z}^{S_I}$ write $c_+ = (\max(c_s, 0))_s$ and set

$$\mathcal{Z}_2(S_I) = \left\{ c \neq 0 : \sum_s c_s = 0, \quad \sum_s (c_s)_+ \leq 2 \right\}, \quad R_c(T) = \sum_{s \in S_I} \frac{c_s}{T - s}. \quad (8.1)$$

This is a finite set of integer vectors: at most four entries are nonzero, and each entry has absolute value at most two. For $A \subset S_I$ put $D_A = \text{span}\{(T - s)^{-2} : s \in A\}$.

Theorem 8.1 (Polar classification). *The detector space $\mathcal{F}$ is regularly separating on I if and only if*

$$R_c \notin E \quad \text{for every } c \in \mathcal{Z}_2(S_I), \quad (8.2)$$

$$E \cap D_A = \{0\} \quad \text{for every } A \subset S_I, \quad 1 \leq |A| \leq 2. \quad (8.3)$$

Consequently regular admissibility is decidable by finitely many coefficient-space membership and rank tests, with no search over unknown action pairs. The first test classifies exact identification; the second classifies differentiable local inversion.

Proof. Differentiating a nontrivial secant difference gives the rational function with signed residue measure

$$\delta_x - \delta_y - \delta_{x'} + \delta_{y'}.$$

If it belongs to E , every noncancelled pole belongs to S_I . Its residue vector, when nonzero, belongs to $\mathcal{Z}_2(S_I)$. If the vector is zero, equality of the signed measures $\delta_x - \delta_y = \delta_{x'} - \delta_{y'}$, with both ordered pairs off the diagonal, implies $x = x'$ and $y = y'$. Since constants belong to $\mathcal{F}$, membership of the derivative in E is equivalent to membership of the original secant difference in $\mathcal{F}$: subtract an antiderivative in $\mathcal{F}$, leaving a constant.

Conversely, every vector in (8.1) is a secant residue vector. If its positive mass is two, list its positive atoms, with multiplicity, as p_1, p_2 and its negative atoms as n_1, n_2 . Take $(x, y) = (p_1, n_1)$ and $(x', y') = (n_2, p_2)$. Both pairs are nondegenerate because positive and negative supports are disjoint. If its positive mass is one, write $c = \delta_p - \delta_n$ and choose $z \in I \setminus \{p, n\}$. The pairs (p, z) and (n, z) realize c . This proves (8.2) in both directions.

The derivative in T of a quotient tangent is

$$\left(-\frac{u}{T - x} + \frac{v}{T - y} \right)' = \frac{u}{(T - x)^2} - \frac{v}{(T - y)^2}.$$

Distinct double poles cannot cancel. Every pole with nonzero coefficient must therefore lie in S_I if this function belongs to E . Conversely any nonzero element of $E \cap D_A$, $|A| \leq 2$, is such a derivative; in the one-pole case choose an arbitrary different second action with coefficient zero. Again constants remove the integration ambiguity. These observations prove (8.3) and the theorem. $\square$

The tests concern subspace membership, not only the presence of a pole somewhere in a basis. For instance a coupled simple-pole direction can fail the integer-residue test even when each individual coordinate is absent. This distinction is essential for nonsaturated spaces. No sign condition on a sampled annihilator occurs in the classification.

8.1. Maximal spaces under principal-part closure. Fix an ambient finite-dimensional rational space with prescribed pole orders and prescribed polynomial degree. Call a subspace *principal-part saturated* if it is invariant under projection to each individual real partial-fraction coordinate, to each conjugate real block for nonreal poles, and to the polynomial part. In the polynomial part no admissibility restriction is necessary. The next statement is within this fixed ambient space; it does not posit a largest finite-dimensional detector space without an ambient bound.

Corollary 8.2 (Maximal saturated enlargements). *For a principal-part saturated derivative space E , regular separation is equivalent to the following two conditions: no double-pole direction $(T - s)^{-2}$ with $s \in I$ is present, and simple-pole directions $(T - s)^{-1}$ are present at at most one point of I . Every polynomial direction, every principal part at a pole outside I , and every direction of order at least three at a pole in I is admissible.*

Among saturated subspaces of the fixed ambient space, the maximal regularly separating ones contain all these admissible directions, contain no forbidden double-pole direction, and contain exactly one of the available simple-pole directions in I when any is available. If none is available there is one maximal saturated subspace. Detector spaces are obtained by adjoining constants to antiderivatives.

Proof. Saturation makes membership coordinatewise. Two available simple-pole directions give R_c for $c = \delta_s - \delta_t$, whereas a single simple-pole coordinate cannot support a nonzero vector of total residue zero. A double-pole coordinate violates the tangent test already with $|A| = 1$. All other coordinates are disjoint from the supports of the two obstruction families. Theorem 8.1 therefore gives the first assertion. Adding any missing safe coordinate preserves both tests. Omitting all available simple-pole coordinates is not maximal; including two is inadmissible. This proves the maximality statement, including all alternatives. $\square$

Thus the closure operation has a precise answer. In particular a third-order pole in the derivative, whose primitive is a second-order rational detector, need not destroy regularity; the forbidden order for regular inversion is specifically two. Regularity and mere set-theoretic identification have different polar boundaries.

8.2. One pole budget for uniform clock designs. Let Q_E be the monic least common denominator of E , with degree D , and take $Q_E = 1$ if E consists of polynomials. Let $\ell(E)$ be the largest order at infinity of a nonzero member of E : a function with leading term aT^j has order j . Put

$$b(E) = \max\{\ell(E), -2\},$$

with $\ell(\{0\}) = -\infty$. Both D and $b(E)$ are read from the same coefficient space as the polar tests.

Theorem 8.3 (Polar clock budget). *If the tests of Theorem 8.1 hold, then any*

$$N = D + 6 + b(E) \tag{8.4}$$

distinct clocks in J_0 identify $e \in \mathcal{F}$ and the ordered unequal actions, and the sampled map is an immersion. Its inverse is locally smooth and uniformly Lipschitz on each compact separated action class, in the sense of (7.3). The count is sufficient, not in general minimal.

Proof. For either a scalar collision or a scalar tangent, call the associated function H . Its derivative is the sum of a member of E and a rational function with at most four moving pole factors, counted with multiplicity, which is $O(T^{-2})$ at infinity. Multiply H' by Q_E and those four factors, omitting unnecessary factors if fewer are present. The result is a polynomial of degree at most $D + 4 + b(E)$. If H vanishes at the N clocks, Rolle's theorem gives $N - 1 = D + 5 + b(E)$ distinct zeros of H' in the clock interval, where the denominator never vanishes. The numerator is zero. Thus H is constant, and a sampled zero makes it identically zero. The secant test then gives identical action pairs and zero nuisance difference; the tangent test gives a zero parameter tangent. This proves global injectivity and full differential rank. Smooth local inversion follows from a nonsingular coordinate minor. The normalized-difference and normalized-tangent compactness proof of Theorem 7.2 applies to this fixed clock set and proves its compact-class estimate. $\square$

For protected logarithms and polynomial log-detectors of degree q , $D = k$ and $b(E) = q - 1$ for $q \geq 1$, while $b(E) = -1$ for $q = 0$ and $k > 0$. This recovers the $q + k + 5$ sufficient count. For $\mathcal{F} = \text{span}\{1, \log((T - a)/(T - b))\}$, $D = 2$ and $b(E) = -2$, giving six clocks. The improved count is the cancellation of the total residue at infinity, not positivity of an annihilator. The sharp polynomial count in Theorem 6.3 is stronger than (8.4); it uses the additional oriented-interval structure. For several action thresholds, the identical argument uses $4n$ moving factors and the integer-residue balance underlying Theorem 7.7. A common action divisor is then a projective gauge, rather than a failure of the partial-fraction test.

8.3. Quantitative conventions and comparison. Throughout the function-space design statements, $|a - a'|$ denotes the Euclidean norm on $\mathbb{R}^2$. A compact $K \subset \mathcal{A}$ has $\inf_K |x - y| > 0$ and positive distance from the clock singularities. For rational-derivative spaces, compact conditioning constants also depend on the fixed coefficient basis and on the separation of moving actions from the fixed poles when such separations enter that class. An exact residue identity is not a uniform conditioning assertion as these quantities degenerate.

The closest general predecessors to the finite-design and analytic incidence arguments are the analytic experiment-count theory of Sontag [38] and finite-measurement stability results of Alberti-Santacesaria [35]. The collision incidence here has d nuisance and four action coordinates; its dimension is $d + 4$. The normalized tangent family has $d + 1$ coefficient coordinates and two action coordinates, hence dimension $d + 3$. These dimensions explain the sufficient generic count, not a random-design condition number or a new general embedding principle. The new structural conclusion in this section is instead the finite charge and double-pole classification, including maximal spaces under principal-part closure. Partial fractions and rational zero counting are classical tools; Prony-type recovery [29] uses related singular-support data. Here the obstruction vectors classify which unknown detector subspaces preserve the nonlinear action quotient before a clock design is chosen.

9. OBSERVABLE COMPONENTS AND THE EXACT GLOBAL INVERSE

At a fixed endpoint, positivity and full column rank give $B = \text{rank } P_1$. Choose sets I, J of B rows and columns such that $(P_1)_{IJ}$ is invertible. The reduced operators are

$$K_j = (P_j)_{IJ}(P_1)_{IJ}^{-1} = U_I \text{diag}(q_{j,b})U_I^{-1}, \quad q_{j,b} = \pi_{j,b}/\pi_{1,b}. \quad (9.1)$$

Lemma 9.1 (Joint separation and component recovery). *In the polynomial model the vectors $(q_{2,b}, \dots, q_{q+2,b})$ are pairwise distinct. More generally, whenever the available joint vectors are pairwise distinct, their common rank-one projectors determine*

$$E_{j,b} = (P_1)_{:J}(P_1)_{IJ}^{-1}R_b(P_j)_I = \pi_{j,b}U_{:b}V_{:b}^\top. \quad (9.2)$$

Consequently

$$\pi_{j,b} = \mathbf{1}^\top E_{j,b} \mathbf{1}, \quad U_{:b} = E_{1,b} \mathbf{1} / \pi_{1,b}, \quad V_{:b} = E_{1,b}^\top \mathbf{1} / \pi_{1,b}. \quad (9.3)$$

The outputs are independent of the minor and spectral combination up to one simultaneous permutation.

Proof. If two polynomial-model joint vectors were equal, the log ratio of unnormalized weights would be constant at $T_1, \dots, T_{q+2}$. For unequal actions, its $(q+1)$ st derivative is a nonzero constant multiple of $(T - W_b)^{-q-1} - (T - W_c)^{-q-1}$, with a strict constant sign above the actions. Repeated Rolle's theorem forbids those equal values. For equal actions, polynomial interpolation forces all nonconstant coefficient differences to vanish, contrary to the tuple hypothesis.

A real linear combination of the available K_j avoiding finitely many collision hyperplanes has simple spectrum and the desired common projectors. In their common basis, $R_b = U_I e_b e_b^\top U_I^{-1}$ and $(P_1)_{:J} (P_1)_{IJ}^{-1} = U U_I^{-1}$. Substitution gives (9.2); summation and normalization give (9.3). These are the actual component matrices, so all valid choices agree as unordered objects. $\square$

The spectral choice can be made finite and deterministic. Set $L = B(B-1)/2$. For $q \geq 1$ use

$$A_t = \sum_{\ell=0}^q t^\ell K_{\ell+2}, \quad t = 0, 1, \dots, qL. \quad (9.4)$$

Each pair excludes at most q choices, because its eigenvalue difference is a nonzero polynomial of degree at most q . One of the $qL+1$ choices separates all pairs. For $q=0$, K_2 already has simple spectrum. For general detector spaces use the analogous polynomial combination of all available operators; the same finite counting argument applies.

The recovered scalar ratios are

$$\ell_j^{bc} = \log(\pi_{j,b}/\pi_{j,c}) = \log(a_b/a_c) + \sum_{\nu=1}^q (\theta_{b,\nu} - \theta_{c,\nu}) T_j^\nu + L_{W_b}(T_j) - L_{W_c}(T_j). \quad (9.5)$$

A nonzero first contrast in (6.5) selects an unequal pair. Recover it absolutely, use the fixed-anchor inverse for the other components, and centre all coefficients as in (5.3).

Proof of exact identification and sharpness in Theorem 5.1. Rank, component recovery and scalar inversion give every centred parameter. They also give uniqueness against another full-rank representation with unknown visible count. The common C^m gauge (5.4) multiplies all weights at a clock by a common positive factor and is the entire remaining continuous ambiguity.

Define the fibre over z to be the unordered recovered columns $U_{:b}(z)$. They are distinct by full rank. Every spectral neighbourhood gives B disjoint local continuous sections, defining a covering $\mathcal{C} \rightarrow Q$, and the inverse formulas define Ξ on it. On connected overlaps the transition permutation is constant. This alone does not give a global enumeration. If Q is simply connected, path lifting from a chosen base fibre is independent of path and gives one; two such enumerations differ by a constant permutation.

For sharpness, fix positive full-rank two-component channels. The $q+2$ -clock scalar fibre keeps every log probability ratio fixed, hence every normalized matrix, while changing an action. An interior segment preserves strict probability and action constraints. The common retention baseline may first be decreased to satisfy a strict upper probability bound on the finite clock interval. Thus the lower bound lies in the same physical observation class. $\square$

10. A REFERENCE-LOCAL NOISY INVERSE OVER THE WHOLE ENDPOINT DOMAIN

Globality over Q is distinct from uniform recovery across rank or covering strata. In this section the reference rank and covering type are fixed. The finite-record theorem is a different

statement: it first learns the covering from data under additional spatial sampling and signal assumptions.

10.1. Quantitative local charts. Fix q, m, u, v, B , the clocks, Q and its finite coordinate atlas. On the reference exact field bound the C^m norms of the centred parameters and observations by L_0 , and suppose

$$\begin{aligned} r \leq T_j - W_b \leq R, \quad \pi_{j,b} \geq p_*, \quad \sigma_B(P_1) \geq \sigma_*, \\ \min_{b \neq c} \max_{2 \leq j \leq q+2} |q_{j,b} - q_{j,c}| \geq \delta, \quad \max_b W_b - \min_b W_b \geq \gamma > 0. \end{aligned} \quad (10.1)$$

Selected minors satisfy $\sigma_{\min}((P_1)_{IJ}) \geq \mu > 0$. Fix a finite refinement of the charts and a subordinate partition of unity before perturbing the data. Constants may depend on the finite cover and its partition C^m norms, the atlas, all the displayed margins, the selected spectral gap, inverse minor bounds and the centred coefficient bound. No unobservable common detector polynomial is bounded as a recovered parameter.

These choices can be selected from exact observations: rank gives the minor size, singular values select the minor, candidate spectra select a separating combination, and component contrasts select an anchor. Continuity preserves the choices locally and compactness gives a finite subcover. This is not a certification of interstitial margins from an arbitrary finite noisy mesh.

Lemma 10.1 (Uniform local extension). *On a fixed quantitative chart with selected minor, separating spectrum and unequal anchor, a pointwise smooth extension F_a to locally enumerated centred parameters satisfies*

$$\|F_a(\hat{P}) - F_a(P)\|_{C^m} \leq C_a \max_j \|\hat{P}_j - P_j\|_{C^m}. \quad (10.2)$$

Its target is the ambient affine space of normalized channel columns and centred coefficients. At a simplex boundary the extension may have signed entries; no smooth open-neighbourhood retraction into that simplex is claimed. No spatial derivative above order m is used.

Proof. For $q \geq 1$, let C_V be the largest inverse-Vandermonde infinity norm on any $q + 1$ distinct integers in $\{0, \dots, qL\}$. If a pair's eigenvalue polynomial has magnitude below $\delta/(2C_V)$ at $q + 1$ choices, every coefficient has magnitude below $\delta/2$, contradicting the joint gap. Each pair excludes at most q choices at that threshold, so some combination has all gaps at least $\delta/(2C_V)$. For $q = 0$ use the gap δ . Shrinking the chart preserves half the chosen gap.

Matrix inversion is smooth near the selected minor. At a simple root, the derivative of the characteristic polynomial is the product of the root separations. The implicit function theorem gives smooth real eigenvalue branches: a small real perturbation has one root in each disjoint root disk, and conjugation forces that root to be real. The projectors are

$$R_b = \prod_{c \neq b} \frac{A_t - \lambda_c \text{id}}{\lambda_b - \lambda_c}.$$

Thus (9.2)–(9.3), divisions by positive component masses and logarithms extend smoothly. The perturbed operators need not commute or be jointly diagonalized by these projectors.

Set $a_0 = \gamma/R^{q+2}$. Some exact first contrast has magnitude at least a_0 , being an integral over an action interval of length at least γ with integrand at least R^{-q-2} . If all contrast errors are below $a_0/4$, the largest observed contrast exceeds $3a_0/4$. Accept a pair above $a_0/2$ and freeze it on the chart. Its exact contrast exceeds $a_0/4$, so its action separation is at least $a_0 r^{q+2}/4$. Finite difference coefficients, the probability floor and the preceding spectral estimates control the contrast error. A merely nonzero noisy contrast is not treated as evidence of an anchor.

The anchor inverse is smooth with uniform derivative bounds on the compact separated class. For each remaining component, the fixed-anchor contrast has derivative $w_0(W_b) > 0$, also when

its action equals the anchor. Interpolate its nuisance polynomial on a fixed set of $q + 1$ nodes after subtraction; off the model the unused residual coordinates need not vanish. Centre the resulting coefficients. Product and chain rules through order m give (10.2). The formulas have controlled derivatives through order $m + 1$ as finite-dimensional maps; these are not unknown spatial derivatives of order $m + 1$. $\square$

With a strict entrywise channel margin, small perturbations also keep channel positivity. Without it, column sums remain one but signed entries are allowed in the ambient extension. Exact observation identities are asserted on the exact image, not on arbitrary noisy data.

10.2. Quantitative overlap matching. Fix a finite good cover Q_a and a subordinate partition χ_a . The local outputs are centred before matching. Full rank and compactness give a positive signature margin

$$s_* \leq \inf_{z \in Q} \min_{b \neq c} \|U_{:b}(z) - U_{:c}(z)\|. \quad (10.3)$$

If $\sigma_B(U) \geq s_U$, one may take $s_* = \sqrt{2}s_U$. Comparisons use columns in the original readout space, not eigenvectors from unrelated minor coordinates.

Lemma 10.2 (Synchronization under off-model perturbations). *If every extended channel column is within $e < s_*/4$ of its exact counterpart, nearest-column matching on overlaps is a unique bijection. It is constant on connected overlaps, satisfies the exact permutation cocycle on triple overlaps and agrees with the reference covering. It is unchanged on a sufficiently small data neighbourhood.*

Proof. Two approximations of the same column are within $2e$; approximations of different columns are separated by at least $s_* - 2e > 2e$. Nearest matching is therefore exactly matching to the same true component. The strict separation makes it locally constant, hence constant on connected overlaps. On a triple overlap all matchings identify the same reference column, so their composition equals the direct matching, not just approximately. The strict inequalities persist under a further small perturbation. Reference labels verify this observable nearest-column rule; they are not its input. $\square$

Write g_{ab} for these permutations. They define a covering $\hat{\mathcal{C}} \simeq \mathcal{C}$ and act on the centred local vectors. In chart a put

$$\hat{\Xi}_a = \sum_b \chi_b g_{ab} F_b(\hat{P}). \quad (10.4)$$

Products supported on overlaps are extended by zero. The cocycle $g_{ab}g_{bc} = g_{ac}$ makes these sums transform by the required transition permutation. Thus they form a field on the covering; no unmatched labels or incompatible branch gauges have been averaged.

10.3. The quotient and its quantitative scope. For isomorphic coverings over Q , define

$$d_m([\Xi], [\Xi']) = \min_{\iota: \mathcal{C} \simeq \mathcal{C}'} \|\Xi - \iota^* \Xi'\|_{C^m}. \quad (10.5)$$

The norm uses local lifts of the fixed base atlas. An isomorphism is determined by its action on one fibre because Q is connected, even when the total covering is disconnected. Only finitely many fibre permutations extend: in a fixed base fibre, an automorphism must commute with the monodromy action, and precisely such permutations extend by path lifting. Thus the minimum is finite and attained. On a simply connected base it is the minimum over one constant permutation. The continuous gauge has already been removed. This metric compares the same rank and isomorphic covering types; it does not identify fields in different monodromy or rank strata.

Theorem 10.3 (Reference-local global quotient inverse). *Fix an exact B -component reference field satisfying (10.1)–(10.3), its quantitative finite chart cover, partition and C^m bounds. There is an open C^m neighbourhood $\mathcal{N}$ of its normalized matrices and a reconstruction $\mathcal{R}$ into ambient centred affine fields on its fixed covering stratum, modulo covering isomorphisms. It is exact on nearby exact B -component data in the prescribed class and, for arbitrary real matrix fields $P', P'' \in \mathcal{N}$, satisfies*

$$d_m(\mathcal{R}(P'), \mathcal{R}(P'')) \leq C \max_j \|P'_j - P''_j\|_{C^m}. \quad (10.6)$$

The constant depends on the fixed atlas, cover, partition norms, centred coefficient bounds, residual and probability bounds, inverse minors, spectral and anchor separations and channel-signature margin. After identifying the covering with the reference one, the map is locally Lipschitz between C^m spaces, with no loss of derivatives. Monodromy is fixed on this neighbourhood. This is global over Q but local in the data and the covering stratum, and may be signed at a channel simplex boundary.

Proof. Shrink the neighbourhood so that every frozen minor, separating combination, positive division and scalar inverse remains valid, and all local column errors are below $s_*/8$. The matching lemma gives transition permutations fixed throughout this neighbourhood. Use them in (10.4). On exact model data the matched local outputs are the same centred field, and the partition sums to one, proving exactness. For two arbitrary fields, compare on the common reference covering, differentiate the finite sum through order m and apply the product rule, partition bounds and local Lipschitz estimates. This proves the bound for that covering identification and hence for the minimum in (10.5). The same transition cocycle gives the stated monodromy. $\square$

The ambient neighbourhood may contain models with additional arbitrarily weak components. The fixed-rank extension is not their model-order inverse. On a rectangle, a base-fibre labelling and transport through a spanning tree give a global representative; the cocycle and simple connectedness make it independent of the remaining paths. A local permutation statement alone would not give this conclusion.

Proposition 10.4 (A genuine monodromy example). *There are smooth log-affine observations on a circle with all the strict margins and a component covering with three-cycle monodromy.*

Proof. For $t \in [0, 2\pi]$, $b = 0, 1, 2$, put $\phi_b = t/3 + 2\pi b/3$, $W_b = -C + \cos \phi_b$, $\theta_{b,1} = \sin \phi_b$ and $a_b = 1$. Choose C large enough for the residual bound. For orthonormal e, f in the coordinate-sum-zero plane in $\mathbb{R}^3$, set

$$U_{:b} = V_{:b} = \frac{1}{3}\mathbf{1} + \epsilon(e \cos \phi_b + f \sin \phi_b).$$

Small $\epsilon > 0$ gives positive entries. The three columns are affinely independent in their probability plane and hence linearly independent. The action-rate pairs are distinct points on a circle, and the action coordinates are never all equal. Compactness gives uniform strict margins. At $t = 2\pi$ all tuples are the cyclic permutation of those at zero, so the unordered matrices are smooth on the endpoint circle. Following one column once around moves to the next; no global enumeration exists. Two independent draws from the specified, here identical, channel laws realize conditional independence. $\square$

Corollary 10.5 (Visible rank and error transfer). *With the known signal lower bound σ_* , operator-norm error below $\sigma_*/4$ in P_1 permits recovery of B by threshold $\sigma_*/2$. If joint densities H_j satisfy $f_j = \mathbf{1}^\top H_j \mathbf{1} \geq f_* > 0$ and bounded C^m norms, then C^m error at most ε in those densities gives quotient parameter error at most $C\varepsilon$ after normalization and reconstruction. Additional C^m model error η changes this to $C(\varepsilon + \eta)$ while the reference margins persist.*

Proof. Singular values change by at most the operator error; the zero values stay below the threshold and the nonzero ones above it. Summation and the density floor keep perturbed marginals above $f_*/2$. Product and reciprocal estimates control the normalization error; apply the global quotient estimate and the triangle inequality. $\square$

This corollary is a conditional error-transfer result, not a finite sample rate or unrestricted order test. Theorem 11.3 supplies a separate finite-record experiment, learns the covering type, and states the spatial and signal information needed for that task.

11. COMPLETE CLOCK FIBRES AND FINITE SPATIAL RECORDS

11.1. The entire fibre below the sharp polynomial threshold. The implicit-function lower bound has a global form. This form also specifies which additional information removes the remaining ambiguity.

Theorem 11.1 (Global polynomial ambiguity fibres). *Use $J = q + 2$ clocks, let $I \subset (-\infty, T_1)$ be an open interval, and put*

$$w(s) = \prod_{j=1}^{q+2} (T_j - s)^{-1}, \quad F' = w, \quad d = (-1)^{q+2} [T_1, \dots, T_{q+2}] f.$$

For $d \neq 0$, the complete fibre of $f_j = P(T_j) + \log(T_j - x) - \log(T_j - y)$, with $P \in \mathcal{P}_q$ and $x, y \in I$, is parametrized without repetitions by

$$u \in U_d := \{u : u, u + d \in F(I)\}, \quad y(u) = F^{-1}(u), \quad x(u) = F^{-1}(u + d). \quad (11.1)$$

For each such u there is exactly one interpolating nuisance polynomial P_u . Every nonempty fibre is a connected smooth open interval. Its orientation is fixed by the sign of d . Knowing either action selects at most one point of the fibre. If $d = 0$, there is no unequal-action solution; allowing equal actions gives the diagonal action fibre and the uniquely determined sampled polynomial.

Proof. The unique annihilator of the degree- q polynomial evaluations gives $d = \int_y^x w = F(x) - F(y)$. Since F is strictly increasing, this is equivalent to (11.1). The set U_d is the intersection of two open intervals and is an open interval whenever nonempty. For every parameter u , subtracting the corresponding logarithms from the sampled f gives a vector killed by that annihilator. It therefore lies in the $(q + 1)$ -dimensional polynomial evaluation space and determines exactly one P_u . Conversely every solution has this form. Smoothness follows from $w > 0$ and the fixed interpolation matrix; $y'(u) = 1/w(y(u)) > 0$ proves regularity and absence of repetitions. If $d = 0$, strict monotonicity gives $x = y$. $\square$

For the full interval $I = (-\infty, T_1)$ one may take the explicit primitive

$$F(s) = - \sum_{i=1}^{q+2} \frac{\log(T_i - s)}{\prod_{h \neq i} (T_h - T_i)}. \quad (11.2)$$

Partial fractions show that $F' = w$, $F(-\infty) = 0$ and $F(T_1-) = +\infty$. Thus $U_d = (\max(0, -d), \infty)$. For $q = 0$, with $\Delta = T_2 - T_1$, the parametrization is elementary:

$$F^{-1}(u) = T_1 - \frac{\Delta}{e^{\Delta u} - 1}.$$

For larger q , (11.2) and a single monotone inverse specify the whole fibre, rather than just a neighbourhood supplied by the implicit function theorem. Additional parameter bounds may cut this interval; connectedness is asserted for the declared unrestricted polynomial nuisance class, not for every additional constrained class.

Corollary 11.2 (Finite-record consequence of an exact fibre). *Fix positive full-rank two-component channels and choose two distinct points of a fibre whose ordered action pairs have distance $D > 0$. They give identical projective matrices, and therefore identical laws for any number of independent records drawn from those matrices. Every estimator of the channel-labelled action pair has worst-case expected Euclidean error at least $D/2$ on these two models, and worst-case mean squared error at least $D^2/4$. Two points may be chosen within any sufficiently small strict-margin fibre segment.*

Proof. The fixed log probability ratios determine the two normalized component weights and hence the matrices. Under their common record law, the triangle inequality gives the first bound after taking expectations; the parallelogram identity gives the squared bound. The scalar fibre varies the actions, not merely the common detector gauge. Decreasing a common retention baseline gives a strict upper retention bound on the finite clock interval, and continuity preserves strict constraints on a small segment. $\square$

On the identifiable side, a finite design satisfying (7.3) has a deterministic error guarantee. If both a true scalar model and a fitted model approximate the observed samples within ε , their nuisance and action distance is at most $2\varepsilon/c_K$. With measurement error ε and detector misspecification bounded by η in the sampled supremum norm, the same argument gives $2(\varepsilon + \eta)/c_K$ for any fit with that residual tolerance. This statement uses a specified identifiable compact class; it does not identify an unrestricted detector error.

11.2. Pointwise compact classes. For the following theorem take the polynomial model with $J = q + 3$. Fix q, u, v , residual bounds r, R , centred coefficient bounds and probability, rank, spectral, action and column-signature margins as in (10.1)–(10.3). For every $2 \leq B \leq \min(u, v)$ these inequalities define a compact pointwise parameter class $\mathcal{K}_B$, after bounding all actions and centred coefficients. The channels lie in their probability simplices. Use the Euclidean metric on the centred parameters modulo permutation. Empty classes are omitted.

There is a constant C_0 such that, for any two exact normalized matrix tuples in the same $\mathcal{K}_B$, the quotient distance of their parameters is at most C_0 times their matrix distance. Indeed the exact inverse is injective modulo a finite permutation group and has a smooth local extension at every point, by Theorem 5.1 and Lemma 10.1. A finite subcover gives the bound for nearby parameter pairs. Pairs outside these neighbourhoods have a positive minimum observed separation by compactness and injectivity, which supplies the remaining bound. Taking the maximum over the finitely many B gives C_0 . This constant may depend on the separation modulus of this entire compact pointwise class; it is not estimated by an unverified local Jacobian alone.

For a noisy tuple, first threshold the rank as below and then minimize $(\max_j \|P_j^{\text{model}} - \hat{P}_j\|_F)^2$ over $\mathcal{K}_B$. A minimizer exists by compactness. A fixed measurable tie-breaking rule may be used. If the true tuple is in $\mathcal{K}_B$ and the data error is η , the fitted and true observation tuples differ by at most 2η ; the fitted centred parameter error is consequently at most $2C_0\eta$ modulo permutation. This is an existence construction, not a claim of a polynomial-time optimization algorithm. Crucially it uses the specified class, not a supplied reference covering or a reference set of component names.

Theorem 11.3 (Finite records recover the covering and its monodromy). *Let Q have a fixed finite triangulation of mesh diameter at most h , with N_h vertices. Distances and simplex coordinates are fixed. Assume the exact field takes values in the classes $\mathcal{K}_B$ above, with one unknown constant rank B . Along lifts of each simplex, assume that the channel columns and the full centred field have Lipschitz constants at most L_U and L_Ξ , respectively, uniformly over all local enumerations. These are prior regularity bounds, not quantities certified from arbitrarily sparse data.*

At each vertex and clock observe n independent readout pairs with law P_j at that vertex. Put

$$\eta_n = \sqrt{\frac{uv}{2n} \log \frac{2JuvN_h}{\zeta}}, \quad e_n = 2C_0\eta_n, \quad 0 < \zeta < 1.$$

If

$$\eta_n < \sigma_*/4, \quad 2L_U h + 4e_n < s_*, \quad (11.3)$$

there is a reconstruction from these finite records which, with probability at least $1 - \zeta$, recovers B , the component covering up to isomorphism over Q , and its monodromy. It also gives a continuous piecewise affine centred field on that covering satisfying

$$d_0([\widehat{\Xi}], [\Xi]) \leq e_n + L_{\Xi} h. \quad (11.4)$$

The guarantee is uniform over the stated classes and all covering types satisfying these bounds. The rank threshold uses the supplied signal lower bound σ_* , not unrestricted model-order discovery. No C^m , $m \geq 1$, conclusion is asserted for piecewise affine output.

Proof. For each empirical matrix entry, the bounded-indicator exponential inequality [32] gives error at most $\sqrt{\log(2JuvN_h/\zeta)/(2n)}$ simultaneously at all vertices, clocks and entries, except on an event of probability ζ . The Frobenius error of every matrix is then at most η_n . Singular-value thresholding at $\sigma_*/2$ recovers B at every vertex: the nonzero exact singular values exceed σ_* and the remaining ones vanish. If the observed ranks are inconsistent, the procedure may reject; this does not occur on the stated event. Use the compact-class fit at each vertex and enumerate its columns arbitrarily. Its component error is at most e_n after some matching to the true fibre.

On an edge, transport a true component along that edge. Its true channel moves by at most $L_U h$. The fitted columns for its two ends are thus at distance at most $L_U h + 2e_n$. The distance to a fitted column of a different component at the other end is at least $s_* - L_U h - 2e_n$. Condition (11.3) separates these two bounds strictly. Nearest-column matching therefore gives exactly the true edge transport as a unique bijection, without knowing that transport in advance.

Every triangular face is contractible in Q , so the true transports multiply to the identity around it. The recovered edge permutations therefore satisfy the face relations exactly. Gluing copies of the simplices with these permutations constructs a covering isomorphic to the true covering. Monodromy along any edge loop is its permutation product; the face relations and cellular approximation give the same representation for every loop in Q . This also handles disconnected total covering spaces. A failed face relation may be declared a rejection outside the good event; a small numerical mismatch is not silently substituted for an exact permutation cocycle.

Within each simplex, transport the fitted vertex labels to one of its fibres and interpolate by barycentric coordinates. The edge relations make these definitions agree on common faces. The centred gauge is preserved by this affine interpolation. In the corresponding true trivialization the noise contribution is bounded by e_n and the variation from a vertex by $L_{\Xi} h$. Their sum proves (11.4) on the reconstructed covering and hence in the quotient metric. $\square$

This is a controlled spatial-design experiment with repeated readings at each vertex, not a theorem about conditioning an arbitrary continuous endpoint distribution on a point of probability zero. A random-endpoint experiment requires a density or regression estimation step. On regular d -dimensional meshes, $N_h \asymp h^{-d}$, the displayed bound has order $h + \sqrt{\log(N_h/\zeta)/n}$. With a fixed total budget it gives the familiar bias–variance balance, but no optimality for geometric parameters or derivative loss is inferred from that calculation.

12. FINITE-RECORD ACTION CERTIFICATES WITHOUT A SUPPLIED SIGNAL FLOOR

A uniform inverse constant is useful for a rate theorem but need not be an input to a confidence procedure. We distinguish a compact forward model, on which confidence is valid even at singular channels, from regular subsets, on which its diameter contracts. The distinction will allow finite records to yield a geometric error certificate.

12.1. A closed two-arm model. Fix $q \geq 0$, $J = q + 3$ distinct clocks, $u, v \geq 2$, and apparatus bounds

$$0 < D_0 < \tau_- < \tau_+ < T_1, \quad A > 0. \quad (12.1)$$

Let Δ_{r-1} be the closed probability simplex in $\mathbb{R}^r$. Our parameter set is the explicitly closed compact polytope

$$\mathfrak{K} = (\Delta_{u-1})^2 \times (\Delta_{v-1})^2 \times [0, D_0] \times [\tau_-, \tau_+] \times [-A, A]^{q+1}. \quad (12.2)$$

Write its elements as $\omega = (U, V, x, \tau, c)$, and define

$$\begin{aligned} f_j(\omega) &= \sum_{\nu=0}^q c_\nu T_j^\nu + \log(T_j - x) - \log(T_j - \tau), \\ p_j(\omega) &= \frac{\exp f_j(\omega)}{1 + \exp f_j(\omega)}, \quad \Phi_j(\omega) = U \operatorname{diag}(p_j, 1 - p_j) V^\top. \end{aligned} \quad (12.3)$$

Here c is the relative detector polynomial, including relative throughput. The common detector gauge has disappeared. The first arm is labelled by the known ordering in (12.1), not by an observed channel label. The second action is unknown and may vary independently from one acquisition site to another. No rank, entrywise positivity, spectral gap, or channel-separation lower bound is imposed in (12.2).

Use the ambient Euclidean norm on $\mathfrak{K}$ and $\|\Phi - \Phi'\|_{\text{obs}} = \max_j \|\Phi_j - \Phi'_j\|_F$. A forward Lipschitz constant is known from the apparatus bounds. For example, with $r_0 = T_1 - \tau_+$ one may take

$$L_\Phi = 2 + \frac{1}{2} \max_j \left(\sum_{\nu=0}^q T_j^{2\nu} + 2r_0^{-2} \right)^{1/2}. \quad (12.4)$$

Indeed $p'(f) = p(1 - p) \leq 1/4$, channel columns have norm at most one, and the two channel-matrix differentials have norms at most $\sqrt{2} \|dU\|_F$ and $\sqrt{2} \|dV\|_F$. The coefficient and action differential contributes at most half the displayed gradient norm. Cauchy–Schwarz and integration on the convex set $\mathfrak{K}$ give (12.4). This is a *forward* constant, which does not diverge when the channels lose rank.

12.2. Simultaneous confidence intervals. At sites $z_1, \dots, z_m$, observe n independent readout pairs at each clock. The law in cell (i, j) is $\Phi_j(\omega_i)$ for some $\omega_i \in \mathfrak{K}$. Independence is required within each cell; the different cells need not be mutually independent. Let $\hat{P}_{ij}$ be the empirical probability matrix, and put

$$\eta_n = \left\{ \frac{uv}{2n} \log \frac{2mJuv}{\zeta} \right\}^{1/2}, \quad 0 < \zeta < 1. \quad (12.5)$$

Define compact confidence sets and their action projections by

$$\begin{aligned} \mathfrak{C}_i &= \left\{ \omega \in \mathfrak{K} : \max_j \|\Phi_j(\omega) - \hat{P}_{ij}\|_F \leq \eta_n \right\}, \\ a_i^- &= \min_{\mathfrak{C}_i} x, \quad a_i^+ = \max_{\mathfrak{C}_i} x. \end{aligned} \quad (12.6)$$

An empty set is reported as model inconsistency; no endpoint is then silently assigned. For nonempty sets the extrema are attained. Because the sample spaces are finite for fixed n , fixed choices among minimizers and all reported endpoints are measurable.

Theorem 12.1 (Observable action certificates). *Uniformly over all $\omega_i \in \mathfrak{K}$, including rank-deficient channels, with probability at least $1 - \zeta$ every $\mathfrak{C}_i$ is nonempty and*

$$x_i \in [a_i^-, a_i^+] \quad (1 \leq i \leq m). \quad (12.7)$$

The confidence procedure uses only the finite records and the forward bounds (12.1); in particular it does not use an inverse constant or a supplied channel signal floor.

For $\kappa > 0$ define the regular compact subset

$$\mathfrak{K}_\kappa = \{\omega \in \mathfrak{K} : \sigma_2(U), \sigma_2(V) \geq \kappa\}. \quad (12.8)$$

There are $\epsilon_\kappa > 0$ and $C_\kappa < \infty$, depending only on this subset and the forward class, such that if every true $\omega_i \in \mathfrak{K}_\kappa$ and $\eta_n < \epsilon_\kappa$, then, on the same event,

$$\max_i (a_i^+ - a_i^-) \leq C_\kappa \eta_n. \quad (12.9)$$

The value of κ is not an input to the procedure. It indexes subsets on which its observed certificate contracts at the stated rate.

Proof. Apply the bounded-indicator exponential inequality to every entry of every empirical matrix. A union bound over $mJuv$ entries gives

$$\max_{i,j} \|\hat{P}_{ij} - \Phi_j(\omega_i)\|_F \leq \eta_n$$

outside an event of probability at most ζ . This argument uses no independence between cells. On this event the true parameters belong to their confidence sets, proving nonemptiness and coverage.

For contraction, compactness and (12.1) give a number $p_0 > 0$ such that $p_j, 1 - p_j \geq p_0$ on all of $\mathfrak{K}$. At a true parameter in $\mathfrak{K}_\kappa$,

$$\sigma_2(\Phi_1) \geq p_0 \kappa^2.$$

Every feasible candidate differs from this exact tuple by at most $2\eta_n$. If $2\eta_n < p_0 \kappa^2/2$, its first matrix has second singular value at least $p_0 \kappa^2/2$. Since probability-column matrices have operator norm at most $\sqrt{2}$, the inequalities

$$\sigma_2(\Phi_1) \leq \sqrt{2} \sigma_2(U), \quad \sigma_2(\Phi_1) \leq \sqrt{2} \sigma_2(V)$$

force both candidate channels to have full rank with a uniform lower bound. Thus all feasible candidates and the truth lie in one fixed regular compact subset of $\mathfrak{K}$.

On that subset Theorem 5.1 gives injectivity after ordering the two actions. The smooth ambient extensions of Lemma 10.1 give local inverse Lipschitz bounds. A finite subcover gives such a bound for nearby pairs of parameters. Pairs outside these neighbourhoods have a positive minimum observed separation by compactness and injectivity, so the bound extends to all pairs in that compact subset. Denote it by C'_κ . Every feasible action is therefore within $2C'_\kappa \eta_n$ of the true action. Taking both extrema proves (12.9), after enlarging the constant. This use of an inverse modulus proves a rate; it does not insert that modulus into (12.6). $\square$

The guarantee at singular channels is coverage, not a uniform shrinking radius. A wide interval is an observable lack of precision. This avoids confusing a finite-record certificate with model-order identification at arbitrarily weak signal. The two-arm architecture remains known.

Proposition 12.2 (Finite implementation). *Let $\mathfrak{K}_\delta$ be a finite Euclidean δ -net in $\mathfrak{K}$. At site i , retain its points whose residual is at most $\eta_n + L_\Phi \delta$. Take the smallest and largest retained x coordinates, pad the resulting interval by δ at both ends, and intersect it with $[0, D_0]$. These finite intervals have the simultaneous coverage of Theorem 12.1. On its regular subsets their widths are at most $C_\kappa(\eta_n + \delta)$ when $\eta_n + \delta$ is sufficiently small.*

Proof. A net point within δ of the truth has residual at most $\eta_n + L_\Phi \delta$ and an x coordinate within δ of the true one. Conversely every retained point has observed distance at most $2\eta_n + L_\Phi \delta$ from the truth. The compact regular inverse estimate in the preceding proof applies once this number is small. The two padding terms add 2δ to the width. $\square$

The simplex and interval factors in (12.2) admit explicit rational grids. Hence the procedure terminates after a finite search, rather than only specifying a continuous minimizer. Its size grows as a power of δ^{-1} with exponent equal to the fixed parameter dimension; no dimension-uniform polynomial-time guarantee is claimed. Taking $\delta \leq \eta_n$ preserves the statistical rate.

12.3. Relation to calibration and synchronization. Blind bilinear calibration can sometimes be reduced to linear least squares, with efficient algorithms under the hypotheses of Ling–Strohmer [36]. Our forward map instead has two unknown readout channels and a nonlinear action factor under polynomial log selection. The theorem supplies a finite exhaustive certificate for that compact model, not a replacement for those efficient calibration algorithms. Spectral permutation synchronization from noisy pairwise permutations, including the sharp statistical analysis in Nguyen–Zhang [37], is likewise a different observation experiment. Here scalar action intervals require no spatial permutation synchronization. For component coverings, the nearest-column and face-cocycle reconstruction remains the separate statement of Theorem 11.3.

13. FINITE SELECTIVE RECORDS DETERMINE A METRIC WITH A CERTIFICATE

We now join the statistical experiment to the geometric inverse. The observations are finitely many categorical readout pairs, not exact boundary distances at a continuum of endpoints. The reference delay may vary with the endpoint pair. Neither its values nor the detector coefficients nor either readout channel are calibrated.

13.1. The geometric stability class. Fix a compact d -dimensional manifold M , $d \geq 2$, with identified smooth boundary and a fixed coordinate atlas. We use the following precise geometric input from Stefanov–Uhlmann [39, Theorem 5]. For every $0 < \mu < 1$ there is an integer k such that, near a simple metric g_0 with s-injective geodesic ray transform, metrics with a fixed C^k bound satisfy

$$\|g' - \psi^* g\|_{C^2(M)} \leq C_{\text{geom}} \|d_g - d_{g'}\|_{C^0(\partial M \times \partial M)}^\mu \quad (13.1)$$

for some diffeomorphism ψ fixing the boundary. Here proximity and the constant have exactly the meaning of that theorem: g, g' are in its C^0 neighbourhood of g_0 and have a prescribed C^k bound. S-injectivity means that the kernel of the ray transform on symmetric two-tensors consists of symmetric differentials of vector fields vanishing at the boundary. Analytic simple metrics are examples by [39, Theorem 1].

Fix a compact subset $\mathcal{G}$ of this neighbourhood in the C^k topology, consisting of simple metrics with the same bounds. One may, for example, take the C^k closure of a sufficiently small smooth C^{k+1} -bounded family near g_0 , with uniform ellipticity and a smaller C^k neighbourhood ensuring simplicity. Fix a smooth boundary metric with distance ρ_∂ and use the product distance

$$\rho((x, y), (x', y')) = \rho_\partial(x, x') + \rho_\partial(y, y').$$

Choose L, D_0 such that for every $g \in \mathcal{G}$,

$$\sup d_g < D_0, \quad |d_g(z) - d_g(z')| \leq L\rho(z, z'). \quad (13.2)$$

Such constants follow from a uniform upper bound on the boundary metric and the triangle inequality for d_g . These are geometric prior bounds, not estimates of unobserved detector derivatives. The pullback in (13.1) is indispensable; we do not identify different coordinate representatives as equal tensors.

Let $\Lambda_h \subset \partial M$ be a finite $h/2$ -net. Acquire records at all ordered distinct pairs in Λ_h^2 , denoted $z_1, \dots, z_{m_h}$. At equal pairs the boundary distance is exactly zero and requires no acquisition. These pairs together form an h -net for the product distance. At each acquired pair use the two-arm model (12.3), with

$$x_i = d_g(z_i), \quad \tau_i \in [\tau_-, \tau_+]. \quad (13.3)$$

The remaining parameters are arbitrary members of their factors in (12.2). In particular no continuity of the unknown channels, detector coefficients, or reference delays between acquired pairs is required. The ordering and bounds in (12.1) remain known architectural information.

Theorem 13.1 (Finite noisy geometric rigidity). *In this experiment, form the action intervals of Theorem 12.1, or its finite-grid implementation, with $m = m_h$. Let w_h be their largest width, and set*

$$\widehat{\mathcal{G}} = \{g' \in \mathcal{G} : a_i^- \leq d_{g'}(z_i) \leq a_i^+, 1 \leq i \leq m_h\}. \quad (13.4)$$

With probability at least $1 - \zeta$, this is a nonempty compact set containing the true metric. Every two of its members satisfy, for some boundary-fixing diffeomorphism ψ ,

$$\|g' - \psi^*g\|_{C^2(M)} \leq C_{\text{geom}}(w_h + 2Lh)^\mu. \quad (13.5)$$

The number w_h is obtained from the records; no inverse Lipschitz constant or channel signal floor is needed to produce it.

If the true channels at every acquired pair belong to one regular class (12.8), then, for sufficiently large n ,

$$\|\widehat{g} - \psi^*g\|_{C^2(M)} \leq C \left\{ h + \left(\frac{\log(2m_h Juv/\zeta)}{n} \right)^{1/2} \right\}^\mu \quad (13.6)$$

for every $\widehat{g} \in \widehat{\mathcal{G}}$ on the same event. For finite-grid action intervals assume $\delta \leq \eta_n$. The constants may depend on $\mathcal{G}, \mu$, the apparatus bounds, clocks, and the true regular class; the confidence procedure does not need the last dependence as input.

Proof. Simultaneous action coverage puts the true g in (13.4). The map $g \mapsto d_g$ is continuous on the uniformly elliptic compact class; this also follows by comparing lengths when two metrics are uniformly close. Consequently the inequalities defining (13.4) are closed, and the set is compact.

For two feasible metrics the boundary distances differ by at most w_h at each observed pair, and agree at every diagonal pair. For arbitrary $z \in \partial M \times \partial M$, choose $z_0 \in \Lambda_h^2$ with $\rho(z, z_0) \leq h$. Twice using (13.2) gives

$$|d_g(z) - d_{g'}(z)| \leq |d_g(z) - d_g(z_0)| + |d_g(z_0) - d_{g'}(z_0)| + |d_{g'}(z_0) - d_{g'}(z)| \leq w_h + 2Lh.$$

This reasoning includes the case in which z_0 is on the diagonal. Apply the stated geometric theorem (13.1) to obtain (13.5). Finally Theorem 12.1 bounds w_h by $C_\kappa \eta_n$, or Proposition 12.2 bounds it by $C_\kappa(\eta_n + \delta)$. Substitution proves (13.6). $\square$

No component covering or global enumeration is used in this theorem. The lower action at an individual acquisition site is the propagation distance by the apparatus ordering. The interpolation step uses only the regularity of boundary distance, not a spatial interpolation of

unknown nuisance parameters. Thus rough or independently changed readout channels do not enter its spatial bias term.

Corollary 13.2 (A finite geometric dictionary). *Suppose a finite set $\mathcal{G}_\rho \subset \mathcal{G}$ is supplied whose boundary distances form a ρ -net for those of $\mathcal{G}$ in the uniform norm. Retain its metrics satisfying*

$$a_i^- - \rho \leq d_{g'}(z_i) \leq a_i^+ + \rho \quad \text{for every } i.$$

On the coverage event this finite set is nonempty. Every retained metric differs from the truth, after a boundary-fixing pullback, by at most $C_{\text{geom}}(w_h + \rho + 2Lh)^\mu$. The corresponding bound between two retained metrics uses $w_h + 2\rho + 2Lh$.

Proof. A dictionary metric within boundary-distance error ρ of the truth passes every padded interval. At observed pairs a retained metric and the truth differ by at most $w_h + \rho$. Repeat the proof of Theorem 13.1. For two retained metrics the padding is used twice. $\square$

Compactness guarantees the existence of finite dictionaries. Constructing one with certified distance errors is additional numerical work for a given prior; it is not silently supplied by an infinite-dimensional optimization oracle. With a supplied dictionary and a forward-parameter grid, both searches are finite. The exact compact-set statement does not claim an efficient general-purpose metric reconstruction algorithm.

Corollary 13.3 (Total acquisition budget). *For regular quasiuniform boundary nets on a d -dimensional manifold, $m_h \asymp h^{-2(d-1)}$. With $N = Jm_h n$ total records, the choice $h \asymp (\log(N/\zeta)/N)^{1/(2d)}$, and comparable allocation to all cells, gives*

$$\|\hat{g} - \psi^* g\|_{C^2(M)} \leq C \left(\frac{\log(N/\zeta)}{N} \right)^{\mu/(2d)} \quad (13.7)$$

with probability at least $1 - \zeta$, whenever the regular-class bounds hold uniformly along the sequence of experiments. For surfaces the exponent is $\mu/4$. This is an upper rate for the stated design; no nonparametric minimax assertion for metrics is inferred from it.

Proof. Substitute $n = N/(Jm_h)$ into (13.6) and balance h with $\{h^{-2(d-1)} \log(N/\zeta)/N\}^{1/2}$. Integer rounding changes only constants for sufficiently large budgets. $\square$

13.2. What is used from geometric rigidity. Theorem 18.2 remains an exact continuum theorem for all simple surfaces, with a constant reference arm and observations at every ordered distinct boundary pair. Its last step is the global boundary-distance theorem of Pestov–Uhlmann [33]. The present finite-record theorem instead uses the precise local stability class of [39], allows an endpoint-dependent reference delay, and ends with a quantitative C^2 pullback error from finitely many noisy records. Neither theorem replaces the other’s hypotheses.

The geometric estimate (13.1) is imported explicitly, not reproved by the clock inversion. The new statement is the complete finite-observation implication with a data-derived action certificate and its statistical rate. Inverse problems with finite measurements as in [35] provide important methodological precedent; here the measured variables are uncalibrated categorical records, and the spatial bias is controlled even without interstitial nuisance regularity. A shrinking confidence diameter requires regularity of the true channels, whereas validity of the certificate does not.

14. SHARP GEOMETRIC EXPERIMENTS

The preceding spatial rate does not assert nonparametric optimality. For finite-dimensional geometric priors one can give matching upper and lower statistical bounds. One can also transfer the exact clock fibre into a genuine metric ambiguity. These statements distinguish loss on a geometrically realizable parameter from loss on an arbitrary action field.

14.1. Finite boundary designs for geometric priors. Let $\theta \mapsto g_\theta$ be a C^2 map from a neighbourhood of $0 \in \mathbb{R}^p$ into smooth simple metrics, with g_0 having s-injective geodesic ray transform. Suppose the tangent tensors $h_\ell = \partial_{\theta_\ell} g_\theta|_0$ are linearly independent modulo symmetric differentials of vector fields vanishing on the boundary. The parameter is a chosen coordinate on this geometric family, not an arbitrary coordinate representative modulo an unspecified diffeomorphism.

Lemma 14.1 (A finite geometric design). *There are p ordered distinct boundary pairs $z_1, \dots, z_p$ and a sufficiently small closed Euclidean ball $B \subset \mathbb{R}^p$ about zero such that*

$$|\theta - \theta'| \leq C_D \left| (d_{g_\theta}(z_i) - d_{g_{\theta'}}(z_i))_{i=1}^p \right| \quad (\theta, \theta' \in B). \quad (14.1)$$

In particular these finitely many distances distinguish all metrics in the family on B , including against boundary-fixing isometries.

Proof. For a fixed distinct boundary pair, first variation of the length of its unique geodesic gives

$$\partial_{\theta_\ell} d_{g_\theta}(z)|_0 = \frac{1}{2} \int_{\gamma_{0,z}} h_\ell(\dot{\gamma}_{0,z}, \dot{\gamma}_{0,z}) dt.$$

The variation of the geodesic itself contributes zero by stationarity with fixed endpoints. Simplicity gives smooth dependence near a fixed nondegenerate pair. If a linear combination of these p functions vanished at every boundary pair, s-injectivity would make the corresponding combination of the h_ℓ a potential tensor, contrary to the hypothesis. Their evaluation row vectors therefore span $\mathbb{R}^p$: otherwise a nonzero vector orthogonal to their span would give just such a vanishing combination. Choose p rows forming an invertible matrix A_0 .

Write $D(\theta) = (d_{g_\theta}(z_i))_i$ and $s_0 = \sigma_{\min}(A_0) > 0$. Shrink a convex ball about zero until $\|D'(\theta) - A_0\|_{\text{op}} \leq s_0/2$ there. Integrating on the line segment from θ' to θ gives

$$|D(\theta) - D(\theta')| \geq |A_0(\theta - \theta')| - (s_0/2)|\theta - \theta'| \geq (s_0/2)|\theta - \theta'|.$$

This proves (14.1). Boundary-fixing isometries preserve all the selected distances, so the last assertion follows as well. $\square$

Finite measurement selection from a finite-dimensional prior is a classical principle; see [35]. The lemma identifies its geometric derivative explicitly. Its role here is to combine the geometric prior with uncalibrated finite readout records, rather than to claim a new abstract finite-dimensional inverse theorem.

Choose apparatus bounds covering all the distances of this family on B . At the p selected pairs use $J = q + 3$ clocks, and take n independent pairs per cell, now independently across all cells. The reference delay, detector coefficients and both channels may differ at every site. Define a reconstruction by minimizing the maximum matrix residual over $\theta \in B$ and all nuisance parameters in the compact factors of (12.2), with $x_i = d_{g_\theta}(z_i)$. This compact feasible model is nonempty and a minimizer exists. Fix a deterministic tie-breaking rule.

Theorem 14.2 (Sharp parametric geometric risk). *Suppose the true nuisance parameters belong to a fixed nonempty regular class (12.8). With $N = pJn$ total records, the estimator just defined satisfies*

$$\sup \mathbb{E}|\hat{\theta} - \theta|^2 \leq C/N. \quad (14.2)$$

The supremum is over $\theta \in B$ and all such nuisance parameters. The estimator does not require the value of their rank lower bound.

If the class contains an interior parameter point, strictly positive full-rank channels and interior detector and delay values, then for all sufficiently large N ,

$$c/N \leq \inf_{\tilde{\theta}} \sup \mathbb{E}|\tilde{\theta} - \theta|^2 \leq C/N. \quad (14.3)$$

The constants depend on the fixed geometric family and acquisition design. This is a minimax theorem for that geometrically realizable finite-dimensional prior, not for unrestricted simple metrics.

Proof. Let $e = \max_{i,j} \|\hat{P}_{ij} - P_{ij}\|_F$. The minimizing model has residual no larger than e , so its exact observation tuple differs from the true one by at most $2e$ at each site. For e below a fixed threshold, the regular-compact argument in the proof of Theorem 12.1 gives a distance error at most $C_\kappa e$ at every selected pair, even though the minimization allowed singular candidate channels. Equation (14.1) gives $|\hat{\theta} - \theta| \leq Ce$. For e above that threshold, the same inequality holds after enlarging C , since B has bounded diameter. For a categorical empirical matrix,

$$\mathbb{E} \|\hat{P}_{ij} - P_{ij}\|_F^2 = n^{-1} \sum_{a,b} P_{ij,ab} (1 - P_{ij,ab}) \leq n^{-1}.$$

Bounding the maximum by a sum yields $\mathbb{E} e^2 \leq pJ/n$. Since p, J are fixed and $N = pJn$, this proves (14.2). Measurability is immediate on the finite sample space with fixed choices of minimizers.

For the lower bound fix the nuisance parameters at one of the strict interior points specified in the statement. Move the metric parameter from an interior θ_0 to $\theta_0 + tv$ along a unit vector. All categorical probabilities are bounded below by a positive constant and depend smoothly on t , at the finitely many fixed cells. Therefore

$$\text{KL}(P_{ij,t}, P_{ij,0}) \leq \sum_{a,b} \frac{(P_{ij,t,ab} - P_{ij,0,ab})^2}{P_{ij,0,ab}} \leq Ct^2.$$

Independence gives relative entropy at most CNt^2 for all records. Taking $t = a/\sqrt{N}$ with a fixed sufficiently small $a > 0$ makes the total variation distance between the two record laws at most $1/2$. For any estimator, the test choosing the nearer parameter errs only when its squared error is at least $t^2/4$. The sum of the two testing errors is at least one minus the total variation distance. Thus the larger of its two squared risks is at least $t^2/16$. This proves the lower bound; the upper bound is already established. $\square$

The usual information inequalities used in this two-point argument are stated, for example, in [40]. Unlike the earlier weighted-action risk theorem, the alternatives here are actual metrics in a fixed geometric family. There is no transfer of an arbitrary action perturbation into geometry without a realizability argument.

14.2. The geometric meaning of the missing clock. The endpoint-dependent reference class admits an exact clock threshold, not only a sufficient count. It is important to state the reference class in the lower bound: the following ambiguity need not preserve a constant reference delay.

Theorem 14.3 (Sharp clock threshold for geometric recovery). *For the ordered two-arm class with arbitrary unknown endpoint-dependent reference delay in (12.1), $q + 3$ clocks suffice for the geometric conclusions of Theorems 13.1 and 14.2. With $q + 2$ fixed clocks there are non-boundary-isometric simple metrics, with strictly positive full-rank readout channels and strict apparatus margins, giving identical projective matrices at every boundary pair. They therefore give the same law of any number of records. Such alternatives can be chosen arbitrarily close to an analytic simple metric, and within a one-dimensional smooth geometric family.*

Proof. Sufficiency has been proved. For necessity put $J = q + 2$ and choose an analytic simple metric g_0 with $\sup d_{g_0} < D_0$. Let $g_t = (1 + t)^2 g_0$, for t in a sufficiently small interval about zero; then $x_t(z) = d_{g_t}(z) = (1 + t)d_{g_0}(z)$. These metrics are simple, and distinct values of t have different boundary distances, hence are not related by a boundary-fixing isometry.

Let $w(s) = \prod_{j=1}^{q+2} (T_j - s)^{-1}$ and choose $F' = w$ on $(-\infty, T_1)$. Fix an interior constant delay $\tau_0 \in (\tau_-, \tau_+)$ and an interior relative polynomial P_0 (one may take $P_0 = 0$). Define, at every boundary pair,

$$\tau_t(z) = F^{-1}(F(x_t(z)) - F(x_0(z)) + F(\tau_0)). \quad (14.4)$$

For uniformly small $|t|$ this is well defined and belongs to (τ_-, τ_+) , by compactness of the range of x_0 and positivity of F' on the relevant compact intervals. Moreover

$$F(x_t(z)) - F(\tau_t(z)) = F(x_0(z)) - F(\tau_0).$$

The complete-fibre theorem, Theorem 11.1, now supplies a unique degree-at-most- q polynomial $P_{t,z}$ satisfying, at all $q+2$ clocks,

$$P_{t,z}(T_j) + \log(T_j - x_t(z)) - \log(T_j - \tau_t(z)) = P_0(T_j) + \log(T_j - x_0(z)) - \log(T_j - \tau_0). \quad (14.5)$$

For completeness, the difference of the two logarithmic sample vectors has zero degree- q annihilator by the preceding identity. It therefore lies in the polynomial evaluation subspace and has a unique interpolating polynomial. The fixed interpolation map and compact residual margins imply $P_{t,z} \rightarrow P_0$ uniformly in its coefficient vector, over every boundary pair. Thus the fixed strict coefficient bounds persist. All these quantities are smooth away from the boundary diagonal; no diagonal observations are needed.

Fix strictly positive full-rank probability channels, the same for all t and z . Equation (14.5) keeps the normalized component weights, and hence every projective matrix, exactly fixed. Positive throughputs realizing $P_{t,z}$ are available; decreasing a common retention baseline preserves strict retention bounds without changing any projective record. All apparatus inequalities have uniform strict room. The construction is a small scaling family near g_0 , so it lies in the local geometric stability class after shrinking its parameter interval. This proves the claimed geometric, not merely scalar, ambiguity. $\square$

The theorem exhibits precisely the ambiguity removed by the additional clock. It does not assert the same necessity for a constant-reference subclass, where cross-endpoint information can constrain the scalar fibres. At $q+3$ clocks even an independently varying bounded reference is recoverable pointwise; at $q+2$ the same scalar fibre can conceal a change of metric. This is the common observation mechanism behind the sharp scalar inverse, the confidence certificates, and the geometric experiment.

15. RAW ACQUISITION COSTS AND QUANTITATIVE CONVENTIONS

The budget N in the preceding readout experiment counts retained pairs. A physical acquisition must also count failures. We now give a fixed-attempt experiment, without assuming that its success probability is calibrated or that a positive lower bound is supplied to the confidence procedure.

Theorem 15.1 (Fixed raw attempts with observed acceptance). *At each of mJ cells make B independent raw attempts. An attempt returns either a failure symbol or a readout pair. Its success probability $r_{ij} \in [0, 1]$ is unknown, and the conditional law of a successful pair is $\Phi_j(\omega_i)$ from (12.3). Different cells need not be independent. Let K_{ij} be the observed number of successes. For $K_{ij} > 0$ put*

$$\eta_{ij} = \left\{ \frac{uv}{2K_{ij}} \log \frac{4mJuv}{\zeta} \right\}^{1/2}, \quad (15.1)$$

and let $\hat{P}_{ij}$ be the empirical conditional matrix. Construct action confidence sets by imposing the residual bound η_{ij} at each nonempty cell and imposing no constraint at empty cells. These sets cover all true actions simultaneously with probability at least $1 - \zeta/2$, for every choice of

the success probabilities. In particular they give the metric certificate (13.5) in the geometric experiment.

On a true regular channel class, suppose in addition that $r_{ij} \geq r_* > 0$ at every cell, where r_* need not be known to the procedure. If

$$Br_* \geq 8 \log \frac{2mJ}{\zeta},$$

then, for sufficiently large Br_* , with probability at least $1 - \zeta$ the largest action interval width is at most

$$C_\kappa \left\{ \frac{\log(4mJuv/\zeta)}{Br_*} \right\}^{1/2}. \quad (15.2)$$

The raw budget is exactly mJB , including every failure. In the boundary experiment the resulting metric error is at most

$$C \left\{ h + \left(\frac{\log(4m_h Juv/\zeta)}{Br_*} \right)^{1/2} \right\}^\mu. \quad (15.3)$$

Proof. Conditional on $K_{ij} = k > 0$, the retained categorical observations in that cell have the multinomial law of k independent draws from $\Phi_j(\omega_i)$. Apply the entrywise exponential bound conditional on this count. The radius in (15.1) gives failure probability at most $\zeta/(2mJ)$ for that matrix, uniformly in k . Integrate over K_{ij} and take a union bound over cells. Empty cells impose no constraint and hence cannot exclude the truth. This proves the first assertion without a success floor. A finite-grid implementation uses cellwise tolerances $\eta_{ij} + L_\Phi \delta$ and the same action padding as Proposition 12.2.

For the rate, a binomial lower-tail bound gives

$$\Pr\{K_{ij} < Br_*/2\} \leq \exp(-Br_*/8).$$

One elementary derivation applies Markov's inequality to $e^{-tK_{ij}}$, uses $(1 - r_{ij} + r_{ij}e^{-t})^B \leq \exp\{-Br_{ij}(1 - e^{-t})\}$, and takes $t = \log 2$; the resulting exponent is at most $-Br_{ij}/8$ at half its mean. The event with threshold $Br_*/2$ is smaller when $r_{ij} \geq r_*$. A union bound and the displayed budget condition make the total failure probability at most $\zeta/2$. On the remaining event every radius is bounded by a constant times the square root in (15.2). Repeat the regular compact inverse argument of Theorem 12.1, using the largest cell radius. Theorem 13.1 then gives (15.3). All attempts, including late arrivals, missed detections and rejected readings represented by the failure symbol, have already been counted in mJB . $\square$

The conditional law of a successful readout remains a model assumption. Unmodelled false readings change that law; they cannot be repaired by only charging their cost. At very small success probabilities the certificate remains valid but may stay wide. Thus no uniform raw budget rate is asserted without actual acceptance information in the class. The rate parameter r_* is not a hidden calibration input to the procedure.

15.1. Closed classes for covering reconstruction. For clarity, the compact classes used in Theorem 11.3 can be specified without chart choices. Fix all its margins and an additional bound A_0 . For each $2 \leq B_0 \leq \min(u, v)$ define

$$\mathcal{K}_{B_0} = \{\Xi = (W, U, V, \alpha, \vartheta) : \Xi \text{ satisfies (15.5)}\}. \quad (15.4)$$

The closed constraints in this definition are

$$\begin{aligned}
U &\in (\Delta_{u-1})^{B_0}, & V &\in (\Delta_{v-1})^{B_0}, \\
\sum_b \alpha_b &= 0, & \sum_b \vartheta_{b,\nu} &= 0, \\
\max(\|W\|_\infty, \|\alpha\|_\infty, \|\vartheta\|_\infty) &\leq A_0, \\
r \leq T_j - W_b \leq R, & \quad \pi_{j,b} \geq p_*, & \quad \sigma_{B_0}(P_1) \geq \sigma_*, \\
\min_{b \neq c} \max_{2 \leq j \leq q+2} |\pi_{j,b}/\pi_{1,b} - \pi_{j,c}/\pi_{1,c}| &\geq \delta, \\
\max_b W_b - \min_b W_b &\geq \gamma, & \quad \min_{b \neq c} \|U_{:b} - U_{:c}\| &\geq s_*.
\end{aligned} \tag{15.5}$$

All component and clock inequalities are imposed, and empty classes are omitted. The weights π and matrices P are computed from (5.1) in the centred gauge. The inequalities are continuous on the stated residual and coefficient bounds, so these sets are compact. Selected minors and separating combinations are local tools for proving the inverse theorem, not extra unclosed choices in the definition of the estimator.

With d_{perm} the Euclidean distance modulo simultaneous permutation, a precise admissible inverse constant is any upper bound for

$$\max \left\{ 1, \max_{B_0} \sup_{d_{\text{perm}}(\Xi, \Xi') > 0} \frac{d_{\text{perm}}(\Xi, \Xi')}{\max_j \|P_j(\Xi) - P_j(\Xi')\|_F} \right\}. \tag{15.6}$$

The suprema are over the declared classes. Their finiteness follows from the compact exact inverse proof preceding Theorem 11.3. Thus C_0 is a global prior-class modulus, not a quantity inferred from the finite records or from one local Jacobian. The new two-arm confidence construction does not require this constant as input.

For the experiment in Theorem 11.3, independence is required among the n readings in each vertex–clock cell. Its union bound does not require independence between cells. For the topological step, the edge permutations and triangular-face relations describe the covering on the two-skeleton of the triangulation. Cellular approximation identifies its fundamental group with that of the whole complex, and covering classification extends the same monodromy over all higher-dimensional simplices. No further higher-dimensional cocycle or unobserved permutation relation is being assumed.

16. RETURN LIFTS, PHYSICAL OBSERVATIONS, AND PHASE COVERAGE

The clock inverse recovers absolute functions, not just their differentials. In particular the constant in a generating function is fixed by the zero of $T - W$; it is not chosen later by integrating a one-form. This is essential for return time.

On a canonical domain $X \subset \mathbb{R}^{2h}$ with $\lambda = p dx$, an exact lift is (F, A) with $dA = F^* \lambda - \lambda$. Its operations are

$$(F, A)(G, B) = (F \circ G, B + A \circ G), \quad (F, A)^{-1} = (F^{-1}, -A \circ F^{-1}). \tag{16.1}$$

An absolute type-I sheet $S(x, y)$ represents

$$p = -S_x(x, y), \quad p' = S_y(x, y), \quad A(x, p) = S(x, y). \tag{16.2}$$

It is regular when S_{xy} is invertible. No minimization or convexity of S is required.

Theorem 16.1 (Selective records determine a covered return lift). *Let r, s be coprime positive integers. Suppose finitely many regular endpoint gates, in fixed canonical coordinates, are given for each of g^r, g^s . Their actual absolute generating sheets are the components of Theorem 5.1 with fixed detector degree q . The sheet counts, channels, actions and pairing across gates are not*

supplied. Assume that these sheets are powers of one deterministic exact lift, and that their phase graphs cover the nested domains of a fixed word $(g^r)^u(g^s)^v$, $ur + vs = 1$, including inverse-factor sources, with fixed collars. Domains and factor order have the precise recursive meaning in (18.1).

Then $q + 3$ deadlines on each gate determine g on that covered target, including its absolute primitive. Under the margins of Theorem 10.3, fixed C^{k+2} action bounds and uniform inverse mixed-Jacobian and phase-map inverse bounds, the reconstruction is locally Lipschitz from C^{k+2} normalized matrix data to the C^k lift. Theorem 18.1 gives a separate observable replacement for the common-lift assumption on an observed invariant domain.

Proof. The action theorem gives all unordered absolute sheets, including coincident action values. Their derivatives give the phase graphs. At an initial phase point, regularity solves $p = -S_x(x, y)$ locally for y . Overlapping solutions have the same output and primitive because they represent the same deterministic lift. Equality of initial phase coordinates therefore pairs the recovered graphs, without semantic component names. This gives g^r, g^s on the required domains, and the lift group law gives $g = (g^r)^u(g^s)^v$. Every factor is evaluable on its nested source by the coverage hypothesis; the primitive constants are retained.

For noisy extension fix a finite cover by smaller phase charts whose closures lie inside the implicit-function domains, and a partition of unity. Continue the selected reference sheet and the solution of $p = -S_x$ using the global quotient inverse. Uniform mixed-Jacobian bounds give C^k estimates for the local phase maps and primitives from C^{k+2} sheet errors. The additional derivative bounds their C^{k+1} norms. Exact maps agree on overlaps in the fixed canonical coordinates, so their nearby coordinate-valued extensions and primitives may be averaged with the fixed partition. This is not averaging unmatched sheet indices. For inverse factors continue the local inverse branches on their own fixed source cover; the inverse Jacobian bound gives the corresponding estimates. Compose in the specified order on the nested collars. Product, chain and inverse estimates give the asserted finite-word C^k bound. Off the model, the extension need not be symplectic, nor need independently extended forward and inverse factors be exact inverses. Both properties hold on the exact image. $\square$

Corollary 16.2 (An existing transverse return section). *Let a smooth Reeb flow have a smooth transverse global section, with first-return map F and time $\tau > 0$. If the actual return sheets of two coprime counts satisfy Theorem 16.1, selective records recover (F, τ) and its section-preserving strict contact suspension on the covered return domain. This determines the entire suspension only if the entire relevant section and its return trajectories are covered; otherwise the conclusion is the covered suspension or flow-box germ.*

Proof. Let λ be the contact form restricted to the section. The section flow-box pulls the contact form back to $dt + \lambda$. Differentiating the variable-time return identity gives $F^*\lambda - \lambda = d\tau$. Thus (F, τ) is an exact lift, whose powers have elapsed return times as their absolute primitives. Apply the theorem. The quotient $(x, t) \sim (Fx, t - \tau(x))$ preserves $dt + \lambda$ on the domain where that return identification is observed. $\square$

Supplied structure in Theorem 16.1	Recovered from matrices
Endpoint gates and canonical coordinates	Visible sheet count and component covering
Return counts and their common deterministic lift	Absolute actions and their primitives
Phase domains, inverse sources and collars	Canonical graphs and coordinate pairing
Regularity and a fixed detector family	Relative detector coefficients and weights
Two independent clock-invariant full-rank readouts	Both unknown channel laws
An existing transverse global section for a flow	The return lift and covered suspension

No unobserved section or continuation through an unobserved caustic is inferred. Later we replace the common-lift premise by observable identities and determine word domains from recovered graphs.

16.1. An exact ambiguity outside the sampled orbit tube.

Theorem 16.3 (An unsampled open phase region). *Let $g = (F, A)$ be a smooth exact lift on a canonical phase domain. For finitely many positive n , suppose even the full restrictions $g^n|_{D_n}$ are known, with compact $\overline{D_n}$ and all iterates defined. Set*

$$K = \bigcup_n \bigcup_{j=1}^n F^j(\overline{D_n}).$$

If $O \subseteq X \setminus K$ is open and $F^{-1}(O) \cap D_ \neq \emptyset$ for an open target D_* , there are arbitrarily small smooth exact perturbations with identical lifted observations on every D_n but different values on D_* . They may preserve a strict positive-roof margin on a fixed compact neighbourhood of the target.*

Proof. Choose $z_* \in F^{-1}(O) \cap D_*$. Choose a smooth Hamiltonian with support compactly contained in O and nonzero Hamiltonian vector field at $F(z_*)$. Its sufficiently small-time flow h_ϵ stays in the canonical domain, is the identity outside O , and moves $F(z_*)$ for all small nonzero ϵ . Compact support allows extension by zero to a neighbourhood of the domain, so no boundary escape is hidden in this assertion. With $\iota_{X_H} d\lambda = -dH$, Cartan's formula gives

$$B_\epsilon = \int_0^\epsilon (\lambda(X_H) - H) \circ h_t dt, \quad h_\epsilon^* \lambda - \lambda = dB_\epsilon.$$

Both the displacement and this primitive vanish outside O . Put $g_\epsilon = (h_\epsilon, B_\epsilon)g$. For $z \in D_n$, induction through $j \leq n$ gives $g_\epsilon^j(z) = g^j(z)$ as lifts: each original successive phase image lies in K , where the extra lift is the identity with zero primitive. This preserves accumulated primitives as well as phase maps. The maps differ at z_* . Compact support gives convergence in every fixed finite smooth norm, so a strict positive-roof margin on the chosen compact set persists. $\square$

The result shows an unsampled-orbit freedom in the unrestricted smooth class. It does not claim that sources illuminating additional trajectories remain unchanged, that the construction lies in every billiard subclass, or that every sufficient collar and phase-domain hypothesis is separately necessary.

16.2. Physical common clocks.

Proposition 16.4 (One source and common deadlines). *Suppose the actions on finitely many regular gates lie in $[0, H]$. For $J = q + 3$, choose $h > 0$, a uniform independent release delay on $[0, D]$, where $D = H + (J + 1)h$, and deadlines $T_j = H + jh$. Use one normalized phase source at all clocks. After the time gate, apply unknown branchwise polynomial log-retention of degree at most q on the finite clock interval and two conditionally independent clock-invariant readings.*

The model is (5.1), with residual and unused-delay margins at least h . No component audit is needed to choose the deadlines.

Proof. The release event has probability $(T_j - W_b)/D$, with residual in $[h, D - h]$. For a regular sheet S_b , the change from initial momentum to terminal position gives successful density

$$\frac{\rho(x, -S_{b,x}) |\det S_{b,xy}|}{D} \eta_b \exp \left(\sum_{\nu=1}^q \theta_{b,\nu} T_j^\nu \right) (T_j - W_b).$$

All factors outside the exponential and residual are clock independent and are absorbed in a_b . On the finite interval a sufficiently small positive baseline η_b makes retention strictly below one. Independent conditional randomizations give the product channel laws. A branch-blind detector or exposure only rescales the matrix. $\square$

For $q = 1$, a phase-dependent survival rate supplies such a detector. Conditional independence and deadline invariance are apparatus assumptions; copying a noisy observation does not establish them. The detector family is prescribed, not unrestricted.

16.3. A nonconvex return system and its persistence.

Proposition 16.5 (Three sheets through an action crossing). *There is a nonconvex exact return system with three regular sheets, two having coincident actions along a curve, which satisfies the log-affine hypotheses of Theorem 16.1. These properties persist in a sufficiently small smooth neighbourhood of the lift on slightly smaller fixed phase and endpoint domains.*

Proof. On $\mathbb{R}^2$ let

$$\phi(p) = p^3 - p, \quad G(p) = \frac{1}{4}p^4 - \frac{1}{2}p^2, \quad F(x, p) = (x + \phi(p), p), \quad A(p) = 1 + \frac{3}{4}p^4 - \frac{1}{2}p^2.$$

Then $F^*\lambda - \lambda = p\phi'(p) dp = dA$ and $A \geq 11/12$. The positive-roof quotient preserves $dt + p dx$ and has this return lift. For count n , put $t = (y - x)/n$ and let $p_-(t), p_0(t), p_+(t)$ be the three smooth roots of $\phi(p) = t$ near $-1, 0, 1$. On a small interval about zero, the absolute sheets are

$$S_{n,b}(x, y) = n + p_b(t)(y - x) - nG(p_b(t)) = nA(p_b(t)).$$

They satisfy $-S_{n,b,x} = S_{n,b,y} = p_b(t)$ and $S_{n,b,xy} = -1/(n\phi'(p_b(t))) \neq 0$. On the diagonal the outer actions are both $5n/4$, the central one is n , and the derivative of the outer action difference with respect to $y - x$ is 2. Thus the crossing is transverse and an unequal-action anchor persists on a small compact gate. The central and outer mixed derivatives have opposite signs.

Choose $r(p) = r_0 + r_1 p > 0$ on the compact momentum patches, with $r_1 \neq 0$, and retention $\eta e^{-r(p)(T-T_1)}$, $0 < \eta < 1$. The rates separate the two outer components at their action crossing. For one reading take

$$(1, e^{\beta p}, e^{2\beta p}) / (1 + e^{\beta p} + e^{2\beta p}), \quad \beta \neq 0,$$

and for the other the analogous law with a different unknown nonzero coefficient. Independent conditional draws give two positive full-rank channels: their columns are normalized Vandermonde columns at the three distinct momenta. The observer is not given momenta, rates or response coefficients. A fixed normalized smooth source positive on the compact patches and Proposition 16.4 give the common clocks.

For counts two and three, momentum is unchanged and shifts are bounded on these patches. Choose finitely many endpoint rectangles for the successive input and output neighbourhoods of $g = g^3(g^2)^{-1}$, then smaller target domains with collars. These supply the required nested cover. Under a small exact smooth perturbation on the chosen finite orbit tubes, implicit-function continuation keeps the three regular branches of both iterates. No extra branch is included in this restricted compact source class. Mixed-Jacobian, channel-rank, residual and anchor margins persist, and the nonzero derivative of the outer action difference continues the crossing curve

on smaller gates. Keep the sensors and their independent randomizations fixed as functions of initial momentum; their rank and rate separation persist. Collars permit the same finite gate families over smaller targets and the strict roof bound persists. The fold points $\phi'(p) = 0$ remain outside these domains. $\square$

17. EXACT BOUNDARIES OF THE OBSERVATION THEOREM

Proposition 17.1 (Equal actions and unrestricted detectors). *If every component has one common action w , even arbitrarily many projective clocks do not determine it. If the branch detector can vary freely with deadline, small changes of unequal actions can likewise be compensated exactly.*

Proof. The common factor $T_j - w$ cancels under projectivization. In the second case replace W_b by a nearby $W'_b < T_1$ and the retention by

$$e'_b(T) = e_b(T) \frac{T - W_b}{T - W'_b}.$$

All unnormalized weights remain identical. Strict residual and retention margins preserve admissibility for small changes. The new logarithm need not lie in the prescribed detector family, so this does not contradict the identification theorems. $\square$

Proposition 17.2 (One uncalibrated reading does not suffice). *With one unknown reading, two unequal actions and strict positive full-rank channels need not be identifiable at any number of clocks.*

Proof. Take unit weights and zero detector coefficients. The first model has actions $(0, 1)$ and columns $(.2, .8)^\top, (.8, .2)^\top$; the second has actions $(-1, 2)$ and columns $(.4, .6)^\top, (.6, .4)^\top$. For all $T > 2$, both unnormalized single-reading vectors equal $(T - .8, T - .2)^\top$. The channels are positive and invertible, but the unordered action sets differ. A common scaling gives valid finite-clock success probabilities without changing equality. $\square$

This is an obstruction to one unknown channel over the full class, not to calibrated or otherwise supplemented observations. Enough clock contrasts, independent readout information and an unequal-action anchor are distinct requirements. Phase acquisition is a separate geometric issue. The positive function-space and geometric results above identify larger classes without removing these stated distinctions.

18. OBSERVABLE COMPATIBILITY AND A GEOMETRIC APPLICATION

The common-lift assumption can be replaced by identities of the reconstructed objects, provided one works on an observed invariant domain. We first specify what is to be checked and what domain is actually obtained. An identity of phase maps alone is insufficient: the absolute primitives are part of every identity below.

18.1. Graph consistency and domains of words. For each indicated count n , recover all absolute sheets and form

$$\mathcal{G}_n = \bigcup_{\text{gates, sheets}} \{((x, -S_x(x, y)), (y, S_y(x, y)), S(x, y))\}.$$

Regularity makes each piece a local exact lift. On an open subset of the union of its source projections, check that all graph pieces with the same source have the same target *and the same primitive*. Check likewise that the target projection has only one source. These are equalities and injectivity tests on the recovered graphs. Where they hold, the pieces glue to an exact diffeomorphism between the observed source and target domains; no semantic component

matching is supplied. The tests may fail for inconsistent data, in which case no common deterministic graph is asserted on that region.

For a finite word with factors $a_1, \dots, a_N$, in their order of application, write the observed partial lifts as (F_i, A_i) and let D_i be their observed source domains. For an inverse factor the source is the corresponding observed range and the primitive is $-A \circ F^{-1}$. Define recursively

$$\begin{aligned} \Omega_0 &= X, & Z_0 &= \text{id}, & B_0 &= 0, \\ \Omega_i &= \{z \in \Omega_{i-1} : Z_{i-1}(z) \in D_i\}, \\ Z_i &= F_i \circ Z_{i-1}, & B_i &= B_{i-1} + A_i \circ Z_{i-1} \quad \text{on } \Omega_i. \end{aligned} \tag{18.1}$$

Each Ω_i is open. Induction on the number of factors proves that Ω_N is exactly the domain on which this word can be evaluated using these observed pieces. The formula keeps every primitive constant and resolves the order of negative factors. For example, $g^3(g^2)^{-1}$ first uses the observed inverse of g^2 , then the observed g^3 at its output. It is not enough for both maps to be separately known somewhere on the original target. This is a maximal *word-evaluation* domain, not a claim of maximal recovery by all possible geometric arguments.

Theorem 18.1 (Certification of a common return lift). *Let r, s be coprime positive integers. Suppose graph consistency and source–target tests on the recovered count- r and count- s sheets give exact lifts R, S that are diffeomorphisms of an observed open domain Ω onto itself. There exists an exact lift g on Ω with $g^r = R$ and $g^s = S$ if and only if*

$$RS = SR, \quad R^s = S^r \tag{18.2}$$

as full exact lifts. When these conditions hold, that lift is unique and, for any $ur + vs = 1$,

$$g = R^u S^v.$$

Thus the common-lift premise is testable from the reconstructed graphs and primitives on Ω . The counts, canonical coordinates and apparatus class remain supplied information; a section or unobserved phase region is not inferred by this test.

Proof. Powers of a common lift satisfy (18.2). Conversely, exact lifts form a group under (16.1) on the invariant domain. Commutativity and $R^s = S^r$ give

$$(R^u S^v)^r = R^{ur} (S^r)^v = R^{ur+sv} = R, \quad (R^u S^v)^s = (R^s)^u S^{vs} = S^{ru+vs} = S.$$

These equalities hold in the lift group, so include the primitives. If h is any other common lift, then $h = h^{ur+vs} = R^u S^v$, proving uniqueness and independence of the Bezout pair. All compositions are defined because both lifts preserve Ω . On noninvariant observed domains one uses (18.1) instead and makes no unverified assertion that powers of the resulting partial word extend to a common global lift. $\square$

The source and target domains in this theorem are obtained from the generating sheets, rather than being phase charts supplied with component labels. Their invariance and graph consistency are genuine hypotheses expressed in observable quantities. On compact subsets with the collars and inverse-Jacobian bounds of Theorem 16.1, the word map remains locally Lipschitz under perturbation of its input lifts. A small compatibility residual is not, by itself, asserted to imply a nearby exact common lift.

The orbit-localized ambiguity of Theorem 16.3 explains why an unsampled open orbit region cannot in general be filled in smoothly from these graph tests. It does not show that each particular collar, inverse-domain choice or common-lift certificate is necessary. The positive conclusion here is instead the exact domain recursion and a common-lift criterion that can be checked on recovered objects.

18.2. Selective boundary travel-time observations. The next application uses a familiar geometric rigidity problem, not a detector specifically engineered for the polynomial shear. A two-arm apparatus has a physical propagation arm and a reference arm with an unknown constant delay. Neither the readout channels, relative throughput nor selective detector coefficients are calibrated. The existence of the reference arm and a coarse delay bound are apparatus information; its delay value is recovered from the records.

Theorem 18.2 (Two-arm boundary-distance rigidity). *Let (M, g) be a compact simple Riemannian surface with identified boundary. At each ordered pair $x \neq y$ of boundary points, suppose two-arm observations have the model (5.1) with actions*

$$W_1(x, y) = d_g(x, y), \quad W_2(x, y) = \tau_*,$$

where the constant τ_ is unknown and $\sup_{x,y} d_g(x, y) < \tau_* < T_1$. Both arms have positive throughput, and the two unknown conditionally independent readout channels have full column rank. The branchwise polynomial log-detectors have known degree bound q , with otherwise unknown coefficients that may depend on x, y . Then $q + 3$ projective deadline matrices at every such pair determine the boundary distance, the unknown reference delay and the metric up to a boundary-fixing isometry.*

Proof. The two actions are unequal at every pair. Theorem 5.1 therefore recovers their unordered values and both unknown channels from the projective records. The smaller recovered value is the boundary distance, by the declared delay ordering; this use of order is justified by an apparatus separation, not by sorting sheets across an action crossing. The larger value gives τ_* . Continuity adds the zero boundary distance on the diagonal. The boundary-distance rigidity theorem for simple surfaces [33] then identifies the metric up to a diffeomorphism fixing the boundary. In particular two admissible metrics with the same projective records have identical boundary distances, even if their detector coefficients, throughputs, channels and initially unknown reference delays differ. $\square$

To realize the clock factor, choose a release delay uniform on $[0, D]$ with $D > T_J$, independently of the arm, and use an apparatus bound placing both travel times in $[0, T_1]$. The event $\alpha + W_b \leq T_j$ has probability $(T_j - W_b)/D$. On a compact gate, a positive unknown source and the endpoint change of variables give a positive arm weight. A constant-delay reference path supplies the second weight. Apply the branchwise retention and the two independent readouts as in Proposition 16.4; a small common baseline keeps retention below one. The same source is used at all clocks. This construction uses no sensor response tied to a special formula for the unknown metric.

The geometric last step is the cited boundary-distance theorem, not a new proof of that theorem or a claim for nonsimple surfaces. The new input reduction removes detector and channel calibration while making the required reference acquisition explicit. This application coexists with the nonconvex multiple-sheet example: it does not replace or remove that action-crossing regime. For a flow, reconstructing a suspension on a covered invariant section domain still gives only that suspension; the whole contact manifold follows only when the whole relevant global section and its return trajectories are covered.

19. SHARED CALIBRATION AND QUANTITATIVE PROOF SUPPLEMENTS

The following statements supply details for the shared-apparatus and meromorphic regimes above. They do not replace their hypotheses by those of the projective spectrum experiment.

19.1. Finite obstruction levels and continuity.

Lemma 19.1 (Finite-level exclusion). *Let Q be a nonempty topological space, let $Q_* \subset Q$ be dense, and let $x : Q \rightarrow \mathbb{R}$ be continuous. If $x(Q_*)$ is contained in a finite set S , then $x(Q) \subset S$. If x is nowhere locally constant, the preimage of each finite set is closed with empty interior, and its complement is open and dense. If two continuous real-valued functions agree on that complement, they agree everywhere.*

Proof. Since S is closed, $x^{-1}(S)$ is closed and contains the dense set Q_* , and therefore equals Q . For any $s \in S$, an interior point of $x^{-1}(s)$ would give an open set on which x is constant. Thus every $x^{-1}(s)$ is closed with empty interior. A finite union of such sets again has empty interior: intersecting their open dense complements gives an open dense set, by a finite induction. Finally the equality set of two continuous real-valued functions is closed, so agreement on a dense set implies agreement everywhere. $\square$

In Theorem 2.2, different clock constants would force the true action on the regular set into at most one direct obstruction level and two swapped obstruction levels. The lemma makes the contradiction explicit. After the constants agree, it extends equality of the actions off the finitely many exceptional levels to all of Q . The choice between the two labels may vary from point to point; no continuation of eigenvectors, no local constancy of that choice, and no assumption excluding arbitrarily fine label oscillation is used.

19.2. A quantitative certificate for four shared anchors. The four-site calibration procedure has a finite, computable separation certificate. It displays separately the conditioning of an admissible label choice and the distance to a false calibration.

Fix four sites as in Proposition 2.3, with four distinct actions and noncoincident two-clock profiles. Choose one representative (r_{i1}, r_{i2}) of the recovered odds at each site. For $\sigma \in \{0, 1\}^4$ set

$$s_{ij}^\sigma = \begin{cases} r_{ij}, & \sigma_i = 0, \\ r_{ij}^{-1}, & \sigma_i = 1, \end{cases} \quad A_\sigma = (s_{i1}^\sigma, -s_{i2}^\sigma)_{i=1}^4, \quad b = (T_1 - T_2)\mathbf{1}.$$

Let H be a fixed compact rectangle contained in $(0, \infty)^2$ whose interior contains $h_* = (1/k_1, 1/k_2)$. Define

$$\mathcal{A} = \{\sigma : A_\sigma h_* = b\}, \quad \lambda = \min_{\sigma \in \mathcal{A}} \sigma_{\min}(A_\sigma), \quad d = \min_{\sigma \notin \mathcal{A}} \min_{h \in H} \|A_\sigma h - b\|_2. \quad (19.1)$$

An empty outer minimum in d is interpreted as $+\infty$.

Proposition 19.2 (Certified four-anchor calibration). *The constants λ, d in (19.1) are positive. Suppose all recovered odds and their inverses are bounded by a fixed number M , and suppose perturbed representatives can be matched to them, independently at each site, with entrywise error at most $\varepsilon \leq (2M)^{-1}$. There is $C = C(M, H)$ such that a minimizer*

$$(\hat{\sigma}, \hat{h}) \in \operatorname{argmin}_{\sigma \in \{0,1\}^4, h \in H} \|\hat{A}_\sigma h - b\|_2$$

satisfies

$$C\varepsilon < d/3 \implies \hat{\sigma} \in \mathcal{A}, \quad \|\hat{h} - h_*\|_2 \leq C\varepsilon/\lambda. \quad (19.2)$$

The clock constants $k_j = 1/h_j$ have the same local error order. The procedure does not require the true label choices as input.

Proof. The obstruction argument for four distinct action values proves that any positive solution of any $A_\sigma h = b$ must equal h_* : a different calibration could account for at most three distinct levels. This argument uses positivity, not a prior bound on a competing action range. The true label pattern belongs to $\mathcal{A}$. If an active matrix had a nonzero null vector, an arbitrarily

small movement of h_* in that direction would stay in the interior of H and give another positive solution, a contradiction. Hence $\lambda > 0$. Inactive matrices have no solution on H ; compactness and the finite number of patterns give $d > 0$.

For sufficiently small ε , inversion is Lipschitz on the fixed odds interval, so all sixteen matrix perturbations have operator norm at most $C\varepsilon$. At an active choice and h_* the perturbed residual is at most $C\varepsilon$; the minimum has no larger residual. Since H is bounded, the unperturbed residual of a minimizing pair is therefore at most $C\varepsilon$, after increasing C . If $C\varepsilon < d/3$, this pair is active. Now

$$\lambda \|\hat{h} - h_*\|_2 \leq \|A_{\hat{\sigma}}(\hat{h} - h_*)\|_2 = \|A_{\hat{\sigma}}\hat{h} - b\|_2 \leq C\varepsilon.$$

The reciprocals of the coordinates of H have bounded derivatives, which proves the final assertion. $\square$

The certificate involves only sixteen two-variable constrained least-squares problems once the noiseless odds are fixed. Bounds for nearby exact odds follow by the same finite minimization. For the correct direct labelling, the determinant of two rows of A has absolute value

$$k_1 k_2 |T_2 - T_1| |x_i - x_l|.$$

Thus repeated anchor levels and small residuals of false swapped calibrations are distinct sources of poor conditioning. To start from noisy *matrices*, one must also control the odds-extraction error at the four regular anchors; on the compact regular classes of Proposition 2.3 it is bounded by a constant times the matrix error. This proposition is a finite-anchor certificate, not a claim of uniform Lipschitz recovery through a two-clock profile collision or a minimax theorem for an unrestricted spatial field.

19.3. The binary determinant cancellation, coefficient by coefficient. In the proof of Theorem 3.2, write $e = h' - h$, $C_b = U_{:b} V_b^T$ and $\mathfrak{d} = \det U \det V$. With $r = a/(1+a)$,

$$K(T) = s(T)C_1 + t(T)C_2, \quad s(T) = r(T-x), \quad t(T) = (1-r)(T-\tau).$$

Put $H = (T_1 - h)(T_2 - h)$. The affine interpolant of the true matrices is $Q = q_1 C_1 + q_2 C_2$, where

$$\begin{aligned} q_1(T) &= r [1 + (h-x)(T_1 + T_2 - h - T)/H], \\ q_2(T) &= (1-r) [1 + (h-\tau)(T_1 + T_2 - h - T)/H]. \end{aligned}$$

All four affine scalar polynomials have uniformly bounded coefficients. Since $\det(vC_1 + wC_2) = vw\mathfrak{d}$, the exact difference is

$$\det(K - eQ) - \det K = \mathfrak{d}\{-e(sq_2 + tq_1) + e^2 q_1 q_2\}. \quad (19.3)$$

Every coefficient is at most $C\rho(|e| + e^2)$ in absolute value. The pole estimate $|e| \leq C\delta/\gamma$, the bounded pole interval, and $\rho \leq C\gamma$ make this at most $C\delta$.

For the second perturbation, let an affine matrix be $M_0 + TM_1$. Its determinant coefficients are

$$\det M_0, \quad (M_0)_{11}(M_1)_{22} + (M_1)_{11}(M_0)_{22} - (M_0)_{12}(M_1)_{21} - (M_1)_{12}(M_0)_{21}, \quad \det M_1.$$

On $\|M_0\|_F + \|M_1\|_F \leq R$, the difference of each coefficient under a coefficient perturbation of size $\nu \leq 1$ is at most $C(R+1)\nu$, by subtracting the displayed products. Here $K' - (K - eQ)$ is the interpolant of $(T_j - h')(P'_j - P_j)$, $j = 1, 2$, whose coefficient norm is at most $C\delta$. No inverse channel factor is introduced by this second step. This proves the required $C\delta$ determinant-coefficient bound before division by its leading coefficient of order ρ .

19.4. Finite-cell radii. For binary observations, twelve entries across the three clocks are controlled. If every entry error is at most t , each Frobenius error is at most $\sqrt{4}t = 2t$. Thus the radius $\eta = \sqrt{(2/n) \log(24/\zeta)}$ satisfies

$$\Pr\{\max_j \|\hat{P}_j - P_j\|_F > \eta\} \leq 24 \exp(-n\eta^2/2) = \zeta.$$

This supplies explicitly the norm conversion in Theorem 3.3; independence between different cells is not needed for the union bound.

19.5. Periods and entire branched fibres. In Lemma 4.1, imaginary-period normalization means vanishing real parts of periods on the *punctured* surface. A basis of the $2g$ handle cycles together with small residue loops generates those periods, subject to the usual one relation between puncture loops. A small positive loop around a pole of real residue r has period $2\pi ir$, already imaginary. The holomorphic correction fixes the handle periods; there is no additional omitted condition at a puncture. Multiplication by i converts this convention to real periods.

For Theorem 4.4, the finite set $S_\pi = \pi(\text{supp } D) \cap I_0$ is only a candidate set. At a point p over s , of ramification index e_p , a nonzero secant charge produces a simple pole of $d\pi/(\pi - s)$ with residue e_p , whereas the tangent $d\pi/(\pi - s)^2$ has pole order $e_p + 1$. Consequently the sharper support candidates are

$$S_{\text{sec}} = \{s \in I_0 : (\pi^{-1}(s))_{\text{red}} \leq D\}, \quad S_{\text{tan}} = \{s \in I_0 : \sum_{p \rightarrow s} (e_p + 1)p \leq D\}.$$

They are finite subsets of S_π . The charge test may be restricted to S_{sec} and the tangent test to S_{tan} : outside them membership in E is impossible. The proof is the disjointness of fibres and the displayed local pole orders. These support conditions are necessary, not substitutes for membership in the actual subspace E . They leave the stated sufficient clock budgets unchanged.

COMPANION OBSERVATION REGIMES

The preceding sections contain the selective-detector and shared-apparatus regimes together with their proof supplements. The following sections retain the remaining geometric and acquisition derivations. Their hypotheses are stated separately; they are not additional premises of the principal singular-spectrum theorem.

APPENDIX A1. ACTION RIGIDITY UNDER SELECTIVE DETECTION

The information in an endpoint experiment depends on the detector as well as on the dynamics. A detector common to all trajectories acts by a scalar on a joint readout matrix. A detector which can vary freely with both the trajectory and the deadline can instead compensate a change of action. Between these two regimes lies a natural question: can the action be separated from an unknown, branch-selective detector whose dependence on the deadline belongs to a fixed finite-dimensional family? We answer this question for log-affine detectors, and then for polynomial log-detectors. Neither the detector coefficients nor the readout channels are calibrated.

Fix five distinct deadlines $T_1 < \dots < T_5$, and a compact connected rectangle $Q \Subset \mathbb{R}^d$. At endpoint z , two finite readings are conditionally independent given the latent component. Their probability matrices $U(z) \in \mathbb{R}^{u \times B}$ and $V(z) \in \mathbb{R}^{v \times B}$ have full column rank. Their columns sum to one, and their entries are nonnegative. Both channels may depend on z , but not on the deadline. The positive integer $B \leq \min(u, v)$ is not supplied to the observer. Before arbitrary branch-blind rescaling, the joint matrices are

$$M_j = U \text{diag}\{a_b e^{\kappa_b T_j} (T_j - W_b)\}_{b=1}^B V^\top, \quad a_b > 0, \quad W_b < T_1. \quad (\text{A1.1})$$

Every coefficient in this formula is an unknown function of z . The observations are positive projective classes $[M_j]$, represented without loss by

$$P_j = \frac{M_j}{\mathbf{1}^\top M_j \mathbf{1}} = U \operatorname{diag}(\pi_{j,b}) V^\top, \quad \pi_{j,b} = \frac{a_b e^{\kappa_b T_j} (T_j - W_b)}{\sum_c a_c e^{\kappa_c T_j} (T_j - W_c)}. \quad (\text{A1.2})$$

In particular, failure counts, success masses, endpoint exposures and any common positive detector factor at each deadline may all be absent. Normalization is of the entire joint matrix, not of each row or column.

A physical instance has retention $e_b(T, z) = \eta_b(z) \exp(\kappa_b(z)(T - T_*))$, restricted to the finite clock interval so that it is a probability. The factor $\eta_b e^{-\kappa_b T_*}$ is absorbed in a_b . An exponential survival probability with an unknown nonnegative, phase-dependent rate is an example. The mathematical model does not require an observer to know that rate or to measure the phase point.

Theorem A1.1 (Five-deadline selective-detector rigidity). *Assume (A1.1) on Q with C^m coefficients, where $m \geq 0$ is an integer. At every endpoint assume that the pairs (W_b, κ_b) are pairwise distinct and that the actions are not all equal. Then five projective observations determine the visible component count B , all absolute actions W_b , both channels, the relative weights a_b/a_c , and all differences $\kappa_b - \kappa_c$. They do so up to one simultaneous permutation on the connected gate. The complete continuous ambiguity is*

$$a_b \mapsto h a_b, \quad \kappa_b \mapsto \kappa_b + \rho, \quad h(z) > 0, \quad (\text{A1.3})$$

common to every component. It does not change an action.

The reconstruction is locally Lipschitz in C^m , without loss of spatial derivatives, on the quantitatively bounded classes specified in Proposition A4.1. No component audit, semantic branch catalogue, calibrated channel, or measured exposure is used. Actions of different components may coincide at an endpoint: distinct detector rates separate such components, provided another pair of components has unequal actions at that same endpoint.

The unequal-action condition is pointwise, not merely somewhere on Q . A uniformly separated anchor pair can be chosen locally and is selected from the observations. It is not an extra label supplied by the apparatus.

The theorem concerns the number of full-rank visible components. It does not count invisible duplicates in a rank-deficient representation. Nor is conditional independence obtained by copying one noisy reading. The two readings must arise from independent randomizations conditional on the trajectory and endpoint. These assumptions are part of the observation operator.

A1.1. The principal mechanism and its context. The proof separates two issues. Simultaneous diagonalization first recovers latent probability trajectories from the matrices. A scalar interpolation theorem then separates the absolute action from an unknown exponential detector rate and from all projective normalizations. The second step, rather than a new claim about diagonalization, is the information-rigidity assertion of this revision.

Algebraic identification of latent classes is classical; see Allman–Matias–Rhodes [2]. Observable-operator methods and moment methods are developed in [3]; joint diagonalization in a nonorthogonal basis is treated directly in Bonhomme–Jochmans–Robin [20]. Those results motivate the spectral part below. Here the third experimental coordinate is a prescribed deadline, not another unknown component response: its affine zero is the absolute geometric action, even after unknown selective attenuation. Our scalar proof is elementary and is included in full. No priority claim for the general matrix-pencil or latent-class mechanisms is made.

The conversion of recovered generating functions into canonical graphs uses classical variational calculus [12, 16]. Return-time primitives and contact suspensions have their usual meaning [13, 1].

The new conclusion precedes those constructions: the absolute functions, their count, and their local pairing are recovered despite a selective detector and without semantic branch labels. The subsequent geometric application still states its phase-coverage hypothesis explicitly.

APPENDIX A2. OBSERVABLE COMPONENTS WITHOUT ACTION SEPARATION

Fix an endpoint for the moment. Since the weights in (A1.2) are positive and both channels have full column rank,

$$B = \text{rank } P_1. \quad (\text{A2.1})$$

Choose row and column index sets I, J of size B for which $(P_1)_{IJ}$ is invertible, and put

$$K_j = (P_j)_{IJ}(P_1)_{IJ}^{-1} = U_I \text{diag}(q_{j,b})U_I^{-1}, \quad q_{j,b} = \pi_{j,b}/\pi_{1,b}. \quad (\text{A2.2})$$

Neither K_2 nor K_3 needs to have simple spectrum.

Lemma A2.1 (Joint spectral separation). *If the pairs (W_b, κ_b) are distinct, the vectors $(q_{2,b}, q_{3,b})$ are pairwise distinct. Consequently the matrices K_2, K_3 determine their common rank-one spectral projectors.*

Proof. Suppose two such vectors were equal. The ratio of their unnormalized component weights would take the same value at T_1, T_2, T_3 . Its logarithm, up to a constant, is

$$f(T) = (\kappa_b - \kappa_c)T + \log(T - W_b) - \log(T - W_c).$$

If $W_b \neq W_c$, then $f''(T) = -(T - W_b)^{-2} + (T - W_c)^{-2}$ has a strict constant sign on $T > \max(W_b, W_c)$. A strictly convex or strictly concave function cannot take one value at three ordered points. If $W_b = W_c$, f is a nonconstant affine function because the parameter pairs are distinct, and cannot take one value even at two points. This is a contradiction. A linear combination of K_2, K_3 avoiding the finitely many pairwise eigenvalue coincidences has simple spectrum and the desired common projectors. $\square$

There is a finite deterministic choice in the last sentence. Set $L = B(B - 1)/2$. Among $K_2 + tK_3$, $t = 0, \dots, L$, at least one has simple spectrum: a pair can exclude at most one value of t . Let R_b be its spectral projectors, labelled locally. For every clock, form

$$E_{j,b} = (P_1)_{:J}(P_1)_{IJ}^{-1}R_b(P_j)_{I:}. \quad (\text{A2.3})$$

Lemma A2.2 (Recovery of channels and probability trajectories). *The preceding construction gives*

$$E_{j,b} = \pi_{j,b}U_{:b}V_{:b}^T, \quad \pi_{j,b} = \mathbf{1}^T E_{j,b} \mathbf{1}, \quad U_{:b} = E_{1,b} \mathbf{1} / \pi_{1,b}, \quad V_{:b} = E_{1,b}^T \mathbf{1} / \pi_{1,b}.$$

The result is independent of the invertible minor and of the separating linear combination, up to the same permutation.

Proof. In the common diagonalizing basis, $R_b = U_I e_b e_b^T U_I^{-1}$, and $(P_1)_{:J}(P_1)_{IJ}^{-1} = U U_I^{-1}$. Also $(P_j)_{I:} = U_I \text{diag}(\pi_{j,c}) V^T$. Substitution proves the first identity. Summation proves the remaining identities since each channel column has total mass one. The resulting expressions are intrinsic component matrices and hence do not depend on the construction choices. $\square$

These lemmas explain why a coincidence of actions need not destroy component recovery. Distinct selective rates can separate the observed clock trajectories even when the two actions are equal. They do not resolve two components with identical (W, κ) by merely declaring their channel columns different.

APPENDIX A3. FIVE VALUES SEPARATE ACTIONS FROM DETECTOR RATES

For two recovered components define

$$\ell_j^{bc} = \log \frac{\pi_{j,b}}{\pi_{j,c}} = A_{bc} + K_{bc}T_j + \log(T_j - W_b) - \log(T_j - W_c), \quad (\text{A3.1})$$

where $A_{bc} = \log(a_b/a_c)$ and $K_{bc} = \kappa_b - \kappa_c$. All common exposures have disappeared.

Lemma A3.1 (Five-value rigidity). *Let $x, y, \tilde{x}, \tilde{y} < T_1$, and let $T_1 < \dots < T_5$. If*

$$A + KT_j + \log(T_j - x) - \log(T_j - y) = \tilde{A} + \tilde{K}T_j + \log(T_j - \tilde{x}) - \log(T_j - \tilde{y})$$

for $j = 1, \dots, 5$, and $x \neq y$, then $(A, K, x, y) = (\tilde{A}, \tilde{K}, \tilde{x}, \tilde{y})$.

Proof. Let $H(T)$ be the difference of the two expressions. It has five zeros. Its derivative is

$$H'(T) = k + \frac{1}{T-x} - \frac{1}{T-y} - \frac{1}{T-\tilde{x}} + \frac{1}{T-\tilde{y}}, \quad k = K - \tilde{K}.$$

Multiply by $D(T) = (T-x)(T-y)(T-\tilde{x})(T-\tilde{y})$. The rational part has numerator of degree at most two, because its four residues sum to zero. Thus $DH' = kD + R$, $\deg R \leq 2$. Rolle's theorem supplies four distinct zeros of H' in (T_1, T_5) . If $k \neq 0$, these are all four roots of the quartic $kD + R$. Their sum, by its cubic coefficient, is $x + y + \tilde{x} + \tilde{y} < 4T_1$. Each root exceeds T_1 , a contradiction. If $k = 0$, four zeros force R to vanish identically. Therefore H' is identically zero. Its limit at infinity also gives $k = 0$.

Equality of the rational functions now gives equality of the signed pole multisets $\{x, \tilde{y}\} = \{y, \tilde{x}\}$, with multiplicities. Since $x \neq y$, it follows that $x = \tilde{x}$ and $y = \tilde{y}$. Finally, a value at one deadline gives $A = \tilde{A}$. Repeated poles cause no difficulty in the polynomial argument or in this cancellation argument. $\square$

No optimization algorithm or genericity argument is used in this global uniqueness statement. Five is a sufficient count; the theorem does not claim that it is the smallest global count.

Lemma A3.2 (An observable anchor and the remaining components). *The second divided difference of the three numbers $\ell_1^{bc}, \ell_2^{bc}, \ell_3^{bc}$ is nonzero exactly when $W_b \neq W_c$. Once such a pair has been recovered by Lemma A3.1, every other W_b, A_{bc}, K_{bc} is uniquely determined from three deadlines and the known anchor action W_c , even when $W_b = W_c$.*

Proof. The affine part of (A3.1) has zero second divided difference. For unequal actions the logarithmic part has a second derivative of strict constant sign, as in Lemma A2.1; for equal actions it vanishes. This proves the first assertion and makes selection of an unequal pair an operation on the data.

For the second assertion add $\log(T_j - W_c)$ to the observed values. Two alternative parameter triples would make $A + KT + \log(T - w) - \log(T - \tilde{w})$ vanish at three clocks, with the constants understood as differences. If $w \neq \tilde{w}$ this function has second derivative of strict constant sign, which is impossible. Thus the actions agree, and two clocks determine the affine coefficients. $\square$

Proof of the exact part of Theorem A1.1. Rank gives B , and Lemmas A2.1 and A2.2 recover the channels and all $\pi_{j,b}$. There is an observable unequal-action anchor pair by hypothesis. Lemmas A3.1 and A3.2 recover every action and every relative coefficient. Relative coefficients determine the weights and rates precisely modulo (A1.3); conversely that gauge plainly leaves every projective observation unchanged. This proves both identification and completeness of the continuous ambiguity. Any alternative full-rank representation has the same B and must pass through the same reconstruction. Local permutations agree on overlaps after matching the recovered component matrices. Since distinct parameter pairs cannot exchange continuously, two

continuous global representations differ by one constant permutation on the connected gate. No sorting by action is required at a crossing. $\square$

APPENDIX A4. FINITE-ORDER STABILITY AND THE NUMBER OF CLOCKS

A local inverse is not obtained by applying a global nonlinear solver and assuming its success. We exhibit its nonsingular local coordinates. For an unequal-action anchor write $\Phi_j(A, K, x, y) = A + KT_j + \log(T_j - x) - \log(T_j - y)$. On any four distinct deadlines its Jacobian has rows

$$(1, T_j, -(T_j - x)^{-1}, (T_j - y)^{-1}).$$

A direct Vandermonde calculation gives

$$\det D\Phi = \frac{(y - x) \prod_{i < j} (T_j - T_i)}{\prod_j (T_j - x)(T_j - y)}, \quad (\text{A4.1})$$

where both products use these four deadlines. Indeed, multiplication of row j by $(T_j - x)(T_j - y)$ reduces the columns to four polynomials of degrees at most three; their coefficient determinant is $y - x$. The determinant is nonzero. For a remaining component with known anchor, the three local coordinates (A, K, w) have independent derivative columns $1, T, -1/(T - w)$: multiplication by $T - w$ reduces the question to a polynomial of degree at most two at three distinct points.

Proposition A4.1 (Quantitative local inverse). *Fix m, u, v, B , the deadlines, a finite gate cover with fixed smooth gluing functions, and bounds for the C^m norms of P_j, U, V, W_b and the relative logarithmic weights and rates. Fix positive lower bounds for the residuals $T_1 - W_b$, all $\pi_{j,b}$, the least singular value of each chosen $(P_1)_{IJ}$, and*

$$\min_{b \neq c} \max\{|q_{2,b} - q_{2,c}|, |q_{3,b} - q_{3,c}|\}.$$

On each member of the cover fix an anchor with $|W_r - W_s| \geq \gamma > 0$. Also fix upper bounds for $|W_b|$ and the residuals. Then the inverse, after a local permutation and gauge choice, extends to a neighborhood of the exact normalized matrix data and satisfies

$$\|\widehat{W} - W\|_{C^m} + \|\widehat{U} - U\|_{C^m} + \|\widehat{V} - V\|_{C^m} + \|\widehat{A} - A\|_{C^m} + \|\widehat{K} - K\|_{C^m} \leq C \max_j \|\widehat{P}_j - P_j\|_{C^m}. \quad (\text{A4.2})$$

Here A, K are relative to one fixed component on each local chart. The constant depends only on the displayed bounds, margins, cover and clock gaps, and not on unknown derivatives above order m .

Proof. Matrix inversion on the fixed minor charts is smooth with controlled derivatives. If the displayed joint gap is at least δ , one can choose $t \in \{0, \dots, L\}$ with every gap of $K_2 + tK_3$ at least $\delta/(4L + 4)$, where $L = B(B - 1)/2$. To see this, a given pair can have a gap smaller than that quantity at at most one integer t : small gaps at two distinct integers would force both its q_3 difference and its q_2 difference below δ . There are at most L pairs. A finite refinement makes the same choice valid on each local chart.

For a simple real eigenvalue, the characteristic-polynomial derivative is the product of its eigenvalue gaps. The implicit function theorem therefore gives smooth real eigenvalues on a neighborhood with uniformly bounded derivatives of every fixed order. Realness follows also by conjugation and uniqueness in a small disk. Spectral projectors are polynomials in the chosen matrix divided by gap products. Thus (A2.3), summation, positive division and logarithms all have controlled smooth local extensions.

Equation (A4.1) gives a smooth local inverse for the anchor parameters using four clocks. The fifth clock supplies global uniqueness on exact data by Lemma A3.1; it need not be inverted a second time locally. The three-coordinate inverse above recovers the other components, including equal-action ones. Repeated chain and product rules through order m establish (A4.2). Fixed

smooth gluing functions combine the local extensions. Exact outputs agree on overlaps after their recovered permutation and relative-gauge transitions. Perturbed outputs need not satisfy all latent-model identities; they obey the error bound. This proves a local inverse on an open neighborhood, not an assertion that arbitrary data lie in the model. $\square$

On any compact parameter class with full-rank channels, positive weights, uniformly distinct (W, κ) pairs and positive action range, the needed finite covers and margins follow by compactness. They deteriorate as a component becomes invisible or all actions coalesce. Thresholding singular values recovers the visible count if the observation error is less than a fixed fraction of the least nonzero singular value. Without such a margin, counting an arbitrarily weak component is not stable. The norm in (A4.2) is a norm of normalized observations; converting an absolute error for unnormalized matrices to it requires a lower bound on their total observed mass.

Proposition A4.2 (Three clocks cannot identify this class). *There are two-component, full-rank models with unequal actions and strict probability margins for which three projective deadlines have an action-changing one-parameter ambiguity. Four clocks give a local inverse for the anchor scalar parameters. Five clocks give the global inverse proved above; no global optimality claim for five is made.*

Proof. Fix strictly positive full-rank channels and unequal $x, y < T_1$. At three clocks the map $(A, K, x, y) \mapsto (\Phi_1, \Phi_2, \Phi_3)$ has rank three. The columns $1, T, 1/(T - x)$ are already independent, by multiplying a putative relation by $T - x$. The implicit function theorem gives a local one-dimensional level set. Its tangent cannot have both action coordinates zero, since $1, T$ are independent. Along a sufficiently small nonconstant part of the level set the actions therefore vary. For two components each probability vector is determined by its log ratio, so the entire three normalized matrices are unchanged. Positivity and all strict finite margins persist. Baseline weights or retention probabilities can be decreased beforehand to keep all raw success probabilities strictly below one. The four-clock assertion is (A4.1). $\square$

Proposition A4.3 (The equal-action obstruction). *If all actions equal w , arbitrary projective deadline observations cannot determine w , even when the rates are distinct and both channels have full column rank.*

Proof. The common factor $T_j - w$ cancels from (A1.2). Any other common action below the first deadline gives the same data. Distinct rates can still identify different components; they do not supply the missing common action. This is an exact nonlinear obstruction, not a failure of an estimator. $\square$

A4.1. Polynomial detector families. The preceding dichotomy extends beyond a single unknown rate. Suppose

$$M_j = U \operatorname{diag}\{a_b \exp(P_b(T_j))(T_j - W_b)\} V^T, \quad P_b(T) = \sum_{\nu=1}^q \theta_{b,\nu} T^\nu, \quad (\text{A4.3})$$

with a fixed known degree bound $q \geq 1$. Constants in P_b are absorbed in a_b .

Theorem A4.4 (Finite-degree detector rigidity). *Assume positive weights, full-rank clock-invariant channels, pairwise distinct tuples $(W_b, \theta_{b,1}, \dots, \theta_{b,q})$, and at least two unequal actions at each endpoint. Any $q + 5$ distinct projective deadlines determine all actions, channels, relative weights and relative polynomial coefficients. The remaining gauge is a common positive weight multiplier and addition of one common degree- q polynomial. The inverse is locally Lipschitz in C^m under the corresponding compact margins. For an anchor pair, $q + 3$ clocks give a local inverse, whereas $q + 2$ clocks admit an action-changing ambiguity in a two-component class. For $q = 1$, Theorem A1.1 improves the sufficient global count from six to five.*

Proof. For a component pair, its log weight ratio is a degree- q polynomial plus $\log(T - x) - \log(T - y)$. If $x \neq y$, the derivative of order $q + 1$ of the logarithmic part has strict constant sign on $T > \max(x, y)$, since it is a nonzero constant times $(T - x)^{-q-1} - (T - y)^{-q-1}$. Such a function cannot take one constant value at $q + 2$ distinct clocks, by repeated Rolle's theorem. If $x = y$, a nonconstant polynomial of degree at most q cannot do so either. Thus the joint eigenvalue vectors from the first $q + 2$ clocks are pairwise distinct. A separating real linear combination exists: the excluded combinations form finitely many proper hyperplanes. Local finite choices follow by compactness. The same component-matrix formula as (A2.3) recovers the channels and weights. An unequal-action pair is detectable by a divided difference of order $q + 1$.

Two alternatives for its log ratio differ by a polynomial Q of degree at most q and four signed logarithms. Multiplication of the derivative by the four-pole denominator gives $Q'D + R$, $\deg R \leq 2$, of degree at most $q + 3$. If the difference vanishes at $q + 5$ clocks, its derivative has at least $q + 4$ distinct zeros and this polynomial vanishes identically. The rational identity has no nonzero polynomial part at infinity, so $Q' = 0$; pole cancellation then identifies the two unequal actions. One value gives the constant. With an anchor known, the same argument, or repeated Rolle applied to two logarithms, identifies every other component and its relative polynomial.

For the local anchor inverse the derivative columns are $1, T, \dots, T^q, -1/(T - x), 1/(T - y)$. Multiplication by $(T - x)(T - y)$ turns a relation into a polynomial of degree at most $q + 2$. Vanishing at $q + 3$ clocks forces it to vanish identically, and its poles and polynomial part force all coefficients to vanish. The remaining components use $q + 2$ local coordinates with one action denominator. Smooth finite-order stability follows exactly as in Proposition A4.1. At $q + 2$ clocks, the anchor Jacobian has rank $q + 2$ already on columns $1, T, \dots, T^q, 1/(T - x)$. Its one-dimensional level set cannot vary only the polynomial coefficients, since their values at these clocks have full column rank. Thus an action changes while the two-component observations stay fixed. $\square$

A degree bound is substantive. With J fixed clocks and arbitrary branchwise log-detectors, any change of admissible actions can be compensated at those clocks by polynomial interpolation of degree at most $J - 1$. The finite-degree theorem therefore does not imply recovery for an unrestricted detector by increasing the degree after seeing the data. It specifies an identifiable observation class and its exact gauge before acquisition.

APPENDIX A5. RETURN DYNAMICS AND A NONCONVEX CROSSED-ACTION EXAMPLE

We spell out the passage to geometry without treating full iterates as observed group elements. On a fixed canonical phase domain let $\lambda = p dx$. An exact lift is a pair $g = (F, A)$ with $dA = F^*\lambda - \lambda$. Its product and inverse are

$$(F, A)(G, B) = (F \circ G, B + A \circ G), \quad (F, A)^{-1} = (F^{-1}, -A \circ F^{-1}). \quad (\text{A5.1})$$

An absolute regular type-I sheet $S(x, y)$ gives

$$p = -S_x(x, y), \quad q = S_y(x, y), \quad A(x, p) = S(x, y), \quad (\text{A5.2})$$

with invertible S_{xy} .

Corollary A5.1 (Unpaired return sheets with selective detection). *Let $g = (F, A)$ be an exact lift and let r, s be coprime positive integers. For each iterate assume a finite family of regular endpoint gates in fixed canonical coordinates. On each gate the actual absolute generating sheets form the latent components of Theorem A1.1. Their count, channel columns and cross-gate pairing are not given. Assume that the resulting phase graphs cover all domains required by a fixed Bezout word, including its inverse factors, with fixed collars. Then five projective deadlines per gate determine g on the covered target. With fixed C^{k+2} action bounds, inverse mixed-Jacobian*

and phase-map inverse bounds, the inverse from C^{k+2} normalized matrix data to the C^k lift is locally Lipschitz.

Proof. Theorem A1.1 recovers the unordered absolute functions on each gate, including at action coincidences. Formula (A5.2) recovers their canonical graphs and absolute primitives. At an initial phase point the local equation for y is regular. Any overlapping representations have the same image and primitive, because the observation class consists of sheets of the same deterministic lift. This gives g^r, g^s on the required domains without a supplied matching of sheet indices. Choose integers u, v with $ur + vs = 1$. Since these are powers of one lift,

$$g = (g^r)^u (g^s)^v.$$

The domain hypothesis allows every factor to be evaluated, and (A5.1) preserves the additive primitive constants. This proves uniqueness.

For stability, the first step has no derivative loss. Solving $p = -S_x$ by the quantitative implicit function theorem gives C^k map estimates from C^{k+2} actions. The same bounds control the additional C^{k+1} norms needed for composition and inverse estimates. A finite product on fixed nested collars gives the asserted C^k bound. Fixed local extensions handle perturbed data without asserting their exact symplectic realizability. The elementary Bezout identity is not claimed as a new group-theoretic theorem; its inputs have just been reconstructed from weaker observations. $\square$

For a pre-existing Reeb flow with a smooth transverse global section, the first-return map and time satisfy $F^*\lambda - \lambda = d\tau$. Indeed the section flow-box map pulls back the contact form to $dt + \lambda$, and differentiation of $\phi_{\tau(x)}(x) = F(x)$ gives that identity. Hence the corollary, when its actual return sheets are covered, recovers (F, τ) and its section-preserving strict contact suspension. It does not infer the existence of the section or of the phase cover from endpoint matrices.

Proposition A5.2 (Common deadlines and a fixed source). *Suppose actual return actions on the gates satisfy $0 \leq W_b \leq H$. For $h > 0$ take a common external release delay, uniform on $[0, D]$ with $D = H + 6h$, and deadlines $T_j = H + jh$, $1 \leq j \leq 5$. Use the same normalized phase source ρ at every setting. Then a trajectory is accepted by the time gate with probability $(T_j - W_b)/D$, with both residual and unused-delay margins at least h . A subsequent log-affine selective retention produces (A1.1). No true-component audit is needed to set these clocks.*

Proof. For an independent delay α , success means $\alpha + W_b \leq T_j$. The stated inequalities keep this event strictly inside the uniform-delay interval. On a regular sheet, change initial momentum to terminal coordinate. Its successful density before the extra readouts is

$$\frac{\rho(x, -S_{b,x}) |\det S_{b,xy}|}{D} \eta_b e^{\kappa_b(T_j - T_*)} (T_j - W_b).$$

The factors outside $e^{\kappa_b T_j} (T_j - W_b)$ are independent of the setting and are absorbed in a_b . Independent readings give the product columns; an additional branch-blind detector acts by projective scaling. This proves the assertion using a literally fixed source density. For J deadlines replace $6h$ by $(J + 1)h$, so the polynomial-detector protocol also has common windows. $\square$

Proposition A5.3 (Three regular sheets through an action crossing). *The preceding observation and return hypotheses have a nonconvex three-sheet realization which includes an action coincidence, not merely separated abstract mixture components.*

Proof. On $\mathbb{R}^2$ put

$$\phi(p) = p^3 - p, \quad F(x, p) = (x + \phi(p), p), \quad A(p) = 1 + \frac{3}{4}p^4 - \frac{1}{2}p^2.$$

Then $F^*\lambda - \lambda = p\phi'(p) dp = dA$ and $A \geq 11/12$. Its positive-roof quotient preserves $dt + p dx$ and has F as return map with return time A . For a fixed return count n , write $t = (y - x)/n$ and let $p_-(t), p_0(t), p_+(t)$ be the three solutions of $\phi(p) = t$ near $-1, 0, 1$. They exist on an interval containing zero. With $G(p) = p^4/4 - p^2/2$, their generating sheets are

$$S_{n,b}(x, y) = n + p_b(t)(y - x) - nG(p_b(t)) = nA(p_b(t)).$$

They satisfy $-S_{n,b,x} = S_{n,b,y} = p_b(t)$ and $S_{n,b,xy} = -1/(n\phi'(p_b(t))) \neq 0$. At $x = y$, the two outer actions equal $5n/4$, while the central action is n . Thus there is always an unequal-action anchor on a sufficiently small compact rectangle through the diagonal, although action separation of all three sheets fails. The central and outer mixed derivatives have opposite signs, so this is not a globally convex minimization model.

Use a selective detector with survival rate $r(p) = r_0 + r_1 p > 0$ on these compact momentum patches, where $r_1 \neq 0$. Its retention is $\eta \exp[-r(p)(T - T_1)]$, with $0 < \eta < 1$. The unknown rates $\kappa_b = -r(p_b)$ separate the two outer components at their action coincidence. For the first readout take the probability vector $(1, e^{\alpha p}, e^{2\alpha p})/(1 + e^{\alpha p} + e^{2\alpha p})$, and for the second use the analogous vector with an unknown $\beta \neq 0$, where also $\alpha \neq 0$. Draw the readings independently given the phase point. The three columns form normalized Vandermonde matrices at distinct momenta, hence have full rank and positive entries. The inverse is not supplied α, β, r_0, r_1 or the response curves. This is a physical measurement mechanism, not an observed momentum coordinate disguised as a branch label.

A fixed smooth normalized source positive on all three compact phase patches gives positive weights. Compactness gives all required margins, and Proposition A5.2 gives common clocks. For return counts two and three, choose finite compact endpoint rectangles containing the input and output patches needed for $g = g^3(g^2)^{-1}$; the momentum is unchanged and the horizontal shifts are bounded. This supplies the finite covered target in Corollary A5.1. The example includes the diagonal crossing but not the fold points $\phi'(p) = 0$. No caustic-crossing claim is made. $\square$

APPENDIX A6. OBSERVATION ERRORS AND THE BOUNDARY OF THE CONCLUSION

The inverse acts on conditional joint readout matrices at each endpoint. For a finite-sample implementation let $H_j(z)$ be the observed joint density of endpoint and the two readings, and write $f_j(z) = \mathbf{1}^\top H_j(z) \mathbf{1}$. Raw and retained-record budgets are different; either can estimate its corresponding H_j .

Corollary A6.1 (Error transfer without a calibration sample). *On a gate with $f_j \geq f_* > 0$ and bounded C^m densities, suppose $\max_j \|\hat{H}_j - H_j\|_{C^m} \leq \varepsilon$. If the true model has the margins of Proposition A4.1 and ε is sufficiently small, normalization of $\hat{H}_j$ followed by the selective-detector inverse gives $C\varepsilon$ error in actions, channels and relative detector coefficients. The same conclusion holds with an additional C^m model perturbation η , replacing ε by $\varepsilon + \eta$. There is no calibration-sample term.*

Proof. The sum of the estimated cell densities differs from f_j by at most $uv\varepsilon$ in C^m and has floor $f_*/2$ for small error. The C^m product and reciprocal estimates bound $\hat{H}_j/(\mathbf{1}^\top \hat{H}_j \mathbf{1}) - H_j/f_j$ by $C\varepsilon$. Apply Proposition A4.1. A perturbation adds by the triangle inequality. Threshold failures can be assigned a fixed reference output to define an estimator everywhere; this statement concerns the event on which the displayed errors and margins hold. It makes no minimax assertion about the density-estimation rate or about geometric derivative loss. $\square$

The fixed-source compensation construction in the retained physical analysis allows a different retention function for every component and deadline. It remains valid in that larger apparatus class. It does not contradict the present result: the exact compensator need not have an affine,

or a fixed-degree polynomial, logarithm in the deadline. Conversely, restricting the detector does not make all geometry unconditionally observable. A single common action is still hidden, independent readouts still need rank, and the return theorem still needs a covered phase domain. These are explicit mathematical hypotheses, not statements that an unavailable experiment has been performed.

The full revision retains the earlier two-row and three-projective clock inverses, conditional-law information theory, calibrated-label analysis, fixed-source physical lower bound, convex regularity proofs, and marked billiard reconstruction. The marked billiard theorem still uses its word/lift roles, scalar-coordinate registrations and reference atlas. The latent components recovered here are not silently renamed as those semantic marks. The strengthened result is a single observation-rigidity theorem with a specified selective detector family, its exact gauges and obstructions, and its consequences for unpaired regular generating sheets. The companion regimes retain their own assumptions and proofs.

APPENDIX A7. THE OBSERVATION QUOTIENT AND THE PRINCIPAL THEOREM

An endpoint law may conceal both its generating action and the identity of the trajectory which produced it. A calibrated classifier separates these two problems only by supplying the component identities in another experiment. We instead use two noisy readings of the same retained trajectory. Neither reading is calibrated, and neither has a prescribed interpretation as a reflection word. Variation of the deadline separates the hidden components. A third deadline removes even the scalar normalization of each endpoint law.

The relevant observation is a matrix, not a collection of separately normalized label laws. Let $Q \subseteq \mathbb{R}^d$ be a compact rectangle and let $T_1 < T_2 < T_3$. For two finite readout alphabets of sizes u, v , write $M_j(z) \in \mathbb{R}^{u \times v}$ for the joint successful density of endpoint z and the two readings. Our model is

$$M_j(z) = U(z) \operatorname{diag}\{a_b(z)(T_j - W_b(z))\}_{b=1}^B V(z)^\top. \quad (\text{A7.1})$$

Here $a_b > 0$, $W_b < T_1$, and the columns of U, V are probability vectors. The latent number B is not given; $B \leq \min(u, v)$ and both matrices have full column rank. Conditional on (b, z) , the two readings are independent. Their conditional laws may depend arbitrarily and smoothly on z , but are the same at every deadline. This last invariance is a hypothesis, not a consequence of using two sensors.

The more coarsened observation is

$$N_j(z) = c_j(z) M_j(z), \quad c_j(z) > 0, \quad (\text{A7.2})$$

where none of the functions c_j is known. This includes a common branch-blind detector depending on endpoint and deadline, unknown exposures, normalized endpoint/readout laws, and their combinations. It also includes observing only the conditional distribution of the *pair* of readings at each endpoint: then $c_j(z) = (\mathbf{1}^\top M_j(z) \mathbf{1})^{-1}$. Frequencies within the joint readout matrix are retained; normalizing its rows separately is not allowed.

Theorem A7.1 (Blind clock-pencil rigidity). *Assume the preceding model on a connected gate, with C^m coefficients, $m \geq 0$, and distinct actions $W_b(z)$ at every endpoint.*

From two matrices M_1, M_2 , the observations determine B , the absolute actions, the weights, and both readout channels, up to one simultaneous permutation. No true-component audit or list of structural branch identities is supplied.

If $B \geq 2$, three projective matrices N_1, N_2, N_3 determine B , the absolute actions, both channels and all relative weights $a_b / \sum_c a_c$. Only a common positive weight multiplier remains unobservable. Thus the actions survive arbitrary positive branch-blind detector changes at each endpoint and deadline, even when all success masses are discarded.

On compact classes with bounded C^m norms, positive weight and residual floors, positive clock gaps, full-rank margins and separated actions, these reconstructions are locally Lipschitz in C^m . In the projective case impose in addition bounds and positive floors for c_j , or the equivalent observed scale and two-plane margins specified below. The inverse loses no spatial derivatives.

The proof is given in Sections A8 and A9. The simultaneous permutation can be fixed by increasing action on each connected gate. This is an intrinsic ordering of recovered real functions, not a supplied word or lift label. It cannot manufacture the semantic names needed by a marked billiard experiment. It is unnecessary when one reconstructs a canonical graph as an unordered union of generating sheets.

The new observation assumption is substantive. It replaces one calibrated readout and a ground-truth audit by two conditionally independent, deadline-invariant readouts. It permits endpoint-dependent channels and learns the number of visible components. It does not assert that one may obtain independent information by copying a noisy label or randomly relabelling a single sensor output. Conditional independence given the hidden branch and endpoint has to come from the acquisition mechanism. Rank, separation and a finite sensor capacity are also essential to the uniform theorem.

A7.1. Relation to earlier inverse mechanisms. Spectral identification of latent variables is classical. Kruskal-type identifiability, observable-operator methods, repeated measurements and nonclassical measurement error provide important precedents [2, 3, 4, 5]. We do not claim a new general uniqueness theorem for tensor decompositions. In the present model the known affine dependence on deadline identifies an *absolute generating action* rather than an unnamed eigenvalue. More importantly, three projective observations recover this affine dependence without measured exposures or a calibrated channel. That step makes a branch-blind physical detector an exact nuisance symmetry instead of a first-order error term.

The deterministic conversion of action to a canonical relation is classical generating-function calculus [12, 16]; the conversion of return action to a Reeb suspension uses the usual return-time primitive [13, 1]. The additional conclusion here is upstream of those constructions: their local absolute action sheets, including the sheet count, can be recovered from unpaired, uncalibrated projective records. Coverage of the required phase domain is still a hypothesis.

For billiards, endpoint actions on a regular gate determine local covectors by differentiation. Consequently they contain local canonical relation information, not merely lengths of periodic trajectories. A finite endpoint atlas does not determine an entire global scattering relation unless a coverage theorem supplies the missing trajectories. Conversely, a scattering relation recorded in Euclidean boundary coordinates supplies information absent from abstract scalar labels. We make no unconditional information ordering between these different experiments. Marked spectral rigidity results such as [7, 11] and the planar lens theorem [9] start from other data and impose other geometric hypotheses. The marked/reference-atlas reconstruction retained later in this article is not relabelled as a canonical unmarked billiard theorem.

APPENDIX A8. A BLIND INVERSE FOR TWO AFFINE MATRIX OBSERVATIONS

Put $\Delta = T_2 - T_1$ and

$$D = \frac{M_2 - M_1}{\Delta} = U \operatorname{diag}(a_b) V^\top. \quad (\text{A8.1})$$

Since U, V have full column rank and the weights are positive, $\operatorname{rank} D = B$. This determines the latent dimension without knowing either channel. Fix index sets I, J of B rows and columns for which D_{IJ} is invertible. Define the reduced observed operator

$$K = (M_1)_{IJ} (D_{IJ})^{-1}. \quad (\text{A8.2})$$

The invertibility of $D_{IJ} = U_I \text{diag}(a_b) V_J^\top$ implies that both square factors U_I, V_J are invertible. Therefore

$$K = U_I \text{diag}(T_1 - W_b) U_I^{-1}. \quad (\text{A8.3})$$

In particular its eigenvalues are positive and determine the absolute actions, with multiplicities, by $W_b = T_1 - \lambda_b(K)$. This identity holds even when some actions coincide. Distinctness is used for separating the channels and for the stated smooth inverse.

Lemma A8.1 (Observable rank-one components). *For distinct eigenvalues $\lambda_b = T_1 - W_b$, let*

$$P_b = \prod_{c \neq b} \frac{K - \lambda_c \text{Id}}{\lambda_b - \lambda_c}, \quad E_b = D_{:J} (D_{IJ})^{-1} P_b D_{I:}. \quad (\text{A8.4})$$

Then $E_b = a_b U_{:b} V_{:b}^\top$, independently of the chosen invertible minor. In particular

$$a_b = \mathbf{1}^\top E_b \mathbf{1}, \quad U_{:b} = E_b \mathbf{1} / a_b, \quad V_{:b} = E_b^\top \mathbf{1} / a_b. \quad (\text{A8.5})$$

Proof. Equation (A8.3) gives $P_b = U_I e_b e_b^\top U_I^{-1}$. Also $D_{:J} (D_{IJ})^{-1} = U U_I^{-1}$ and $D_{I:} = U_I \text{diag}(a_b) V^\top$. Substitution proves the rank-one formula. Summing its entries and its rows or columns proves (A8.5), because both channel columns have total mass one. The resulting expression depends only on the true component and not on I, J . $\square$

This proves the exact two-clock part of Theorem A7.1. It also proves uniqueness against alternatives with a different unknown number of components, provided the alternative satisfies the same full column-rank hypothesis. Splitting a component into invisible duplicate columns is outside that hypothesis and cannot be excluded from data.

Proposition A8.2 (Uniform finite-order inverse). *Fix the dimensions, m , clock values, a bound L on the C^m norms of the coefficients and observations, and positive constants a_*, r_*, s_*, γ . Assume*

$$a_b \geq a_*, \quad T_1 - W_b \geq r_*, \quad \sigma_B(D) \geq s_*, \quad |W_b - W_c| \geq \gamma \quad (b \neq c).$$

On a finite cover by minor charts with $\sigma_{\min}(D_{IJ}) \geq \mu_ > 0$, formulas (A8.2)–(A8.5) admit local extensions to nearby real matrix data. After fixing the action ordering, their C^m errors are bounded by C times the C^m observation error. Here C depends only on the displayed constants, the finite cover, its fixed gluing functions and Δ^{-1} . No unknown higher spatial derivative is used in this assertion.*

Proof. Matrix inversion is smooth on the set of matrices with smallest singular value at least $\mu_*/2$. The product and reciprocal inequalities in C^m bound the reduced operator and its perturbation. For its characteristic polynomial p , the derivative at a simple root has magnitude $|p'(\lambda_b)| = \prod_{c \neq b} |\lambda_b - \lambda_c| \geq \gamma^{B-1}$. The finite-dimensional implicit function theorem makes each root a smooth real function of the real matrix entries in a neighborhood of the exact operator. Its derivatives of every fixed order are bounded on the compact set under consideration. Realness also follows from conjugation and uniqueness of the root in a small disk about each distinct real root. Composition with C^m matrix functions, and repeated differentiation through order m , gives the claimed root estimate. The projector formula and positive denominators in (A8.5) give the other estimates.

Compactness and rank provide finitely many minor charts. One can take fixed smooth gluing functions subordinate to a slightly smaller cover. For exact data the local inverses agree by Lemma A8.1. For perturbed nonmodel data the glued outputs need not be an exact latent model, but have the same error bound. Sorting the separated real actions consistently on the connected gate prevents a spatially varying permutation. This construction supplies an inverse on an open neighborhood, rather than only on the model image. $\square$

Remark A8.3 (What can be counted stably). Exact rank determines B , not the number of arbitrary hidden causes in a larger rank-deficient representation. If a matrix perturbation has operator norm at most $e < s_*/4$, singular-value thresholding at $s_*/2$ recovers the rank of the reference model. This follows from the singular-value perturbation bound. An extra component whose entire observed contribution has norm at most e cannot in general be counted reliably. The theorem learns visible components above a signal margin; it does not assert exact open-world discovery at zero signal.

APPENDIX A9. THREE PROJECTIVE OBSERVATIONS AND UNKNOWN DETECTORS

The matrices in (A7.2) need not have comparable masses, so their difference is not the slope D . Nevertheless the known deadlines identify their correct relative scales. Set

$$A = U \operatorname{diag}(a_b) V^\top, \quad B_0 = U \operatorname{diag}(a_b W_b) V^\top; \quad M_j = T_j A - B_0.$$

The matrices A, B_0 are linearly independent whenever not all W_b are equal. Indeed $B_0 = tA$ would, after multiplication by left and right inverses of $U, V^\top$, imply $W_b = t$ for every b . Thus any two M_j , and hence N_1, N_2 , are linearly independent.

Lemma A9.1 (Clock deprojectivization). *Suppose $N_j = c_j(T_j A - B_0)$, where $c_j > 0$ and A, B_0 are linearly independent. There are uniquely determined scalars α, β with $N_3 = \alpha N_1 + \beta N_2$. Put*

$$\alpha_0 = \frac{T_2 - T_3}{\Delta}, \quad \beta_0 = \frac{T_3 - T_1}{\Delta}, \quad \widetilde{M}_1 = \frac{\alpha}{\alpha_0} N_1, \quad \widetilde{M}_2 = \frac{\beta}{\beta_0} N_2. \quad (\text{A9.1})$$

Then $\widetilde{M}_j = c_3 M_j$ for $j = 1, 2$. In particular their matrix-pencil inverse gives the original absolute actions and channels.

Proof. Affine interpolation in the known variable T gives $M_3 = \alpha_0 M_1 + \beta_0 M_2$. Hence $\alpha = \alpha_0 c_3 / c_1$ and $\beta = \beta_0 c_3 / c_2$. Uniqueness follows from linear independence. Both scale factors in (A9.1) are positive. Their substitution proves the claim. The common positive multiplier c_3 cancels from (A8.2); it multiplies each E_b by c_3 and hence cancels from the recovered channels and relative weights as well. $\square$

To make the computation explicit, vectorize the matrices and form $H = (\operatorname{vec} N_1 \ \operatorname{vec} N_2)$. Then

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = (H^\top H)^{-1} H^\top \operatorname{vec} N_3. \quad (\text{A9.2})$$

For exact data the projection residual vanishes. The same expression is defined on a neighborhood of exact data, even when the residual is nonzero. If $\sigma_{\min}(H) \geq \chi_* > 0$, the observed norms are bounded, and the two rescaling factors have a positive floor, it is a uniformly smooth map there. Product and reciprocal estimates in C^m followed by Proposition A8.2 prove the projective stability assertion. These are the observed scale and two-plane margins referred to in Theorem A7.1.

The dimension can be recovered before rescaling: since $T_1 - W_b > 0$, $\operatorname{rank} N_1 = B$. Equivalently it is the rank of the rescaled slope. The equality of these ranks is a model consistency check. At least two distinct actions are needed pointwise in the projective argument. A single constant action with arbitrary unknown exposures is not identified by any number of such scalar projective observations.

Corollary A9.2 (Exact immunity to a common detector). *Suppose a retention decision made before the two readouts has probability $e_j(z) > 0$ conditional on endpoint and deadline, independently of the latent component. The functions e_j are unknown, may differ between clocks, and need not be close to one another. Under the projective hypotheses of Theorem A7.1, they cause*

no ambiguity in any absolute action or recovered channel. The same is true if the failure counts or the total exposure at each deadline are discarded.

Proof. The detector multiplies every entry of the matrix by the same scalar $e_j(z)$. Unknown exposure and normalization add further positive scalar factors. They are all absorbed in c_j in (A7.2). Apply Lemma A9.1 and the blind affine inverse. $\square$

Thus the common-detector model has an exact nonlinear quotient, not an $O(\epsilon)$ efficiency floor. This does not contradict a lower bound for independently branch-dependent retention: that larger apparatus class does not act by scalar multiplication of the observed matrices. The distinction will be made physical in Section A12.

Proposition A9.3 (Two projective clocks are insufficient). *Even with two independent full-rank readouts, distinct smooth actions and strict margins, two projective clocks do not generally determine absolute actions. There is a nontrivial one-parameter ambiguity in the same class.*

Proof. Let $r_b = (T_2 - W_b)/(T_1 - W_b) > 1$. For $q > 0$ sufficiently close to one, put

$$W_b^{(q)} = \frac{T_2 - qr_b T_1}{1 - qr_b}, \quad a_b^{(q)} = a_b \frac{T_1 - W_b}{T_1 - W_b^{(q)}}.$$

Then $a_b^{(q)}(T_1 - W_b^{(q)}) = a_b(T_1 - W_b)$ and $a_b^{(q)}(T_2 - W_b^{(q)}) = qa_b(T_2 - W_b)$. Keep U, V unchanged, replace c_2 by c_2/q and keep c_1 fixed. Both observations are unchanged. Choose the initial weights small enough that the total raw success masses are strictly below one. Compactness then preserves the raw probability constraints, positivity, distinctness and every strict margin for q close to one. The fractional map is not the identity unless $q = 1$, so the action tuple changes. This obstruction is in the unrestricted generating-action observation class; no claim that every transformed action is a billiard action is required. $\square$

Proposition A9.4 (The joint second readout carries information). *After discarding the second reading, even known raw masses at arbitrarily many deadlines need not identify the action tuple in an uncalibrated one-readout model. Full column rank of the unknown one-readout channel does not repair this.*

Proof. Take a strictly positive full-column-rank stochastic matrix U , positive a , and distinct W . The observed vector is $g_j = U(T_j a - aW)$, where aW denotes the componentwise product. Let R be an invertible matrix close to the identity with $\mathbf{1}^\top R = \mathbf{1}^\top$, not a permutation. Put $U' = UR$, $a' = R^{-1}a$, and $a'W' = R^{-1}(aW)$. For a sufficiently small nontrivial perturbation, U' remains strictly positive and stochastic, $a' > 0$, and the actions remain distinct and admissible. All vectors g_j agree for every clock. One may choose R so that at least one action changes, for example by mixing two unequal action components. Two independently noisy readings are therefore not merely two names for one observation. $\square$

APPENDIX A10. A NONCONVEX RETURN SYSTEM WITH THREE UNRESOLVED SHEETS

The blind hypotheses can occur on actual generating sheets, not only on abstract positive matrices. The following example has no globally convex generating function and uses no true-component audit.

Proposition A10.1 (A three-sheet shear). *On $\mathbb{R}^2$ with $\lambda = p dx$, put*

$$\phi(p) = p^3 - p, \quad F(x, p) = (x + \phi(p), p), \quad A(p) = 1 + \frac{3}{4}p^4 - \frac{1}{2}p^2.$$

Then (F, A) is an exact action lift with $A \geq 11/12$. There exist compact endpoint rectangles on which its canonical graph has three regular type-I sheets with positive, pairwise separated

absolute actions. Two uncalibrated, conditionally independent three-valued readouts can have strictly positive full-rank channels on all three sheets. The resulting clock experiment satisfies Theorem A7.1, including its projective version.

Proof. The inverse shear is $(x, p) \mapsto (x - \phi(p), p)$, and $F^*\lambda - \lambda = p\phi'(p) dp = dA$. Minimizing the quadratic polynomial in p^2 gives $A \geq 11/12$. Write $G(p) = p^4/4 - p^2/2$ and $s = y - x$. For sufficiently small $|s|$, the equation $\phi(p) = s$ has three smooth solutions $p_-(s), p_0(s), p_+(s)$ continuing $-1, 0, 1$. Define

$$S_b(x, y) = 1 + p_b(s)s - G(p_b(s)).$$

Differentiation gives $\partial_s S_b = p_b(s)$, so $-\partial_x S_b = \partial_y S_b = p_b(s)$ and $\partial_{xy}^2 S_b = -1/\phi'(p_b(s)) \neq 0$. Substitution of $s = \phi(p_b)$ gives $S_b = A(p_b)$, the actual absolute primitive. At $s = 0$, $S_0 = 1$ and $S_- = S_+ = 5/4$. The derivative of $S_+ - S_-$ there is 2. Therefore on a sufficiently small compact interval $0 < s_0 \leq s \leq s_1$, all three actions are separated. Choose an endpoint product rectangle with $y - x$ in this interval. The middle sheet has mixed derivative of opposite sign to the two outer sheets. These are regular parts of a folded endpoint projection, not three minimizers of a globally convex action.

For any unknown nonzero constants κ, ν , let the first reading conditional on momentum p have probabilities

$$\frac{(1, e^{\kappa p}, e^{2\kappa p})^\top}{1 + e^{\kappa p} + e^{2\kappa p}},$$

and give the second reading the analogous probabilities with ν . Use independent randomizations conditional on the phase point. Given an endpoint and its sheet, the momentum is fixed, so the two readings are conditionally independent. The channel columns at the three momenta form a normalized Vandermonde matrix in $e^{\kappa p_b}$, respectively $e^{\nu p_b}$. Their determinants are nonzero; compactness gives positive singular-value margins. Every label has positive probability on every sheet. No reported label identifies a true sheet, and the inverse need not know either constant or either response curve. This is a genuine extra measurement mechanism, not a relabelling of one common noisy output.

Choose one smooth normalized initial phase density positive on the three compact phase patches, and a fixed positive retention. The positive-roof suspension of (F, A) has actual first-return time A . The delayed experiment of Proposition A12.1 then supplies common clock windows and the matrix model on the chosen rectangle. All its action, weight and channel margins are positive. $\square$

This example proves nonvacuity in a nonconvex multi-sheet system. It does not assert recovery through the fold points where $\phi'(p) = 0$, nor does it remove the independent-readout hypothesis. In particular it does not serve as a table-independent billiard acquisition theorem.

Geometric conventions for the billiard lower bound. For this lower bound fix an integer $k \geq 2$. With fixed scalar parametrizations of the labelled quotient boundaries R_a and a marked lattice matrix L , use the distance

$$d_k(\mathcal{T}, \tilde{\mathcal{T}}) = \inf_{O \in O(2), v \in \mathbb{R}^2} \max \left\{ \max_a \|\tilde{R}_a - (OR_a + v)\|_{C^k}, \|\tilde{L} - OL\| \right\}.$$

The fixed raw sources are smooth and positive on the compact initial states feeding the observed gates. Regularity bounds and positive floors refer to these gates and larger fixed launch charts. A change of the physical embedding of the same scalar phase chart is a change of geometry, not a change of its fixed source density. In a lower-bound statement, a fixed post-retention readout means a fixed Markov kernel of the retained component and scalar endpoint, including fixed calibrated label channels. It does not mean an additional sensor which measures momentum,

flight time or other geometric information not contained in that retained statistic. Such an additional observation need not preserve the lower-bound equivalence.

APPENDIX A11. UNPAIRED GENERATING SHEETS AND EXACT RETURN SYSTEMS

The blind inverse recovers functions without structural names. For an exact symplectic map, this is enough to recover its graph: the initial covector identifies a phase point and hence pairs local sheets on an overlap. We give the precise domain assumption, since a missing phase cover cannot be replaced by an algebraic identity.

Work in a fixed canonical phase space $X \subset \mathbb{R}^{2h}$ with $\lambda = p dx$. An action lift is a pair $g = (F, A)$ with $dA = F^* \lambda - \lambda$. Its product is

$$(F, A)(G, B) = (F \circ G, B + A \circ G), \quad (F, A)^{-1} = (F^{-1}, -A \circ F^{-1}). \quad (\text{A11.1})$$

The chain rule proves closure; associativity follows by substitution. An absolute type-I generating sheet $S(x, y)$ represents a part of a lift by

$$p = -S_x(x, y), \quad q = S_y(x, y), \quad A(x, p) = S(x, y). \quad (\text{A11.2})$$

The mixed derivative S_{xy} is invertible on a regular sheet. Several such sheets may occupy the same endpoint gate. They need not minimize an action, and they need not be convex.

Definition A11.1 (Unpaired regular observation cover). For each of two prescribed return counts $r = m, n$, finitely many endpoint gates are specified in the fixed canonical coordinates. On each gate the actual iterate has a finite collection of regular generating sheets, whose number and pairing with other gates are not supplied. Their phase graphs cover every point at which the iterates or their inverses are used in a specified finite reconstruction word. The union is the graph of the actual deterministic iterate, with its actual absolute primitive. All endpoint gates and intermediate phase domains have fixed collars. This is a coverage and realizability hypothesis, not observed sheet labels or observed phase covectors.

Theorem A11.2 (Reconstruction from unpaired projective records). *Let $g = (F, A)$ be an exact lift and let m, n be relatively prime positive integers. Suppose the two iterates have an unpaired regular observation cover. On each gate acquire the matrices of Theorem A7.1, with the latent actions equal to the absolute generating sheets of that iterate.*

Two raw clocks on every gate determine g on the covered target. Three projective clocks suffice instead on gates having at least two separated actions. Single-sheet gates may use the two-row protocol. Neither calibrated channels, true-component audits, a supplied sheet count, nor cross-gate sheet pairing is required. The conclusion includes the absolute primitive, not only the canonical transformation.

For a C^k target estimate, fixed C^{k+2} bounds on the generating sheets, inverse mixed-Jacobian margins, fixed nested phase domains and the spectral margins of Theorem A7.1 give a locally Lipschitz inverse from C^{k+2} matrix data to the C^k lift.

Proof. Recover the unordered functions $S_{r,b}$ on each endpoint gate by the blind inverse. Their derivatives in (A11.2) give the union of local canonical graphs and their absolute primitives. For any initial phase point in the covered domain, regularity gives local solutions of $p = -S_x(x, y)$. All solutions representing that phase point have the same output and primitive: the data arise from the graph of one actual action lift. Thus overlapping recovered sheets agree as lifts. This reconstructs g^m, g^n on the needed domains without an external matching between sheet indices. It does not choose a root among unrelated stationary functions which are not known to represent the same deterministic graph.

Choose integers s, t with $sm + tn = 1$. Powers of one lift commute, so

$$g = (g^m)^s (g^n)^t. \quad (\text{A11.3})$$

The cover assumption includes all domains used by this finite word and its inverse factors. Equation (A11.1) evaluates the right side and recovers the primitive including its constant. Any alternative system with the same observations yields the same local actions, the same two lifted iterates and hence the same right side.

For stability, the first stage yields C^{k+2} action errors. Implicit differentiation of $p = -S_x$ gives C^k estimates for the local maps and primitives; bounded C^{k+2} actions also supply the additional bounded C^{k+1} local maps needed for composition estimates. Inverting the maps uses their inverse-Jacobian bounds on the fixed collars. A finite cover and fixed smooth local extension and gluing operators give nearby outputs even for data not lying exactly on the model. The finite products and inverses in (A11.3) are locally Lipschitz in C^k under these bounds. A finite number of such estimates proves the assertion. We do not assert that arbitrary perturbed outputs satisfy every exact symplectic identity. $\square$

The group identity (A11.3) is elementary. The content of the theorem is that its inputs are recovered from unpaired projective endpoint matrices, not handed to the observer as full group elements. The phase cover remains assumed. The theorem neither crosses an unobserved caustic nor discovers a global section.

Corollary A11.3 (Return time in an existing contact flow). *Let a complete Reeb flow on (M, α) have a smooth transverse global section with smooth first-return map F and positive return time τ . On canonical subdomains of the section, suppose the two return counts satisfy Theorem A11.2. Clocked endpoint observations of those actual returns determine (F, τ) on the covered target and therefore its section-preserving strict contact suspension there.*

Proof. For the section inclusion i put $\lambda = i^*\alpha$. The Reeb identities $\mathcal{L}_R\alpha = 0$ and $\alpha(R) = 1$ give $\Psi^*\alpha = dt + \lambda$ for $\Psi(x, t) = \phi_t(i(x))$. Differentiating $i(Fx) = \phi_{\tau(x)}(i(x))$ gives $F^*\lambda - \lambda = d\tau$. Hence $g = (F, \tau)$ is an action lift, and the elapsed time of the r th return is its r th absolute action. The global section identifies the flow with $(x, t) \sim (Fx, t - \tau(x))$, preserving $dt + \lambda$. Apply Theorem A11.2. The clock realization used here is proved in the next proposition. $\square$

APPENDIX A12. COMMON DEADLINES AND THE BOUNDARY OF DETECTOR INVARIANCE

There are two acquisition mechanisms in the article. The original billiard flow-box experiment requires a deadline inside both adjacent free flights. Its branchwise admissibility does not imply simultaneous admissibility for a mixture. An externally delayed, fixed-return-count experiment has a different window and supplies common deadlines without knowing the hidden word or sheet count.

Proposition A12.1 (A common-clock construction without branch auditing). *Fix a return or collision count r and a class-wide upper bound H_r for its elapsed action on the observed regular gates. Assume these actions are nonnegative. Choose $h > 0$, a uniform external release delay on $[0, D]$ with $D = H_r + 4h$, and deadlines*

$$T_j = H_r + jh, \quad j = 1, 2, 3.$$

For every branch in every such gate, independently of its name or number, $h \leq T_j - W_b \leq D - h$. With one normalized phase source shared by the settings, the resulting endpoint subdensities before readout have the form $a_b(T_j - W_b)$, with a_b independent of j . The same settings work throughout the bounded class.

For unit-speed finite-horizon billiards with free-flight bound H , one may take $H_r = rH$ for a segment containing at most r flights. No grouping by the unknown reflection word is used in this clock construction.

Proof. Prepare an initial phase point from the same normalized source at each setting, wait an independent uniform delay $\alpha \in [0, D]$, and record the endpoint at the prescribed return count only when $\alpha + W_b \leq T_j$. Conditional on that trajectory, the success probability is $(T_j - W_b)/D$, because it lies strictly between zero and one. No numerical action is stored in the record; the event is an ordinary elapsed-time threshold. On a regular type-I sheet, change initial momentum to terminal coordinate. If the initial phase density is ρ , the pre-readout density is

$$D^{-1}e_b(z)\rho(x, -\partial_x W_b(z))|\det \partial_{xy}^2 W_b(z)|(T_j - W_b(z)).$$

Here e_b is any baseline retention common to the clocks. A further branch-blind detector contributes the factor $c_j(z)$ of the projective model. The asserted inequalities follow immediately from $0 \leq W_b \leq H_r$. Summing at most r flights proves the billiard bound. The argument requires neither a branch classifier nor a count of the regular endpoint sheets. $\square$

This proposition proves simultaneous windows for the delayed protocol; it does not silently change the physical meaning of the older adjacent-flight experiment. It does not prove that all return paths are regular or have separated actions. The spectral inverse still requires a regular gate, full-rank independent readouts and the stated separation. Thus a common-window obstruction has been removed without claiming a universal observation cover.

For the original flow-box experiment, a useful precise sufficient condition is the following. On a common compact gate, suppose $\ell_b(z) = \min(\tau_b^-(z), \tau_b^+(z))$ and

$$\max_b \sup_Q W_b + 4h < \min_b \inf_Q (W_b + \ell_b). \quad (\text{A12.1})$$

Then $T_j = \max_b \sup_Q W_b + jh$, $j = 1, 2, 3$, are simultaneously admissible with positive margins. Compactness makes this condition persistent on a reference neighborhood. The proof is substitution. Absent this or another simultaneous-window condition, the old calibrated-block theorem has an additional acquisition hypothesis; branchwise admissibility alone is insufficient.

Theorem A12.2 (A fixed-source branch-selective efficiency floor). *Fix a smooth periodic dispersing table with a finite regular identifying atlas and strict margins. Fix, once, normalized raw source densities in common scalar phase coordinates, supported in compact launch charts which remain physical under small deformations. These source functions, the lattice, endpoint labels and deadlines are held fixed.*

If independently branch- and deadline-dependent retention is permitted, then for all sufficiently small $\epsilon > 0$ there exist noncongruent tables at geometric distance at least $c\epsilon$ with identical full raw laws and retention log changes of C^{k+2} size at most ϵ . This holds for any fixed readout mechanism applied after retention. Consequently any estimator, at any sample size, has worst-case expected geometric error at least $c'\epsilon$ in this branch-selective apparatus class. The source law need not vary between the two parameter points.

Proof. Choose a nonzero nonnegative smooth normal displacement φ on a small boundary arc of one obstacle. Let the displacement be $t\varphi$, $t \geq 0$, keeping other obstacles and the lattice fixed. For small t , strict convexity and disjointness persist; the obstacle expands, so it cannot open a new free corridor. Every margin of the finite atlas persists by smooth dependence. The area derivative is $\int \varphi ds > 0$. Area is isometry invariant and locally Lipschitz in the bounded C^1 boundary embeddings. Hence the geometric distance modulo Euclidean isometry is at least $c_0 t$.

Here is a source choice which is literally independent of t . For each launch chart choose a larger compact set of scalar phase coordinates (s, p) contained in the common physical domain for all small t . This is possible by nongrazing and compactness. Choose one smooth normalized density on that fixed coordinate domain, positive on the required initial states, and one fixed delay or free-time density. Push forward this same probability distribution to each physical table

by its scalar phase parametrization. Its coordinate density and total normalization do not change. The physical map from a scalar phase coordinate to position and velocity does change; this is part of the geometric parameter, not a change in the experimental source law.

Let $W_b^{(t)}$ be the resulting action and $J_b^{(t)}$ the endpoint Jacobian times this fixed source evaluated at the corresponding initial momentum (and the fixed delay factor). In the scalar billiard case,

$$J_b^{(t)}(s, y) = D^{-1} \rho(s, -\partial_s W_b^{(t)}(s, y)) |\partial_{sy}^2 W_b^{(t)}(s, y)|$$

for the delayed protocol; omit D^{-1} for the original flow box. Finite regular stationary equations give smooth C^{k+2} dependence of these positive functions on t , after imposing the corresponding finite derivative budget on the physical family.

Choose strictly interior baseline probabilities e_b . Define

$$e_{j,b}^{(t)} = e_b \frac{J_b^{(0)} T_j - W_b^{(0)}}{J_b^{(t)} T_j - W_b^{(t)}}. \quad (\text{A12.2})$$

Positive denominator margins and smooth dependence imply that these remain in $(0, 1)$ and that $\max_{j,b} \|\log(e_{j,b}^{(t)}/e_b)\|_{C^{k+2}} \leq C_0 t$. Multiplying (A12.2) by $J_b^{(t)}(T_j - W_b^{(t)})$ proves equality of every successful joint measure. The complementary failure mass is also equal. A fixed post-retention readout preserves equality of laws. Taking $t = \epsilon/C_0$ proves the geometric separation. Under the two parameter values any estimator has the same distribution; the triangle inequality bounds the sum of its two expected errors below by the parameter distance. This gives the claimed lower bound. $\square$

Theorem A12.2 and Corollary A9.2 describe different detector classes. In the latter, one function $e_j(z)$ multiplies every latent component; in the former, $e_{j,b}(z)$ may vary with the hidden branch. The compensator (A12.2) is generally of the second kind. It cannot be inserted into the first class to deduce an impossibility theorem there. This is a nonlinear identifiable versus nonidentifiable distinction with the non-efficiency source fixed, not a universal claim about all physical detectors.

APPENDIX A13. FINITE OBSERVATIONS WITHOUT A CALIBRATION SAMPLE

We state a sampling theorem for the new observation operator. Its purpose is to propagate actual readout and endpoint error through the blind inverse; a sharp geometric minimax exponent is not asserted. Let $Q_0 \subseteq Q_+$ be nested rectangles in $\mathbb{R}^d$. There are J settings and at most uv joint readout cells. At each setting obtain N independent raw trials, a failure contributing zero, or N independent retained records with their joint conditional density. These are distinct sampling budgets. In the retained-record version we do not count N as the number of raw preparations.

Theorem A13.1 (Finite-record blind action reconstruction). *Assume the uniform hypotheses of Theorem A7.1 on Q_+ , including the observed scale, rank and separation margins. Suppose the observed joint densities have bounded C^{m+s} norms, with integers $m \geq 0$, $s \geq 1$. Recorded endpoint errors have norm at most δ . Let η bound, in C^m , an additional joint-matrix perturbation away from an exact admissible model. It is not an arbitrary total-variation contamination budget.*

There is a locally defined estimator of the visible count and unordered actions such that, with probability at least $1 - \zeta$, the count is correct above the fixed signal threshold and

$$\max_b \|\widehat{W}_b - W_b\|_{C^m(Q_0)} \leq C \left\{ \left(\frac{\ell}{N} \right)^{s/[2(m+s)+d]} + \delta^{s/(m+s+d+1)} + \eta \right\}, \quad \ell = \log \frac{C_1 J uv (N+2)}{\zeta}, \quad (\text{A13.1})$$

when the right side is sufficiently small. Neither a calibration sample size nor a true-component audit probability appears. A common detector, however different at the clocks, belongs to the exact model and contributes no η term.

For Theorem A11.2, use $m = k + 2$ in this bound, its action regularity hypotheses, and C^{k+2+s} bounds on the observed densities. If those density bounds are derived from physical generating actions rather than assumed, C^{k+4+s} action bounds and C^{k+2+s} source, retention, channel and exposure bounds are sufficient on the fixed larger domains with the stated nondegeneracy margins. The resulting bound controls the C^k recovered lift with the same braces. The constant also includes its finite phase-cover and inverse-Jacobian margins.

Proof. Estimate each observed matrix entry by a compactly supported smooth kernel of order at least s :

$$\hat{N}_{j,ab}(z) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\text{readout pair } (a, b)\} K_h(z - Z_i).$$

Failures give zero in the raw version. The retained version estimates the joint conditional density, not a separately normalized density for each readout pair. For $|\alpha| \leq m$, the derivative summand has envelope $Ch^{-d-|\alpha|}$, variance at most $Ch^{-d-2|\alpha|}$ and spatial Lipschitz bound $Ch^{-d-|\alpha|-1}$. Taylor expansion gives C^m bias at most Ch^s . Bernstein's inequality on a polynomially fine grid, followed by the Lipschitz bound between grid points, gives simultaneously

$$\|\hat{N} - N\|_{C^m(Q_0)} \leq C \left(h^s + \sqrt{\frac{\ell}{Nh^{2m+d}}} + \frac{\ell}{Nh^{m+d}} + \delta h^{-(m+d+1)} \right). \quad (\text{A13.2})$$

The last term follows directly from the mean value theorem for the derivative of the kernel at the perturbed endpoint. A sufficiently fine fixed polynomial grid makes its interpolation remainder smaller than N^{-2} . The classical kernel ingredients may be compared with [8].

Take $h = \max\{(\ell/N)^{1/[2(m+s)+d]}, \delta^{1/(m+s+d+1)}\}$, small enough for the fixed collar. The decreasing stochastic and coordinate terms are bounded by a constant times the first two terms of (A13.1); the bias is bounded by their sum. Add η by the triangle inequality. Singular-value thresholding first fixes the visible rank. Equations (A9.2) and (A9.1), when needed, then correct the exposures, and the blind inverse gives the displayed bound by Proposition A8.2.

Use fixed chart choices on a small model neighborhood and define a fixed reference output if the positivity, rank or separation tests fail. This supplies a measurable estimator everywhere and the stated locally smooth reconstruction on the high-probability event. For the lift, apply Theorem A11.2. This proof does not infer a minimax sampling rate from a deterministic Lipschitz estimate. $\square$

The constants in (A13.1) depend on d, m, s, J, u, v , the kernel, the gate collar, density bounds, clock values and gaps, the minor and two-plane singular-value margins, action separation, weight floors, residual floors, observed rescaling floors and the fixed gluing cover. The geometric specialization additionally depends on return counts, phase collars and inverse mixed derivatives. Deterioration near an invisible branch or an action collision is not suppressed in a “uniform” constant.

A small extra component is handled as a matrix perturbation only when its contribution is small in the norm just specified. New components above the rank margin can instead be discovered by the exact unknown-count theorem, provided the enlarged channel pair remains full rank. An arbitrary omitted component, or an arbitrary singular contamination, is not covered by the smooth perturbation claim.

APPENDIX A14. THE RETAINED GEOMETRIC AND CONDITIONAL-LAW DEVELOPMENTS

The rest of the full manuscript retains the calibrated inverse, conditional-law quotient and action-risk theory, prefix cancellation, distance certificates, finite coverage, convex-suspension inverse, regularity budgets and finite-record billiard reconstruction. These results serve different observation regimes. The new projective multi-sheet theorem does not replace the single-sheet conditional inverse, nor does it delete a marked geometric theorem merely because its mark structure is stronger.

A14.1. Hypotheses for the calibrated companion. Every calibrated theorem below uses a *known finite exhaustive catalogue*, with structural branch identities attached to the columns. Its channels C_j are independent of endpoint conditional on the true component, and may differ between deadlines. The new blind theorem instead allows endpoint-dependent channels but requires their invariance across clocks and requires two independent readouts. Neither set of hypotheses is silently transferred to the other.

For every audited-calibration statement below impose the following explicit sampling design. Calibration raw trials are independent of the scientific trials and of one another. Conditional on a trial having passed the physical stages with true component b , its reported label has column law $C_j(:, b)$ including rejection. The audit-availability indicator is conditionally independent of this reported label given b . A sufficient design is independent Bernoulli selection for audit after retention, using a coin independent of endpoint and reported label; its probability may depend on b . Every component is available with raw probability at least p_c . Then, conditional on the audited component counts, reported labels are independent samples from the correct columns. Chernoff and Hoeffding bounds used in the companion are consequently unbiased. Selecting audits by the reported label without inverse-probability correction does not satisfy this design.

The original calibrated billiard blocks also require common transported endpoint coordinates and simultaneous windows. Condition (A12.1) is sufficient for the original protocol; Proposition A12.1 supplies a different explicit protocol at a fixed return count. Neither an audit nor branchwise admissibility proves those conditions by itself.

A14.2. Data dependence.

Observation regime	Information and further assumptions
Two raw, blind matrices	Two independent clock-invariant readouts; unknown visible count; no audit; absolute actions and channels.
Three projective matrices	The same, with at least two separated actions; unknown branch-blind detector and exposures; relative weights.
Two raw, calibrated vectors	One endpoint-independent calibrated channel per clock; known catalogue; true audit or known matrix.
Three conditional, one branch	One specified nonconstant action, common weight; no success masses; conditional quotient inverse.
Unpaired phase gates	Blind recovered actions plus regular coverage of the required lifted iterates; no supplied sheet pairing.
Marked billiard atlas	Word/lift roles and scalar overlaps remain; actions feed prefix, distance and registration certificates.
Finite observations	Density smoothness and signal margins in addition to the relevant deterministic assumptions; no automatic minimax claim.

In particular the new latent indices are not automatically reflection words. In the retained billiard theorem those semantic roles are still part of the marked experiment. A universal

table-independent stopping rule, arbitrary caustic resolution, arbitrary missing components and geometric sampling optimality are not consequences of the present proofs. The positive new conclusion is the blind projective recovery of the action sheets and the exact common-detector quotient; the retained assumptions on downstream geometry remain visible.

APPENDIX A15. INTRODUCTION

An endpoint experiment does not directly measure an action. Its successful records are selected by the source, the clock and the detector; its branch labels may themselves be wrong. A geometric inverse must distinguish these losses of information before using a generating function. We study this problem for reflected trajectories and exact symplectic return systems. The central observation is a raw endpoint measure whose density depends affinely on a known deadline,

$$r_T(z) = a(z)(T - W(z)). \quad (\text{A15.1})$$

Here W is an absolute action, a is an unknown positive weight, and all unsuccessful preparations are retained as failure symbols. The affine dependence is obtained by integrating a physical free-time coordinate or an independent release delay. It is not an extra numerical value supplied with each endpoint.

There are two distinct inverse problems. An archive containing only the conditional densities $f_T = r_T / \int r_T$ has discarded the success masses. Three deadlines recover a nonconstant action in this archive; two do not, because an unknown normalizing factor leaves a nontrivial action gauge. By contrast, a raw archive contains the masses as well. Writing $\Delta = T_2 - T_1$, its two successful subdensities give

$$a = \frac{r_{T_2} - r_{T_1}}{\Delta}, \quad W = \frac{T_1 r_{T_2} - T_2 r_{T_1}}{r_{T_2} - r_{T_1}}. \quad (\text{A15.2})$$

This formula also recovers constant actions, has no anchor condition, and loses no spatial derivatives. The distinction matters whenever a preparation budget is stated in raw launches: the failures being charged contain information and need not be discarded by the estimator.

We combine this distinction with a finite calibrated channel for incorrect positive branch labels. On a common endpoint gate let $r_{j,b}$ be the true raw subdensity of branch b , and let $g_{j,a}$ be the observed joint subdensity of reported label a . A class-dependent classifier has

$$g_j = C_j r_j. \quad (\text{A15.3})$$

A calibrated left inverse recovers the component subdensities *before* any normalization. The clock inverse then recovers the absolute branch actions. This allows false word or lift labels on scientific records; it does not assume that every positive label is correct. It does require an exhaustive finite catalogue and a calibrated full-rank channel. These are measurable experimental conditions, not conclusions inferred from a small fitting residual.

Theorem A15.1 (Calibrated action rigidity). *Consider a finite catalogue of regular branches on specified scalar endpoint gates. At two admissible deadlines on each acquisition block, suppose the true raw endpoint measures have the form $r_{j,b} = a_b(T_j - W_b)$ with unknown common positive weights, and the reported labels have known full-column-rank channels C_j as in (A15.3). The full raw laws determine every absolute branch action and raw weight, with a locally Lipschitz inverse in every fixed C^m norm on bounded classes with positive weight, clock-gap and channel singular-value margins.*

When the catalogue supplies a finite identifying atlas of regular prefixes and one-flight extensions for a periodic dispersing billiard, these laws determine the entire marked table and its lattice Gram matrix up to a common Euclidean isometry. When instead they supply a phase-graph cover for two relatively prime iterates of an exact return map, they determine its one-step map and

absolute action on every compact region covered by the associated finite inverse word. The latter assertion needs neither convexity nor minimizing stationary paths.

For the uniformly convex negative-twist class (A27.1), four fixed raw deadlines, two at each of any prescribed consecutive return counts $n, n+1$, determine the unknown two-variable generating function on a prescribed compact square. The gates, source phase box, release interval and deadlines depend only on the class bounds. For C^k reconstruction, fixed C^{k+4} bounds on the generating function and C^{k+2} bounds on source and efficiency suffice for the uniform C^{k+2} to C^k estimate.

Proof. The raw inverse and channel estimate are Theorems A18.1 and A18.2. The absolute actions are the inputs of the prefix, distance and registration proof in Sections A23–A26; no conditional normalizing constants enter that deterministic proof. The exact-map assertion is Theorem A19.3. The class-wide construction and its explicit derivative budget are proved in Theorem A28.2. $\square$

A15.1. Statistical and physical stability. The geometric consequence persists when the channel must be estimated and the clock weights are only approximately shared. On a bounded reference-atlas neighborhood, let N be the scientific raw-launch budget at each of J settings. Let N_c be the raw calibration budget, with true-component audit probability at least p_c . For target C^k geometry put $m = k + 2$, and assume $s \geq 1$ additional density derivatives for smoothing. The estimator in Theorem A33.3 has, with high probability,

$$d_k(\hat{\mathcal{T}}, \mathcal{T}) \leq C \left\{ \left(\frac{\ell}{N} \right)^{s/[2(m+s)+2]} + \delta^{s/(m+s+3)} + \sqrt{\frac{\ell}{N_c p_c}} + \epsilon \right\}, \quad (\text{A15.4})$$

where ℓ is logarithmic in the budgets and inverse confidence, δ bounds coordinate errors, and ϵ bounds clock-dependent log-efficiency errors in C^{k+2} . The number of raw preparations is $J(N + N_c)$, including unsuccessful calibration attempts. The constants explicitly depend on branch weight and channel conditioning. The estimator never obtains an exact true branch label on its scientific trials.

The term ϵ has a physical, not only an abstract, interpretation. A strictly interior endpoint-retention probability admits every small smooth log perturbation as an actual Bernoulli retention rule (Proposition A32.1). Along a genuine smooth deformation of a billiard, a ratio of its source–Jacobian factors and its clock residuals produces valid retention probabilities for which *all raw record laws coincide* (Theorem A32.2). An outward deformation of one obstacle gives noncongruent tables separated by order ϵ . Thus Corollary A32.3 proves that the linear physical efficiency term in (A15.4) is unavoidable for this apparatus class. This is not a claim that the sampling exponent for geometric loss is minimax optimal.

The conditional-law archive has a complementary, fully retained theory. Common positive reweighting gives an ambient Banach quotient, and a nonconstant two-anchor chart splits it smoothly into action and functional residual. The normal space is infinite-dimensional; the assertion is a complemented split, not finite codimension. The residual tests transverse misspecification but not perturbations exactly confounded with another action. On the specified unrestricted Hölder weighted-action class, the conditional-record minimax C^m action risk is

$$\asymp (\log n/n)^{(\beta-m)/(2\beta+d)} + \epsilon, \quad \beta > m.$$

The theorem and both lower-bound arguments remain in Section A22. It has a different observation operator and parameter class from the physical geometric lower bound.

A15.2. Deterministic content and comparison with prior work. The exact part of the reconstruction has two levels. At the phase level, an exact map together with its absolute

primitive forms an action lift. Relatively prime iterates determine this lift by Bézout’s identity in the lift group. This observation separates the deterministic inverse from the clock experiment and allows several nonminimizing endpoint sheets when their phase graphs are covered. At the billiard level, determinism identifies a prefix from equality of initial covectors. Its positive stationary Hessian gives a unique downward matching zero, and subtraction yields a last-flight distance kernel. A finite rank certificate reconstructs boundary arcs and registers them through labelled overlaps. The entire smooth, nonanalytic whole-table proof is included, not replaced by the phase-level identity.

Generating-function composition, discrete Euler–Lagrange equations, and symplecticity are classical ingredients [12, 16]. Likewise positive exact actions have classical contact realizations [13]. Proposition A19.1 explains the converse interpretation needed here: for an existing Reeb flow with the stated global section, its return time is its exact symplectic action, and the flow is recovered as that suspension. The class-wide convex example is a separate application in which the unknown suspension varies with L . Neither the elementary composition identity nor construction of a contact suspension is claimed as a new existence theorem.

In inverse billiards, the periodic and exterior observations are materially different from the scalar gates used here. Bálint–De Simoi–Kaloshin–Leguil [10] extract dynamical and geometric information from marked lengths of open dispersing billiards. De Simoi–Kaloshin–Leguil [11] establish analytic marked-length determination under their symmetry hypotheses. Noakes–Stoyanov [9] treat planar scattering and travelling-time data. Finamore–Leguil [7] prove finite-horizon Sinai rigidity from an enriched marked length spectrum. Our exact comparison with canonical data is Proposition A20.2: a scalar canonical graph and one absolute action value determine a conditional clock-law orbit, and conversely. No global ordering of the different marks is inferred.

Calibrated matrix correction for class-dependent errors is also not a new principle [17]. Mutual-contamination models with unknown components and mixing proportions require additional identifying hypotheses; see [18]. Here the matrix acts on *raw joint subdensities*, so correction preserves precisely the masses needed for the two-deadline inverse. The theorem couples that correction to absolute action and geometric reconstruction, with explicit calibration cost and a physical retention floor. It does not assume that more deadlines separate an arbitrary unknown mixture. The rank counterexample in Proposition A18.3 records the exact distinction.

For clarity, the main comparisons can be stated at the level of inputs and conclusions rather than journal categories.

Input and prior mechanism	Classical conclusion used	Conclusion established here
Exact generating functions [12, 16]	Composition and canonical maps	Absolute lifts from calibrated raw endpoint measures; covered coprime returns need no minimizing hypothesis.
Positive action and contact return [13]	Contact realization of a positive action	Four class-wide raw deadlines for an unknown convex twist suspension; a separate inverse for covered existing Reeb returns.
Calibrated class-dependent label noise [17]	Linear correction with a known channel	Joint subdensity correction before normalization, with raw calibration budget and action-to-geometry error.
Marked length and scattering data [11, 9, 7]	Rigidity for their geometric data and hypotheses	Smooth periodic geometry for the stated scalar reference atlas, allowing calibrated false word and lift labels.
Uniform density estimates and robust risk [8, 14, 19]	Sampling and contamination bounds in their models	A physical retention tangent, indistinguishable noncongruent billiards, and a matching linear efficiency term.

The rank-two and rank-five distance certificate used below is an explicit sufficient reconstruction rule. General generic rigidity of complete bipartite frameworks is treated in [6]. Neither failure of our certificate on some conic arcs nor the presence of richer scalar labels proves a necessity statement about geometric rigidity.

A15.3. Scope and arrangement. The structural scalar atlas, word catalogue, coordinate transports and lift conventions remain part of the data design. For billiards the uniform statement is on bounded neighborhoods of a reference atlas, not on all unknown tables with a universal acquisition rule. The class-wide fixed protocol is proved for the explicitly bounded convex twist class. The Reeb statement requires a genuine global section and the declared phase-graph coverage. These hypotheses distinguish three geometric consequences of the same raw action inverse.

Sections A18 and A19 establish the raw observation and deterministic action principles. Sections A20–A22 retain the conditional quotient and sharp action-risk theory. Sections A23–A26 give the full billiard proof. The convex suspension, its regularity budget, and the mass–potential subfamily follow. The last sections combine raw costs, physical nuisance and calibrated finite records. The exact-label conditional results are special cases with a different archive, not assumptions silently imposed on the calibrated scientific records.

The periodic boundary-law inverse remains a separate companion. Its relative determinant normalization, positive quadratic inverse, signed jets, smooth envelope, finite-preparation consequences and flat smooth perturbations are retained without changes. They are not invoked as an unproved bridge between periodic boundary data and the endpoint experiment studied here.

APPENDIX A16. MARKED PHYSICAL GATES AND A GEOMETRIC CERTIFICATE

We give the exact-labelled physical framework first. Its scalar geometric catalogue remains in the calibrated theorem; correct labels on every retained scientific trajectory do not. All observations in this section are at the law level. The later raw and finite-record statements specify what information is retained before conditioning.

A periodic table is the complement of disjoint translates $K_a + L\lambda$, $1 \leq a \leq q$, $\lambda \in \mathbb{Z}^2$, where L is invertible and each obstacle has a smooth strictly convex boundary with positive curvature. A scalar boundary atlas, obstacle and lift labels, and regular reflection words are part of the observation convention. An atlas coordinate is realized by an unknown regular embedding $R_I : I \rightarrow \mathbb{R}^2$. Its Euclidean position, speed, tangent and normal are not observations. The speed of the moving particle is one. One can instead fix, once for the whole class, the abstract label space $\{1, \dots, q\} \times (\mathbb{R}/\mathbb{Z}) \times \mathbb{Z}^2$ and its standard charts; then chart transitions are fixed by this label space, not chosen from an unknown Euclidean realization.

A branch γ has first and last scalar labels (s, t) and stationary length $W_\gamma(s, t)$. The observer knows its word and its endpoint gate, but records only (s, t) on retention. An unsuccessful attempt has a failure symbol. In the flow box before the first collision, the source has an unknown smooth positive density $\rho_\gamma(s, p)$, independent of the time α to that collision. The unknown retention probability $e_\gamma(s, t) > 0$ is common to the clocks of this branch. It need not separate in s, t , and different branches may have different preparations and retention laws. All three conditional laws are normalized on the same gate. Their restrictions to a smaller gate are not renormalized.

Let $\tau_\gamma^-, \tau_\gamma^+$ be the adjacent free flights. Three absolute times $T_1 < T_2 < T_3$ are admissible on a gate if, with strict margins,

$$0 < T_j - W_\gamma(z) < \min(\tau_\gamma^-(z), \tau_\gamma^+(z)). \quad (\text{A16.1})$$

This is a restriction on the physical class. It is not a measured length function. The resulting densities are

$$f_{\gamma,j}(z) = \frac{a_\gamma(z)(T_j - W_\gamma(z))}{\int_{Q_\gamma} a_\gamma(u)(T_j - W_\gamma(u)) du}, \quad a_\gamma = e_\gamma \rho_\gamma(s, -W_{\gamma,s}) |W_{\gamma,st}|. \quad (\text{A16.2})$$

The flow-box proof is given in Lemma A17.1. For raw-launch acquisition with classification, the clock-independent classification probability is an additional factor of a_γ ; Definition A31.1 and Proposition A31.2 specify its meaning and cost without renormalizing away branch rarity.

A boundary is *locally nonconic* when no nonempty open subarc is contained in the zero set of a nonzero polynomial of degree at most two. This is a sufficient condition for the finite certificate, not a necessity claim for rigidity.

Definition A16.1 (Geometric certificate). A finite identifying atlas consists of finitely many regular prefixes and their one-flight extensions, fixed scalar gates and admissible clock triples. The final-flight patches obtained by matching their common initial states cover the target. Each patch has the rank-two and rank-five distance certificate of Proposition A25.1. The graph registering the patches by scalar overlaps is connected. For a whole periodic table, the cover includes complete quotient boundaries and common intervals on one boundary and its two marked basis translates. Positive matching margins, certificate anchors and overlap triples are witnesses to this definition, not additional observed marks. The local construction in Proposition A23.2 supplies such witnesses. Theorem A24.2 recovers matching roots without receiving them.

Theorem A16.2 (Marked statistical-action rigidity). *For a finite identifying atlas, the branch-resolved three-clock endpoint laws determine every target boundary embedding up to a common Euclidean isometry. No root-selection bracket, Euclidean contact point, tangent frame, chart speed, absolute branch length, source weight or lattice Gram matrix is supplied. In the whole-table case they determine the entire marked periodic table and its lattice Gram matrix.*

The laws determine their absolute generating actions and the normalized nuisance shapes separately. Modulo simultaneous positive reweighting at the three clocks, their exact information is precisely that of the absolute actions. Matching of a prefix to its one-flight extension is unique on

a connected common initial interval whenever it exists. The matching domain itself is determined by signs of recovered functions.

Every finite-horizon periodic dispersing table with locally nonconic boundaries admits such a finite protocol, with prefix lengths at least any prescribed finite J . A single protocol works on a neighborhood of each such table. On bounded neighborhoods with positive certificate margins it has a uniform finite-order stable inverse. There is also an explicit finite-record reconstruction by smoothing, root solving, linear algebra and patch blending; it does not select from an unknown dense enumeration of possible tables.

The theorem compares any two realizations of the same admissible data; its uniqueness assertion is not a local inverse-function assertion in the unknown geometry. Uniform budgets, in contrast, are asserted on the bounded neighborhoods specified in Section A30. The existence of a finite protocol at each table is not a universal acquisition rule for all unknown tables. Exact word and lift labels are hypotheses of this conditional-law statement. In Theorem A33.3 the same structural catalogue is retained, but its per-record labels pass through a calibrated channel. That extension requires common admissible deadlines within each confusion block; it does not claim that an arbitrary collection of unrelated branches has such a common window.

APPENDIX A17. FINITE PHYSICAL LAWS AND ABSOLUTE ACTION RECOVERY

Lemma A17.1 (Physical law with an unknown joint multiplier). *On a regular finite branch and an admissible clock gate, the retained conditional density is exactly*

$$f_j(s, t) = Z_j^{-1} a(s, t) \{T_j - W(s, t)\}, \quad a(s, t) = e(s, t) \rho(s, -W_s(s, t)) |W_{st}(s, t)| > 0, \quad (\text{A17.1})$$

where Z_j is the integral of the numerator over the common gate. The same multiplier a occurs at all three clocks. It is not required to factor into a function of s and a function of t .

Proof. Write $R'(s) = v(s)\mathbf{t}(s)$, $v(s) > 0$, and let φ be the angle with the inward-to-free-space normal. The tangential covector is $p = v(s) \sin \varphi$. In coordinates before the first collision, the unit-speed phase-volume element is

$$v(s) \cos \varphi \, ds \, d\varphi \, d\alpha = ds \, dp \, d\alpha.$$

Indeed the spatial Jacobian of a boundary point and distance along its incoming ray is $v \cos \varphi$; specular reflection preserves the tangential covector. The first variation of stationary length gives $p = -W_s$. The change from p to the last endpoint t has Jacobian $|W_{st}|$, which is nonzero by Lemma A23.1.

If the first collision occurs at time α , the last occurs at $\alpha + W$. The no-previous-collision and no-additional-collision conditions read

$$0 < \alpha < \tau^-, \quad 0 < T - \alpha - W < \tau^+.$$

Under (A16.1), the allowed interval is exactly $0 < \alpha < T - W$. Integrating the source density, which is independent of α in this box, gives $\rho(s, -W_s) |W_{st}| (T - W)$. Multiplication by the retention probability and normalization give (A17.1). The parametrization is one-to-one on the selected regular branch, so no unrecorded multiplicity has been inserted. $\square$

Lemma A17.2 (Clock design from coarse brackets). *Suppose on a gate and a prescribed class one knows $w_- \leq W \leq w_+$ and $\tau^\pm \geq \lambda > 0$, with $w_+ - w_- < \lambda/4$. Then*

$$T_j = w_+ + j\lambda/4, \quad j = 1, 2, 3,$$

is admissible, with margins on both sides. Such brackets, with rational endpoints and strict inequalities, exist after shrinking a gate about any regular finite trajectory. A finite family of these choices persists under small finite-smooth perturbations of the table.

Proof. The lower residual is at least $\lambda/4$; the upper is at most $3\lambda/4 + w_+ - w_- < \lambda$. For a single regular branch, continuity of its stationary length permits a gate on which its oscillation is arbitrarily small. The adjacent flights have a positive lower bound on a smaller closed gate. Choose rational enclosing bounds with room in the inequalities. The implicit construction in Lemma A23.1 and continuity of the clear flights preserve all strict inequalities on an open neighborhood. No exact value of W is subsequently supplied to the inverse. $\square$

A fixed design therefore needs coarse admissibility information, not calibrated branch lengths. The finite-atlas theorem is an existence and persistence theorem for such designs. It does not identify the finite selection of suitable words and clocks with a universal finite adaptive search over all unknown tables. Scalar charts, words, rational subgates and rational clock triples form a countable acquisition family containing every finite design constructed in that theorem.

A17.1. Elimination of normalization and joint weights. The following statement is independent of the dynamics. Let Q be a connected rectangle and let the positive densities

$$f_j(z) = Z_j^{-1} a(z)(T_j - W(z)), \quad j = 1, 2, 3,$$

be given on a neighborhood of Q . Assume $T_1 < T_2 < T_3$, $a > 0$, $T_j - W > 0$, and W nonconstant. Fix $z_0 \in Q$, and set

$$g_j(z) = \frac{f_j(z)}{f_j(z_0)}, \quad (c_1, c_2, c_3) = (T_2 - T_3, T_3 - T_1, T_1 - T_2). \quad (\text{A17.2})$$

Theorem A17.3 (Three-clock absolute action inverse). *There exists $z_* \in Q$ with $D_* := \sum_j c_j g_j(z_*) \neq 0$. For any such point the data determine*

$$W_0 = \frac{\sum_j c_j T_j g_j(z_*)}{\sum_j c_j g_j(z_*)}, \quad d_j = T_j - W_0, \quad (\text{A17.3})$$

$$Q_{12}(z) = \frac{g_1(z)}{g_2(z)}, \quad W(z) = W_0 + \frac{d_1 d_2 (1 - Q_{12}(z))}{d_2 - d_1 Q_{12}(z)}. \quad (\text{A17.4})$$

In these formulas $W_0 = W(z_0)$. Thus the three laws determine the absolute action, without a supplied anchor value or any supplied multiplier. They also determine $a(z)/a(z_0)$.

For each integer $m \geq 0$, the displayed inverse is locally Lipschitz in $C^m(Q)$ on bounded sets with a fixed usable anchor z_* and positive floors for its scalar denominator and all density denominators. It extends as a map on an open neighborhood of each such set, without requiring perturbed functions to satisfy the model.

Proof. Put $w_0 = W(z_0)$ and $b(z) = a(z)/a(z_0)$. Then

$$(T_j - w_0)g_j(z) = b(z)(T_j - W(z)). \quad (\text{A17.5})$$

Since $\sum c_j = \sum c_j T_j = 0$, taking the weighted sum gives $\sum c_j T_j g_j = w_0 \sum c_j g_j$. Moreover

$$\sum_j c_j g_j(z) = b(z)(w_0 - W(z)) \sum_j \frac{c_j}{T_j - w_0}.$$

The last sum equals

$$-\frac{(T_2 - T_1)(T_3 - T_1)(T_3 - T_2)}{(T_1 - w_0)(T_2 - w_0)(T_3 - w_0)},$$

which is nonzero. Nonconstancy supplies z_* , and (A17.3) follows. With $H = W - w_0$,

$$Q_{12} = \frac{d_2(d_1 - H)}{d_1(d_2 - H)}, \quad d_2 - d_1 Q_{12} = \frac{d_2(d_2 - d_1)}{d_2 - H} > 0.$$

Solving gives (A17.4). Finally $b = g_1 d_1 / (T_1 - W)$.

Point evaluation, multiplication and restriction are bounded in C^m ; reciprocal maps are locally Lipschitz on bounded sets bounded away from zero. Apply these facts to the successive formulas. The last displayed identity supplies the remaining denominator floor from clock separation and bounded positive residuals. All floors persist for a sufficiently small nonmodel perturbation. No differentiation of the observations is used at this stage. $\square$

For physical branches, $W_{st} \neq 0$ ensures nonconstancy. Usable anchors can be specified by strict tests on the functions g_j ; they are scalar labels, not known spatial contact points. In a finite atlas one fixes such anchors and retains a positive margin after shrinking the class. An arbitrary additive constant in an action is not harmless: the three absolute clock values are what determine it.

For comparison, when $W(z_0)$ is known, the second formula needs only two clocks. Conversely, in the unrestricted weighted-function model two unanchored clocks leave a fractional-linear freedom. This is an algebraic explanation of the third clock, not an assertion that every fractional-linear change of an action is a realizable billiard action.

The shared-weight hypothesis has a precise role. A multiplier that varies freely with the clock could absorb a change of W . A source varying with α would instead give $\int_0^{T-W} \rho(s, p, \alpha) d\alpha$. Neither alteration is made in Theorem A16.2. The allowed multiplier is, however, an arbitrary positive function of both retained endpoints.

APPENDIX A18. RAW ENDPOINT MEASURES AND CALIBRATED BRANCH LABELS

A conditional law and a raw record are different data. A raw trial has an outcome in $Q \cup \{\dagger\}$, where $\dagger$ includes every unsuccessful launch. Its successful part is a subprobability measure, not a probability measure. If p_j is the success probability and f_j the conditional density, the observable subdensity is $r_j = p_j f_j$. The normalization of the source is fixed before any gate, time, or branch selection. Changing the deadline does not renormalize this source.

Theorem A18.1 (Two raw deadlines). *Let $T_1 < T_2$, $\Delta = T_2 - T_1$, and let $a, W \in C^m(Q)$ satisfy $a > 0$ and $W < T_1$. Suppose the successful subdensities of two raw experiments are*

$$r_j = a(T_j - W), \quad j = 1, 2. \quad (\text{A18.1})$$

Then the full raw laws determine both a and the absolute action W :

$$a = \frac{r_2 - r_1}{\Delta}, \quad W = \frac{T_1 r_2 - T_2 r_1}{r_2 - r_1}. \quad (\text{A18.2})$$

This includes constant actions and requires no clock anchor. On bounded C^m classes with $a \geq a_- > 0$ and $\Delta \geq \Delta_- > 0$, the inverse is uniformly locally Lipschitz in C^m . One raw deadline does not identify W when the positive weight is unknown, even locally in a strictly admissible class.

Proof. Subtracting the two identities in (A18.1) gives $r_2 - r_1 = \Delta a$; eliminating aW gives the second formula. For C^m functions on a compact rectangle, multiplication is continuous and the reciprocal map is smooth on the positive cone. The lower bound $r_2 - r_1 \geq \Delta_- a_-$ proves the stated stability, including its extension to a neighborhood of the exact data. For one deadline and any sufficiently small nonzero $h \in C^m(Q)$, put $V = W + h$ and $\tilde{a} = r_1/(T_1 - V)$. These remain admissible and have exactly the same subdensity and hence the same failure mass. $\square$

Known exposure factors may be divided out before applying the theorem. Unknown deadline-dependent exposure factors may not. In particular, applying (A18.2) to conditional densities f_j in place of r_j is incorrect. Proposition A20.3 exhibits precisely the ambiguity left by that loss of masses. Thus the two-raw and three-conditional inverse statements concern genuinely different observation operators.

For three or more raw deadlines an exact model has a testable affine residual. With $\theta_j = (T_j - T_1)/\Delta$, set

$$E_j = r_j - (1 - \theta_j)r_1 - \theta_j r_2. \quad (\text{A18.3})$$

On the open set where (A18.2) gives $a > 0$ and $W < \min_j T_j$, all these residuals vanish if and only if all subdensities have the same weight and action. Indeed the first two subdensities already give the unique affine pencil. Additional clocks validate this pencil; they are not necessary to fix its absolute scale.

A18.1. A finite channel for false positive labels. We now allow incorrect positive branch labels. In one acquisition block there are B possible true branches, with the same recorded endpoint coordinate $z \in Q$, and $A \geq B$ reported labels. These branches are an *exhaustive* list of the components which can produce a reported record in the block. A branch may represent a reflection word, a lift, or a regular sheet of a return map. Known coordinate transports can first put its endpoint labels in this common gate. The channel below corrupts the branch label, not the endpoint coordinate.

For deadline j , let C_j be an A by B nonnegative matrix with column sums at most one. Conditional on the true branch b of a record which has passed the physical gate and retention stage, its reported label is a with probability $(C_j)_{ab}$, independently of z . The missing column mass is reported as failure. This is a class-dependent error model; it does not assume that a reported label is usually correct. Calibration specifies which column corresponds to which structural branch. It is not an unlabelled mixture model.

If the true raw subdensities are $r_{j,b}$, the observed joint subdensities are

$$g_j(z) = C_j r_j(z). \quad (\text{A18.4})$$

Both the frequency of each reported label and its conditional endpoint law are used in g_j . Rowwise normalization before inversion would change (A18.4) and discard necessary information.

Theorem A18.2 (Calibrated recovery with false positive labels). *Suppose C_1, C_2 have full column rank and the true subdensities satisfy*

$$r_{j,b} = a_b(T_j - W_b), \quad a_b > 0, \quad W_b < T_1.$$

Two raw deadlines determine every a_b, W_b , including their absolute scales. An explicit inverse is

$$r_j = C_j^\dagger g_j, \quad C_j^\dagger = (C_j^T C_j)^{-1} C_j^T, \quad (\text{A18.5})$$

followed by (A18.2) in each component.

For a quantitative statement fix m , bounded C^m weights and actions, $a_b \geq a_- > 0$, positive residual and clock-separation bounds, and $\sigma_{\min}(C_j) \geq \sigma > 0$. Allow actual subdensities $r_{j,b} = a_b(T_j - W_b)e^{\eta_{j,b}}$ with $\max_{j,b} \|\eta_{j,b}\|_{C^m} \leq \epsilon$. If $\|\hat{C}_j - C_j\|_{\text{op}} \leq \delta_C \leq \sigma/2$ and $\max_j \|\hat{g}_j - g_j\|_{C^m} \leq e$, the same inverse, using hats, has, for sufficiently small $e + \delta_C + \epsilon$,

$$\max_b (\|\hat{W}_b - W_b\|_{C^m} + \|\hat{a}_b - a_b\|_{C^m}) \leq C(e + \delta_C + \epsilon). \quad (\text{A18.6})$$

The constant depends only on the displayed bounds, dimensions and clock values; in particular it is allowed to deteriorate as σ or a_- tends to zero. No spatial derivatives are lost.

Proof. A left inverse in (A18.5) recovers the true raw components, and Theorem A18.1 proves exact recovery. Singular-value perturbation gives $\|\hat{C}_j^\dagger\|_{\text{op}} \leq 2/\sigma$. Writing $\hat{r}_j = \hat{C}_j^\dagger \hat{g}_j$ and using $\hat{C}_j^\dagger \hat{C}_j = I_B$ gives the useful exact identity

$$\hat{r}_j - r_j = \hat{C}_j^\dagger (\hat{g}_j - g_j) + \hat{C}_j^\dagger (C_j - \hat{C}_j) r_j. \quad (\text{A18.7})$$

Since these matrices do not depend on z , this identity holds after every spatial derivative of order at most m . Its norm is at most $C\sigma^{-1}(e + \delta_C \max_j \|r_j\|_{C^m})$. The C^m algebra estimate gives

$$\|a_b(T_j - W_b)e^{\eta_{j,b}} - a_b(T_j - W_b)\|_{C^m} \leq C\epsilon.$$

The two-row inverse is defined on an open neighborhood of the exact pencils; its denominator is bounded below there by $\Delta a_-/2$. Applying its local Lipschitz estimate proves the result. Intermediate estimated subdensities need not be normalized or projected onto a statistical model. $\square$

For example the two-class channel with correct-label probability $1 - q$ and incorrect-label probability q has singular values 1 and $|1 - 2q|$. Both $q < 1/2$ and $q > 1/2$ are identifiable when the channel is calibrated. The critical issue is separation of its columns, not whether the names attached to the columns are more often right than wrong.

Proposition A18.3 (Rank is an exact obstruction for raw action pencils). *Suppose the same channel C is used at all deadlines and has nontrivial kernel. In the unrestricted positive action-pencil class, arbitrarily many raw deadlines need not determine the branch actions. This remains true with smooth strictly positive subdensities and strict residual margins.*

Proof. Take $0 \neq v \in \ker C$, positive smooth a_b , and strictly admissible W_b . For a sufficiently small real t define

$$W_b^{(t)} = W_b + t v_b / a_b.$$

Then $r_{j,b}^{(t)} = r_{j,b} - t v_b$ for every deadline. Thus $C r_j^{(t)} = C r_j$, including equality of the total reported masses and of the failure probabilities. Positivity and smoothness persist for small $|t|$. At least one action changes. Choosing the original subdensities to have total mass strictly below one also preserves the raw probability constraint before classification. $\square$

The proposition is a sharp boundary for this channel model, not a claim that every rank-deficient channel destroys every special geometric subclass. It explains why adding clocks cannot replace branch separation. Conversely, Theorem A18.2 removes exact per-record word and lift classification while retaining a calibrated finite catalogue. It does not supply that catalogue, identify an unknown number of branches, or absorb an omitted background component. Those distinctions will remain explicit in the geometric theorem.

APPENDIX A19. RETURN ACTIONS AND RECONSTRUCTION WITHOUT CONVEXITY

The deterministic part of the inverse can be stated without clocks. We first identify which action is intrinsic to a return system, and then separate an algebraic reconstruction of the dynamics from the problem of observing its actions. The symplectic composition law used below is classical; see [12, 16]. Its role here is to specify exactly what is recovered, including the otherwise invisible additive constants.

Proposition A19.1 (The action of a Reeb return is its elapsed time). *Let a complete Reeb flow ϕ_t on a contact manifold (M, α) have a smooth transverse global section $i : \Sigma \rightarrow M$. Suppose every orbit meets the section in both time directions, the first-return map F is a diffeomorphism, and its first-return time $\tau \geq \tau_- > 0$ is smooth. Put $\lambda = i^* \alpha$. Then*

$$F^* \lambda - \lambda = d\tau. \tag{A19.1}$$

The flow on M is the suspension of (F, τ) , with contact form $dt + \lambda$ in flow coordinates and identification $(x, t) \sim (Fx, t - \tau(x))$. In particular the roof is fixed by the contact form and

section; it is not an additional freely chosen observable. Its r th return action is

$$A_r(x) = \sum_{i=0}^{r-1} \tau(F^i x).$$

Proof. The Reeb equations imply $\mathcal{L}_R \alpha = 0$ and $\alpha(R) = 1$. For $\Psi(x, t) = \phi_t(i(x))$ one consequently has $\Psi^* \alpha = dt + \lambda$. Differentiating the variable-time identity $i(Fx) = \phi_{\tau(x)}(i(x))$ and evaluating α gives $F^* \lambda(v) = \lambda(v) + d\tau(v)$ for every $v \in T_x \Sigma$. This proves (A19.1). Every point has a unique representation with $0 \leq t < \tau(x)$, apart from the endpoint identification. Transversality and the smooth return map give compatible local flow charts at those endpoints. These charts identify the suspension smoothly with M and preserve the displayed contact form. Summing successive first-return times proves the last formula. The hypotheses that this is a global section, not merely a local transversal, are used in this identification. $\square$

The proposition is the standard geometric relation between contact return time and exact symplectic action; compare [13]. It applies to an existing Reeb system with the stated section. In the convex-twist construction of Section A27, the unknown system is the particular suspension associated with L and a fixed offset c . Its manifold and contact form vary with L . These are two uses of the same relation, not an assertion that the ordinary time of an unsuspended discrete map is an action.

Definition A19.2 (An exact map with its absolute action). On a connected exact symplectic manifold $(\Sigma, d\lambda)$, an *action lift* is a pair $g = (F, A)$ with F a diffeomorphism and $dA = F^* \lambda - \lambda$. The function A , not just its differential, is part of the pair. Composition and inversion are

$$(F, A)(G, B) = (F \circ G, B + A \circ G), \quad (F, A)^{-1} = (F^{-1}, -A \circ F^{-1}). \quad (\text{A19.2})$$

The chain rule verifies that these are again action lifts. Associativity follows either by substitution or from composition of the maps $(x, t) \mapsto (Fx, t - A(x))$. Thus the lifts form a group. Constant actions above the identity are retained in this group; discarding them would discard absolute elapsed time.

Theorem A19.3 (Coprime return actions determine the one-step lift). *Let $g = (F, A)$ be an action lift. If m, n are relatively prime positive integers, the two lifts g^m and g^n determine g uniquely. For any integers u, v satisfying $um + vn = 1$,*

$$g = (g^m)^u (g^n)^v. \quad (\text{A19.3})$$

In particular, absolute type-I generating functions for the two iterates, on regular charts covering their phase graphs, determine F and A . No convexity, minimizing property, or uniqueness of an endpoint stationary path is required.

The same conclusion holds on a compact target if the supplied charts cover every intermediate state needed by the finite word on the right of (A19.3), including its inverse factors. On nested chart domains with fixed collars, uniform inverse-Jacobian bounds and bounded C^{k+2} generating functions, this reconstruction is locally Lipschitz from C^{k+2} action data to C^k lifts.

Proof. Powers of a single group element commute, and the group law gives $(g^m)^u (g^n)^v = g^{um+vn} = g$. Hence any other lift with the same two powers must give the same right side. This elementary identity also specifies a reconstruction, rather than only uniqueness.

On a type-I chart with initial and terminal coordinates (x, y) , a known absolute generating function $S_r(x, y)$ gives

$$p = -\partial_x S_r(x, y), \quad q = \partial_y S_r(x, y), \quad A_r(x, p) = S_r(x, y(x, p)).$$

The nonzero determinant of $\partial_{xy}^2 S_r$ gives the local phase map and the action by the implicit function theorem. Several stationary sheets can occupy the same endpoint gate: they are

distinguished by their phase covectors. The assumption that the charts cover the graph of the actual deterministic iterate makes these phase maps agree on overlaps. It does not select arbitrary roots of unrelated stationary functions. They therefore specify g^r . Apply the group formula.

For the compact statement only finitely many chart restrictions along the prescribed word are used. Differentiating the implicit equations through order $k + 1$ bounds these maps and their inverses by the stated C^{k+2} bounds and inverse-Jacobian margins. Subtracting two implicit equations gives their C^k local Lipschitz estimates by induction on k . Products, inverses and compositions on the fixed nested domains are locally Lipschitz in C^k when their C^{k+1} norms are bounded. Applying these estimates to the finite word proves the claim. The collars ensure that nearby data are evaluated inside the same charts. $\square$

For $m = n + 1$ this reduces to $F = F^{n+1}(F^n)^{-1}$ together with subtraction of the corresponding primitive. This identity is not a new group-theoretic rigidity phenomenon. Its observational content in this paper is that the *absolute* lifts, rather than only their canonical graphs, can be obtained from raw endpoint records with unknown weights and incorrect positive labels.

Proposition A19.4 (The arithmetic restriction cannot be dropped). *If $d = \gcd(m, n) > 1$, there exist distinct positive-roof exact symplectic return systems whose m th and n th action lifts agree. The example can be taken on a disk, with regular local type-I charts for the two observed iterates.*

Proof. On a disk in the (x, p) plane put $\lambda_0 = (p dx - x dp)/2$ and $h = xp/2$, so $\lambda = p dx = \lambda_0 + dh$. Rotations F_θ preserve λ_0 . Therefore

$$g_\theta = (F_\theta, c + h \circ F_\theta - h)$$

is an action lift, and $g_\theta^r = (F_{r\theta}, rc + h \circ F_{r\theta} - h)$. The lifts for θ and $\theta + 2\pi/d$ have identical m th and n th powers but different first maps. Choose c larger than the oscillation of h on the disk; both roofs are strictly positive and have the contact suspensions of Proposition A27.2, whose proof applies equally on this invariant disk. Choosing θ outside the finite forbidden angle sets makes the mixed-coordinate Jacobians of both observed iterates nonzero. They then admit the claimed type-I charts. No claim about a particular billiard subclass is needed for this example. $\square$

Corollary A19.5 (Four raw laws for a covered Reeb return system). *In Proposition A19.1, suppose the section has specified canonical charts with $\lambda = p dx$, and the two relatively prime return counts have a regular phase-graph cover as in Theorem A19.3. At each chart use two deadline trials with a common normalized source and a common retention rule, with an independent uniform external release delay. Suppose the delay and deadlines have strict residual margins on the chart. If branch labels are reported through the calibrated full-rank channels of Theorem A18.2, the four raw endpoint laws per pair of charts determine the return map and its absolute return time on the covered target. They therefore determine the section-preserving strict contact suspension there.*

Proof. Write the r th return action in endpoint coordinates as $S_r(x, y)$. Changing the initial momentum to the terminal coordinate contributes $|\det \partial_{xy}^2 S_r|$. Integrating a uniform external delay of length D contributes $(T_j - S_r)/D$ on an admissible gate. Thus the raw subdensity is

$$D^{-1} e_r(x, y) \rho_r(x, -\partial_x S_r(x, y)) |\det \partial_{xy}^2 S_r(x, y)| (T_j - S_r(x, y)).$$

No action is evaluated by the apparatus; this is the density of an elapsed-time event in the given Reeb flow. Deconvolving the tags and using two raw deadlines recovers S_r absolutely on every chart. Theorem A19.3 and Proposition A19.1 complete the reconstruction. $\square$

The number four refers to two observations at each of two prescribed return counts, not to a universal lower bound on every dynamical inverse problem. For example, when the first return

itself is observed on a complete regular cover, its two raw laws already determine its action lift. The corollary does not manufacture a regular cover across an unobserved caustic, nor does it replace a local transversal by a global section. It does allow several nonminimizing stationary sheets whenever the stated phase-graph coverage is available.

APPENDIX A20. THE EXACT OBSERVATION QUOTIENT

Fix a compact connected rectangle Q , a point $z_0 \in Q$, and three distinct clocks. Let $\mathcal{W}$ consist of nonconstant smooth functions W with $W < T_1$ on a neighborhood of Q . For a positive smooth function a , write

$$\mathcal{F}(W, [a])_j = \frac{a(T_j - W)}{\int_Q a(T_j - W)}, \quad [a] = \{ca : c > 0\}.$$

The brackets denote only multiplication by a constant, not arbitrary reweighting. Let $\mathcal{L}$ be the image of this map. The group of positive smooth functions modulo constants acts on $\mathcal{L}$ by

$$(b \cdot f)_j = \frac{bf_j}{\int_Q bf_j}. \quad (\text{A20.1})$$

This operation changes the source shape while preserving the clocked action. It is defined on laws, not on individual recorded endpoints.

Theorem A20.1 (Complete invariant of common reweighting). *The map $(W, [a]) \mapsto \mathcal{F}(W, [a])$ is a bijection onto $\mathcal{L}$. The absolute action is a complete invariant of the reweighting action: two law triples are in the same orbit if and only if their actions agree. Consequently*

$$\mathcal{L} / \{\text{common positive reweightings}\} \simeq \mathcal{W}.$$

The displayed orbit identification is set-theoretic here. The quotient topology and a complete metric are defined in Proposition A21.1, and its smooth action charts in Theorem A21.2. At the representative level, with the denominator margins of Theorem A17.3, recovery of W and $a/a(z_0)$ is locally Lipschitz in each fixed C^m norm.

Proof. The absolute inverse in Theorem A17.3 recovers W from the three laws, and then recovers

$$\frac{a(z)}{a(z_0)} = \frac{f_1(z)}{f_1(z_0)} \frac{T_1 - W(z_0)}{T_1 - W(z)}.$$

This proves injectivity, and surjectivity holds by the definition of $\mathcal{L}$. Formula (A20.1) sends $\mathcal{F}(W, [a])$ to $\mathcal{F}(W, [ba])$, so each orbit has one fixed action. Conversely, if the actions agree, the multiplier $b = \tilde{a}/a$ carries one triple to the other. The displayed formulas and the reciprocal estimates in the clock theorem give the stability assertion. No statistical sufficiency or finite-sample Markov kernel is asserted by this quotient construction. $\square$

Proposition A20.2 (Local comparison with canonical data). *Suppose $W_{st} \neq 0$ on a rectangle. Its three-clock law orbit determines the marked local canonical relation*

$$\mathcal{C}_W = \{((s, -W_s(s, t)), (t, W_t(s, t))) : (s, t) \in Q\}$$

and the absolute value $W(z_0)$. Conversely, this relation, parametrized by the same endpoint labels, together with $W(z_0)$ determines the law orbit. With the extra nuisance shape $[a]$ it determines the laws themselves. The canonical relation alone determines W only up to an additive constant.

Proof. Recovering W and differentiating gives the first assertion. Conversely write the covectors read from the relation as $p(s, t), q(s, t)$. For realizable canonical data the one-form $-p ds + q dt$ is dW . On the simply connected rectangle its integral from z_0 is independent of the path. Adding $W(z_0)$ recovers W and hence its law orbit. The forward formula recovers the particular laws if $[a]$ is specified. Two functions with the same differential on a connected rectangle differ by a

constant; adding any admissible constant leaves this canonical relation unchanged. Thus the anchor has an actual mathematical role. $\square$

The momenta in this proposition are canonical covectors in scalar coordinates, not measured Euclidean angles. Once a boundary embedding is recovered, its speed converts them to the usual tangential components of unit velocity. This is a local comparison on the given branches; it does not identify the marks or observation domain with those of an exterior lens problem or of an enriched periodic length spectrum.

Proposition A20.3 (Two-clock ambiguity in the weighted-action model). *For two clocks and unknown weight, every nonconstant action with positive residuals admits nontrivial nearby weighted-action pairs giving the same two normalized laws.*

Proof. Set $r = (T_1 - W)/(T_2 - W)$ and, for $\lambda > 0$ near one, set

$$W_\lambda = \frac{T_1 - \lambda T_2 r}{1 - \lambda r}, \quad a_\lambda = a \frac{T_2 - W}{T_2 - W_\lambda}.$$

Compactness and strict residual margins make these smooth and admissible for sufficiently small $|\lambda - 1|$. Direct substitution gives

$$a_\lambda(T_2 - W_\lambda) = a(T_2 - W), \quad a_\lambda(T_1 - W_\lambda) = \lambda a(T_1 - W).$$

Normalization erases the factor λ . Since $r > 0$ and the clocks are distinct, $W_\lambda \neq W$ when $\lambda \neq 1$. $\square$

This proposition quantifies a gauge in the stated weighted-function space. It is not a construction of two noncongruent billiards having the same physical two-clock experiment. The three-clock sufficiency theorem does not require such a construction. Equally, a branch mixture is not removed by additional clocks in this model: whenever all residuals are positive,

$$\sum_\gamma a_\gamma(T - W_\gamma) = \left(\sum_\gamma a_\gamma \right) \left(T - \frac{\sum_\gamma a_\gamma W_\gamma}{\sum_\gamma a_\gamma} \right).$$

The branch-resolution assumption is therefore explicit in the title, observation convention and main theorem, rather than hidden in notation.

APPENDIX A21. A SPLIT QUOTIENT OF ARBITRARY CLOCKED OBSERVATIONS

The exact inverse has a useful extension to observations outside the shared-weight model. It is important here to quotient the whole observation space, not only its model image. Throughout this section $Q \subset \mathbb{R}^d$ is a compact rectangle with nonempty interior, $m \geq 0$ is an integer, and $T_1 < \dots < T_J$, $J \geq 3$, are fixed. The C^m norm is the maximum of the suprema of the derivatives of order at most m . Functions are understood as restrictions from a neighborhood of Q . Equivalent standard Banach norms give the same statements.

Let $\mathcal{D}_m$ be the space of J strictly positive C^m probability densities on Q . Give it the relative product C^m topology. The group G_m of strictly positive C^m functions modulo positive constants acts by simultaneous reweighting and normalization, as in (A20.1). Fix a scalar point $z_0 \in Q$ and put

$$Y_j(f)(z) = \log \frac{f_j(z)}{f_1(z)} - \log \frac{f_j(z_0)}{f_1(z_0)}, \quad 2 \leq j \leq J. \quad (\text{A21.1})$$

Write $C_0^m(Q) = \{h \in C^m(Q) : h(z_0) = 0\}$ and $\mathcal{Y}_m = (C_0^m(Q))^{J-1}$.

Proposition A21.1 (The ambient observation quotient). *The map Y induces a homeomorphism from the topological quotient $\mathcal{D}_m/G_m$ onto the Banach space $\mathcal{Y}_m$. In particular*

$$d_m([f], [\tilde{f}]) = \max_{2 \leq j \leq J} \|Y_j(f) - Y_j(\tilde{f})\|_{C^m} \quad (\text{A21.2})$$

is a complete metric inducing that quotient topology. The quotient has the corresponding global Banach coordinate, independently of any shared-weight assumption.

Proof. Common reweighting changes f_j/f_1 by a constant depending on j ; anchoring removes that constant. If $Y(f) = Y(\tilde{f})$, let $b = \tilde{f}_1/f_1$. Then $\tilde{f}_j$ is a positive constant multiple of $b f_j$ for every j . Their integrals are one, so $\tilde{f} = b \cdot f$. Conversely this relation implies equality of Y . For arbitrary $y \in \mathcal{Y}_m$ a representative is

$$f_1 = |Q|^{-1}, \quad f_j = \frac{e^{y_j}}{\int_Q e^{y_j}}.$$

This is a continuous section of Y . The map Y is continuous by positivity, compactness and continuity of logarithms on bounded positive ranges. The induced bijection on the quotient and the section are continuous in opposite directions. This proves the homeomorphism, and completeness follows from completeness of $\mathcal{Y}_m$. The assertion does not put a uniform lower bound on all representatives of an orbit; representative-level estimates still require such bounds. $\square$

For an action $W < T_1$, set

$$k_j(w) = \log(T_j - w) - \log(T_1 - w), \quad \Gamma_j(W) = k_j(W) - k_j(W(z_0)). \quad (\text{A21.3})$$

Thus $Y(\mathcal{F}(W, [a])) = \Gamma(W)$ independently of a . Constant actions all give $\Gamma(W) = 0$ and are excluded from the inverse charts. Choose a second scalar point $z_1 \in Q$, distinct from z_0 , and put

$$\mathcal{W}_{01}^m = \{W \in C^m(Q) : \max_Q W < T_1, W(z_1) \neq W(z_0)\}.$$

The choice specifies evaluation labels, not an action value.

For arbitrary $y \in \mathcal{Y}_m$ define $c_j = e^{y_j}$, $\tau_j = T_j - T_1$, and, whenever the following expressions exist,

$$D(y) = c_2(z_1) - c_3(z_1), \quad (\text{A21.4})$$

$$u_0(y) = \frac{(1 - c_2(z_1))/\tau_2 - (1 - c_3(z_1))/\tau_3}{D(y)}, \quad (\text{A21.5})$$

$$u_y(z) = \frac{c_2(z)(1 + \tau_2 u_0(y)) - 1}{\tau_2}, \quad \mathcal{R}(y)(z) = T_1 - \frac{1}{u_y(z)}. \quad (\text{A21.6})$$

Let $\mathcal{U}_{01}^m$ be the open subset of $\mathcal{Y}_m$ on which $D(y) \neq 0$ and $\min_Q u_y > 0$. Finally let

$$\mathcal{N}_{01}^m = \{r \in \mathcal{Y}_m : r_2 \equiv 0, r_3(z_1) = 0\}. \quad (\text{A21.7})$$

No constraint beyond anchoring is placed on r_j for $j \geq 4$.

Theorem A21.2 (Global split coordinates on an inverse chart). *The map*

$$y \mapsto (\mathcal{R}(y), \mathcal{S}(y)), \quad \mathcal{S}(y) = y - \Gamma(\mathcal{R}(y)), \quad (\text{A21.8})$$

is a C^∞ diffeomorphism

$$\mathcal{U}_{01}^m \longrightarrow \mathcal{W}_{01}^m \times \mathcal{N}_{01}^m.$$

Its inverse is $(W, r) \mapsto \Gamma(W) + r$. In particular $\Pi = \Gamma\mathcal{R}$ is a smooth retraction onto the shared-weight model in this chart, and

$$y \text{ is represented by shared-weight laws} \iff \mathcal{S}(y) = 0.$$

The inverse loses no spatial derivatives. Its fixed-order bounds are uniform on bounded subsets with positive lower bounds for $|D|$ and u_y and with bounded u_y .

Proof. For a model action put $u = (T_1 - W)^{-1}$ and $u_0 = u(z_0)$. Then

$$e^{\Gamma_j(W)(z)} = \frac{1 + \tau_j u(z)}{1 + \tau_j u_0}. \quad (\text{A21.9})$$

Evaluating at z_1 and eliminating $u(z_1)$ gives (A21.5); the second clock gives (A21.6). Moreover

$$D(\Gamma(W)) = \frac{(\tau_2 - \tau_3)(u(z_1) - u_0)}{(1 + \tau_2 u_0)(1 + \tau_3 u_0)} \neq 0.$$

Consequently $\mathcal{R}\Gamma(W) = W$.

For arbitrary $y \in \mathcal{U}_{01}^m$, the anchoring $c_2(z_0) = 1$ gives $u_y(z_0) = u_0(y) > 0$. Formula (A21.6) shows that $\Gamma_2(\mathcal{R}(y)) = y_2$ everywhere. The definition of u_0 is exactly the equation

$$\frac{c_2(z_1)(1 + \tau_2 u_0) - 1}{\tau_2} = \frac{c_3(z_1)(1 + \tau_3 u_0) - 1}{\tau_3}.$$

It follows that $\Gamma_3(\mathcal{R}(y))(z_1) = y_3(z_1)$. Also $u_y(z_1) \neq u_0$: equality would force both $c_2(z_1)$ and $c_3(z_1)$ to equal one, contradicting $D \neq 0$. Thus $\mathcal{R}(y) \in \mathcal{W}_{01}^m$ and $\mathcal{S}(y) \in \mathcal{N}_{01}^m$.

Conversely, adding $r \in \mathcal{N}_{01}^m$ to $\Gamma(W)$ changes neither the second contrast nor the third contrast at z_1 . All quantities in (A21.4)–(A21.6) are therefore unchanged. Hence $\mathcal{R}(\Gamma(W) + r) = W$, $\mathcal{S}(\Gamma(W) + r) = r$, and the resulting y is in the open chart. This proves both inverse identities globally on the displayed sets. Point evaluations, products and exponentials are smooth maps of C^m spaces, and reciprocals are smooth on the positive open cone. This proves the smoothness and uniform local estimates. In particular no derivative of y is used to recover an action derivative of the same order beyond the derivatives already present in its C^m norm. $\square$

A different pair of anchors gives another chart with a smooth transition map on its overlap. These charts cover all nonconstant model actions. They supply the topology and quantitative meaning of the exact orbit identification in Theorem A20.1; they are not a claim of statistical sufficiency for finite records.

Corollary A21.3 (Residuals measure distance from the model). *On a sufficiently small bounded chart neighborhood, let $\text{dist}_m(y, \Gamma(\mathcal{V}))$ denote the distance to model points in a slightly larger fixed chart neighborhood $\mathcal{V}$ containing $\mathcal{R}(y)$. There is a constant C such that*

$$\text{dist}_m(y, \Gamma(\mathcal{V})) \leq \|\mathcal{S}(y)\|_{C^m} \leq C \text{dist}_m(y, \Gamma(\mathcal{V})).$$

Already three clocks have a functional model-validation residual: r_3 away from z_0, z_1 . Each additional clock contributes another unconstrained anchored residual function.

Proof. The point Πy is an allowed model point, giving the first inequality. The formulas in the theorem make Π Lipschitz on the indicated neighborhood, say with constant L . For any allowed $\Gamma(V)$,

$$\|y - \Pi y\| \leq \|y - \Gamma(V)\| + \|\Pi\Gamma(V) - \Pi y\| \leq (1 + L)\|y - \Gamma(V)\|.$$

Take the infimum. The description of residual coordinates is precisely (A21.7). $\square$

A21.1. Approximate sharing and exact confounding. Consider the larger observation model

$$f_j = \frac{a(T_j - W)e^{\eta_j}}{\int_Q a(T_j - W)e^{\eta_j}}, \quad a > 0, \quad (\text{A21.10})$$

where clock-dependent nuisance functions η_j are now allowed. Define their observable contrasts by

$$E_j = (\eta_j - \eta_1) - (\eta_j(z_0) - \eta_1(z_0)). \quad (\text{A21.11})$$

Then $Y(f) = \Gamma(W) + E$ exactly. This identity separates normalization, common nuisance, and clock-dependent nuisance before an estimate is made.

Theorem A21.4 (Robust reconstruction and complete ambiguity formula). *Fix a bounded inverse chart with strict margins. For $\|E\|_{C^m}$ sufficiently small the reconstructed action $W^\dagger = \mathcal{R}(Y(f))$ satisfies*

$$\|W^\dagger - W\|_{C^m} + \|\mathcal{S}(Y(f))\|_{C^m} \leq C\|E\|_{C^m}. \quad (\text{A21.12})$$

In particular weights $a(1 + \epsilon_j)$ with $\max_j \|\epsilon_j\|_{C^m} \leq \epsilon$ and $\max_j \|\epsilon_j\|_\infty \leq 1/2$ induce action bias at most $C\epsilon$.

The laws in (A21.10) coincide with exact shared-weight laws for another action $V < T_1$ if and only if there exist a common function b and constants c_j such that

$$\eta_j(z) = b(z) + c_j + \log \frac{T_j - V(z)}{T_j - W(z)}, \quad 1 \leq j \leq J. \quad (\text{A21.13})$$

For V and W in an inverse chart this is the complete ambiguity: there are no other off-model nuisance functions whose residual vanishes. The formula holds for any number of clocks.

Proof. The first assertion is the Lipschitz estimate for $\mathcal{R}$ and $\mathcal{S}$ at $\Gamma(W)$, where their values are W and zero. For multiplicative errors take $\eta_j = \log(1 + \epsilon_j)$; the C^m algebra and reciprocal bounds give $\|E\|_{C^m} \leq C\epsilon$.

If the normalized laws equal those generated by $(V, \tilde{a})$, their unnormalized densities differ by positive constants. Taking logarithms gives (A21.13) with $b = \log(\tilde{a}/a)$. Conversely this formula cancels the action ratio, leaving $ae^b(T_j - V)$ times a clock constant; normalization removes the latter. Finally, Theorem A21.2 proves that zero residual is equivalent to an exact shared-weight representation, so the equivalence just established is exhaustive in that chart. $\square$

Thus extra clocks test departures transverse to the model, not all clock-dependent efficiencies. A departure of the form (A21.13) is indistinguishable from a genuine change of action even with exact laws. This statement concerns the weighted-action experiment; no pair of noncongruent physical billiards is asserted by it.

Proposition A21.5 (Tangent and normal components of nuisance). *At a model point let $L_W = D\mathcal{R}_{\Gamma(W)}$. Then*

$$\begin{aligned} (D\Gamma_W h)_j &= k'_j(W)h - k'_j(W(z_0))h(z_0), \\ L_W D\Gamma_W &= \text{Id}, \\ P_W &= D\Gamma_W L_W, \\ \mathcal{Y}_m &= \text{ran } D\Gamma_W \oplus \mathcal{N}_{01}^m. \end{aligned}$$

Here P_W is the bounded projection onto the first summand and $\ker L_W = \mathcal{N}_{01}^m$. Uniformly on bounded smaller charts,

$$\mathcal{R}(\Gamma(W) + E) - W = L_W E + O(\|E\|_{C^m}^2), \quad (\text{A21.14})$$

$$\mathcal{S}(\Gamma(W) + E) = (\text{Id} - P_W)E + O(\|E\|_{C^m}^2). \quad (\text{A21.15})$$

Proof. Differentiate the two inverse identities of Theorem A21.2. Differentiation in the residual variable gives $L_W r = 0$ for $r \in \mathcal{N}_{01}^m$, while differentiating the whole inverse gives $\text{Id} = P_W + D\mathcal{S}_{\Gamma(W)}$ and the asserted direct sum. The displayed derivative of Γ follows from the chain rule. The smooth formulas have bounded second derivatives on the indicated sets; the integral Taylor remainder proves the two quadratic estimates. $\square$

The residual therefore identifies the normal component to first order, whereas the tangential component changes the reconstructed action. Neither component is inferred from the mere smallness of a fitting error; the split and the ambiguity formula state what is actually observable.

APPENDIX A22. FINITE RECORDS, VALIDATION, AND SHARP ACTION RISK

The quotient identifies a function from exact laws, whereas records require estimation of those laws. In this section the loss is imposed on the action itself. The subsequent passage to billiard geometry has additional conditioning and derivative loss, and its optimal risk is not inferred from the result below.

Proposition A22.1 (A finite-record residual test). *Suppose the true contrast y lies in a fixed bounded smaller inverse chart and an estimator $\hat{y}$ satisfies*

$$\Pr\{\|\hat{y} - y\|_{C^m} \leq \rho\} \geq 1 - \zeta.$$

For ρ sufficiently small, put $\hat{r} = \mathcal{S}(\hat{y})$ when the chart formulas are defined. There is a fixed C for which rejecting the shared-weight model when $\|\hat{r}\|_{C^m} > C\rho$ has type-I error at most ζ . It rejects with probability at least $1 - \zeta$ whenever $\|\mathcal{S}(y)\|_{C^m} > 2C\rho$. Undefined chart outputs may be declared rejections. They occur only on the exceptional event under the stated chart and error bounds.

Proof. On the stated event the chart margins persist and the Lipschitz bound for $\mathcal{S}$ gives $\|\hat{r} - \mathcal{S}(y)\|_{C^m} \leq C\rho$. Under the null the true residual is zero. Under the separated alternative the reverse triangle inequality proves rejection. The chart persistence statement follows from openness with uniform positive margins. No assertion of optimal testing separation is used. $\square$

Three clocks suffice for this functional test, since their third contrast has a residual away from the two anchor points. Additional clocks supply further residual coordinates but do not overcome the exact confounding in (A21.13).

A22.1. The statistical class. Fix rectangles $Q_0 \Subset Q_+ \subset \mathbb{R}^d$ with nonempty interiors, anchors $z_0, z_1 \in \text{int } Q_0$, and an integer $m \geq 0$. Let $\beta > m$ be nonintegral. The C^β norm denotes the usual Hölder norm, with the Hölder seminorm on derivatives of order $\lfloor \beta \rfloor$. A fixed class $\mathcal{A}_\beta(\epsilon)$ consists of triples $(W, a, (\eta_j)_{j=1}^J)$ satisfying (A21.10), normalized on Q_+ , with the following bounds. The $C^\beta(Q_+)$ norms of W, a, η_j are bounded by a fixed A ; a is bounded above and below by fixed positive constants; $T_j - W$ belongs to a fixed positive compact interval; and $|W(z_1) - W(z_0)|$ is bounded below by a fixed positive number. Finally $\max_j \|\eta_j\|_{C^m(Q_+)} \leq \epsilon$. All constants, clocks, rectangles, m, β and J are independent of n, ϵ . For small fixed ϵ_0 , these bounds imply uniform positive density and inverse-chart margins on Q_0 . Contrasts of density restrictions can be used without renormalization, since all normalization constants disappear from (A21.1).

For the lower bound we assume a nondegenerate class, in the precise sense that it contains an exact member $(W_*, 1, 0)$ with strict room in all the above inequalities and in its C^β norm bound. Choose an open rectangle $B \Subset \text{int } Q_0$ disjoint from the two anchors. We require this strict room on B as well. Thus small C^β perturbations of W_* supported in B are in the class. This condition rules out a singleton class described by saturated bounds; it is satisfied, for example, by a nonconstant affine action with sufficiently generous fixed bounds.

Theorem A22.2 (Sharp local risk with clock-dependent misspecification). *Observe n independent draws from each of the J conditional laws, independently across clocks. For the class just specified there are constants $0 < c < C < \infty$ and $\epsilon_0 > 0$ such that, for sufficiently large n and $0 \leq \epsilon \leq \epsilon_0$,*

$$c \left\{ \left(\frac{\log n}{n} \right)^{\frac{\beta-m}{2\beta+d}} + \epsilon \right\} \leq \inf_{\hat{W}} \sup_{\mathcal{A}_\beta(\epsilon)} \mathbb{E} \|\hat{W} - W\|_{C^m(Q_0)} \leq C \left\{ \left(\frac{\log n}{n} \right)^{\frac{\beta-m}{2\beta+d}} + \epsilon \right\}. \quad (\text{A22.1})$$

The infimum is over measurable estimators with $C^m(Q_0)$ values. The result is for the unrestricted weighted-action class on the stated chart, not a minimax theorem for geometrically realizable billiard actions.

Proof. We give both statistical and misspecification bounds. The underlying kernel and packing methods are classical; see [8, 14, 15]. The point here is to carry them through a fully specified quotient and its nuisance class.

For the upper bound use a smooth compactly supported kernel with integral one and vanishing moments through order $\lfloor \beta \rfloor$. Such kernels are obtained by a finite linear combination of dilates of a smooth kernel, solving the finite moment system. Estimate the densities on Q_0 using samples on Q_+ . For sufficiently small bandwidth h the support of each kernel remains inside Q_+ . Taylor expansion, or convolution with each derivative of f , gives bias at most $Ch^{\beta-m}$ in $C^m(Q_0)$. Uniform C^β bounds and density floors follow from the class definition and the product, exponential and reciprocal estimates on bounded sets.

For a multi-index α with $|\alpha| \leq m$, the centered kernel summand has envelope $Ch^{-d-|\alpha|}$, variance $Ch^{-d-2|\alpha|}$, and spatial Lipschitz bound $Ch^{-d-|\alpha|-1}$. A grid of mesh $n^{-q}h^{d+m+1}$ on Q_0 , with fixed sufficiently large q , reduces the supremum to polynomially many pointwise inequalities when h is a fixed power of n times a logarithmic factor. Bernstein's inequality, a union bound over the grid, derivatives and clocks, and the Lipschitz bound between grid points give

$$\max_j \|\hat{f}_j - f_j\|_{C^m(Q_0)} \leq C \left\{ h^{\beta-m} + \sqrt{\frac{\log n}{nh^{2m+d}}} + \frac{\log n}{nh^{m+d}} + n^{-q} \right\} \quad (\text{A22.2})$$

except on an event of probability at most $n^{-q'}$, where q' can be made arbitrarily large by increasing the constant. Choose $h = (\log n/n)^{1/(2\beta+d)}$. The square-root and bias terms then have size $r_n = (\log n/n)^{(\beta-m)/(2\beta+d)}$; the envelope term is smaller, since its exponent is $(2\beta-m)/(2\beta+d)$.

On this event density floors persist, hence the anchored logarithms and the reconstruction are defined. Theorems A21.2 and A21.4 give error at most $C(r_n + \epsilon)$. To define an estimator on all samples, test the density and chart margins and a fixed C^m output bound larger than the class bound; when a test fails return a fixed C^m function. The output is then uniformly bounded. Suprema of continuous derivatives on compact rectangles are measurable (they equal suprema on a countable dense set), so these tests and the formulas give a measurable estimator. On the good event all tests pass for large n and small ϵ ; on its complement the bounded error contributes only $O(n^{-q'})$ to expectation. This proves the upper bound.

For the statistical lower bound take $a = 1$ and $\eta_j = 0$. In B choose $M \asymp h^{-d}$ disjoint cubes of side comparable to h . For a fixed nonzero smooth bump ψ supported in the unit cube set

$$W_l(z) = W_*(z) + a_0 h^\beta \psi((z - z_l)/h), \quad 1 \leq l \leq M.$$

Choose ψ with a nonzero derivative of some order m (for $m = 0$ use its nonzero value). The fixed number $a_0 > 0$ can be chosen small enough that every W_l is in the class for all small h . The anchor values are unchanged. Disjoint supports imply pairwise $C^m(Q_0)$ separation at least $ca_0 h^{\beta-m}$.

Let P_l be the law of all Jn records and P_* the corresponding reference law. The exact densities are $f_{j,W} = (T_j - W) / \int_{Q_+} (T_j - W)$. Positivity and uniform normalizer bounds give

$$\|f_{j,W_l} - f_{j,W_*}\|_{L^2(Q_+)} \leq C \|W_l - W_*\|_{L^2(Q_+)}, \quad \text{KL}(f_{j,W_l}, f_{j,W_*}) \leq Ca_0^2 h^{2\beta+d}.$$

For the first inequality, subtract the two normalized numerators and bound the normalizer difference by $|Q_+|^{1/2} \|W_l - W_*\|_2$. For the second use $\text{KL}(f, g) \leq \int (f - g)^2 / g$, valid for probability densities with $g > 0$. Independence gives $\text{KL}(P_l, P_*) \leq C J n a_0^2 h^{2\beta+d}$.

For clarity, the information inequality used here is the following. For a uniform index on $\{1, \dots, M\}$, every classifier has error probability at least $1 - (M^{-1} \sum_l \text{KL}(P_l, P_*) + \log 2) / \log M$. Indeed the mutual information is at most that average relative entropy; conditional entropy is at most $\log 2 + p_e \log M$, and subtracting it from $\log M$ proves the bound. Taking $h = c_0(\log n/n)^{1/(2\beta+d)}$ with sufficiently small fixed c_0 makes the error probability bounded below by a positive constant. Nearest-neighbor classification of any function estimate would be correct whenever its error is less than half the pairwise separation. Thus its maximal expected C^m error is at least cr_n .

For the misspecification lower bound fix a nonzero smooth function ϕ supported in B and compare W_* with $W_1 = W_* + c_1 \epsilon \phi$. Under W_1 choose

$$a = 1, \quad \eta_{1,j} = \log \frac{T_j - W_*}{T_j - W_1}.$$

For small fixed c_1 these functions have C^m norm at most ϵ and satisfy the uniform C^β bounds. The observed densities are exactly the reference densities at every clock, but the actions are separated by $c\epsilon$ in C^m . The triangle inequality under this common data law gives maximal risk at least $c\epsilon/2$. The maximum of the statistical and misspecification lower bounds is at least half their sum. This completes (A22.1). $\square$

The bound shows that approximate sharing is not a discontinuous identification assumption on these charts: the action error changes linearly with its violation. At the same time that linear term cannot be removed by more data. Both statements use the exact normalized observation law, rather than a likelihood written after eliminating an uncharged nuisance parameter.

APPENDIX A23. ELIMINATING A SHARED REFLECTION PREFIX

We first record the finite variational facts, including the sign and nondegeneracy needed to compare two actual experiments. No periodicity is used in this section.

Lemma A23.1 (Regular finite actions). *Let a finite trajectory in a dispersing table have clear flights of positive length and nongrazing reflections on positive-curvature boundaries. Use arbitrary regular scalar coordinates on its successive boundary patches. After restricting the patches, its stationary trajectory is uniquely determined near the reference trajectory by the two endpoint coordinates. Its length W is smooth and $W_{st} \neq 0$. The corresponding initial and final covectors satisfy*

$$p = -W_s(s, t), \quad q = W_t(s, t).$$

All these constructions and their fixed-order derivatives depend continuously on the boundary embeddings on smaller closed domains.

Proof. Write $r_i = R_i(x_i)$, $e_i = (r_{i+1} - r_i)/\ell_i$ and $\ell_i = |r_{i+1} - r_i|$. For a variation $v_i = R'_i(x_i)h_i$, the second differential of the i th chord is

$$\ell_i^{-1} |P_{e_i^\perp}(v_{i+1} - v_i)|^2 + e_i \cdot \{R''_{i+1} h_{i+1}^2 - R''_i h_i^2\}.$$

At an interior reflection the coefficient from the second terms is $(e_{i-1} - e_i) \cdot R''_i h_i^2$. With outward obstacle normal n_i , reflection gives $e_{i-1} - e_i = -2\nu_i n_i$, $\nu_i > 0$, and $n_i \cdot R'_i = -\kappa_i |R'_i|^2$. Thus, with endpoints fixed,

$$\delta^2 \sum_i \ell_i = \sum_i \ell_i^{-1} |P_{e_i^\perp}(v_{i+1} - v_i)|^2 + \sum_{i \text{ interior}} 2\kappa_i \nu_i |R'_i|^2 h_i^2 > 0 \quad (\text{A23.1})$$

for every nonzero interior variation. Tangential terms caused by the choice of parameter vanish by stationarity. The interior Hessian is therefore positive definite. The implicit function theorem

constructs the unique nearby stationary path; continuity preserves clearance and reflection, hence it remains a physical path.

The mixed derivative of an individual chord is

$$(\ell_i)_{x_i x_{i+1}} = -\ell_i^{-1} (e_i^\perp \cdot R'_i) (e_i^\perp \cdot R'_{i+1}). \quad (\text{A23.2})$$

Both factors are nonzero by nongrazing. The interior Hessian is tridiagonal. If there are interior sites, the mixed derivative of the stationary value is the negative product of its two endpoint couplings and the opposite-corner entry of the inverse interior Hessian. The cofactor of that corner is, up to sign, the product of all intervening off-diagonal entries. Each factor is nonzero by (A23.2), and the determinant is positive by (A23.1). Hence $W_{st} \neq 0$. With no interior site, this is just (A23.2).

The first variation contains only the endpoint terms, giving the two covector formulas. The implicit-function and cofactor arguments apply with parameters and can be differentiated at any fixed order. On a compact strictly regular domain their inverses have finite bounds. $\square$

Suppose γ goes from I to J and its extension γe goes from I to K . Write

$$U(s, t) = W_\gamma(s, t), \quad V(s, v) = W_{\gamma e}(s, v), \quad \ell(t, v) = |R_K(v) - R_J(t)|.$$

The stationary composition identity on the common physical branch is

$$V(s, v) = U(s, t(s, v)) + \ell(t(s, v), v), \quad U_t(s, t(s, v)) + \ell_t(t(s, v), v) = 0. \quad (\text{A23.3})$$

The covector is unchanged by reflection at J . This explains why the same stationarity equation joins the prefix to its last chord even though the prefix record stops at J .

Proposition A23.2 (Nonlinear prefix cancellation). *Near any such extended regular trajectory there are closed coordinate intervals I_0, J_0, K_0 on which the equation*

$$H(s, t, v) := V_s(s, v) - U_s(s, t) = 0 \quad (\text{A23.4})$$

has a unique solution $s = \sigma(t, v) \in \text{int } I_0$ for $(t, v) \in J_0 \times K_0$. The domains can be chosen with $|H_s| \geq \mu > 0$ and a strict endpoint sign bracket. The absolute last-flight distance is then determined by the two actions:

$$\ell(t, v) = V(\sigma(t, v), v) - U(\sigma(t, v), t). \quad (\text{A23.5})$$

There is no additive ambiguity. On bounded classes with these margins, for every $k \geq 0$ the map $(U, V) \mapsto \ell$ is locally Lipschitz from C^{k+2} on slightly larger fixed domains to $C^k(J_0 \times K_0)$.

Proof. For a fixed initial state, the common prefix in the two experiments is the same actual path. Eliminating the interior coordinates of the prefix in the full second variation leaves a strictly positive Schur complement

$$B = U_{tt} + \ell_{tt} > 0$$

at the matched path, by (A23.1). Differentiate the stationarity equation in (A23.3) at fixed v . It gives $t_s = -U_{st}/B$. Also $V_s = U_s(s, t(s, v))$, so at a matched triple

$$H_s = V_{ss} - U_{ss} = U_{st} t_s = -\frac{U_{st}^2}{B} < 0. \quad (\text{A23.6})$$

The implicit function theorem therefore solves for s as a function of (t, v) . Restrict to a product box about the reference triple on which $H_s < 0$. Choose its two s endpoints on opposite sides of the root and then shrink J_0, K_0 so that the endpoint signs are uniform. Compactness gives the stated strict margins. Monotonicity gives uniqueness on this entire specified box, not just formal uniqueness of a derivative at the reference point.

Equation (A23.4) equates initial covectors at the same initial scalar label. A nongrazing ray is determined by this covector and the choice of the outgoing half-plane. The branch labels select

that half-plane and the same first reflection. Determinism and local uniqueness of the physical path identify the full common prefix. Subtracting its length proves (A23.5). Equivalently the same conclusion follows by substituting the root in (A23.3).

For stability, retain the strict root bracket and $|H_s| \geq \mu$ on a slightly larger compact box. For a nearby pair $(\tilde{U}, \tilde{V})$ these margins persist. The mean value theorem first gives

$$\|\tilde{\sigma} - \sigma\|_{C^0} \leq C\|\tilde{H} - H\|_{C^0}.$$

Differentiate $H(\sigma(t, v), t, v) = 0$. At every order, the highest derivative of σ is multiplied by H_s ; the remaining terms involve lower derivatives and derivatives of H . Subtracting the two equations and inducting gives

$$\|\tilde{\sigma} - \sigma\|_{C^k} \leq C\|\tilde{H} - H\|_{C^{k+1}} \leq C(\|\tilde{U} - U\|_{C^{k+2}} + \|\tilde{V} - V\|_{C^{k+2}}).$$

The extra derivative controls composition at shifted roots. Products, composition on compact boxes and the mean value theorem applied to (A23.5) prove the claimed bound. The order two loss is a sufficient bound, not an optimality assertion. $\square$

In symplectic notation this argument is $\mathcal{F}_e = \mathcal{F}_{\gamma_e} \circ \mathcal{F}_\gamma^{-1}$, where $\mathcal{F}_W : (s, -W_s) \mapsto (t, W_t)$. The explicit root formula also recovers the additive constant in its generating function. It does not use any measured momentum: derivatives of recovered actions supply all covectors in the argument. Different source weights for the two branches were already eliminated separately by their clock triples.

The local root bracket is a witness used in constructing a scalar gate. It need not be supplied to the observer: the next section recovers the matching domain from the actions. An arbitrary choice among roots of unrelated stationary functions is not justified. Formula (A23.6) proves that suitable brackets exist for every genuine shared-prefix pair. The finite-atlas theorem uses precisely these physical boxes.

APPENDIX A24. MATCHING WITHOUT A SUPPLIED ROOT MARK

The local prefix argument proves more than local solvability once one uses determinism at every zero of the matching equation. The following one-dimensional fact is useful.

Lemma A24.1 (Downward zeros). *Let $h \in C^1([a, b])$. If $h'(x) < 0$ at every zero in (a, b) , then there is at most one such zero. If the endpoint values are nonzero, a zero exists exactly when $h(a) > 0 > h(b)$.*

Proof. A zero is isolated, positive immediately to its left and negative immediately to its right. If there were two zeros, take a point just after the first with negative value and a point just before the second with positive value. Between them the zero set is compact and discrete, and hence finite. Every crossing there is from positive to negative, which cannot carry a negative starting value to a positive ending value. This is a contradiction. The assertion about endpoints follows from the intermediate value theorem and the signs on either side of the unique zero. $\square$

Theorem A24.2 (Root-selection-free prefix reconstruction). *Let $U(s, t)$ and $V(s, v)$ be single-valued actions of actual regular physical trajectories, for one fixed reflection word and its one-flight extension, on their entire product domains. Every trajectory on these domains has the specified itinerary, clear positive flights and nongrazing reflections. The two actions use the same initial boundary embedding and scalar coordinate, the same unit speed and the same outgoing half-plane. They are not merely continuations of stationary values outside the physical branch. Suppose their products have a common connected initial interval $[a, b]$, including its endpoints. For fixed (t, v) put*

$$H(s, t, v) = V_s(s, v) - U_s(s, t).$$

Every interior zero corresponds to the same physical prefix and obeys

$$H_s = -\frac{U_{st}^2}{B} < 0, \quad B = U_{tt} + \ell_{tt} > 0. \quad (\text{A24.1})$$

There is at most one interior root. On the data-determined open set

$$\Omega_{a,b} = \{(t, v) : H(a, t, v) > 0 > H(b, t, v)\}$$

it exists, is smooth, and determines

$$\ell(t, v) = V(\sigma(t, v), v) - U(\sigma(t, v), t).$$

No root label, small root bracket or measured initial momentum is needed. Every regular extended trajectory with initial coordinate in (a, b) and terminal coordinates in the interiors of the two gates belongs locally to such an open set after restriction within the original gates.

Proof. At a zero, the two trajectories start at the same boundary point and have the same initial tangential covector. The prescribed outgoing side and unit speed determine the outgoing velocity uniquely. Determinism of nongrazing billiard motion therefore identifies the entire prefix. In particular, every zero is physical, not just the zero chosen near a reference trajectory. The stationary composition identity and positive interior Hessian from Lemma A23.1 apply there. Eliminating prefix variables gives the positive Schur complement B ; differentiation gives (A24.1).

Apply Lemma A24.1 to $s \mapsto H(s, t, v)$. It is unnecessary, and in general unjustified, to assert $H_s < 0$ at points which are not roots. The endpoint-sign criterion, the implicit function theorem at the unique root, and the stationary composition identity give all the stated conclusions. For a regular matched trajectory, take a, b strictly on its two sides in the original common gate. The signs hold there and persist for nearby terminal coordinates. Rational choices are possible by openness. These are constructed from recovered actions; they are not extra observations. $\square$

Corollary A24.3 (Data-derived finite certificates). *In the finite-atlas theorem, supplied matching brackets and supplied clock-anchor points can be omitted. For an exact admissible law tuple, usable clock anchors, root domains, distance anchors and overlap triples can be selected by strict tests on the recovered functions and the fixed scalar labels. A finite collection suffices whenever an identifying atlas exists.*

Proof. Nonconstancy and Theorem A17.3 give a nonzero anchor denominator on an open set, so a countable dense list of pairs of scalar points contains a usable pair. Recover the actions. Enumerate rational common initial subintervals and rational terminal rectangles with closures inside the open sets $\Omega_{a,b}$ of the preceding theorem. Each true local matching patch has such a restriction. On it recover ℓ . The nonzero finite determinants in Proposition A25.1 are open evaluation tests, as are noncollinearity of overlap triples. Rational parameters therefore suffice. The scalar cover and its graph are combinatorial tests on these intervals. The finite cover from Theorem A26.3 ensures that one finite selection passes all tests.

This is a law-level selection statement. With computable C^r names and certified derivative bounds, each strict inequality on a compact rectangle can be semidecided by a rational grid and its derivative error bound; dovetailing these tests then terminates on the stated class. Without a representation of functions, exact laws must not be called a finite computational input. No test here certifies physical admissibility of an arbitrary clock request outside the assumed class. $\square$

A24.1. A principle for scalar generating systems. The argument is not specific to the Euclidean chord. Suppose a scalar deterministic twist system has smooth generating actions U, V for a word and its extension, with nonzero U_{st} and V_{sv} . Suppose equality of their initial covectors identifies the common trajectory, and the stationary composition

$$V(s, v) = \text{stat}_t \{U(s, t) + L(t, v)\}$$

has $B = U_{tt} + L_{tt} > 0$ at every match. Supply three normalized clocked laws for each action with an unknown positive common weight.

Theorem A24.4 (Clocked generating-action reconstruction). *Under these hypotheses, the laws determine the absolute one-step generating function L on every matched open domain. They also determine the corresponding local canonical transformation, without a momentum observation or a root-selection mark. On bounded nested domains with positive clock and matching margins, recovery of L is locally Lipschitz from C^{k+2} laws to C^k functions for every finite k .*

Proof. The clock theorem recovers U, V and their constants. At any zero of $V_s - U_s$, the deterministic prefix property makes the stationary composition applicable. Differentiating it gives $V_{ss} - U_{ss} = -U_{st}^2/B$. Lemma A24.1 proves uniqueness on a connected common interval. Subtraction at the root recovers L , including its constant. Differentiation yields its canonical relation. The implicit differentiation and composition estimate in Proposition A23.2 uses only these margins, so proves the stated finite-order bound without invoking Euclidean geometry. $\square$

APPENDIX A25. DISTANCE CERTIFICATES AND EUCLIDEAN REGISTRATION

Let $d(s, t) = |A(s) - B(t)|$ be known on a product of scalar intervals; A and B are unknown regular strictly convex boundary patches. We give a finite certificate fixing their common Euclidean gauge and then recover their entire smooth parametrizations. The certificate is only a sufficient condition; in particular it is not a necessity theorem for circular or conic patches.

Choose six parameters $s_0, \dots, s_5$ and three parameters t_0, t_1, t_2 , all in the interiors of their intervals. Put

$$D_{ij} = d(s_i, t_j)^2, \quad X_{ij} = -\frac{1}{2}(D_{ij} - D_{i0} - D_{0j} + D_{00}), \quad 1 \leq i \leq 5, 1 \leq j \leq 2. \quad (\text{A25.1})$$

Regard the i th row as a vector $x_i \in \mathbb{R}^2$, set $x_0 = 0$, and write $y_0 = 0, y_1 = (1, 0)^\top, y_2 = (0, 1)^\top$. Define

$$\mathcal{M}_i = (x_{i1}^2, 2x_{i1}x_{i2}, x_{i2}^2, -2x_{i1}, -2x_{i2}), \quad 1 \leq i \leq 5. \quad (\text{A25.2})$$

Proposition A25.1 (A finite certificate for a whole distance patch). *If X has rank two and the square matrix $\mathcal{M}$ is invertible, the kernel d determines both parametrized boundary patches, up to one common Euclidean isometry. Their relative translation, angle and scale are determined. On the image of actual patches and on fixed bounded classes with positive singular-value margins, this inverse is locally Lipschitz from C^k distance kernels to C^k parametrized patches for every finite $k \geq 0$.*

Proof. Write $a_i = A(s_i)$ and $b_j = B(t_j)$. The cross differences are

$$X_{ij} = (a_i - a_0) \cdot (b_j - b_0).$$

Rank two implies that $C = (b_1 - b_0 \quad b_2 - b_0)$ is invertible. In the translation gauge $a_0 = 0$, set $P = C^{-\top}$, $G = P^\top P > 0$, and $h = P^\top b_0$. Then

$$a_i = Px_i, \quad b_j = P^{-\top}(h + y_j), \quad D_{i0} - D_{00} = x_i^\top Gx_i - 2h^\top x_i. \quad (\text{A25.3})$$

The last five equations are a linear system with coefficient matrix $\mathcal{M}$ for the five entries $(G_{11}, G_{12}, G_{22}, h_1, h_2)$. They uniquely determine G, h . For realizable data G is positive definite. Choosing the positive square root $P = G^{1/2}$ in (A25.3) gives a canonical realization. Every other invertible matrix with $P^\top P = G$ is an orthogonal left multiple, so the anchors are unique up to that same orthogonal transformation and the initial translation. Realizable data also satisfy $D_{00} = h^\top G^{-1}h$ and all the remaining distance identities; these identities are not extra observed coordinates.

For an arbitrary s , subtract the squared distances to b_0 from those to b_1, b_2 :

$$2(b_j - b_0) \cdot A(s) = |b_j|^2 - |b_0|^2 - (d(s, t_j)^2 - d(s, t_0)^2), \quad j = 1, 2. \quad (\text{A25.4})$$

The two coefficient vectors are independent. This recovers all of $A(s)$. Select three of its now known noncollinear anchors and use the transposed formula to recover all of $B(t)$. Such anchors exist because X has rank two. Thus the conclusion concerns full functions, not a finite list of values.

For stability, the linear solve, positive square root on a compact positive-definite set, and inversion of the two trilateration matrices are locally Lipschitz. Slice evaluation of d is bounded in C^k , and squared-distance differences use products of bounded functions. Apply these observations to each of the explicit formulas. They also define a local reconstruction for nearby nonmodel kernels, although such outputs need not satisfy every unused distance equation. $\square$

Lemma A25.2 (Availability of the certificate). *If an open arc of A is not contained in any conic, one can choose the six parameters there so that (A25.2) is invertible, for any three distinct parameters on a strictly convex open arc of B . The choices and the inequalities persist under small C^0 perturbations of the finitely many anchor positions.*

Proof. On the first arc the six functions $1, A_1, A_2, A_1^2, A_1 A_2, A_2^2$ are linearly independent; otherwise a nonzero polynomial of degree at most two would vanish there. Their evaluation vectors therefore span $\mathbb{R}^6$. Choose six vectors forming a basis. Subtract the row at the first point from the other five, translate that point to the origin, and make the invertible linear coordinate change $x = C^\top(a - a_0)$. The six-point evaluation matrix is invertible exactly when the five-column matrix (A25.2) is invertible. Strict convexity ensures that three distinct points on the second arc are noncollinear, hence C is invertible. Invertibility and positive singular values are open conditions on these finite matrices. $\square$

A25.1. Registration by overlaps. Recover each distance patch in its own canonical Euclidean frame. Whenever two copies overlap on the same lifted obstacle, select three distinct scalar parameters in the overlap and evaluate their two reconstructed parametrizations, using the given overlap map. The three points are noncollinear: a line meets the boundary of a strictly convex body at at most two points. There is exactly one Euclidean isometry carrying the ordered first triple to the ordered second triple. For realizable data it carries the entire common arc to that arc.

Choose a spanning tree of the connected registration graph and propagate these isometries from a root frame. This places every patch. Non-tree overlaps are consistency equations; for realizable data they hold because the actual table supplies one placement. This is not an assumption of uniqueness: uniqueness follows from the finite distance certificates and the uniquely determined triple matches along the tree. Changing the root frame changes all patches by the same element of $E(2)$.

In the periodic case, let R_0, R_1, R_2 be the placed versions of the same labelled interval on a representative obstacle and its translates by the two specified abstract basis elements. At any common scalar parameter s_* ,

$$l_j = R_j(s_*) - R_0(s_*), \quad j = 1, 2, \quad L = (l_1 \ l_2), \quad G_\Lambda = L^\top L. \quad (\text{A25.5})$$

These differences are constant on the common interval for actual data. They are independent because the table has rank-two lattice. Complete boundary representatives together with these vectors determine every translated obstacle and hence the whole periodic table.

This proves the uniqueness part of Theorem A16.2. Equality of scalar laws first gives equality of absolute actions and of distance patches, even if source weights differ. Their canonical

reconstructions and all tree registrations agree. No supplied curvature signature, spatial origin, lattice Gram form, or analytic continuation is used.

APPENDIX A26. FINITE COVERAGE BY LONG EXPERIMENTS

The conclusion of Theorem A16.2 becomes a whole-table statement only when the experiment genuinely covers the boundary. We prove the existence of such experiments on an explicit class of periodic smooth dispersing tables. An upper bound on a collision-free flight will be called a finite-horizon bound. Obstacle separation and positive curvature are strict throughout this section.

Lemma A26.1 (Regular extension of a fan). *Fix a boundary point x in a finite-horizon dispersing table and an open interval of nongrazing directions entering the free region. For every integer J , this interval contains a direction whose forward trajectory is regular for its first J collisions. The same statement holds for backward trajectories. In particular, a clear nongrazing one-flight patch can be restricted about a point on either endpoint boundary so that it has a regular prefix of any prescribed finite length, together with positive adjacent free-time margins.*

Proof. At the first flight, only finitely many obstacle lifts are within the finite-horizon bound. From a fixed exterior point a strictly convex body has at most two tangent rays. Removing these directions leaves open regular branches. On each such branch the outgoing family is a diverging ray fan. Here is the transversality fact needed to iterate this argument. Write Y for transverse separation of neighboring rays and $\mathcal{B}$ for wave-front curvature. Free propagation over length τ and a dispersing reflection give respectively

$$\mathcal{B}_{\text{free}} = \frac{\mathcal{B}}{1 + \tau\mathcal{B}}, \quad \mathcal{B}_{\text{reflected}} = \mathcal{B}_{\text{incoming}} + \frac{2\kappa}{\nu}. \quad (\text{A26.1})$$

These formulas follow by differentiating the direction of the normal rays to a wave front and then the specular reflection identity. A point fan has $Y = \tau$ at its first positive distance, in the initial-angle normalization; thereafter free propagation multiplies a positive separation by $1 + \tau\mathcal{B}$, and reflection preserves its nonvanishing, with the corresponding incidence-coordinate factor. Thus Y never vanishes before a singular collision. This is also the no-conjugacy calculation encoded by the positive form (A23.1).

At a tangent encounter with a possible next obstacle, let F be a smooth local defining function for that obstacle and let $r(\theta, t)$ be the ray family. At tangency, $F_t = 0$ and $F_{tt} \neq 0$ by positive curvature. Solving $F_t = 0$ for the location of closest approach, the derivative of its signed distance with respect to the fan parameter is a nonzero multiple of Y : the normal to the obstacle is perpendicular to the tangent ray. Hence a tangent direction is an isolated zero on every previously regular parameter branch. There are countably many such branches, and finitely many next obstacle lifts on each bounded flight. At each fixed collision order only a countable set of additional directions is excluded. A finite union of these sets cannot exhaust an open interval. Finite horizon prevents a regular ray from escaping without a next collision. This proves the forward assertion; reversal proves the backward assertion.

For a given visible patch and fixed point x at its initial end, directions reaching its other open endpoint arc contain an open interval. At x , reflection converts them smoothly to backward directions. Choose one avoiding the first $J + 1$ backward singularities. The first J flights supply a prefix and the extra backward flight supplies the initial free-time margin. The terminal outgoing free segment has a positive length. Restricting the regular trajectory to compact small coordinate boxes preserves all these margins and keeps x inside the final-flight patch. $\square$

Lemma A26.2 (Connected visibility graph of the lifted obstacles). *The graph whose vertices are all obstacle lifts and whose edges are clear nongrazing one-flight patches is connected.*

Proof. Suppose its vertices were partitioned into a connected component and its nonempty complement. The minimum distance between a body in one part and a body in the other is attained. Indeed, start with any cross-part pair at distance D ; modulo the finite list of obstacle orbits there are only finitely many relative lattice translations whose body distance is at most D . Their distances form a finite nonempty list, even if the partition is not periodic.

Let $[x, y]$ realize the least cross-part distance. If another body met its open segment, that body would be in one of the two parts and would be strictly closer to the endpoint body in the other part, a contradiction. The open segment is therefore clear. At the minimizing endpoints it is normal to both smooth strictly convex boundaries, so it is nongrazing. Compactness and local finiteness give clearance from all other obstacles, and small endpoint perturbations produce a one-flight patch. This is an edge between the two parts, contradicting the choice of a graph component. $\square$

Theorem A26.3 (Finite identifying experiments and their persistence). *Let a finite-horizon periodic dispersing table have locally nonconic C^∞ boundaries. For every integer $J \geq 1$ it admits a finite identifying atlas for its complete periodic table, in which every measured prefix has at least J flights and every measured extension one additional flight. Its scalar coordinate domains, branch and lift labels, anchor parameters and three absolute clocks per branch can be fixed on a nonempty finite-smooth open neighborhood of the table. Every realization of the same admissible atlas data is congruent to it, whether or not that realization is close to it.*

Proof. Choose one lift of each quotient obstacle and, in addition, the two basis translates of one of them. Lemma A26.2 provides finitely many graph paths connecting this finite set. Add the finitely many intermediate lifts on these paths to the set of target bodies. Choose one clear patch on each path edge. Slight perturbation within a patch makes the backward prefix regular for the required length by Lemma A26.1, without changing its two obstacle labels or destroying its clearance.

For each point of each target boundary choose a nongrazing outgoing ray with a regular next collision and a regular backward prefix of length at least J . Such rays exist by Lemma A26.1. Lemmas A23.1 and Proposition A23.2 give an open final-flight rectangle whose first boundary interval contains that point, together with two branch gates and a unique scalar matching bracket. The initial final-flight interval is nonconic by hypothesis. Lemma A25.2 supplies six and three interior anchors for that rectangle. Compactness of the finitely many target boundaries selects finitely many such intervals covering them. Choose slightly smaller open covering intervals with closures in the constructed rectangles, preserving the anchors in the larger patches as necessary.

A finite open cover of a connected closed boundary has a connected overlap graph after redundant disconnected components are removed: otherwise the unions of its graph components would separate the boundary into disjoint nonempty open sets. In each nonempty open overlap choose three distinct parameter values. The independently reconstructed patches therefore register all pieces of each target boundary. The previously selected path-edge patches connect these registrations across all target bodies. Their second endpoint patches and any further auxiliary patches attach to an already covered target piece and do not disconnect the graph. On the two translated copies include a common scalar subinterval from the atlas transported by the deck labels. The resulting registration graph and target satisfy Definition A16.1.

There are finitely many branch gates. Shrink them, when needed, before choosing the finite subcover and anchors, and apply Lemma A17.2 to choose clock triples with strict margins. Nonconstancy of each action supplies a usable pair of scalar anchors in Theorem A17.3. All choices are finite. In particular no limiting sequence of patch radii is being used to turn density of visited points into coverage.

The positive Hessians, nonzero twists, clock and root brackets, noncollinear overlap triples, and rank certificates all have positive margins on these finitely many compact domains. The finite implicit construction and continuity preserve them on a C^r neighborhood, with r chosen large enough for the finitely stipulated derivatives. The scalar covering intervals and overlap maps are fixed and still cover the perturbed boundary parametrizations. Thus the same design is an identifying atlas on this open class. The last assertion is Theorem A16.2, whose uniqueness argument uses the certificates and not nearness of the candidates. $\square$

This theorem establishes existence and persistence of a finite design. Its constants and the number of settings may depend on the table and on J . It does not replace the known scalar boundary atlas or exact branch labels by conclusions. Nor does it assert that finitely many sampled points determine an arbitrary smooth boundary: each clock setting here specifies a whole conditional law.

A26.1. An explicit open family with nonanalytic members.

Proposition A26.4 (Nonempty smooth whole-table classes). *There are finite-horizon periodic tables to which Theorem A26.3 applies, and their persistent identifying atlas classes contain nonanalytic smooth tables.*

Proof. Use the triangular lattice generated by $(1, 0)$ and $(1/2, \sqrt{3}/2)$. Choose R, ε with

$$\frac{\sqrt{3}}{4} < R - |\varepsilon|, \quad R + |\varepsilon| < \frac{1}{2}, \quad 8|\varepsilon| < R, \quad \varepsilon \neq 0.$$

At each lattice point place the strictly convex body with support function

$$h(\theta) = R + \varepsilon \cos(3\theta). \tag{A26.2}$$

Its radius of curvature is $h + h'' = R - 8\varepsilon \cos(3\theta) > 0$. It contains the centered disk of radius $R - |\varepsilon|$ and is contained in the disk of radius $R + |\varepsilon|$, so its translates are disjoint.

We verify finite horizon for the contained disks. An irrational direction on the torus gives a dense straight line and hence meets a disk interior. In a rational direction parallel to a primitive lattice vector v , the rows of lattice centers have spacing $\sqrt{3}/(2|v|) \leq \sqrt{3}/2$. Every parallel line is within $\sqrt{3}/4$ of one row and meets a disk interior since the disk radius is larger. Thus no full line avoids the disk interiors. If there were arbitrarily long free segments, translate their midpoints into a fixed cell and pass to a subsequence of points and directions. The limit would be a complete line avoiding all disk interiors, a contradiction. The larger bodies therefore have finite horizon as well.

The boundary in (A26.2) is analytic, has threefold rotational symmetry and is not a circle. It cannot lie in a conic: a conic containing a closed strictly convex curve must be an ellipse, and an ellipse invariant under a rotation by $2\pi/3$ is a circle. If any open boundary arc lay in a conic, analyticity of the parametrization would make the corresponding quadratic polynomial vanish on the whole boundary. Thus the boundary is locally nonconic. Theorem A26.3 applies.

Fix one of its finite identifying atlases. Modify the support function by a sufficiently small C^∞ compactly supported bump which is flat but not identically zero at an endpoint of its support. Its size can be arbitrarily small in any prescribed finite C^r norm. Positivity of $h + h''$, disk containment, separation, and all finite atlas margins persist. The new support function is not analytic, while the same finite experiment still determines its complete boundary by actual function inversion. No analytic continuation is applied to this perturbed table. $\square$

APPENDIX A27. CONVEX TWIST SUSPENSIONS: SIX CONDITIONAL LAWS AND FOUR RAW LAWS

We next verify the generating-action hypotheses on a class in which the unknown is a function of two variables. In particular the argument is not restricted to kinetic energy plus a potential. The observed clock will be the return time of an explicitly specified contact suspension. This construction distinguishes a genuine clock process from a device that is instructed to evaluate an unknown stationary action.

Fix positive constants $\lambda, \Lambda, b, A_0, A_1$. Consider all $L \in C^\infty(\mathbb{R}^2)$ satisfying

$$\lambda I_2 \preceq D^2 L(x, y) \preceq \Lambda I_2, \quad -L_{xy}(x, y) \geq b, \quad |L(0, 0)| \leq A_0, \quad |DL(0, 0)| \leq A_1. \quad (\text{A27.1})$$

The class is assumed nonempty; the inequalities themselves constrain the possible constants. Bounds for higher derivatives will be imposed only when quantitative stability is asserted. The scalar configuration coordinate and its canonical covector are fixed coordinates of this mechanical experiment, unlike the unknown Euclidean embeddings in the billiard application.

Lemma A27.1 (Intrinsic determinism, convex composition, and twist). *The equations*

$$p = -L_x(x, y), \quad q = L_y(x, y)$$

define a global smooth exact symplectic diffeomorphism $F_L : (x, p) \mapsto (y, q)$ of $\mathbb{R}^2$. For $n \geq 1$ the function

$$S_n(s, t) = \min_{x_1, \dots, x_{n-1}} \sum_{i=0}^{n-1} L(x_i, x_{i+1}), \quad x_0 = s, \quad x_n = t, \quad (\text{A27.2})$$

is a unique smooth stationary action and is the generating function of F_L^n . Its mixed derivative satisfies

$$|(S_n)_{st}| \geq u_n := b \left(\frac{b}{2\Lambda} \right)^{n-1}. \quad (\text{A27.3})$$

At every match between S_n and S_{n+1} the stationary Schur complement is positive and at most 2Λ . Equality of initial covectors identifies the entire physical prefix, and the matching function has derivative at most $-u_n^2/(2\Lambda)$ at each zero.

Proof. For fixed x , the derivative of $y \mapsto -L_x(x, y)$ is at least b . This map is strictly increasing and onto $\mathbb{R}$, since its values grow at least linearly at both ends. It has a unique smooth inverse. For fixed y , the map $x \mapsto L_y(x, y)$ is strictly decreasing, with derivative at most $-b$, and is likewise onto. These two inverses construct F_L and its smooth inverse. With $\theta = p dx$ and $\ell_L(x, p) = L(x, y(x, p))$, the first variation gives

$$F_L^* \theta - \theta = d\ell_L. \quad (\text{A27.4})$$

Taking an exterior derivative proves symplecticity.

The second variation of the sum in (A27.2), with endpoints fixed, is bounded below by

$$\lambda \sum_{i=0}^{n-1} (h_i^2 + h_{i+1}^2) = 2\lambda \sum_{i=1}^{n-1} h_i^2.$$

Coercivity and strict convexity give a unique minimum. Its Hessian is tridiagonal, with diagonal entries at most 2Λ and neighboring entries $L_{xy} \leq -b$. The implicit function theorem gives smooth dependence on (s, t) . The corner-cofactor formula, including the empty determinant for $n = 1$, gives

$$|(S_n)_{st}| = \frac{\prod_{i=0}^{n-1} |L_{xy}(x_i, x_{i+1})|}{\det H_{n-1}} \geq \frac{b^n}{(2\Lambda)^{n-1}},$$

where the determinant bound is Hadamard's inequality for the positive Hessian. The Euler-Lagrange equations identify consecutive covectors, so this minimizer follows F_L . Conversely an

orbit with the given endpoints is stationary and hence equals the unique minimizer. This also proves the stated generating-function interpretation.

A prefix match is therefore a match of actual trajectories, not merely of stationary-value continuations. The last-site Schur complement of the full positive Hessian is positive and no greater than that site's diagonal entry 2Λ . Differentiating stationary composition gives $H_s = -(S_n)_{st}^2/B$, exactly as in (A24.1). The lower bound for the twist proves the final assertion. $\square$

A27.1. An elapsed-time realization. The elementary contact construction below is part of the classical relation between exact symplectic action and return time; compare the contact realization of action functions in [13]. We include the construction to specify the observation process and its clock, not to claim a new construction of contact suspensions.

Proposition A27.2 (Positive action as a return-time roof). *Let F be a global exact symplectic diffeomorphism of $(\mathbb{R}^2, d\theta)$, $\theta = p dx$, with $F^*\theta - \theta = d\ell$. Suppose $r = c + \ell \geq r_0 > 0$. The quotient of $\mathbb{R}^2 \times \mathbb{R}$ by*

$$G(x, p, z) = (F(x, p), z - r(x, p)) \quad (\text{A27.5})$$

is a smooth contact manifold with contact form induced by $\alpha = dz + \theta$. Its complete Reeb flow has a global section $z = 0$, first-return map F , and first-return time r . For $F = F_L$ in Lemma A27.1, the time of its n th return from (s, p) with terminal coordinate t is

$$nc + S_n(s, t). \quad (\text{A27.6})$$

Proof. The identity $G^*(dz + \theta) = dz - dr + F^*\theta = dz + \theta$ proves invariance. Also $\alpha \wedge d\alpha = dz \wedge dp \wedge dx$ never vanishes. The integer action is free and properly discontinuous: under G^k , $k > 0$, the z coordinate decreases by at least kr_0 ; under negative iterates it increases by at least $|k|r_0$. Thus only finitely many iterates can meet two fixed compact subsets. The quotient is a smooth manifold and the invariant form descends.

On the covering space the Reeb field is ∂_z , since $\alpha(\partial_z) = 1$ and $d\alpha(\partial_z, \cdot) = 0$. It is invariant under G and its translation flow is complete. Starting at $(x, p, 0)$, the first time its class returns to the section is $r(x, p)$, at which $(x, p, r(x, p))$ is identified with $(F(x, p), 0)$. Positivity of the roof excludes any earlier section crossing; negative powers correspond to past crossings. The section is an embedded copy of $\mathbb{R}^2$, since two of its points cannot be related by a nonzero iterate. Summing successive roofs gives $nc + \sum_{i=0}^{n-1} L(x_i, x_{i+1})$ in the convex twist case. The orbit is the unique minimizer for its endpoints by the preceding lemma, proving (A27.6). $\square$

The clock in this proposition is elapsed Reeb time on the stated contact manifold. It is not asserted to be ordinary elapsed time of an unsuspended discrete mechanical map. A choice of the primitive's additive constant changes the roof, and is therefore physically visible to this observation. The known deadlines fix that constant.

Theorem A27.3 (Class-wide six-deadline reconstruction). *Fix a target radius $R > 0$, a finite integer $n \geq 1$, and the constants in (A27.1). There are a fixed initial phase box, fixed endpoint gates, a fixed release-delay interval, and three fixed deadlines for each of the return counts n and $n + 1$, all independent of L , with the following property.*

In the contact suspension of F_L , launch a state on the section after an independent uniform external release delay. At the specified return count retain the record only if it is in the endpoint gate and has arrived by the specified deadline, with a positive endpoint-dependent efficiency. The source density and efficiency may be unknown and may differ between the two return counts, but are shared by the three deadlines of each count. Record only the initial and terminal scalar positions upon retention, and a failure symbol otherwise. The six conditional endpoint laws determine the full function $L|_{[-R, R]^2}$, including its additive constant.

No root-selection mark, observed covector, individual return time, or action evaluation is part of the retained data. On subclasses with a fixed C^{k+4} bound on L on the enlarged compact configuration square, fixed C^{k+2} bounds on source and efficiency on the enlarged phase and endpoint domains, and positive source and efficiency floors, reconstruction is uniformly locally Lipschitz from C^{k+2} laws to $C^k([-R, R]^2)$ for every finite k . For clock-dependent log-weight errors of C^{k+2} size ϵ , the same reconstruction has additional error at most $C_k \epsilon$.

Proof. We construct all settings from the class constants. The stationarity equation at a configuration value x is

$$L_y(x_-, x) + L_x(x, x_+) = 0.$$

For fixed x, x_+ its left side is strictly decreasing in x_- with derivative at most $-b$ and has one zero. At $x_- = 0$ its absolute value is at most $2A_1 + 2\Lambda|x| + \Lambda|x_+|$, using the Hessian bound and the bound on $DL(0, 0)$. Consequently

$$|x_-| \leq \frac{2A_1 + 2\Lambda|x| + \Lambda|x_+|}{b}. \quad (\text{A27.7})$$

Set $r_0 = r_1 = R + 1$ and

$$r_{j+1} = (2\Lambda/b)r_j + (\Lambda/b)r_{j-1} + 2A_1/b, \quad S = r_{n+1} + 1, \quad I = [-S, S], \quad J_0 = [-R - 1, R + 1].$$

The ratios in this recurrence are nonnegative and $\Lambda \geq b$, so its bounds increase. Starting from any final pair $(t, v) \in J_0^2$, backward solution through n sites gives an initial coordinate in the interior of I , with a unit buffer. Its whole path is stationary; strict convexity identifies it with both actions $S_n(s, t)$ and $S_{n+1}(s, v)$ on the fixed gates $I \times J_0$. It therefore supplies a matching root for every $(t, v) \in J_0^2$. Lemma A27.1 and the downward-zero lemma show that this root is unique, without specifying it to the observer.

Put

$$C_0 = A_0 + \frac{A_1^2}{2\lambda}, \quad M = A_0 + \sqrt{2}A_1S + \Lambda S^2.$$

The quadratic lower bound for L gives $L \geq -C_0$ globally. A straight interpolating configuration path for either endpoint gate stays in $[-S, S]$ and has each term at most M . Thus for $q \in \{n, n+1\}$ both actions satisfy the fixed bounds

$$w_- = -(n+1)C_0 \leq S_q \leq w_+ = (n+1)M.$$

Take $c = C_0 + 1$ in Proposition A27.2, so every roof is at least one. Set

$$T_j = w_+ + j, \quad A = w_+ - w_- + 4, \quad D_{q,j} = qc + T_j, \quad j = 1, 2, 3. \quad (\text{A27.8})$$

Then $1 \leq T_j - S_q \leq A - 1$ on the fixed gates.

The source phase box can be fixed as well. The inequality

$$L(x, y) \geq \frac{\lambda}{4}(x^2 + y^2) - C_1, \quad C_1 = A_0 + A_1^2/\lambda,$$

follows by absorbing the linear term in the quadratic lower bound. At a minimizing path of length $q \leq n + 1$ it implies

$$\sum_{i=1}^{q-1} x_i^2 \leq \frac{2q}{\lambda}(M + C_1).$$

Hence all its coordinates lie in $[-K, K]$ for

$$K = \max \left\{ S, \sqrt{2(n+1)(M + C_1)/\lambda} \right\}.$$

The initial covector has magnitude at most $A_1 + \sqrt{2}\Lambda K$. Choose a strictly larger fixed phase box and a normalized smooth source positive on the required compact subset. Such a source can even

be fixed across the entire class; the theorem allows any unknown source with the stated positivity and sharing properties. Different full source laws are not confused with different acquisition rules.

Let α_0 be the external release delay, uniform on $(0, A)$. Stopping at return q and comparing the clock with $D_{q,j}$ requires only counting section crossings and reading a clock. By (A27.6), the event is exactly

$$\alpha_0 + qc + S_q(s, t) < D_{q,j} \iff \alpha_0 < T_j - S_q(s, t).$$

The delay is external: it is not an assertion that a long interval before a section crossing contains no earlier crossings. Changing from the initial p to the terminal t has the nonzero Jacobian $|(S_q)_{st}|$. Integrating the independent delay therefore yields unnormalized density

$$\frac{1}{A} e_q(s, t) \rho_q(s, -(S_q)_s(s, t)) |(S_q)_{st}(s, t)| \{T_j - S_q(s, t)\}.$$

Normalization gives precisely the shared-weight model, separately for $q = n$ and $q = n + 1$. The observer is not instructed to evaluate S_q ; this expression is the mathematical derivation of a clock event in the specified flow.

Recover S_n, S_{n+1} and their absolute constants from their triples. At the unique match subtract the prefix to obtain $L(t, v)$ on all of J_0^2 , by Theorem A24.4. This recovers the required compact square. For quantitative stability, the positive interior Hessian has inverse norm at most $(2\lambda)^{-1}$; Proposition A28.1, with $m = k + 2$, gives the required action and density derivative bounds on the fixed compact region using exactly the orders in the theorem. The twist estimate and the Schur bound control the matching derivative. The unit initial buffer, restriction to the smaller target square, and these derivative bounds give uniform root collars and sign margins. For example, on a collar where $|H_s| \geq u_n^2/(4\Lambda)$, the value of H at its two endpoints has the corresponding fixed positive magnitude. Bounded next derivatives make a common collar possible, with the explicit radius in Theorem A28.2. The action-anchor bound in Proposition A30.2 supplies a usable anchor among finitely many corners. Composing the clock inverse with the root and subtraction estimates proves the C^{k+2} to C^k bound. Finally Theorem A21.4 gives a C^{k+2} action bias at most $C\epsilon$; the same root estimate propagates it to L . $\square$

Remark A27.4 (Breadth and the retained mechanical subfamily). For example

$$L_*(x, y) = \frac{\mu}{2}(y - x)^2 + \frac{\kappa}{2}(x^2 + y^2)$$

has Hessian eigenvalues κ and $2\mu + \kappa$ and mixed entry $-\mu$. Adding a sufficiently small, nonseparable smooth bump supported in a compact subset of $\mathbb{R}^2$ preserves strict versions of (A27.1). The bump may be flat at its support boundary and hence nonanalytic. The inverse recovers this entire unknown two-variable perturbation on a prescribed square, not merely a finite jet or a parameter vector.

The original family $L_{\mu, P}(x, y) = \mu(y - x)^2/2 + P(x)$ with (A29.2) is also contained in a class of the form (A27.1). Indeed its determinant is $\mu P''$, its trace is $2\mu + P''$, and its mixed entry is $-\mu$; one may take

$$\lambda = \frac{\mu - \kappa}{2\mu_+ + \Lambda}, \quad \Lambda_{\text{joint}} = 2\mu_+ + \Lambda, \quad b = \mu_-.$$

Here the two occurrences of Λ in the fraction refer to the potential-curvature bound in (A29.2). The mass and potential formulas, their sharper elementary bounds, and their full proofs are retained in the next section. For that family the suspension construction supplies a return-clock realization of its action-threshold experiment. It does not change the time interpretation of the unsuspended mechanical map.

APPENDIX A28. A QUANTITATIVE REGULARITY BUDGET FOR STATIONARY ACTIONS

We make explicit the finite-order hypotheses for the convex-twist inverse. All configuration and phase domains in this section are the fixed compact domains constructed in the proof of Theorem A27.3, enlarged by a fixed positive collar. The number of iterates is fixed. Constants do not depend on the particular generating function within the declared class.

Proposition A28.1 (Propagation of finite-order bounds). *Fix $q \geq 1$, $m \geq 0$ and the convexity constants in (A27.1). Let the minimizing paths for endpoints in a fixed compact gate have all coordinates in $[-K, K]$. Suppose*

$$\|L\|_{C^{m+2}([-K-1, K+1]^2)} \leq M_L.$$

Then the interior minimizing coordinates X_q have a uniform C^{m+1} bound, and the stationary action S_q has a uniform C^{m+2} bound on the gate. The constants depend only on q, m, K, M_L, λ and the gate. For raw source and efficiency bounds

$$\|\rho_q\|_{C^m} \leq M_\rho, \quad \|e_q\|_{C^m} \leq M_e$$

on their fixed larger domains, the weight

$$a_q = D^{-1}e_q(s, t)\rho_q(s, -(S_q)_s(s, t))(S_q)_{st}(s, t) \quad (\text{A28.1})$$

and the raw clock subdensities have uniform C^m bounds. Positive source and efficiency floors give a uniform positive weight floor using (A27.3). Conditional densities have the same regularity when their normalizing masses have a positive lower bound.

Proof. The case $q = 1$ has no interior variables and follows directly. For $q > 1$ set

$$\mathcal{A}(z, X) = \sum_{i=0}^{q-1} L(x_i, x_{i+1}), \quad z = (s, t), \quad G = \partial_X \mathcal{A}.$$

The stationarity equation is $G(z, X_q(z)) = 0$, and its X derivative H satisfies $H \succeq 2\lambda I$. Hence $\|H^{-1}\| \leq (2\lambda)^{-1}$. Derivatives of G through order $m+1$ are bounded by constants times qM_L .

Here is an explicit recursive form of the implicit bound. Put $P_0 = \sqrt{q-1}K$. At first order use $DX_q = -H^{-1}G_z$. At order $j \geq 2$, differentiation of the stationarity equation gives

$$HD^j X_q = -\mathcal{B}_j,$$

where $\mathcal{B}_j$ is the finite chain-rule sum indexed by partitions of the j differentiations, after removing the single term $HD^j X_q$. Every remaining term contains a derivative of G of order at most j and derivatives of X_q of orders strictly less than j . Let $c_{q,j}$ be the sum of the absolute integer coefficients of that finite multivariable chain-rule sum, enlarged if necessary to cover its coordinate norms and the first-order formula. The recursive choice

$$P_j = \frac{c_{q,j}qM_L}{2\lambda} \left(1 + \sum_{i=1}^{j-1} P_i \right)^j, \quad 1 \leq j \leq m+1, \quad (\text{A28.2})$$

bounds $\|D^j X_q\|_\infty$. This specifies constants using only finite differentiation and the displayed parameters.

The envelope identity is $DS_q = \partial_z \mathcal{A}(z, X_q(z))$. Differentiating it through order $m+1$ uses only the already bounded derivatives of X_q and of L through order $m+2$. Together with the bound for S_q itself this proves the asserted C^{m+2} estimate. Composition and the C^m product inequality now apply to (A28.1): the momentum composition uses derivatives of S_q through $m+1$, and its Jacobian uses derivatives through $m+2$. The mixed derivative has a fixed nonzero sign, so taking its absolute value creates no loss of regularity. The lower bound is $D^{-1}e_{-\rho_-}u_q$. Multiplication by $T_j - S_q$ and division by a positive scalar mass prove the remaining assertions. $\square$

Theorem A28.2 (Uniform budget and four-deadline convex reconstruction). *Fix R, n and the class constants of (A27.1). Use the phase box, gates, roof offset c , release interval and the first two deadlines at each of the counts $n, n+1$ constructed in Theorem A27.3. The four raw endpoint laws of the unknown contact suspension determine $L|_{[-R, R]^2}$, including its additive constant. All settings are independent of L .*

For a target C^k estimate, $k \geq 0$, it suffices to impose fixed C^{k+4} bounds on L on the enlarged configuration square and fixed C^{k+2} bounds on the raw source and efficiency on the enlarged phase and endpoint domains, with positive source and efficiency floors. Under these explicit bounds the inverse is uniformly locally Lipschitz from C^{k+2} raw subdensities to $C^k([-R, R]^2)$. Clock-dependent log-efficiency errors of C^{k+2} size ϵ add at most $C\epsilon$ to the bound. Full-rank calibrated branch channels and their errors can be included as in Theorem A18.2 whenever the declared finite sheets form an exhaustive acquisition block.

Proof. The class-wide construction gives true raw densities $a_q(T_j - S_q)$ for $q = n, n+1$, with the same normalized source at the two deadlines of each count. Theorem A18.1 recovers both absolute actions. The unique physical prefix match of Lemma A27.1 then gives

$$L(t, v) = S_{n+1}(\sigma(t, v), v) - S_n(\sigma(t, v), t)$$

on the whole prescribed square. This is the same deterministic inverse as in the six-conditional-law theorem; success masses have replaced its third clocks.

Apply Proposition A28.1 with $m = k + 2$. It gives exactly the action and raw-density bounds required for the C^{k+2} inverse and its weight floor. The matching derivative at a root has magnitude at least $\mu = u_n^2/(2\Lambda)$. The resulting uniform C^3 action bound gives a uniform Lipschitz constant M_3 for this derivative in the initial coordinate. Every true root has the unit initial buffer supplied by the backward construction. A collar of radius at most $\min(1/2, \mu/(2(1 + M_3)))$ therefore preserves $|H_s| \geq \mu/2$; its endpoints have opposite signs separated from zero by this derivative bound times the collar radius. These are uniform root collars, not unspecified higher-derivative assumptions. The implicit and composition estimate in Proposition A23.2 proves the asserted C^k bound. The raw nuisance estimate and, when present, the calibrated channel estimate supply the two extensions. $\square$

For density smoothing with an additional s derivatives, replace the bounds in this theorem by C^{k+4+s} for L and C^{k+2+s} for source and efficiency. These are sufficient budgets; optimality of the derivative loss is not claimed. The six-conditional-law result is retained because a conditional-law archive may no longer contain the success masses needed by the four-raw-law theorem.

APPENDIX A29. A FIXED PROTOCOL FOR AN UNKNOWN DISCRETE MECHANICAL SYSTEM

We give an application in which the acquisition can be fixed on an entire infinite-dimensional class, rather than on an unknown object's local certificate. Let

$$L_{\mu, P}(x, y) = \frac{\mu}{2}(y - x)^2 + P(x), \quad \mu > 0, \quad (\text{A29.1})$$

where both the mass μ and the smooth potential P are unknown. Assume, for fixed constants,

$$0 < \mu_- \leq \mu \leq \mu_+, \quad 0 < \kappa \leq P'' \leq \Lambda, \quad |P(0)| \leq A_0, \quad |P'(0)| \leq A_1. \quad (\text{A29.2})$$

No analyticity is required. For $n \geq 1$ define the stationary action

$$S_n(s, t) = \min_{x_1, \dots, x_{n-1}} \sum_{i=0}^{n-1} L_{\mu, P}(x_i, x_{i+1}), \quad x_0 = s, \quad x_n = t. \quad (\text{A29.3})$$

For $n = 1$ this is $L_{\mu, P}$ itself. The endpoint coordinates here are the known scalar configuration coordinates, not unknown planar boundary embeddings.

Lemma A29.1 (Global twist and finite conditioning). *The minimum in (A29.3) exists uniquely and is smooth in (s, t) . Its mixed derivative never vanishes. Put $D = 2\mu_+ + \Lambda$. Then*

$$|(S_n)_{st}| \geq u_n := \mu_- \left(\frac{\mu_-}{D} \right)^{n-1}. \quad (\text{A29.4})$$

At a prefix match between S_n and S_{n+1} , the positive Schur complement is at most D , and consequently the matching derivative has magnitude at least u_n^2/D .

Proof. The interior Hessian is tridiagonal, with diagonal $2\mu + P''(x_i)$ and neighboring entries $-\mu$. Its quadratic form is the sum of the nonnegative kinetic second variation and $\sum P''(x_i)h_i^2$, and is positive definite. Strong convexity and coercivity give a unique minimum. The implicit function theorem gives smooth dependence. The corner-cofactor formula yields

$$|(S_n)_{st}| = \frac{\mu^n}{\det H_{n-1}},$$

with empty determinant one. Hadamard's inequality gives $\det H_{n-1} \leq D^{n-1}$, proving the bound. Eliminating earlier interior coordinates in S_{n+1} gives a positive Schur complement at x_n which is no larger than its original diagonal entry $2\mu + P''(x_n) \leq D$. Formula (A24.1) completes the argument. $\square$

The associated scalar canonical map is deterministic: from an initial covector $p = -L_x(x, y)$ one obtains $y = x + (p + P'(x))/\mu$. The stationary equations are

$$x_{i-1} = 2x_i - x_{i+1} + P'(x_i)/\mu. \quad (\text{A29.5})$$

Thus equality of initial covectors has exactly the common-trajectory meaning used in Theorem A24.4.

A29.1. The threshold experiment. Take a source with positive density $\rho(s, p)$ on a relevant initial phase domain, independent of an auxiliary variable α on a fixed interval $(0, A)$. Retain a trajectory of n steps when $\alpha + S_n(s, t) < T$, with an additional positive efficiency $e(s, t)$. Record its endpoints. On a gate where $0 < T - S_n < A$, the change $p = -(S_n)_s$ and integration over α give

$$f_{n,T}(s, t) = Z_{n,T}^{-1} a_n(s, t) (T - S_n(s, t)), \quad a_n = e(s, t) \rho(s, -(S_n)_s) |(S_n)_{st}|.$$

The source and efficiency are held fixed across the three thresholds of one branch, but need not agree between n and $n + 1$. A cutoff outside a larger phase domain makes the preparation a probability measure; the resulting scalar factor disappears on conditioning. This is an action-threshold measurement, not an assertion that S_n is elapsed physical time. The threshold device supplies a binary retention event, not an individual action value.

Theorem A29.2 (Six laws recover mass and a smooth potential). *Fix $R > 0$, an integer $n \geq 1$, and the constants in (A29.2). There are fixed endpoint gates, three fixed thresholds and one fixed auxiliary interval $(0, A)$, independent of (μ, P) in this class, such that the six normalized endpoint laws for S_n and S_{n+1} determine μ and the entire function $P|_{[-R, R]}$. They do so with unknown positive joint weights for each branch, and without a supplied root bracket or absolute action anchor. The result applies for every prescribed finite n .*

On bounded finite-smooth subclasses with positive source and retention margins, the inverse is uniformly locally Lipschitz from C^{k+2} laws to μ and $C^k([-R, R])$ potential for each finite k .

Proof. We first choose gates uniformly. Put $r_0 = r_1 = R + 1$ and recursively

$$r_{j+1} = (2 + \Lambda/\mu_-)r_j + r_{j-1} + A_1/\mu_-.$$

For a prescribed final pair $|t|, |v| \leq R + 1$, the backward recurrence (A29.5) gives $|x_{n-j}| \leq r_{j+1}$, because $|P'(x)| \leq A_1 + \Lambda|x|$. In particular its initial coordinate lies in $[-r_{n+1}, r_{n+1}]$. Take

$$I = [-r_{n+1} - 1, r_{n+1} + 1], \quad J = [-R - 1, R + 1],$$

and observe S_n, S_{n+1} on the fixed gates $I \times J$. There is a matching root in the interior of I for every $(t, v) \in J^2$: backward recurrence constructs its stationary path and strict convexity identifies it with the minima defining these two actions. Theorem A24.4 will therefore recover $L(t, v)$ on this entire target rectangle.

The preparation phase domain can also be fixed. Write $S = r_{n+1} + 1$ and $K = \max(S, A_1/\kappa)$. Every minimizing interior coordinate on these gates lies in $[-K, K]$. Indeed, at a maximum larger than K , the stationarity equation $\mu(2x_i - x_{i-1} - x_{i+1}) + P'(x_i) = 0$ has a nonnegative first term and a strictly positive second term; a minimum below $-K$ is excluded in the same way. Consequently the initial covector has magnitude at most $2\mu_+K + A_1 + \Lambda K$. A fixed larger phase rectangle therefore contains all required preparations, independently of the unknown system. This maximum principle also supplies a uniform compact domain for every derivative bound used below.

We also choose thresholds uniformly. Set $C_- = A_0 + A_1^2/(2\kappa)$. Taylor's lower bound implies $P(x) \geq -C_-$ for every x . A linear interpolating trial path with endpoints in $I \times J$ stays in $[-S, S]$. Hence both actions, of length $m \in \{n, n+1\}$, lie between the fixed bounds

$$w_- = -(n+1)C_-, \quad w_+ = 2\mu_+S^2 + (n+1)(A_0 + A_1S + \Lambda S^2/2).$$

Indeed the total trial kinetic term is $\mu(t-s)^2/(2m) \leq 2\mu_+S^2$. Set $T_j = w_+ + j$ for $j = 1, 2, 3$ and $A = w_+ - w_- + 4$. Then $1 \leq T_j - S_m \leq A - 1$ on both gates. Thus all six laws have the required common-weight representation with fixed thresholds and fixed auxiliary interval. The acquisition does not depend on the unknown table, potential, mass, action range or root.

Recover S_n, S_{n+1} by the three-clock inverse and recover L by root-selection-free subtraction. Choose any fixed t_*, v_0, h with $h > 0$ and $v_0 - h, v_0, v_0 + h \in J$. The exact formulas

$$\mu = \frac{L(t_*, v_0 + h) - 2L(t_*, v_0) + L(t_*, v_0 - h)}{h^2}, \quad (\text{A29.6})$$

$$P(t) = L(t, v_0) - \frac{\mu}{2}(v_0 - t)^2, \quad |t| \leq R, \quad (\text{A29.7})$$

recover the unknowns, including the additive constant of the potential. These formulas use full functions, not their jets or analyticity.

For stability, all gate minimizers stay in the fixed bounded region just proved, and all matched paths also satisfy the backward bounds. Positive interior Hessians and finite derivative bounds control the required action derivatives there. Lemma A29.1 supplies a uniform root derivative, and the unit buffer in I supplies a compact interior collar. Positive source bounds and the quantitative anchor argument of Proposition A30.2 give a uniform usable anchor among finitely many corners. Apply the finite-order generating-action estimate and then the bounded evaluation and subtraction operations in (A29.6)–(A29.7). $\square$

For example, $P(x) = cx^2/2 + \varepsilon b(x)$, with $\kappa < c < \Lambda$ and a nonzero compactly supported smooth bump b flat at the edge of its support, belongs to this class for sufficiently small ε . Thus the theorem recovers nonanalytic smooth potentials and distinguishes them from the quadratic potential on intervals meeting the bump. The explicit factor u_n^2/D also records the conditioning cost of a long prefix. Neither the number of clocks nor the exact inversion removes that cost.

APPENDIX A30. UNIFORM CONSTANTS AND AN EXPLICIT FINITE-RECORD INVERSE

The finite-cover theorem gives a protocol for each table. We now separate its finite design parameters from the estimates of the inverse. This makes the long-prefix cost visible and permits a constructive finite-record output. It does not assert a bound for the number of patches on every positive-curvature table from curvature alone.

A30.1. Primitive geometric bounds. Suppose every flight in the finite protocol has length in $[\ell_-, \ell_+]$, every incidence cosine is at least $\nu_- > 0$, every scalar chart speed belongs to $[v_-, v_+]$, and curvature belongs to $[\kappa_-, \kappa_+]$, with positive lower endpoints. Let N_* be the largest number of flights in any measured branch. Set

$$\begin{aligned} b_* &= v_+^2/\ell_-, & b_0 &= v_-^2\nu_-^2/\ell_+, & h_0 &= 2\kappa_-\nu_-v_-^2, \\ D &= 2v_+^2/\ell_- + 2\kappa_+v_+^2, & u_* &= b_0(b_0/D)^{N_*-1}, & U_* &= \max(b_*, b_*^2/h_0). \end{aligned}$$

All intervals and derivatives are taken on fixed closed gates with slightly larger regular domains.

Proposition A30.1 (Explicit twist and acceptance margins). *On this class every measured action satisfies*

$$u_* \leq |W_{st}| \leq U_*. \quad (\text{A30.1})$$

At a prefix match the positive Schur complement is at most D , and the magnitude of the matching derivative is at least u_^2/D . If the normalized preparation density is at least $\rho_- > 0$, retention is at least $e_- > 0$, the smallest clock residual is $r_- > 0$, and the smallest gate area is $q_- > 0$, then the retained-success probability of every attempted setting is at least*

$$p_* = e_- \rho_- u_* r_- q_- > 0. \quad (\text{A30.2})$$

Proof. An edge coupling has absolute value between b_0 and b_* by (A23.2). The interior Hessian is bounded below by $h_0 I$ by its second-variation formula. Its diagonal entries are at most D : the two projected kinetic terms contribute at most $2v_+^2/\ell_-$ and the curvature term at most $2\kappa_+v_+^2$. Parameter acceleration along the tangent cancels at an interior stationary reflection.

For a branch of N flights the corner-cofactor identity gives $|W_{st}| = \prod_{i=0}^{N-1} |b_i| / \det H_{N-1}$. Hadamard's inequality bounds the determinant by D^{N-1} . Since $b_0/D \leq 1$, the stated lower bound follows for $N \leq N_*$. For $N \geq 2$, the endpoint Schur formula and $\|H_{N-1}^{-1}\| \leq h_0^{-1}$ give the upper bound b_*^2/h_0 ; for $N = 1$ use b_* . The last-site Schur complement of the extended Hessian is positive and no larger than its original diagonal entry, hence at most D . Apply (A24.1). Finally the actual retained mass is the integral of $e\rho|W_{st}|(T-W)$ over the gate, by the physical flow-box calculation. Integrating the indicated lower bounds proves (A30.2). Here ρ is a normalized preparation density; an arbitrary unnormalized weight is not silently turned into a success probability. $\square$

Proposition A30.2 (An anchor from three corner tests). *Suppose $|W_{st}| \geq u > 0$ on a rectangle of side lengths d_s, d_t , $a_- \leq a \leq a_+$ there, and $0 < r_- \leq T_j - W \leq r_+$. Take z_0 to be one corner. Among the other three corners one has the three-clock anchor denominator bounded below by*

$$d_* := \frac{a_-}{a_+} \frac{ud_s d_t}{3} \frac{(T_2 - T_1)(T_3 - T_1)(T_3 - T_2)}{r_+^3} > 0. \quad (\text{A30.3})$$

The corner with largest observed denominator is therefore usable. The rectangle can lie strictly inside the original normalization gate.

Proof. The mixed derivative has constant sign on the rectangle. Its integral is the four-corner difference of W and has magnitude at least $ud_s d_t$. Writing this difference as a signed sum of

the other three corner values minus $W(z_0)$ shows that one has $|W(z) - W(z_0)| \geq ud_s d_t / 3$. The identity in the proof of Theorem A17.3 gives

$$\left| \sum_j c_j g_j(z) \right| = \frac{a(z)}{a(z_0)} |W(z) - W(z_0)| \frac{(T_2 - T_1)(T_3 - T_1)(T_3 - T_2)}{\prod_j (T_j - W(z_0))}.$$

The bounds prove the claim. Restricting functions without changing their original normalizations leaves all ratios unchanged. $\square$

A30.2. One finite design on a bounded neighborhood. Fix a reference table and one finite atlas constructed in Theorem A26.3. The acquisition list consists only of its branch words, scalar gates, and clocks. Clock anchors, root subbrackets, distance anchors and registration witnesses may be used in the proof of validity of the class, but are not provided afresh for an unknown member. The procedure uses the recovered actions and strict tests to find these witnesses.

Theorem A30.3 (Reference-uniform finite protocol). *For every such reference atlas there is a positive finite-smooth neighborhood on which its one fixed acquisition list reconstructs all tables. After imposing bounds for the finitely required derivatives of geometry and source and positive source and retention floors, the same list has fixed $M, N_*, u_*, p_*, d_* > 0$, fixed distance and overlap singular-value margins, and a uniform inverse error bound*

$$d_k(\mathcal{T}, \tilde{\mathcal{T}}) \leq C_k \max_{\gamma, j} \|f_{\gamma, j} - \tilde{f}_{\gamma, j}\|_{C^{k+2}} \quad (\text{A30.4})$$

for sufficiently close realizable law tuples. Here d_k is the maximum boundary C^k error and lattice error modulo one common Euclidean motion. This statement is uniform on the prescribed neighborhood, not on the union of all such neighborhoods.

Proof. The strict inequalities in a reference finite atlas are finitely many compact-domain clearance, incidence, residual, root and rank conditions. Finite-branch implicit dependence and continuity preserve them in a finite-smooth neighborhood. If $\eta_i > 0$ are their reference margins and L_i bound their variations per unit change of the embeddings on a larger bounded neighborhood, then any radius

$$0 < \delta < \min_i \eta_i / [2(1 + L_i)]$$

preserves half of every margin. Such finite L_i follow by implicit differentiation using the interior Hessian inverse, followed by the finite matrix and evaluation operations defining the certificates. They can be bounded recursively from finite derivative bounds and the Hessian lower bound. The reference certificate, rather than a universal numerical atlas bound, specifies M and N_* . The explicit triangular support-function family in Proposition A26.4 supplies nonempty neighborhoods of this kind with nonanalytic smooth members.

The preceding propositions now give uniform twist, success and clock anchor bounds. The fixed distance and overlap matrices retain their singular-value margins. Recover actions without differentiation; recover roots and distances with two derivatives of loss; recover boundary patches by the finite metric solve and trilateration; and register along a fixed finite spanning tree. Each operation has a uniform local bound on these sets. The root operation may be implemented on its uniquely matched domain without a supplied choice among roots. Composing the bounds gives (A30.4), including the lattice columns recovered by differences of translated intervals. This is a composition of explicit inverses, not an appeal to continuity of an unspecified compact inverse. $\square$

In particular the matching inverse retains the possible exponential factor D/u_*^2 in N_* . Formula (A30.2) charges the corresponding possible rarity of successes. Distance rank margins and

registration depth are also retained. The theorem neither removes grazing or near-degenerate finite certificates nor asserts nonidentifiability of conic arcs.

A30.3. From estimated laws to an actual table. For a whole-table protocol use the fixed scalar-circle label convention. Choose a fixed smooth partition of unity on each circle, subordinate to slightly smaller intervals of the finite cover. Each registered reconstructed patch is a smooth function on the larger interval. Blend the patches with this partition to obtain a closed curve for each quotient obstacle. Use the recovered two lattice vectors to generate all its translates. All registration operations use polar orthogonalization of the invertible maps determined by overlap triples, so are defined smoothly near the exact data even when noisy triples are not exactly congruent.

Theorem A30.4 (Constructive finite-record reconstruction). *Fix a bounded neighborhood as in Theorem A30.3, an integer $k \geq 2$, and $s \geq 1$. Suppose the conditional densities have bounded C^{k+2+s} norms on larger gates. Make B independent attempts at each of the M settings and charge every attempt. Retained coordinates may have arbitrary errors of magnitude at most δ . There is a reconstruction using kernel smoothing, explicit clock formulas, scalar root solving, finite linear algebra and patch blending such that, for sufficiently large Bp_* and small recording error,*

$$d_k(\widehat{\mathcal{T}}, \mathcal{T}) \leq C \left\{ \left(\frac{\log(C_0 MB/\zeta)}{Bp_*} \right)^{s/[2(k+2+s)+2]} + \delta^{s/(k+s+5)} \right\} \quad (\text{A30.5})$$

with probability at least $1 - 2\zeta$. On this event the output is a smooth strictly convex disjoint periodic table, not merely a formal list of inconsistent distance kernels. The total preparation budget is MB . For $k = 0, 1$ the same constructive conclusion follows from the $k = 2$ bound; no sharper low-order rate is inferred from this argument.

Proof. Set $m = k + 2$ and $n = \lfloor Bp_*/2 \rfloor$. A Chernoff bound and union bound show that every setting has at least n retained successes except on an event of probability at most $Me^{-Bp_*/8}$. The first n successes in each independent preparation sequence are coupled to n independent draws from the conditional law. This couples the capped experiment without assuming independence conditional on meeting its cap. For $Bp_* \geq C \log(M/\zeta)$, cap failure is at most ζ .

Use a smooth compactly supported order- s kernel on each larger gate. For each derivative of order at most m , Taylor expansion gives bias Ch^s ; the summand has envelope Ch^{-m-2} , variance Ch^{-2m-2} , and Lipschitz bound Ch^{-m-3} . Bernstein's inequality on a grid of mesh $n^{-2}h^{m+3}$, followed by a union bound over settings and derivatives, therefore gives, for $h \geq n^{-1}$,

$$e_n \leq C \left(h^s + \sqrt{\frac{L_n}{nh^{2m+2}}} + \frac{L_n}{nh^{m+2}} + n^{-2} + \frac{\delta}{h^{m+3}} \right), \quad L_n = \log(C_1 Mn/\zeta),$$

with probability $1 - \zeta$. The last term is the deterministic mean value bound for moving every kernel center by at most δ ; the recording errors need not be independent. The derivative bound is on inner gates, so no smooth zero extension at the boundary is assumed. This is the elementary argument underlying the retained density lemma, not a new optimal kernel rate; see also [8]. Choosing

$$h \asymp \max\{(L_n/n)^{1/[2(m+s)+2]}, \delta^{1/(m+s+3)}\}$$

makes all displayed terms bounded by the radius in (A30.5).

For this fixed-class procedure, retain a finite list of rational certificate candidates with strict margins on the reference neighborhood. Use only candidates passing half of those margins. This list is fixed once in the reconstruction rule; it is not a new mark supplied for an unknown member. It prevents an unrestricted certificate search from selecting an arbitrarily ill-conditioned matrix.

Apply the clock inverse on a usable corner chart. Small error preserves its denominator; choosing among the three tested corners does not require an unknown true anchor. On compact matched patches the true root is simple with a uniform margin. Small C^2 action error preserves the endpoint signs and a strictly decreasing neighborhood of that root. On the compact complement of a small root neighborhood the exact matching function is bounded away from zero; after shrinking the reference class this separation is uniform. Small perturbations therefore introduce no additional roots there. Bisection and implicit differentiation give the smooth continuation. The finite distance solves, positive matrix square roots, trilateration and overlap registration likewise extend to sufficiently small nonmodel perturbations. The resulting registered patches are within Ce_n in C^k of the true patches in one common gauge. Blending is a bounded linear operation in these fixed charts, and for exact patches it returns the true closed boundary. Thus its output has the same C^k error. Lattice recovery is a bounded difference of registered point evaluations.

For completeness, actual geometric validity is open in this norm. A compact smooth embedded reference curve has a tubular neighborhood; sufficiently small C^1 perturbations remain embedded and regular, and small C^2 perturbations preserve positive curvature. The lattice remains invertible. Disjointness of the finitely many nearby translated obstacles follows from their positive separation. For all remaining translates, a common diameter bound and a lower lattice singular value exclude intersection. These margins can be made uniform on a smaller reference neighborhood. Hence the blended output is an actual periodic table when e_n is small. In the explicit finite-horizon family the contained-disk margin also preserves finite horizon. Off the stated event the procedure may return a fixed reference table whenever a required inverse is undefined; no off-event accuracy is claimed.

The construction never enumerates candidate smooth tables or minimizes distance to an unspecified observation image. Root solving may be done to any certified tolerance, adding its propagated numerical tolerance to the displayed error. No polynomial arithmetic-time bound or minimax claim is made. Combining the cap and estimation events proves the probability assertion. $\square$

APPENDIX A31. RAW PREPARATION COSTS AND ROBUST GEOMETRIC OUTPUT

The preceding finite-record theorem counts every attempt at a setting. We now specify precisely what such an attempt contains when branch resolution is performed after launch. This also makes explicit how approximate cross-clock sharing enters the geometric error bound.

Definition A31.1 (Raw-launch branch-resolved acquisition). A setting is a prescribed source, clock, and post-launch classification rule. One attempt is one draw from its normalized raw phase source, followed by the trajectory and classification, whether or not it reaches the desired gate or branch. The apparatus either supplies the exact prescribed word and lift labels or returns failure. There are no false positive labels. On the desired branch the probability of successful classification is $c_\gamma(s, t) > 0$, independent of the initial free-time coordinate and common across the clocks, as is the subsequent retention efficiency $e_\gamma(s, t)$. Both factors can be unknown. The raw source density in the physical flow box is $\rho_\gamma^{\text{raw}}(s, p)$, independent of that free-time coordinate. It is the restriction of the normalized source for all raw launches, not a density renormalized after selecting the desired branch.

This is an apparatus hypothesis about available marks. Euclidean contact positions, tangents, chart speeds and lattice vectors remain unknown. The supplied scalar charts and their overlap maps are fixed labels, not geometric observations. A finite identifying atlas remains attached to a table and persistent on its neighborhood; no universal unmarked billiard acquisition theorem is implied by this convention.

Proposition A31.2 (Acceptance with classification fully charged). *Under Definition A31.1, the retained raw density is*

$$c_\gamma e_\gamma \rho_\gamma^{\text{raw}}(s, -W_{\gamma,s}) | W_{\gamma,st} | (T_j - W_\gamma).$$

Consequently, on the bounded reference-atlas classes of Theorem A30.3, a raw-success floor is

$$p_{\text{raw}} = c_- e_- \rho_-^{\text{raw}} u_* r_- q_- > 0. \quad (\text{A31.1})$$

If a physical preselection stage accepts a raw launch with probability q_{entry} and the density subsequently used in the model is a normalized conditional density ρ^{cond} , then $\rho^{\text{raw}} = q_{\text{entry}} \rho^{\text{cond}}$ on that component; its acceptance cost must be included. When the density is already raw, this factor must not be counted twice.

Proof. Apply the change of coordinates and free-time integration of Lemma A17.1 to the normalized raw source and multiply by the two conditional success probabilities. Integration on the endpoint gate gives the actual probability of a successful raw launch. Inserting all pointwise floors and the gate area gives (A31.1). The last assertion is the elementary factorization of the unconditional mass of a selected component into its selection probability and conditional probability density. $\square$

All branch hits that are missed, all rejected labels, and all late arrivals are failures already paid for in B . Unmodelled false positive labels would create branch mixtures; they are not corrected simply by adding clocks. The mixture identity after Proposition A20.3 explains why this distinction is mathematical, not only experimental terminology.

Theorem A31.3 (Robust finite-record billiard reconstruction). *Fix the bounded reference-atlas neighborhood, derivative bounds and positive certificate margins in Theorem A30.4, with $k \geq 2$ and $s \geq 1$. Suppose the true conditional laws on every branch have the form*

$$f_{\gamma,j} \propto a_\gamma (T_j - W_\gamma) e^{\eta_{\gamma,j}}, \quad \max_{\gamma,j} \|\eta_{\gamma,j}\|_{C^{k+2}} \leq \epsilon,$$

on the needed larger gates. Assume also the uniform C^{k+2+s} density bounds required for smoothing. Make B independent raw launches at every one of M settings, with a common actual success floor $p > 0$ for these possibly perturbed settings. Such a floor follows from (A31.1) if the corresponding raw factors retain the displayed floors. Allow coordinate errors of magnitude at most δ .

The explicit reconstruction has, for sufficiently small total error,

$$d_k(\hat{\mathcal{T}}, \mathcal{T}) \leq C \left\{ \left(\frac{\log(C_0 M B / \zeta)}{B p} \right)^{s/[2(k+2+s)+2]} + \delta^{s/(k+s+5)} + \epsilon \right\} \quad (\text{A31.2})$$

with probability at least $1 - 2\zeta$. On this event the output is a smooth strictly convex disjoint periodic table. Its raw preparation cost is MB . This is a sufficient geometric bound, not a claim that the geometric statistical exponent or derivative loss is optimal.

Proof. The cap and kernel argument in Theorem A30.4 applies to the actual conditional densities and the actual raw-success floor. With $m = k + 2$ it bounds the C^m estimation error by the first two terms in braces, on an event of probability at least $1 - 2\zeta$. The misspecification theorem gives an additional C^m action error $C\epsilon$ relative to the true physical actions. The finite corner choice remains possible because the true anchor margins are fixed; small errors preserve the tests on a finite candidate list.

The root, distance, positive square-root, trilateration and registration maps used in the geometric theorem are defined on an open neighborhood of the exact physical data. They need not map intermediate noisy data to exact stationary actions. Their stated local bounds require only closeness to the true actions and persistence of the denominators and root collars. Applying these bounds and the fixed partition-of-unity blending gives (A31.2). The open C^2 geometric

validity conditions proved there apply to the same blended output when the total radius, now including ϵ , is small enough. The cap uses raw launches, so the budget is exactly MB . $\square$

The residual test in Proposition A22.1 can be applied before this reconstruction. Its acceptance does not certify a billiard model or exact source sharing: action-confounded efficiencies can have zero residual. Conversely a detected residual need not destroy useful geometric accuracy when the independent misspecification bound is small. Validation and robust inversion are separate conclusions, both needed to interpret the experiment.

APPENDIX A32. PHYSICALLY REALIZABLE RETENTION NUISANCE

An abstract clock-dependent weight is not automatically a perturbation of a physical experiment. In this section the dynamics, source, retention stage and classifier are specified separately. The source is normalized at the raw-launch level. On a regular branch let $J_W(z) > 0$ denote the endpoint Jacobian times the raw source density and, when applicable, the known uniform-delay factor. Before classification the retained subdensity is

$$r_j(z) = e_j(z)J_W(z)(T_j - W(z)). \quad (\text{A32.1})$$

A retention stage means an actual Bernoulli decision, conditional on the endpoint record, with probability $e_j(z)$. It requires neither a measured momentum nor a measured trajectory length in the retained record. The statement that a particular function is an admissible retention probability is a description of the statistical class, not an instruction to an observer to compute an unknown action.

Proposition A32.1 (The physical retention tangent). *Fix a regular physical branch and its normalized raw source. Suppose the baseline retention probability satisfies $e_- \leq e(z) \leq 1 - e_-$ on a compact gate, with $0 < e_- < 1/2$. Every sufficiently small tuple $\xi_j \in C^m(Q)$ is physically realizable, with the dynamics and source unchanged, by*

$$e_j = e \exp(\xi_j). \quad (\text{A32.2})$$

The resulting log-weight perturbation in the clock model is exactly ξ_j . Consequently the tangent space of independently variable endpoint-retention probabilities is the full tuple $(C^m(Q))^J$. For conditional laws the corresponding scores are $\xi_j - \int_Q \xi_j f_j$. For raw laws the successful-density variation is $\dot{r}_j = r_j \xi_j$ and the failure-mass variation is $-\int_Q r_j \xi_j$. If retention is constrained to be shared by all clocks, only the common-function subspace is available.

Proof. Choose $\|\xi_j\|_\infty$ small enough that $e_- \exp(-\|\xi_j\|_\infty) > 0$ and $(1 - e_-) \exp(\|\xi_j\|_\infty) < 1$. Then (A32.2) is a legitimate Bernoulli probability. Substitution in (A32.1) proves the exact log-weight assertion. Differentiating the raw density and its complementary mass gives the raw variations. Differentiating its normalization gives the conditional score. Conversely every differentiable path of strictly interior positive retention probabilities has a log derivative of this form. The statement about a shared stage follows by imposing $e_1 = \dots = e_J$ before differentiation. $\square$

This proposition identifies a nuisance cone inside the physical apparatus class, rather than inferring physical realizability from an algebraic fit. It does not say that every laboratory has an arbitrary endpoint-dependent detector. When the apparatus restricts retention, its allowed tangent is correspondingly smaller. General robust contamination models address a different perturbation mechanism; see, for example, [19]. Here the disturbance is a smooth retention probability attached to actual trajectories.

Theorem A32.2 (Exact compensation along physical deformations). *Consider finitely many regular billiard branches, or regular return charts, with fixed gates and deadlines. Let a smooth one-parameter family of actual systems have actions $W_b^{(t)}$ and raw Jacobian-source factors $J_b^{(t)} > 0$ on those gates. Suppose the gates and time windows retain strict margins, these factors and actions depend smoothly on t in C^m , and the baseline efficiencies e_b lie strictly between zero and one. There are physically admissible clock-dependent retention probabilities*

$$e_{j,b}^{(t)}(z) = e_b(z) \frac{J_b^{(0)}(z) T_j - W_b^{(0)}(z)}{J_b^{(t)}(z) T_j - W_b^{(t)}(z)} \quad (\text{A32.3})$$

for all sufficiently small $|t|$. They satisfy $\|\log(e_{j,b}^{(t)}/e_b)\|_{C^m} \leq C|t|$ and make the full raw laws, not merely their conditional parts, exactly equal to their values at $t = 0$, at every specified clock. A fixed label channel, including false positive labels, preserves this equality.

Proof. All denominators have fixed positive lower bounds on the finite gates. The reciprocal and logarithm maps are smooth there. At $t = 0$ the multiplier in (A32.3) is one, so uniform continuity and the strict efficiency bounds keep it an admissible Bernoulli probability for small $|t|$. Differentiating on this compact family and applying the mean value theorem gives the C^m log bound. Direct substitution gives

$$e_{j,b}^{(t)} J_b^{(t)} (T_j - W_b^{(t)}) = e_b J_b^{(0)} (T_j - W_b^{(0)}).$$

Every successful joint measure is therefore unchanged. The total mass of their complement, which includes all missed launches, is unchanged as well. Applying a fixed stochastic channel to equal measures preserves equality. The argument also applies to audited calibration records which reveal the true branch only after this retention stage; it does not cover an extra sensor that supplies independent geometric measurements before retention. $\square$

Corollary A32.3 (A sharp physical efficiency floor for billiard geometry). *Fix a smooth periodic dispersing reference table with a finite identifying atlas and strict certificate margins. In a sufficiently small bounded neighborhood of this atlas, allow independent clock-dependent endpoint efficiencies with C^{k+2} log deviations at most ϵ , where $k \geq 2$. For every sufficiently small $\epsilon > 0$, there are two noncongruent physical tables in this neighborhood, at geometric distance at least $c\epsilon$, whose full raw record laws agree at all the finitely many prescribed settings. This remains true for any fixed calibrated label channels. Consequently any estimator from arbitrarily many such records has worst-case expected d_k error at least $c'\epsilon$ on this class. The linear dependence on ϵ in Theorem A33.3 is therefore sharp for this physical retention class. No optimality of its sampling exponent is asserted.*

Proof. Deform one obstacle on a small open boundary arc by outward normal displacement $t\varphi$, where $\varphi \geq 0$ is smooth, supported inside the arc and not identically zero. Keep the lattice and all other obstacles fixed. For sufficiently small $t \geq 0$, smooth strict convexity and disjointness persist. The obstacle expands, so a finite-horizon reference table acquires no new free corridor. All the finitely many clear-flight, nongrazing, chart, root and distance certificate margins persist by continuous dependence. Hence the same identifying atlas remains valid; no assertion about constructing a new universal protocol is used.

The first variation of the area of this obstacle is $\int \varphi ds > 0$. On the bounded C^1 family, area is locally Lipschitz in the boundary embeddings and invariant under a common Euclidean isometry. The geometric distance, which controls these embeddings modulo that isometry, therefore satisfies $d_k(\mathcal{T}_t, \mathcal{T}_0) \geq c_0 t$ for sufficiently small t . In particular the two tables are not congruent.

Choose a smooth family of normalized raw sources in the common scalar phase charts, positive on a larger compact neighborhood of all required initial states and constant in the free-time

coordinate on the required windows. Extend them outside these compact flow boxes and normalize before selection. The same source is used at every deadline of a branch; its normalization is included in $J_b^{(t)}$. The finite regular-action implicit equations give smooth C^{k+2} dependence of $W_b^{(t)}$ and $J_b^{(t)}$ on t . Choose strictly interior baseline retention probabilities. Theorem A32.2 supplies admissible probabilities with identical full raw laws and log deviations bounded by $C_0 t$. Taking $t = \epsilon/C_0$ proves the asserted separation after decreasing the permitted range of ϵ if necessary.

Under these two parameter values any estimator has the same output law, for any number of trials. The triangle inequality gives the sum of its two expected errors at least $d_k(\mathcal{T}_t, \mathcal{T}_0)$. The larger expected error is at least half that distance. This is a two-point physical lower bound, not an inference from an abstract weighted-function pair. $\square$

In the convex contact class an even simpler physical action example is $L_t = L_0 + t$: the return map is unchanged but its roof changes. Formula (A32.3) compensates the resulting return actions at the chosen counts. Thus absolute action has a physical linear efficiency floor there as well. This example does not replace the noncongruent billiard deformation in the preceding corollary.

APPENDIX A33. FINITE RAW EXPERIMENTS WITH AUDITED CALIBRATION

We now estimate the quantities used by the raw inverse. This section counts unsuccessful preparations and audited calibration trials. It does not grant an exact classifier on the scientific sample. Throughout, $Q_0 \in Q_+ \subset \mathbb{R}^d$ are fixed rectangles; all reconstruction gates lie inside Q_0 , and smoothing uses records on Q_+ . There are finitely many acquisition blocks and two deadlines per block; write J for the total number of settings.

Definition A33.1 (Audited raw calibration). At each setting collect N independent scientific raw trials and, independently, N_c calibration raw trials from the same acquisition mechanism. A calibration audit reveals the true component b of a record *after* the physical gate and retention stages, as well as its reported classifier label $a \in \{0, 1, \dots, A\}$. The label 0 is classifier rejection. Trials not passing the physical stages are also counted and return no component. Every component has probability at least $p_c > 0$ of being available to this audit. The only additional audit output used is this true component, not a flight time, position in Euclidean space, or other geometric measurement.

The audit is an explicit additional measurement capability. It can be replaced by a known channel matrix. Without an audit or another identifying calibration condition, a channel matrix is not treated as known merely because it appears in a model. Since the channel is independent of the endpoint given the true component, its column can be estimated from these audited trials without knowing the component's endpoint distribution.

Lemma A33.2 (Calibration and subdensity estimation). *Suppose the joint subdensities $g_{j,a}$ are bounded in $C^{m+s}(Q_+)$, where $m \geq 0$ and $s \geq 1$ are integers. Estimate C_j by the reported-label frequencies within each audited true component, including rejection in the denominator. Estimate the joint subdensities by*

$$\hat{g}_{j,a}(z) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{\text{reported label } a\} K_h(z - Z_i),$$

where a failed trial contributes zero. Here K is a smooth compactly supported kernel of order at least s , and recorded endpoints have errors at most δ . Put $\ell = \log(C_0 J(N + N_c + 2)/\zeta)$, with C_0 depending on the fixed dimensions. If $N_c p_c \geq C\ell$, then, with probability at least $1 - \zeta$,

simultaneously at all settings,

$$\max_j \|\widehat{C}_j - C_j\|_{\text{op}} \leq C \sqrt{\frac{\ell}{N_c p_c}}, \quad (\text{A33.1})$$

$$\max_{j,a} \|\widehat{g}_{j,a} - g_{j,a}\|_{C^m(Q_0)} \leq C \left\{ h^s + \sqrt{\frac{\ell}{N h^{2m+d}}} + \frac{\ell}{N h^{m+d}} + \delta h^{-(m+d+1)} \right\}. \quad (\text{A33.2})$$

The bandwidth is sufficiently small for the fixed collar and may be chosen polynomially in N as below. An arbitrarily small deterministic grid remainder is absorbed in the displayed bound.

Proof. The number of calibration trials having any fixed available true component is binomial with mean at least $N_c p_c$. Chernoff's inequality and a union bound show that all these denominators exceed $N_c p_c/2$, except on an event of probability at most $\zeta/4$. Conditional on the denominators and available components, reported labels have the corresponding channel-column law. Hoeffding's inequality and a union bound over the finitely many entries prove (A33.1), after conversion from entrywise to operator norm. The endpoint-independence assumption is used here.

For subdensity estimation no random success denominator is introduced. For $|\alpha| \leq m$, a derivative-kernel summand has envelope $Ch^{-d-|\alpha|}$, variance at most $Ch^{-d-2|\alpha|}$, and spatial Lipschitz bound $Ch^{-d-|\alpha|-1}$. Failures contribute zero and satisfy the same bounds. Taylor expansion of $D^\alpha g$ and the kernel moments give bias at most Ch^s . On a sufficiently fine polynomial grid in Q_0 , Bernstein's inequality and a union bound control all derivatives, settings and labels by the second and third terms of (A33.2). Interpolation between grid points uses the Lipschitz bound; the grid can be chosen to make this remainder smaller than N^{-2} , then absorbed. These steps are the usual uniform-kernel argument; compare [8]. Finally the mean value theorem for the derivative of K_h bounds the change caused by each coordinate error by $C\delta h^{-d-m-1}$. Averaging cannot increase this deterministic bound. Combining the exceptional events proves the lemma. $\square$

Theorem A33.3 (Calibrated finite-record geometric rigidity). *Fix a bounded neighborhood of a finite identifying billiard atlas, with the scalar labels, overlaps and all geometric margins specified in Theorem A30.4. Partition the possible reported records into finitely many acquisition blocks of the kind in Section A18. Across these blocks the catalogues include every prefix and extension branch required by the atlas. Within each block the catalogue is exhaustive, and its two fixed deadlines are admissible on the larger gates for every component. A prefix and its extension need not belong to the same block or share deadlines. The normalized raw source and baseline endpoint retention are shared by these two deadlines. Allow false positive word or lift labels through class-dependent matrices with $\sigma_{\min}(C_j) \geq \sigma > 0$. Endpoint labels are otherwise observed to coordinate accuracy δ .*

Set $m = k + 2$, with $k \geq 2$, and let $s \geq 1$. Suppose all component weights have a positive raw floor a_- , the relevant actions, weights and actual joint subdensities have the bounded regularity required above, in particular C^{m+s} for smoothing, and

$$r_{j,b} = a_b(T_j - W_b)e^{\eta_{j,b}}, \quad \max_{j,b} \|\eta_{j,b}\|_{C^m} \leq \epsilon.$$

Use the scientific and calibration budgets of Definition A33.1. For sufficiently small total error there is an explicit estimator satisfying, with probability at least $1 - \zeta$,

$$d_k(\widehat{\mathcal{T}}, \mathcal{T}) \leq C \left\{ \left(\frac{\ell}{N} \right)^{s/[2(m+s)+d]} + \delta^{s/(m+s+d+1)} + \sqrt{\frac{\ell}{N_c p_c}} + \epsilon \right\}, \quad d = 2. \quad (\text{A33.3})$$

Its output on this event is a smooth strictly convex disjoint periodic table. The number of raw preparations is exactly $J(N + N_c)$; every failed launch and every failed calibration attempt is

included. The constant depends on the fixed atlas, smoothness bounds, raw weight floor, clock gaps and σ . The calibration term and its cost are omitted when the channel is known.

This is recovery with imperfect labels in a calibrated reference-atlas experiment. It is not a table-independent unmarked protocol, an unsupervised estimate of the catalogue, or a minimax sampling-rate claim. The dependence on physical efficiency uncertainty is sharp by Corollary A32.3.

Proof. Choose

$$h = \max \left\{ (\ell/N)^{1/[2(m+s)+d]}, \delta^{1/(m+s+d+1)} \right\}.$$

For sufficiently small total error this bandwidth fits inside the gate collar. At this choice the square-root and linear Bernstein terms in (A33.2) are bounded by a constant times the first two terms on the right of (A33.3). Indeed they decrease as h increases, while h^s is bounded by the sum of those two terms; and the coordinate term is at most h^s . The lemma supplies simultaneous density and channel errors. Smallness of the calibration term ensures $\|\hat{C}_j - C_j\| \leq \sigma/2$.

Apply (A18.5) to the estimated *joint* subdensities, and then the two-row formula to each recovered component. Theorem A18.2 gives C^{k+2} action errors bounded by the braces in (A33.3). No exact positive label was used on a scientific trial. The error estimate compares the reconstruction with the true physical actions, not with a possibly misleading zero-residual fit.

Use the root and distance reconstruction of Proposition A23.2, the finite distance certificate, trilateration and overlap registration, and the fixed patch blending of Theorem A30.4. These maps are defined and locally Lipschitz on an open neighborhood of the true finite-atlas data; their intermediate inputs need not satisfy exact stationary identities. Their fixed margins persist under the small C^{k+2} action error. They give (A33.3). The same open C^2 conditions as in that theorem preserve strict convexity, disjointness and the lattice in the blended output. Defining a fixed reference output outside the admissible reconstruction neighborhood gives a measurable estimator everywhere. Finally the sample count follows from the raw-trial definitions, before any selection or rejection. $\square$

Corollary A33.4 (Finite records for the convex suspension). *In Theorem A28.2, impose bounded C^{k+4+s} generating functions and bounded C^{k+2+s} source and efficiency functions on the fixed larger domains, with positive floors. The same raw estimator, using $m = k + 2, d = 2$, has the bound (A33.3) with $d_k(\hat{\mathcal{T}}, \mathcal{T})$ replaced by $\|\hat{L} - L\|_{C^k([-R, R]^2)}$. Without label corruption it uses four settings and $4N$ raw trials. Calibration adds only the explicitly declared settings and raw trials.*

Proof. Proposition A28.1, with m replaced by $m + s$, supplies the actual subdensity regularity and positive weight floors. Apply the raw density lemma, the two-deadline inverse, and the uniform prefix bound in Theorem A28.2. When a calibrated channel is present, insert its left inverse as in the preceding theorem. No geometric sampling lower bound is invoked. $\square$

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